

MiNID

Miniature Programmable Network
Interface Device

Version 2.6



Your Network's Edge®

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For further information, contact RAD at the address below, or contact your **local business partner**.

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Limited Warranty

RAD warrants to DISTRIBUTOR that the hardware in the MiNID to be delivered hereunder shall be free of defects in material and workmanship under normal use and service for a period of twelve (12) months following the date of shipment to DISTRIBUTOR.

If, during the warranty period, any component part of the equipment becomes defective by reason of material or workmanship, and DISTRIBUTOR immediately notifies RAD of such defect, RAD shall have the option to choose the appropriate corrective action: a) supply a replacement part, or b) request return of equipment to its plant for repair, or c) perform necessary repair at the equipment's location. In the event that RAD requests the return of equipment, each party shall pay one-way shipping costs.

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This Agreement shall be construed and governed in accordance with the laws of the State of Israel.

Safety and Disposal (English)

General Safety Instructions

The following instructions serve as a general guide for the safe installation and operation of telecommunications products. Additional instructions, if applicable, are included inside the manual.

This equipment is not suitable for use in locations where children are likely to be present.

Safety Symbols



Warning

This symbol may appear on the equipment or in the text. It indicates potential safety hazards regarding product operation or maintenance to operator or service personnel.



Danger of electric shock! Avoid any contact with the marked surface while the product is energized or connected to outdoor telecommunication lines.



Protective ground: the marked lug or terminal should be connected to the building protective ground bus.



Some products may be equipped with a laser diode. In such cases, a label with the laser class and other warnings as applicable is attached near the optical transmitter. The laser warning symbol may be also attached.

Please observe the following precautions:

- ***Before turning on the equipment, make sure that the fiber-optic cable is intact and is connected to the transmitter.***
- ***Do not attempt to adjust the laser drive current.***
- ***Do not use broken or unterminated fiber-optic cables/connectors or look straight at the laser beam.***
- ***The use of optical devices with the equipment increases eye hazard.***
- ***Use of controls, adjustments, or performing procedures other than those specified herein may result in hazardous radiation exposure.***

ATTENTION: The laser beam may be invisible!

In some cases, the users may insert their own SFP laser transceivers into the product. Users are alerted that RAD cannot be held responsible for any damage that may result if non-compliant transceivers are used. In particular, users are warned to use only agency approved products that comply with the local laser safety regulations for Class 1 laser products.

Always observe standard safety precautions during installation, operation, and maintenance of this product. Only qualified and authorized service personnel should carry out adjustment, maintenance or repairs to this product. No installation, adjustment, maintenance, or repairs should be performed by either the operator or the user.

Handling Energized Products

General Safety Practices

Do not touch or tamper with the power supply when the power cord is connected. Line voltages may be present inside certain products even when the power switch (if installed) is in the OFF position or a fuse is blown. For DC-powered products, although the voltages levels are usually not hazardous, energy hazards may still exist.

Before working on equipment connected to power lines or telecommunication lines, remove jewelry or any other metallic object that may come into contact with energized parts.

Unless otherwise specified, all products are intended to be grounded during normal use. Grounding is provided by connecting the mains plug to a wall socket with a protective ground terminal. If a ground lug is provided on the product, it should be connected to the protective ground at all times, by a wire of

diameter 18 AWG or wider. Rack-mounted equipment should be mounted only in grounded racks and cabinets.

Always make the ground connection first and disconnect it last. Do not connect telecommunication cables to ungrounded equipment. Make sure that all other cables are disconnected before disconnecting the ground.

Some products may have panels secured by thumbscrews with a slotted head. These panels may cover hazardous circuits or parts, such as power supplies. These thumbscrews should therefore always be tightened securely with a screwdriver after both initial installation and subsequent access to the panels.

Connecting AC Mains

Make sure that the electrical installation complies with local codes.

Always connect the AC plug to a wall socket with a protective ground.

The maximum permissible current capability of the branch distribution circuit that supplies power to the product is 16A (20A for USA and Canada). The circuit breaker in the building installation should have high breaking capacity and must operate at short-circuit current exceeding 35A (40A for USA and Canada).

Always connect the power cord first to the equipment and then to the wall socket. If a power switch is provided in the equipment, set it to the OFF position. If the power cord cannot be readily disconnected in case of emergency, make sure that a readily accessible circuit breaker or emergency switch is installed in the building installation.

In cases when the power distribution system is IT type, the switch must disconnect both poles simultaneously.

Connecting DC Power

Unless otherwise specified in the manual, the DC input to the equipment is floating in reference to the ground. Any single pole can be externally grounded.

Due to the high current capability of DC power systems, care should be taken when connecting the DC supply to avoid short-circuits and fire hazards.

Make sure that the DC power supply is electrically isolated from any AC source and that the installation complies with the local codes.

The maximum permissible current capability of the branch distribution circuit that supplies power to the product is 16A (20A for USA and Canada). The circuit breaker in the building installation should have

high breaking capacity and must operate at short-circuit current exceeding 35A (40A for USA and Canada).

Before connecting the DC supply wires, ensure that power is removed from the DC circuit. Locate the circuit breaker of the panel board that services the equipment and switch it to the OFF position. When connecting the DC supply wires, first connect the ground wire to the corresponding terminal, then the positive pole, and last the negative pole. Switch the circuit breaker back to the ON position.

A readily accessible disconnect device that is suitably rated and approved should be incorporated in the building installation.

If the DC power supply is floating, the switch must disconnect both poles simultaneously.

Connecting Data and Telecommunication Cables

Data and telecommunication interfaces are classified according to their safety status.

The following table lists the status of several standard interfaces. If the status of a given port differs from the standard one, a notice is given in the manual.

Ports	Safety Status
V.11, V.28, V.35, V.36, RS-530, X.21, 10BASE-T, 100BASE-T, 1000BASE-T, Unbalanced E1, E2, E3, STM, DS-2, DS-3, S-Interface ISDN, Analog voice E&M xDSL (without feeding voltage), Balanced E1, T1, Sub E1/T1, POE Input DC Voltage up to 60 VDC	ES1 Electrical energy source class 1 Ports which do not present a safety hazard. Usually up to 30 VAC or 60 VDC.
FXS, FXO Input DC Voltage up to 72 VDC	ES2 Electrical energy source class 2
AC power source declared	ES3 Electrical energy source class 3

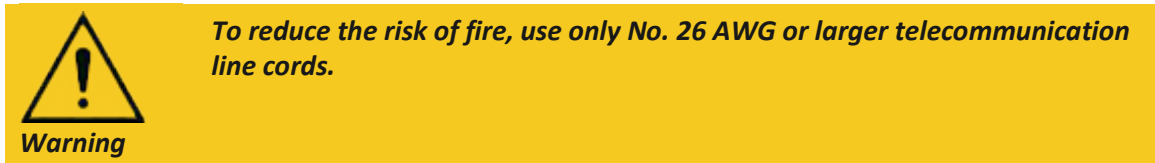
Always connect a given port to a port of the same safety status. If in doubt, seek the assistance of a qualified safety engineer.

Always make sure that the equipment is grounded before connecting telecommunication cables. Do not disconnect the ground connection before disconnecting all telecommunication cables.

Some SELV and non-SELV circuits use the same connectors. Use caution when connecting cables. Extra caution should be exercised during thunderstorms.

When using shielded or coaxial cables, verify that there is a good ground connection at both ends. The grounding and bonding of the ground connections should comply with the local codes.

The telecommunication wiring in the building may be damaged or present a fire hazard in case of contact between exposed external wires and the AC power lines. In order to reduce the risk, there are restrictions on the diameter of wires in the telecom cables, between the equipment and the mating connectors.



Some ports are suitable for connection to intra-building or non-exposed wiring or cabling only. In such cases, a notice is given in the installation instructions.

Do not attempt to tamper with any carrier-provided equipment or connection hardware.

Electromagnetic Compatibility (EMC)

The equipment is designed and approved to comply with the electromagnetic regulations of major regulatory bodies. The following instructions may enhance the performance of the equipment and provide better protection against excessive emission and better immunity against disturbances.

A good ground connection is essential. When installing the equipment in a rack, make sure to remove all traces of paint from the mounting points. Use suitable lock-washers and torque. If an external grounding lug is provided, connect it to the ground bus using braided wire as short as possible.

The equipment is designed to comply with EMC requirements when connecting it with unshielded twisted pair (UTP) cables with the exception of 1000BaseT ports that must always use shielded twisted pair cables of good quality (CAT 5E or higher). However, the use of shielded wires is always recommended, especially for high-rate data. In some cases, when unshielded wires are used, ferrite cores should be installed on certain cables. In such cases, special instructions are provided in the manual.

Disconnect all wires which are not in permanent use, such as cables used for one-time configuration.

The compliance of the equipment with the regulations for conducted emission on the data lines is dependent on the cable quality. The emission is tested for UTP with 80 dB longitudinal conversion loss (LCL).

Unless otherwise specified or described in the manual, TNV-1 and TNV-3 ports provide secondary protection against surges on the data lines. Primary protectors should be provided in the building installation.

The equipment is designed to provide adequate protection against electrostatic discharge (ESD). However, it is good working practice to use caution when connecting cables terminated with plastic connectors (without a grounded metal hood, such as flat cables) to sensitive data lines. Before connecting such cables, discharge yourself by touching ground or wear an ESD preventive wrist strap.

FCC-15 User Information

This equipment has been tested and found to comply with the limits of the Class A digital device, pursuant to Part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the Installation and Operation Manual, may cause harmful interference to the radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at his own expense.

Canadian Emission Requirements

This Class A digital apparatus meets all the requirements of the Canadian Interference-Causing Equipment Regulations.

Warning per EN 55032 (CISPR 32)



Warning

This equipment is compliant with Class A of CISPR 32. In a residential environment, this equipment may cause radio interference.

Product Disposal



To facilitate the reuse, recycling and other forms of recovery of waste equipment in protecting the environment, the owner of this RAD product is required to refrain from disposing of this product as unsorted municipal waste at the end of its life cycle. Upon termination of the unit's use, customers should provide for its collection for reuse, recycling, or other form of environmentally conscientious disposal.

Sécurité et élimination (français)

Instructions générales de sécurité

Les instructions suivantes servent de guide général d'installation et d'opération sécurisées des produits de télécommunications. Des instructions supplémentaires sont éventuellement indiquées dans le manuel.

Cet équipement ne convient pas pour une utilisation dans des endroits où des enfants sont susceptibles d'être présents.

Symboles de sécurité



Avertissement

Ce symbole peut apparaître sur l'équipement ou dans le texte. Il indique des risques potentiels de sécurité pour l'opérateur ou le personnel de service, quant à l'utilisation du produit ou à sa maintenance.



Danger de choc électrique ! Evitez tout contact avec la surface marquée tant que le produit est sous tension ou connecté à des lignes externes de télécommunications.



Mise à la terre de protection : la cosse ou la borne marquée devrait être connectée à la prise de terre de protection du bâtiment.



Certains produits peuvent être équipés d'une diode laser. Dans de tels cas, une étiquette indiquant la classe laser (ainsi que d'autres avertissements le cas échéant) sera jointe près du transmetteur optique. Le symbole d'avertissement laser peut aussi être joint.

Veuillez observer les précautions suivantes :

- ***Avant la mise en marche de l'équipement, assurez-vous que le câble de fibre optique est intact et qu'il est connecté au transmetteur.***
- ***Ne tentez pas d'ajuster le courant de la commande laser.***
- ***N'utilisez pas des câbles ou connecteurs de fibre optique cassés ou sans terminaison et n'observez pas directement un rayon laser.***
- ***L'usage de périphériques optiques avec l'équipement augmentera le risque pour les yeux.***
- ***L'usage de contrôles, ajustages ou procédures autres que celles spécifiées ici pourrait résulter en une dangereuse exposition aux radiations.***

ATTENTION : Le rayon laser peut être invisible !

Les utilisateurs pourront, dans certains cas, insérer leurs propres émetteurs-récepteurs Laser SFP dans le produit. Les utilisateurs sont avertis que RAD ne pourra pas être tenue responsable de tout dommage pouvant résulter de l'utilisation d'émetteurs-récepteurs non conformes. Plus particulièrement, les utilisateurs sont avertis de n'utiliser que des produits approuvés par l'agence et conformes à la réglementation locale de sécurité laser pour les produits laser de classe 1.

Respectez toujours les précautions standards de sécurité durant l'installation, l'opération et la maintenance de ce produit. Seul le personnel de service qualifié et autorisé devrait effectuer l'ajustage, la maintenance ou les réparations de ce produit. Aucune opération d'installation, d'ajustage, de maintenance ou de réparation ne devrait être effectuée par l'opérateur ou l'utilisateur.

Manipuler des produits sous tension

Règles générales de sécurité

Ne pas toucher ou altérer l'alimentation en courant lorsque le câble d'alimentation est branché. Des tensions de lignes peuvent être présentes dans certains produits, même lorsque le commutateur (s'il est installé) est en position OFF ou si le fusible est rompu. Pour les produits alimentés par CC, les niveaux de tension ne sont généralement pas dangereux mais des risques de courant peuvent toujours exister.

Avant de travailler sur un équipement connecté aux lignes de tension ou de télécommunications, retirez vos bijoux ou tout autre objet métallique pouvant venir en contact avec les pièces sous tension.

Sauf s'il en est autrement indiqué, tous les produits sont destinés à être mis à la terre durant l'usage normal. La mise à la terre est fournie par la connexion de la fiche principale à une prise murale équipée d'une borne protectrice de mise à la terre. Si une cosse de mise à la terre est fournie avec le produit, elle devrait être connectée à tout moment à une mise à la terre de protection par un conducteur de diamètre 18 AWG ou plus. L'équipement monté en châssis ne devrait être monté que sur des châssis et dans des armoires mises à la terre.

Branchez toujours la mise à la terre en premier et débranchez-la en dernier. Ne branchez pas des câbles de télécommunications à un équipement qui n'est pas mis à la terre. Assurez-vous que tous les autres câbles sont débranchés avant de déconnecter la mise à la terre.

Certains produits peuvent avoir des panneaux sécurisés par des vis papillons avec une tête fendue. Ces panneaux peuvent couvrir des circuits ou des composants dangereux, tels que les alimentations électriques. Ces vis papillons devront par conséquent être solidement serrées avec un tournevis après chaque installation initiale et chaque accès ultérieur aux panneaux.

Connexion au courant du secteur

Assurez-vous que l'installation électrique est conforme à la réglementation locale.

Branchez toujours la fiche de secteur à une prise murale équipée d'une borne protectrice de mise à la terre.

La capacité maximale permise en courant du circuit de distribution de la connexion alimentant le produit est de 16A (20A aux États-Unis et Canada). Le coupe-circuit dans l'installation du bâtiment devrait avoir une capacité élevée de rupture et devrait fonctionner sur courant de court-circuit dépassant 35A (40A aux États-Unis et Canada).

Branchez toujours le câble d'alimentation en premier à l'équipement puis à la prise murale. Si un commutateur est fourni avec l'équipement, fixez-le en position OFF. Si le câble d'alimentation ne peut pas être facilement débranché en cas d'urgence, assurez-vous qu'un coupe-circuit ou un disjoncteur d'urgence facilement accessible est installé dans le bâtiment.

Le disjoncteur devrait déconnecter simultanément les deux pôles si le système de distribution de courant est de type IT.

Connexion d'alimentation CC

Sauf s'il en est autrement spécifié dans le manuel, l'entrée CC de l'équipement est flottante par rapport à la mise à la terre. Tout pôle doit être mis à la terre en externe.

A cause de la capacité de courant des systèmes à alimentation CC, des précautions devraient être prises lors de la connexion de l'alimentation CC pour éviter des courts-circuits et des risques d'incendie.

Assurez-vous que l'alimentation CC est isolée de toute source de courant CA (secteur) et que l'installation est conforme à la réglementation locale.

La capacité maximale permmissible en courant du circuit de distribution de la connexion alimentant le produit est de 16A (20A aux Etats-Unis et Canada). Le coupe-circuit dans l'installation du bâtiment devrait avoir une capacité élevée de rupture et devrait fonctionner sur courant de court-circuit dépassant 35A (40A aux Etats-Unis et Canada).

Avant la connexion des câbles d'alimentation en courant CC, assurez-vous que le circuit CC n'est pas sous tension. Localisez le coupe-circuit dans le tableau desservant l'équipement et fixez-le en position OFF. Lors de la connexion de câbles d'alimentation CC, connectez d'abord le conducteur de mise à la terre à la borne correspondante, puis le pôle positif et en dernier, le pôle négatif. Remettez le coupe-circuit en position ON.

Un disjoncteur facilement accessible, adapté et approuvé devrait être intégré à l'installation du bâtiment.

Le disjoncteur devrait déconnecter simultanément les deux pôles si l'alimentation en courant CC est flottante.

Connexion de câbles de données et de télécommunications

Les interfaces de données et de télécommunications sont classées selon leur niveau de sécurité.

Le tableau suivant liste les statuts de plusieurs interfaces standards. Si le statut d'un port donné diffère d'un standard, une notification sera fournie dans le manuel.

Ports	Niveau de sécurité
V.11, V.28, V.35, V.36, RS-530, X.21, 10BASE-T, 100BASE-T, 1000BASE-T, asymétrique E1, E2, E3, STM, DS-2, DS-3, S-Interface ISDN (RNIS), Voix analogique E&M xDSL (sans tension d'alimentation), symétrique E1, T1, Sub E1/T1, POE Tension d'entrée DC jusqu'à 60 VDC	ES1 Source d'énergie électrique de classe 1 Ports qui ne présentent pas un danger pour la sécurité. Généralement jusqu'à 30 VAC (courant alternatif) ou 60 VDC (courant continu).
FXS, FXO Tension d'entrée DC jusqu'à 72 VDC	ES2 Source d'énergie électrique de classe 2
Source d'énergie CA déclarée	ES3 Source d'énergie électrique de classe 3

Toujours connecter un port donné à un port de même niveau de sécurité. En cas de doute, solliciter l'assistance d'un ingénieur de sécurité qualifié.

Toujours s'assurer que l'équipement est relié à la terre avant de connecter des câbles de télécommunications. Ne pas déconnecter la connexion à la terre avant la déconnexion de tous les câbles de télécommunications.

Certains circuits SELV et non-SELV utilisent les memes connecteurs. Soyez prudents lors de la connexion des câbles. Une extrême prudence est requise en cas d'orages.

En cas d'utilisation de câbles blindés ou coaxiaux, vérifier qu'il y a bien une connexion à la terre aux deux extrémités. Le raccordement à la terre et la liaison à la prise de terre doivent être conformes à la réglementation locale.

Il se peut que le câblage de télécommunications dans le bâtiment soit endommagé ou présente un danger d'incendie en cas de contact entre des câbles externes dénudés et les lignes électriques AC (courant alternatif). Afin de réduire le risque, il y a une limitation du diamètre des fils dans les câbles de télécommunications, entre l'équipement et les connecteurs homologues.



Avertissement

Pour réduire les risques d'incendie, utiliser seulement des cordons de télécommunications 26 AWG ou de section supérieure.

Certains ports sont uniquement adaptés à une connexion à un câblage interne ou à un câblage non exposé. Dans ce cas, une notification sera fournie dans les instructions d'installation.

Ne pas tenter de démonter l'équipement ou le matériel de connexion.

Compatibilité Electromagnétique (CEM)

L'équipement est conçu et approuvé pour se conformer aux réglementations électromagnétiques des principaux organismes de réglementation. Les instructions suivantes peuvent améliorer les performances de l'équipement et fournir une meilleure protection contre les émissions excessives et une meilleure immunité contre les perturbations.

Une bonne connexion à la terre est essentielle. Lors de l'installation de l'équipement dans un rack, veillez à éliminer toute trace de peinture des points de montage. Utilisez des rondelles de blocage et un couple appropriés. Si une cosse de mise à la terre externe est fournie, connectez-la au bus de terre à l'aide d'un fil tressé aussi court que possible.

L'équipement est conçu pour répondre aux exigences CEM lors de la connexion avec des câbles à paires torsadées non blindées (UTP), à l'exception des ports 1000BaseT, qui doivent toujours utiliser des câbles à paires torsadées blindés de bonne qualité (CAT 5E ou supérieure). Cependant, l'utilisation de câbles blindés est toujours recommandée, en particulier pour les données à haut débit. Dans certains cas, lorsque des câbles non blindés sont utilisés, des noyaux en ferrite doivent être installés sur certains câbles. Dans ce cas, des instructions spéciales sont fournies dans le manuel.

Débranchez tous les câbles qui ne sont pas utilisés de manière permanente, tels que les câbles utilisés pour une configuration unique.

La conformité de l'équipement à la réglementation en matière d'émission conduite sur les lignes de données dépend de la qualité du câble. L'émission est testée pour des câbles UTP avec un affaiblissement de conversion longitudinale (LCL) de 80 dB.

Sauf indication contraire ou décrite dans le manuel, les ports TNV-1 et TNV-3 offrent une protection secondaire contre les surtensions sur les lignes de données. Des protections primaires doivent être fournies dans l'installation du bâtiment.

L'équipement est conçu pour fournir une protection adéquate contre les décharges électrostatiques (DES). Toutefois, il est recommandé de faire preuve de prudence lors du raccordement de câbles munis de connecteurs en plastique (sans capot métallique mis à la terre, tels que des câbles plats) sur des lignes de données sensibles. Avant de connecter ces câbles, déchargez-vous en touchant le sol ou portez un bracelet antistatique.

FCC-15 Information Utilisateur

Cet équipement a été testé et déclaré conforme aux limites d'un appareil numérique de classe A, définies à la section 15 du règlement de la FCC. Ces limites sont conçues pour fournir une protection raisonnable contre les interférences nuisibles lorsque l'équipement est utilisé dans un environnement commercial. Cet équipement génère, utilise et peut émettre de l'énergie de fréquence radio, s'il n'est pas installé et utilisé conformément au Manuel d'Installation et d'Utilisation, il peut provoquer des interférences nuisibles aux communications radio. L'utilisation de cet équipement dans une zone résidentielle est susceptible de provoquer des interférences nuisibles, dans ce cas, l'utilisateur sera tenu de corriger les interférences à ses frais.

Exigences d'émissions canadiennes

Cet appareil numérique de Classe A répond à toutes les exigences de la réglementation canadienne sur les équipements causant des interférences.

Avertissement: EN 55032 (CISPR 32)



Avertissement

Cet appareil est conforme à la Classe A de la CISPR 32. Dans un environnement résidentiel, il peut provoquer des interférences radio.

Élimination du produit



Afin de faciliter la réutilisation, le recyclage ainsi que d'autres formes de récupération d'équipement mis au rebut dans le cadre de la protection de l'environnement, il est demandé au propriétaire de ce produit RAD de ne pas mettre ce dernier au rebut en tant que déchet municipal non trié, une fois que le produit est arrivé en fin de cycle de vie. Le client devrait proposer des solutions de réutilisation, de recyclage ou toute autre forme de mise au rebut de cette unité dans un esprit de protection de l'environnement, lorsqu'il aura fini de l'utiliser.

Sicherheit und Entsorgung (Deutsch)

Allgemeine Sicherheitsanleitung

Die folgenden Anleitungen dienen als allgemeiner Leitfaden für die sichere Installation und Bedienung von Telekommunikationsprodukten. Zusätzliche Anleitungen sind im Nutzerhandbuch vorhanden.

Dieses Gerät ist nicht für die Verwendung an Orten geeignet, an denen sich Kinder aufhalten können.

Sicherheitssymbole



Achtung

Dieses Symbol kann auf ihren Geräten oder im Text auftauchen. Es weist den Nutzer oder das Servicepersonal auf mögliche Gefahren bei der Bedienung der Geräte hin.



Gefahr eines elektrischen Schlages! Vermeiden Sie jeglichen Kontakt mit der gekennzeichneten Oberfläche während das Gerät unter Spannung steht oder an außenliegende Telekommunikationsleitungen angeschlossen ist.



Schutzerdung: Die gekennzeichnete Mutter oder das Terminal müssen an den Anschluss der Haupterdung des Gebäudes angeschlossen sein.



Einige Produkte können mit einer Laserdiode ausgestattet sein. In solchen Fällen muß ein Aufkleber mit der Laserklasse und entsprechenden Warnungen neben dem optischen Transmitter angebracht sein. Das Warnsymbol für Laser kann zusätzlich angebracht sein.

Bitte beachten Sie die folgenden Vorsichtsmaßnahmen:

- ***Vor der Inbetriebnahme des Gerätes, vergewissern Sie sich, daß das optische Glasfaserkabel unbeschädigt ist und an den Transmitter angeschlossen ist.***
- ***Versuchen Sie nicht, den durch den Laser fließenden Strom zu regulieren.***
- ***Verwenden Sie keine gebrochenen oder anderweitig unvollständige Glasfaserkabel oder Stecker. Blicken Sie nicht in den Laserstrahl.***
- ***Die Benutzung optischer Komponenten zusammen mit Ihrem Gerät erhöhen die Gefahr für Ihre Augen.***
- ***Die Benutzung von Bedienelementen, die Geräteeinstellung oder die Ausführung von Prozessen, die hier nicht aufgeführt sind, können zu gefährlicher Strahlung führen.***

ACHTUNG: Der Laserstrahl kann unsichtbar sein!

In einigen Fällen werden Nutzer eigene SFP-Lasertransceiver in das Gerät einführen. Nutzer sind darauf hingewiesen, dass RAD nicht verantwortlich zeichnet für Beschädigungen, die von nicht kompatiblen Transceivern herrühren. Nutzer seien ferner darauf hingewiesen, daß ausschließlich amtlich zugelassene Produkte eingesetzt werden sollten, die den ortsüblichen Sicherheitsbestimmungen für Lasergeräte der Laserklasse 1 entsprechen.

Beachten Sie ferner die üblichen Sicherheitsmaßnahmen während der Installation, des Betriebs, der Wartung oder der Reparatur des Gerätes. Installationen, Einstellungen und Reparaturen sollten weder vom Nutzer oder dem zuständigen Operator durchgeführt werden.

Umgang mit Geräten unter Spannung

Grundlegende Sicherheitsmaßnahmen

Berühren oder verändern Sie das Netzteil nicht wenn das Stromkabel angeschlossen ist. Einige Bauteile im Gerät können auch dann unter Spannung stehen, wenn der Ein/Aus-Schalter auf Aus steht (sofern vorhanden) oder eine Sicherung defekt ist. Für Produkte, die unter Gleichstromspannung (DC) stehen, besteht ebenfalls die Gefahr eines elektrischen Schlages, auch wenn die angelegte Spannung in der Regel nicht gefährlich ist.

Legen Sie Schmuck oder sonstige Metallobjekte ab, bevor Sie mit Geräten arbeiten, die an das Netz oder Telekommunikationsleitungen angeschlossen sind, um zu verhindern, daß dies mit spannungsgeladenen Bauteilen in Berührung kommen.

Falls nicht anders angegeben, sollten alle Produkte bei normalem Gebrauch geerdet werden. Die Erdung erfolgt durch den Anschlss an eine Steckdose mit Schutzerdung. Wenn das Gerät mit einer Erdungslasche ausgestattet ist, sollte diese immer an die Schutzerde angeschlossen sein mit einem Kabel, das einen Durchmesser von mindestens 18 AWG aufweist. Geräte für die Rack-Montage sollten ausschließlich in geerdeten Racks oder Schränken montiert werden.

Schließen Sie grundsätzlich zuerst die Schutzerde an und klemmen Sie diese zuletzt ab. Schließen Sie keine Telekommunikationskabel an nicht geerdete Geräte an. Stellen Sie sicher, dass alle anderen Kabel abgeklemmt sind, bevor Sie die Erdung abklemmen.

Die Frontpaneele einiger Geräte sind mit Flügelschrauben mit Schlitz gesichert. Diese Paneele decken gefährliche Schalkreise oder Teile, wie zum Beispiel Netzteile ab. Diese Flügelschrauben sollten daher immer mittels eines Schraubenziehers sicher angezogen werden nach der Erstinstallation und jedem späterem Zugriff auf die Paneele.

Anschluss an eine Wechselstromquelle (AC)

Stellen Sie sicher, daß die elektrische Installation den örtlichen Bestimmungen entspricht.

Stecken Sie den Stecker immer in eine Steckdose mit Schutzerdung ein.

Der maximal mögliche Stromfluss im Bereich des Verteilerstromkreis, der die Stromversorgung des Gerätes sicherstellt, ist 16 A (20A in den USA und in Kanada). Der Schutzschalter in der Gebäudeinstallation muss starke Ströme unterbrechen können und muss den Stromfluss bei 35A (40A in den USA und Kanada) unterbrechen.

Schließen Sie das Netzkabel zuerst an das Gerät und dann an die Steckdose an. Falls ein Ein/Aus-Schalter zur Verfügung steht, schalten Sie diesen auf AUS (OFF). Falls das Netzkabel im Notfall nicht schnell herausgezogen werden kann, stellen Sie sicher, daß ein Schutzschalter oder Notschalter Bestandteil der elektrischen Installation des Gebäudes ist.

Falls die Stromversorgung über einen IT Netz-Verteiler erfolgt, muss der Schalter die Stromversorgung zu beiden Polen gleichzeitig unterbrechen.

Anschluss an eine Gleichstromquelle (DC)

Falls im Benutzerhandbuch (Manual) nicht anderweitig beschrieben, schwankt die Gleichstromzufuhr relativ zur Erdung. Jeder einzelne Pol kann von aussen geerdet werden.

Aufgrund der Fähigkeit, hohe Stromflüsse zu verarbeiten, muss sorgfältig vorgegangen werden beim Anschluss der Gleichstromquelle, um Kurzschlüsse und Brände zu vermeiden.

Stellen Sie sicher, daß Gleichstromquellen (DC) von Wechselstromquellen (AC) isoliert sind und daß die Installation den örtlichen Richtlinien entspricht.

Der maximal mögliche Stromfluss im Bereich des Verteilerstromkreis, der die Stromversorgung des Gerätes sicherstellt, ist 16 A (20A in den USA und in Kanada). Der Schutzschalter in der Gebäudeinstallation muss starke Ströme unterbrechen können und muss den Stromfluss bei 35A (40A in den USA und Kanada) unterbrechen.

Vor dem Anschluss der Gleichstrom-Speisekabel ist sicher zu stellen, daß kein Strom über den Gleichstromkreis fließt. Finden Sie den Schutzschalter an der Schalttafel, die das Gerät bedient, und schalten Sie ihn auf AUS (OFF). Wenn Sie die Gleichstromdrähte anschließen, schliessen Sie zuerst den Erdungsdraht an das zugehörige Terminal an, dann den Pluspol und zuletzt den Minuspol. Schalten Sie den Schutzschalter zurück auf AN (ON).

Ein verfügbares nicht angeschlossenes Gerät, das ordnungsgemäß genehmigt und abgenommen wurde, sollte in die bestehende Installation eingebaut werden.

Falls die Gleichstromspannung schwankt, muss der Schalter beide Pole gleichzeitig trennen.

Anschluss von Daten- und Telekommunikationskabeln

Daten- und Telekommunikationsschnittstellen sind gemäß ihrem Sicherheitsstatus klassifiziert.

Verschiedene Standardschnittstellen sind zusammen mit ihrem jeweiligen Sicherheitsstatus in der folgenden Tabelle aufgeführt. Auf eventuelle Abweichungen vom Standardsicherheitsstatus wird im Benutzerhandbuch (Manual) gesondert hingewiesen.

Schnittstellen	Sicherheitsstatus
V.11, V.28, V.35, V.36, RS-530, X.21, 10BASE-T, 100BASE-T, 1000BASE-T, Unsymmetrisches E1, E2, E3, STM, DS-2, DS-3, S-Schnittstelle ISDN, Analogsprache E&M xDSL (ohne einspeisende Spannung), symmetrisches E1, T1, Sub-E1/T1, POE Eingangs-Gleichspannung bis zu 60 VDC	ES1 Elektrische Energiequelle Klasse 1 Anschlüsse, die kein Sicherheitsrisiko darstellen, normalerweise bis zu 30 VAC oder 60 VDC.
FXS, FXO Eingangs-Gleichspannung bis zu 72 VDC	ES2 Elektrische Energiequelle Klasse 2

Schnittstellen	Sicherheitsstatus
AC-Spannungsquelle deklariert	ES3 Elektrische Energiequelle Klasse 3

Verbinden Sie Anschlüsse, die denselben Sicherstatus aufweisen. Wenn Sie nicht sicher sind, wenden Sie sich bitte an einen qualifizierten Sicherheitsingenieur.

Vergewissern Sie sich immer, daß das Gerät geerdet ist bevor Sie Telekommunikationskabel anschließen. Klemmen Sie die Erdung nie ab, bevor Sie Telekommunikationskabel abklemmen.

Einige SELV und Nicht-SELV-Stromkreise nutzen dieselben Stecker. Seien Sie vorsichtig, wenn Sie Kabel anschließen. Seien Sie besonders vorsichtig während einem Gewitter.

Wenn Sie abgeschirmte -, oder Koaxialkabel nutzen, stellen Sie sicher, daß diese an beiden Enden eine gute Erdung aufweisen.

Wenn außenliegende Kabel und Wechselstromleitungen (AC) in Kontakt kommen, kann die Verkabelung innerhalb des Gebäudes beschädigt werden oder einen Brand auslösen. Um dieses Risiko zu verringern, gibt es Bestimmungen zum Durchmesser von Telekommunikationskabeln zwischen den Geräten und den Anschlüssen.



Achtung

Um das Brandrisiko zu reduzieren, setzen Sie ausschließlich 26 AWG oder dickere Telekommunikationskabel ein.

Einige Anschlüsse eignen sich lediglich für Verbindungen zu gebäude-internen oder nicht außenliegenden Verkabelungen. Auf solche Fälle wird in der Installationsanleitung gesondert hingewiesen.

Versuchen Sie nicht, die vom Carrier erhaltene Ausrüstung oder Verbindungselemente zu manipulieren.

Elektromagnetische Kompatibilität (EMC)

Die Ausrüstung ist ausgelegt und anerkannt für die Erfüllung elektromagnetischer Bestimmungen der Regulierungsbehörden. Die nachfolgenden Anleitungen sind darauf ausgerichtet, die Leistungsfähigkeit der Ausrüstung zu erhöhen und besseren Schutz gegen extreme Emissionen und besseren Schutz gegen Störungen zu gewährleisten.

Eine gute Erdung ist wesentlich. Wenn die Ausrüstung in einem Rack montiert wird, stellen Sie sicher, daß jegliche Farbspuren von den Befestigungspunkten entfernt sind. Benutzen Sie geeignete Sicherungsscheiben und das richtige Drehmoment. Falls eine externe Erdungsmutter zur Verfügung steht, schließen Sie diese an den Erdbus an mittels kürzestmöglichem verdrehten Draht.

Die Ausrüstung ist ausgelegt, um den Anforderungen der EMC zu entsprechen, wenn man sie mit nicht abgeschirmten und verdrehten (UTP) Kabeln anschließt mit Ausnahme von 1000BaseT-Anschlüssen, die grundsätzlich mit abgeschirmten verdrehten Kabeln hoher Qualität (CAT 5E oder besser) erfordern. Im Allgemeinen ist die Verwendung von abgeschirmten Kabeln immer empfohlen, besonders für schnellen Datendurchsatz. Beim Einsatz nicht abgeschirmter Kabel wird in manchen Fällen empfohlen, einen Ferritkern an bestimmten Kabeln anzubringen. In diesen Fällen werden im Benutzerhandbuch gesonderte Anleitungen bereitgestellt.

Klemmen Sie alle Kabel ab, die nicht permanent in Gebrauch sind, wie zum Beispiel solche, die fuer eine einmalige Konfiguration eingesetzt wurden.

Die Einhaltung der Regeln für elektromagnetische Leitungsemissionen an den Datenleitungen hängt von der Kabelqualität ab. Die Emission wurde für UDP mit 80 db Längsumwandlungsdämpfung (LCL) getestet.

Falls im Benutzerhandbuch nicht anders spezifiziert oder beschrieben, bieten TNV-1 und TNV-3 Anschlüsse lediglich sekundären Schutz gegen Überspannungen in den Datenleitungen. Primäre Protektoren müssen innerhalb der Gebäudeinstallation bereitgestellt werden.

Die Ausrüstung ist ausgelegt, ausreichenden Schutz gegen elektrostatische Entladung (ESD) zu bieten. Es ist jedoch empfehlenswert, vorsichtig zu agieren, wenn Kabel mit Plastikanschlüssen (ohne geerdete Metallhalterung wie bei flachen Kabeln) und empfindliche Datenleitungen angeschlossen werden. Vor dem Anschliessen solcher Kabel, entladen Sie sich selbst durch Berührung des Bodens oder durch das Tragen eines ESD-präventiven Bandes um das Handgelenk.

FCC-15 Informationen für Nutzer

Diese Ausrüstung wurde getestet und bewegt sich innerhalb der Grenzwerte für Class A-Digitalgeräte gemäß Artikel 15 der FCC-Regeln. Diese Grenzwerte wurden festgelegt, um angemessenen Schutz gegen schädliche Einflüsse sicherzustellen wenn die Geräte in einer kommerziellen Umgebung betrieben werden. Diese Geräte produzieren, konsumieren und strahlen möglicherweise Energie im Radiofrequenzbereich ab, die schädliche Auswirkungen auf den Funkverkehr haben kann, falls sie nicht gemäß dem Benutzerhandbuch (Installation and Operation Manual) installiert wurden. Es ist wahrscheinlich, daß der Betrieb dieser Geräte in einem Wohngebiet zu Störungen führt, die der Betreiber auf eigene Kosten zu beseitigen hat.

Kanadische Emissionsbestimmungen

Dieses digitale Gerät der Klasse A erfüllt alle Vorgaben der Kanadischen Regulierungen für Geräte, die Störeffekte haben können (Canadian Interference-Causing Equipment Regulation).

EN 55032 (CISPR 32) Warnung



Achtung

Das vorliegende Gerät fällt unter die Funkstörgrenzwertklasse A. In Wohngebieten können beim Betrieb dieses Gerätes Rundfunkstörungen auftreten, für deren Behebung der Benutzer verantwortlich ist.

Entsorgung des Produktes



Um die Wiedernutzung, die Wiederverwertung oder andere Formen der Wiederaufbereitung von stillgelegten Geräten zum Schutz der Umwelt zu gewährleisten, ist der Besitzer des RAD-Produktes verpflichtet, die Entsorgung als unsortierter Abfall am Ende des Lebenszyklus des Produktes zu unterlassen. Wenn das Gerät ausser Betrieb genommen wird, hat der Kunde dieses Gerät einer umweltverträglichen Wiederverwendung, Wiederverwertung oder Entsorgung zuzuführen.

EU Declarations of Conformity

MiNID Declaration of Conformity

SFP-CA.2 Declaration of Conformity

Environmental Compliance Statement

Environmental Compliance Statement

Quick Start Guide

This section describes the minimum configuration needed to prepare MiNID for operation.

Installing the Unit

Installing an SFP in MiNID SFP Sleeve

Insert the required SFP into MiNID SFP sleeve (refer to [Installing SFPs into the SFP Sleeve](#)).

Inserting MiNID SFP Sleeve into a Host Device

Insert MiNID SFP sleeve into SFP-CA.2 or into a host device that is powered on and connected to a network.

Configuring MiNID

Initial Configuration for Management

MiNID SFP sleeve can be configured via the SFP-CA.2 adaptor (provided that your PC has Windows XP SP2), or via a switch.

Configuring via SFP-CA.2

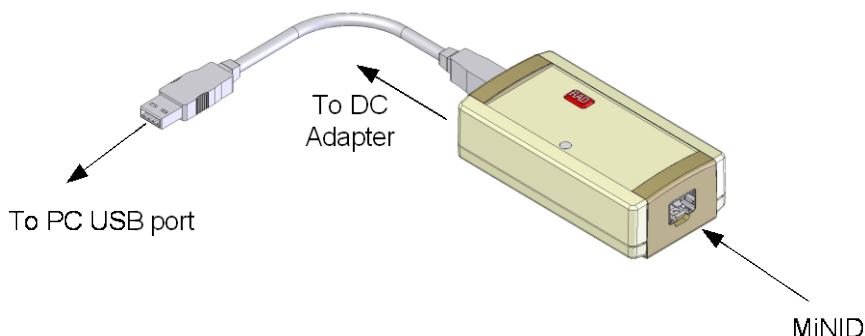
➤ To configure MiNID SFP sleeve via SFP-CA.2:

1. Configure the PC parameters (refer to Configuration via SFP-CA.2 in Chapter 4).
2. Disconnect MiNID from any device (SFP-CA.2 or host).
3. Insert any compatible SFP into MiNID.
4. Set MiNID to Configuration mode, using its DIP switches (refer to [Setting the DIP Switches](#)).

Note While MiNID is in Configuration mode, it responds to the default IP address 192.168.205.1, with no host VLAN, even if a different IP address was configured.

5. Insert MiNID into the SFP socket of the SFP-CA.2 module.
6. Connect SFP-CA.2 to your PC via USB 2.0 port.

The Ready LED on SFP-CA.2 turns on.



SFP-CA.2 Configuration Module

Note If no fiber is connected to the MSA port, ensure that all fault propagation actions are disabled (this is the default setting).

Configuring via Switch

A similar procedure to configuring via SFP-CA.2 can be used to configure MiNID when it is inserted into the SFP port of a switch, as long as connectivity is established to an available Ethernet port.

► To configure MiNID via a switch:

1. Disconnect MiNID from any device (SFP-CA.2 or host).
2. Insert any compatible SFP into MiNID.
3. Set MiNID to Configuration mode, using its DIP switches (refer to [Setting the DIP Switches](#)).

Note While MiNID is in Configuration mode, it responds to the default IP address 192.168.205.1, with no host VLAN, even if a different IP address was configured.

4. Insert MiNID into an available SFP socket of the switch.

5. Connect the available switch port to which connectivity has been established, to your PC, using an IP address in the subnet 192.168.205.XX.

Verifying Connectivity

Ping the MiNID IP address (192.168.205.1).

Preconfigure for Ethernet Service

Using either your browser or Telnet, connect to 192.168.205.1 and log in with your user name (**su** for full configuration and monitoring access) and your password.

Proceed with service configuration (refer to [Traffic Processing](#) for details of different scenarios for provisioning supported services).

Contents

Installing an SFP in MiNID SFP Sleeve	27
Inserting MiNID SFP Sleeve into a Host Device.....	27
Initial Configuration for Management	27
Verifying Connectivity.....	29
Preconfigure for Ethernet Service	29
1 Introduction	39
1.1 Overview.....	39
Product Options	40
Application	41
Features	41
1.2 New in This Version	46
1.3 Physical Description.....	46
MiNID SFP Sleeve	46
MiNID SFP	47
1.4 Functional Description.....	47
1.5 Technical Specifications.....	48
2 Installation and Setup	50
2.1 Safety	50
Laser Safety	51
2.2 Site Requirements and Prerequisites	51
2.3 Package Contents	51
2.4 Required Equipment.....	51
2.5 Setting the DIP Switches.....	52
MiNID SFP Sleeve and SFP	52
2.6 Mounting	52
Installing SFPs into the SFP Sleeve	53
Inserting MiNID SFP Sleeve into a Host Unit.....	54
Inserting MiNID SFP into a Host Unit	56
Inserting MiNID into an SFP-CA.2	57
2.7 Basic Connectivity Test.....	58

3	Operation	59
3.1	Turning On the Unit.....	59
3.2	Indicators	59
3.3	Startup	59
	Configuration and Software Files.....	59
	Loading Sequence	60
3.4	Working with Custom Configuration Files.....	60
	On-net Zero Touch Provisioning	61
	Saving Configuration Changes	62
3.5	Configuration and Management	62
3.6	CLI-Based Configuration	63
	Working with Telnet/SSH.....	63
	Login.....	66
	Using the CLI	66
	Using Scripts.....	72
3.7	Web-Based Configuration	72
	Logging in	72
	Using the Web Interface	74
	Menu Map	77
3.8	SNMP-Based Network Management.....	85
	Working with RADview	85
	Working with Other SNMP-Based NMSs	86
3.9	Configuration of MiNID SFP via SFP-CA.2 (Telnet/SSH/Web)	86
	Preconfiguration for SFP-CA.2	87
	Connecting SFP-CA.2.....	87
3.10	Configuration via Host (Telnet/SSH and CLI/Web Interface/RADview/SNMP)	88
3.11	Turning Off the Unit	88
4	Service Provisioning	89
4.1	Ethernet Service	89
4.2	Service Summary	89
	Benefits	89
	Functional Description	89
	Viewing Service Summary.....	89
5	Ports.....	92

5.1 Physical Ethernet Ports.....	92
Applicability and Scaling	92
Standards Compliance	93
Benefits	93
Factory Defaults	93
Configuring Port Parameters	93
Viewing Port Status.....	95
Viewing Port Statistics	96
5.2 SFP and DDM	98
Applicability and Scaling	99
Benefits	99
Displaying SFP and DDM Details Using the Web Interface	99
Displaying SFP and DDM Details Using the CLI	100
6 Management and Security	101
6.1 Access Control List.....	101
Applicability and Scaling	101
Standards Compliance	101
Benefits	101
Functional Description	101
Factory Defaults	102
Configuring ACL Using the Web Interface	102
Configuring ACL Using the CLI	103
6.2 Authentication via TACACS+ Server	104
Applicability and Scaling	105
Standards Compliance	105
Benefits	105
Functional Description	105
Factory Defaults	106
Configuring Authentication via TACACS+ Server	107
Examples	113
Viewing TACACS+ Statistics.....	114
6.3 Management Access Control.....	117
Applicability and Scaling	117
Functional Description	117
Factory Defaults	118
Configuring Management Access	118
6.4 Network Managers	123
Factory Defaults	123

Configuring Network Managers.....	123
6.5 Router Interface	124
Functional Description	124
Factory Defaults	125
Configuring Router Interface	125
Configuring Router Interface for Management	127
Configuring Static Route	130
Configuring Default Gateway.....	131
Viewing the Configured Router Interface Parameters	132
Viewing Router Interface Status	132
6.6 Session Timeout	134
Factory Default.....	134
Configuring the Session Timeout	134
6.7 SNMP Communities.....	134
6.8 User Access.....	135
7 Resiliency (Fault Propagation)	138
7.1 Standards Compliance.....	138
7.2 Functional Description.....	138
7.3 Factory Defaults.....	139
7.4 Configuring Fault Propagation.....	139
8 Traffic Processing.....	142
8.1 Flows.....	142
Applicability and Scaling	142
Standards Compliance	142
Benefits	142
Functional Description	143
Factory Defaults	148
Flow Configuration Tasks	151
Configuring Port Mode	157
Configuring Flows Using the Web Interface	160
Configuring Flows Using the CLI.....	178
Configuring LBM Reflector.....	185
Configuring L2CP/L2PT.....	187
Examples	194
Configuration Errors.....	196
Viewing Flow Summary.....	197

Viewing Flow Statistics.....	198
Testing Flow Configuration	204
8.2 Quality of Service (QoS).....	205
Applicability and Scaling	205
Standards Compliance	205
Benefits	205
CoS Map Profiles	205
Envelope Bandwidth Profiles	207
Policer Profiles	212
9 Timing and Synchronization	217
9.1 Network Time Protocol (NTP).....	217
Factory Defaults	217
Configuring NTP	217
9.2 Sync-E Clock Source	224
Applicability and Scaling	224
Standards Compliance	224
Benefits	224
Functional Description	224
Factory Defaults	224
Configuring the Sync-E Clock Source	225
10 Administration.....	226
10.1 Device Information	226
Factory Defaults	226
Configuring Device Information.....	226
10.2 File Operations	227
SFTP Application	227
TFTP Application	228
Copying Files	228
Deleting Files.....	232
Displaying File Information	234
10.3 Inventory	235
Standards Compliance	235
Benefits	235
Displaying Inventory Information	235
10.4 Reset	237
Resetting to Factory Defaults	237

Resetting to User Defaults	237
Restarting the Unit.....	238
11 Monitoring and Diagnostics.....	239
11.1 Layer-3 Service Activation Test	239
Applicability and Scaling	239
Standards Compliance	239
Benefits	239
Functional Description	239
Factory Defaults	241
Configuring L3 SAT Entities	242
Viewing L3 SAT Status	250
Viewing L3 SAT Test Reports.....	251
11.2 Microburst Monitoring	253
Applicability and Scaling	254
Benefits	254
Functional Description	254
Configuring Microburst Monitoring.....	254
Viewing Flow Statistics.....	256
11.3 OAM (CFM)	256
Applicability and Scaling	257
Standards Compliance	257
Benefits	257
Functional Description	257
Factory Defaults	264
Configuring OAM (CFM/PM).....	264
Examples	298
Viewing OAM Status	304
Viewing OAM Statistics	309
11.4 OAM (EFM)	321
Applicability and Scaling	321
Standards Compliance	321
Benefits	321
Functional Description	321
Factory Defaults	322
Configuring Link OAM	322
Examples	324
11.5 OAM (TWAMP)	325
Applicability and Scaling	325

Standards Compliance	326
Benefits	326
Functional Description	326
Factory Defaults	328
Configuring TWAMP.....	329
Viewing TWAMP Status	338
Viewing TWAMP Reports.....	339
11.6 Packet Capture	343
Applicability and Scaling	343
Standards Compliance	344
Benefits	344
Functional Description	344
Factory Defaults	345
Configuring Packet Capture	345
Viewing Statistics for Packet Capture	356
11.7 Ping	357
Applicability and Scaling	357
Benefits	357
Factory Defaults	357
Configuring Ping Test	358
Example.....	359
11.8 RFC-2544 Responder	360
Applicability and Scaling	360
Standards Compliance	360
Benefits	360
Functional Description	360
Configuring RFC-2544 Responder	360
11.9 Syslog.....	361
Applicability and Scaling	361
Standards Compliance	361
Benefits	361
Functional Description	361
Factory Defaults	363
Configuring Syslog Parameters	363
Viewing Syslog Statistics	366
Example.....	367
Configuration Errors.....	368
11.10 Detecting Problems	368
Alarms and Traps	368
Statistic Counters	368

11.11 Handling Alarms and Events.....	369
Defining Alarm Types	369
Working with the Event Log.....	370
Alarm, Event and Trap Lists	372
11.12 Performance Monitoring.....	372
Applicability and Scaling	372
Benefits	372
Functional Description	373
Factory Defaults	373
Configuring Performance Monitoring.....	373
11.13 Troubleshooting	374
11.14 Frequently Asked Questions.....	375
11.15 Technical Support.....	375
 A Connection Data	377
A.1 Ethernet Connector.....	377
 B Test Plan.....	378
B.1 Introduction.....	378
B.2 Required Equipment	378
B.3 Preparing the Test Layout	379
Connecting the Test Layout	380
Configuring Devices	381
B.4 Testing Basic Functionality	382
Estimated Duration	382
Setup and Configuration	382
Test Procedure	382
B.5 Testing Flows	383
Estimated Duration	383
Setup and Configuration	383
Test Procedure	383
B.6 Testing OAM (CFM)	386
Estimated Duration	386
Setup and Configuration	387
Test Procedure	387
B.7 Testing OAM (EFM)	387
Estimated Duration	387

Setup and Configuration	388
Test Procedure	388
B.8 Testing Service Loopbacks.....	389
Estimated Duration	389
Setup and Configuration	390
Test Procedure	390
B.9 Testing RFC-2544.....	392
Estimated Duration	392
Setup and Configuration	392
Test Procedure	392
B.10 Testing Sync-E.....	394
Estimated Duration	394
Setup and Configuration	394
Test Procedure	394
B.11 Testing Device Reset	395
Estimated Duration	395
Setup and Configuration	395
Test Procedure	396

1 Introduction

1.1 Overview

MiNID is a miniature programmable network interface device offering MEF 2.0 certified Carrier Ethernet demarcation functionalities. With very low delay, based on cut-through forwarding technology, MiNID offers Hierarchical Quality of Service (HQOS) and SLA verification functionalities as well as fault detection and fault isolation tools.

MiNID can be ordered as a pluggable smart SFP or as an SFP sleeve, hosting standard SFPs.

It can be easily installed at customer sites, mobile base stations, and wholesale interconnection sites in order to establish and monitor business services, mobile backhauling, and wholesale applications. It can be owned and operated by service providers, wholesale providers, and even end customers. Supporting seamless integration with standard SFPs and its “Zero Touch” provisioning capabilities, MiNID offers a real plug-and-play solution, reducing capital expenditure (Capex), operational expenditure (Opex), and total cost of ownership (TCO).

MiNID with Demarcation software load includes comprehensive Ethernet OAM (Operation, Administration, and Maintenance) functionality and SLA monitoring. MiNID SFP sleeve transfers Synchronous Ethernet transparently to ensure highly accurate traffic delivery in packet-based mobile backhaul networks.

MiNID with Platinum software load contains all demarcation capabilities with extra monitoring features.

MiNID can be configured to process customer and network traffic consistently for all incoming traffic (all-in-one bundling), or to classify and map traffic by VLAN ID, VLAN priority, DSCP, MAC source or destination address.

MiNID SFP sleeve form factor is a patent protected design that can be inserted into a standard MSA-compliant SFP port (FE/GbE) on a customer device. On the carrier side, MiNID provides a single SFP port into which standard MSA-compliant SFPs can be inserted.

MiNID can either be managed independently with its own IP address or use a loaned IP mechanism that does not require IP address and uses a dedicated UDP port for management.

MiNID can be managed through the following interfaces:

- Web-based, menu-driven interface
- Telnet/SSH with CLI, enabling configuration duplication in multiple units
- SNMP
- RADview-EMS network management system support, including PM collection of flow utilization counters

Product Options

Two versions of MiNID are available:

- FE (100 Mbps) – fits into FE SFP ports and support 100BASE-FX SFPs
- GbE (1 Gbps) - fits into FE or GE ports and supports 100BASE-FX as well as 1000BASE-FX SFPs

MiNID is available as:

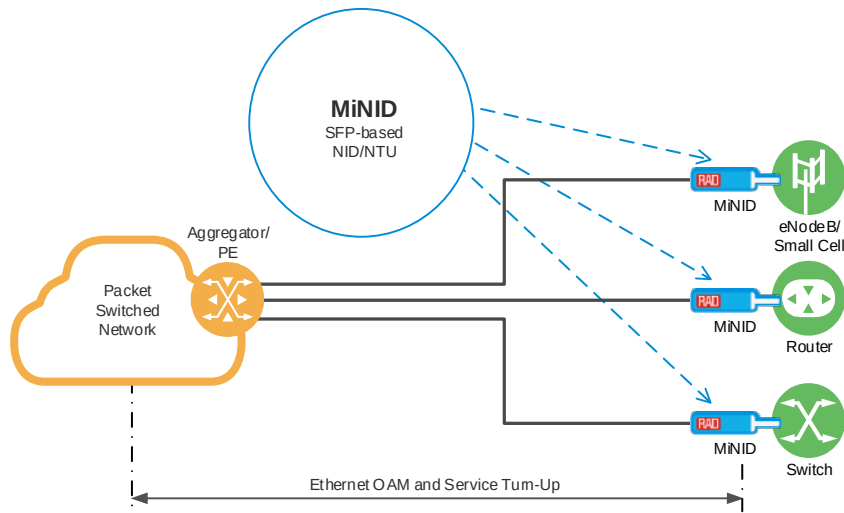
1. SFP sleeve form with the following options:
 - FE (100 Mbps)
 - GbE (1 Gbps)
2. SFP with the following options:
 - 850 nm multimode
 - 1310 nm single mode
3. MiNID can be ordered with Sync-E capability enabled

In all MiNID ordering options, port rates on both ports must be the same (FE or GE). When autonegotiation is enabled, MiNID automatically negotiates through its two ports to achieve the same rate on both ports.

Nonetheless, when the Force Speed option is set to 1000 Mbps in SFP sleeve, the rates on two ports can be different. In such cases, shapers should be used to compensate for the rates difference. For more information on the Force Speed option, refer to [Configuring Port Parameters](#).

Application

MiNID SFP sleeve eliminates the need for standalone demarcation devices, while allowing service providers, mobile operators, and wholesale carriers to receive real-time network/service performance reports with per-CoS SLA definition.



MiNID SFP Sleeve as a Demarcation Device

Features

MiNID includes the following features:

Ethernet

Service Demarcation

MiNID can work in port or flow-based mode. In port-based mode (all-in-one bundling), MiNID processes customer and network traffic data frames consistently for all traffic coming into its single port in each direction. In flow-based mode, MiNID can classify and map frames by any of the following:

- Customer VLAN ID range and P-bit range
- DSCP
- MAC source or destination address
- Ethertype

In flow-based mode, up to 64 unidirectional flows can be configured (no more than 32 flows per direction). You can configure a service name for flows and MAs, to enable network management systems to locate entities related to specific services. For each configured flow, and for default, non-matched (unclassified) traffic, any of the following actions can be defined:

- Drop – Drop the frame
- VLAN – Push (add) for single and double VLANs, pop (remove) for single and double VLANs, or replace a C-tag (Customer VLAN tag) or S-tag (Service VLAN tag), as specified.
- Loopback – The traffic is echoed back through the same port, with or without swapping the source and destination addresses (MAC, and where relevant also IP), as specified.
- Pass through – Pass the traffic through the device without applying any changes.

Priority Mapping

When a flow is configured to push or replace a VLAN, the frame priority is set in the new VLAN priority bits (p-bits). The value of the p-bits can be configured to be set in any of the following ways, per flow:

- Fixed value for the flow
- Copied from the originating frame C-tag p-bit
- According to a configured mapping of the originating frame C-tag p-bit value to p-bit values. All flows configured for C-tag p-bit to S-tag p-bit mapping (not copying) use the same mapping method
- According to a configured mapping of DSCP values to p-bit values. All flows configured for DSCP-p-bit mapping use the same mapping method.

L2CP Handling with L2PT

MiNID can handle Layer-2 control protocol traffic (L2CP) with Layer-2 protocol tunneling (L2PT), on a per-port and/or per-flow basis. L2CP classification is based on destination MAC Address, and L2PT classification is based on destination MAC address and Ethertype. MiNID can pass through, tunnel, discard, or peer (trap to host for protocol processing) L2CP/L2PT packets.

Monitoring and Diagnostics

OAM

MiNID uses its hardware-based processing capabilities to quickly perform OAM and PM measurements with maximum precision. This makes MiNID an effective tool for network failure detection,

troubleshooting, and measuring SLA parameters such as Frame Delay, Frame Delay Variance (jitter), Frame Loss and Availability.

MiNID supports the following OAM standards with MEP/MIP support, for up to 32 MEPs:

- CFM OAM (end-to-end OAM) as per IEEE 802.1ag (draft 8), per-EVC: Continuity check, non-intrusive loopback, and link trace. Maximal rate is 10 frames per second. Maintenance associations can process frames with one or two VLAN tags (user-configurable). MEP per EVC.COS that enables the user to configure MEPs (each one with a unique VLAN + P-Bit) under same MA. In addition, incoming packets are filtered according to the P-Bit configured.
- ETH OAM as per ITU-T Y.1731 (PM), with AIS and RDI support, per EVC: Loss measurement (LMM/LMR, SLM/SLR), delay measurement (DMM/DMR; with hardware-based time stamping), delay variance measurement (hardware timestamping), and availability.
- EFM OAM (link OAM) as per IEEE 802.3-2005 (formerly IEEE 802.3ah) in active or passive mode, for remote management and fault indication, including remote loopback, and dying gasp.

Note

Link OAM is operational only if the L2CP/L2PT action for OAM protocol is set to peer.

Loopback Tests

MiNID supports the following types of loopbacks:

- Intrusive flow loopback (the whole flow is echoed), with or without MAC and IP address swap
- In-service layer-2 loopbacks according to MAC address (source and/or destination), with or without MAC and IP address swap
- In-service layer-3 loopbacks according to destination IP address with MAC and IP address swap
- In-service layer-4 loopbacks according to destination IP address and destination UDP port with MAC and IP address swap
- UDP echo responder
- RFC-2544 responder
- IP-agnostic loops

Resiliency

Fault Propagation

MiNID SFP sleeve provides provider-to-customer fault propagation, in order to enable deactivating affected interfaces when required. You can also configure the direction (MSA to SFP, or SFP to MSA) in which to propagate OAM CFM faults (LOC, Rx AIS).

Dying Gasp

Upon power failure, MiNID finishes transmitting the currently-handled frame, and then sends the following:

- 10 Dying Gasp frames as defined in 802.3ah
- 10 Dying Gasp SNMPv2 traps to up to two configured SNMP servers.

The dying gasp process stops when power is restored.

DDM (Digital Diagnostic Monitoring)

When equipped with an SFP transceiver that support DDM, the MiNID stores DDM statistics, enabling the following:

- Forwarding the parameters to the host device
- Reading the parameters from remote management systems without accessing the MiNID host device.

Management

MiNID can be managed using dedicated IP address or in a Loaned IP mode where an IP address is not required. In case of dedicated IP, the IP address can either be static or obtained from a DHCP server. In loaned IP mode, MiNID uses the IP address of the host device. In this mode, network management systems can communicate with MiNID by sending management packets to the host IP address with a pre-configured UDP port. MiNID responds to the packets or forwards them to the host device, depending on the packet TCP/UDP port number and the access configuration in MiNID.

Note

When MiNID is working in loaned IP mode and is managed via RADview, then in order for automatic discovery to operate properly, there must be no more than one active IP address behind MiNID. Otherwise, MiNID responds to every active IP address behind it, causing incorrect device duplication.

MiNID provides the following management functionality:

- CLI-based management via Telnet / secure Telnet (SSH)
- Menu-based management via Web interface (HTTP)
- SNMP
- Management interfaces for third-party NMS:
 - Configuration via CLI
 - Alarms via SNMPv1/SNMPv2
 - PM collection by retrieving PM file via SFTP/TFTP
- Integration into RADview-EMS for configuration and alarms, as well as support for RADview PM portal
- Zero touch configuration – DHCP-based acquisition of IP address and configuration file

Note *Configuration via SNMP does not cover all MiNID functionalities.*

Four simultaneous Telnet or Web connections can be maintained at any given time.

Access Control List (ACL)

Access control lists are used to flexibly filter by IP address and mark incoming and outgoing network traffic.

The service providers use the ACL to maintain the network security by preventing the malicious traffic from entering the device. The ACL can be used to save the network resources by dropping the unwanted packets. Ten rules can be configured for ACL.

Timing and Synchronization

Transparent Sync-E

MiNID supports Synchronous Ethernet (Sync-E) transparently, transferring the received clock and any ESSM messages from the MSA/Port 2 to the SFP/Port 1 port or from the SFP/Port 1 to the MSA/Port 2 port (user-configurable). This feature is supported by SFP sleeve.

Note *ESSM frames correspond to the OAM protocol, which is peered by default in L2CP/L2PT handling; in this case, ESSM packets are discarded. If OAM protocol action is set to pass through, ESSM packets are delivered but link OAM is not operational.*

MiNID supports the following G.8262 requirements:

- Jitter and wander generation
- Jitter tolerance

1.2 New in This Version

Version 2.6 includes the following new features:

- TACACS+ Authentication
- Dedicated IP addresses for Packet Capture, TWAMP controller, and Layer-3 SAT, when MiNID is managed with loaned IP
- Serial Number display in the inventory
- Free-run timestamping for Packet Capture
- Read-only mode setting for SNMP

1.3 Physical Description

MiNID SFP Sleeve

This figure shows a MiNID with the SFP socket (referred to as the SFP port in this manual) in the lower left corner, and the customer-side SFP connector (referred to as the MSA port in this manual) in the upper right corner.



SFP Sleeve

MiNID SFP

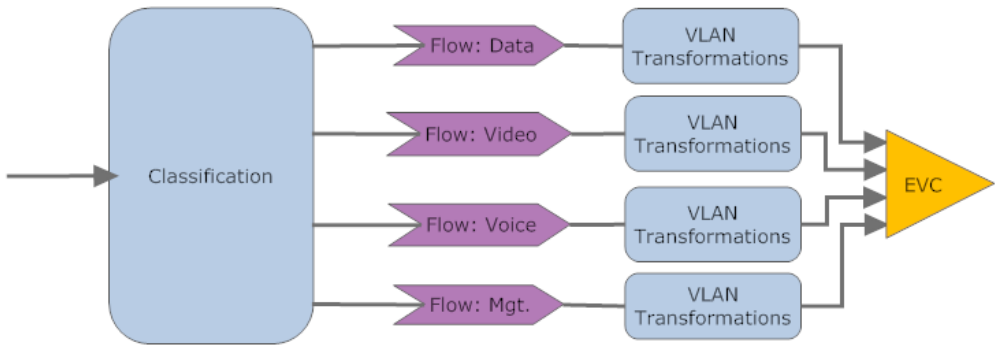
The figure below shows the MiNID SFP enclosure, which is shorter than the SFP sleeve and has no extractable optical module.



MiNID SFP

1.4 Functional Description

The diagram below shows the flow of customer data in the device. The following table describes the traffic handling stages.



Data Flow

Traffic Handling Stages

Processing Stage	Description
Classification	Classifies traffic and defines flows. See Service Demarcation
VLAN Transformations	Push/POP/Change S-VLAN/C-VLAN, sets p-bit. This stage can also loopback or drop traffic.

1.5 Technical Specifications

Ethernet Interface for SFP sleeve	<i>MSA/SFP Connector</i>	SFP-based, MSA-compliant
	<i>SFP Transceivers (optical)</i>	Dual/single, multi mode/single mode fiber: FE: 100BASE-FX/LX/BX GbE: 100/1000BASE-SX/LX/ZX/BX and CWDM
	<i>SFP Transceivers (electrical)</i>	100BASE-T or 1000BASE-T or 100/1000BASE-T multirate
	<i>Compliance</i>	IEEE 802.3
	<i>Max. Frame Size</i>	Traffic packets: 12,000 bytes Management packets: 2,048 bytes
Power	<i>Maximum</i>	SFP sleeve: 1.1W (not including SFP)
Physical	<i>Height</i>	SFP sleeve: 12.7 mm (0.5 in) MiNID SFP: 12.7 mm (0.5 in)

	<i>Width</i>	SFP sleeve: 14.3 mm (0.56 in) MiNID SFP: 14.3 mm (0.56 in)
	<i>Depth</i>	SFP sleeve: 81.1 mm (3.19 in) MiNID SFP: 75.1 mm (2.96 in) Note: SFP sleeve adds 33.5mm (1.32 in) to the depth of the hosting device. SFP adds 27.5mm (1.08in) to the depth of the hosting device.
	<i>Weight</i>	SFP sleeve: 30.0 g (1.0 oz) MiNID SFP: 40.0 g (1.4 oz)
	Environment	
	<i>Operating Case Temperature for SFP sleeve and MiNID SFP</i>	-40 to 85°C (-40 to 185°F)
	<i>Humidity</i>	Up to 90%, non-condensing

2 Installation and Setup

This chapter describes installation and setup procedures for MiNID SFP sleeve and SFP units.

After installing the unit, refer to **Operation** for the operating instructions.

Housed in a Small Form Factor Pluggable (SFP) package, MiNID SFP sleeve complies with the Multi-Source Agreement (MSA) and can be inserted into any MSA-compatible host unit. Furthermore, standard SFPs can be inserted into MiNID.

MiNID is equipped with DIP switches on its underside that enable setting the MiNID unit to various operation modes. Operation modes depend on the desired task and are listed in the table below together with the associated DIP switch settings.

2.1 Safety

Note

Before installing the product, review **Handling Energized Products** at the beginning of the manual.



Warning

The host product must be certified and approved according to EN/IEC/62368-1 or UL/CSA 62368-1 standards.



Avertissement

Le produit hôte doit être certifié et approuvé conformément aux normes EN/IEC/62368-1 ou UL/CSA 62368-1.

Laser Safety



Warning

Third-party SFP optical transceivers must be CDRH-registered and comply with IEC 62368-1 and IEC 60825, as well as with the local laser safety regulations for Class I laser equipment.



Avertissement

Les émetteurs-récepteurs optiques SFP tiers doivent être enregistrés auprès du CDRH et être conformes aux normes IEC 62368-1 et IEC 60825, ainsi qu'aux réglementations locales en matière de sécurité laser pour les équipements laser de classe I.

2.2 Site Requirements and Prerequisites

MiNID SFP sleeve is intended for installation in an MSA-compatible SFP socket of the host equipment. The ambient temperature is 0 to 50°C (32 to 122°F).

2.3 Package Contents

The MiNID package includes the following items:

- MiNID SFP sleeve or MiNID SFP
- Manual download form

Note

The SFP-CA.2 configuration adapter module for the MiNID sleeve can be ordered separately.

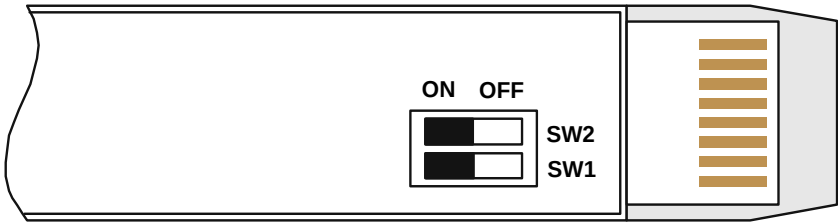
2.4 Required Equipment

MiNID does not require special tools for installation.

2.5 Setting the DIP Switches

MiNID SFP Sleeve and SFP

MiNID SFPs (sleeve and short) include a two-part DIP switch on their underside, used to select the working mode of the device.



MiNID SFP DIP Switch

The working modes are presented in the DIP Switch Settings table below.

DIP Switch Settings

Switch Position		Working Mode
SW1	SW2	
OFF	OFF	Database initialization – Reset the device database to factory default values including the host parameters (IP address, gateway address, VLANs, etc.). The factory default values are taken from the file factory-default-config .
ON	OFF	Normal operation
OFF	ON	For MiNID SFP sleeve: Software download – Download a new software version via YMODEM protocol (can be used only with RAD SFP-CA.2 configuration module)
ON	ON	Configuration –Set the device IP address to the default IP address 192.168.205.1 <i>Note: This DIP switch setting is normally used by field engineers who need to access the device without changing database values.</i>

2.6 Mounting

This section presents instructions for installing MiNID SFP sleeve.

Installing SFPs into the SFP Sleeve

MiNID SFP sleeve provides an MSA-compatible SFP port, for installing standard SFPs.

Note

You must remove MiNID SFP sleeve from the host unit before you insert or remove an SFP from MiNID SFP sleeve (see [Inserting MiNID SFP Sleeve into a Host Unit](#) for instructions on removing MiNID).



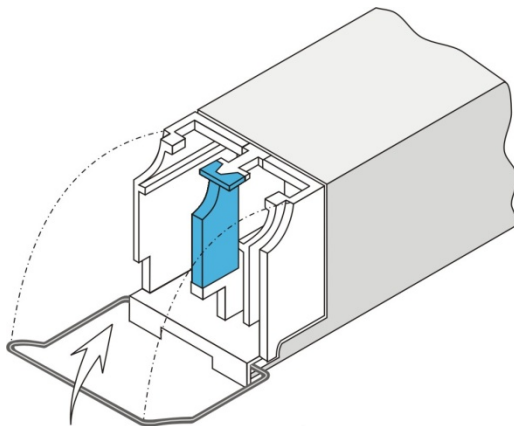
Third-party SFP optical transceivers must be agency-approved, complying with the local laser safety regulations for Class I laser equipment.

► **To install the SFP:**

1. Lock the wire latch of the SFP module by lifting it up until it clicks into place, as illustrated below.

Note

Some SFP models have a plastic door instead of a wire latch.



Locking the SFP Wire Latch

2. Carefully remove the dust covers from the SFP slot.
3. Insert the rear end of the SFP into the socket, and push slowly backwards to mate the connectors until the SFP clicks into place. If you feel resistance before the connectors are fully mated, retract the SFP using the wire latch as a pulling handle, and then repeat the procedure.

Caution *Insert the SFP gently. Using force can damage the connecting pins.*

4. Remove the protective rubber caps from the SFP modules.

➤ **To remove the SFP module:**

1. Disconnect the fiber optic cables from the SFP module.
2. Unlock the wire latch by lowering it downwards (as opposed to locking).
3. Hold the wire latch and pull the SFP module out of the Ethernet port.

Caution *Do not remove the SFP while the fiber optic cables are still connected. This may result in physical damage (such as a chipped SFP module clip or socket), or cause malfunction (e.g., the network port redundancy switching may be interrupted).*

Inserting MiNID SFP Sleeve into a Host Unit

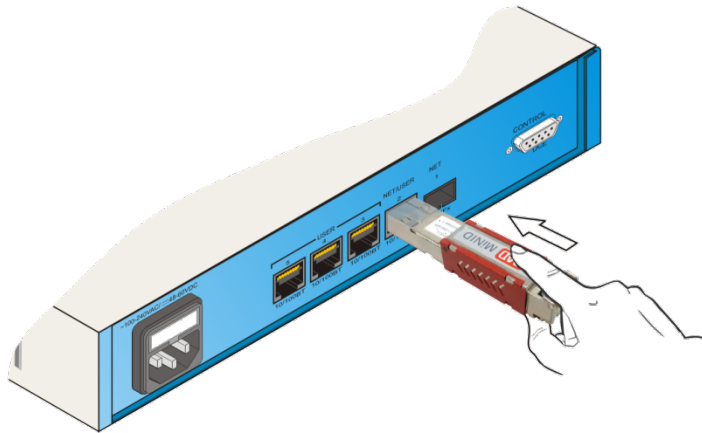
This section explains how to connect/eject MiNID to/from a host device.

You need to insert MiNID into the host device in order to configure it via the host, or in order to operate it. Refer to [Service Provisioning](#) for procedures to configure MiNID via SFP-CA.2 or host device (switch).

Note *You do not have to switch off the host unit when inserting or extracting MiNID.*

➤ **To insert MiNID:**

1. Set the DIP switches to the required operation mode as specified in [Setting the DIP Switches](#).
2. Insert the necessary SFP into MiNID (see [Installing SFPs](#)).
3. Hold MiNID with the label facing up, and insert MiNID into an MSA-compatible SFP socket of the host equipment.



Inserting MiNID

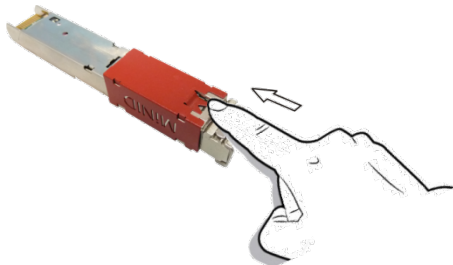
4. Press MiNID firmly into the SFP socket, until MiNID clicks into place. MiNID is ready to operate.

➤ **To eject :**

1. Disconnect any cables attached to MiNID.
2. Push the release latch on MiNID's underside in the direction of the host device.

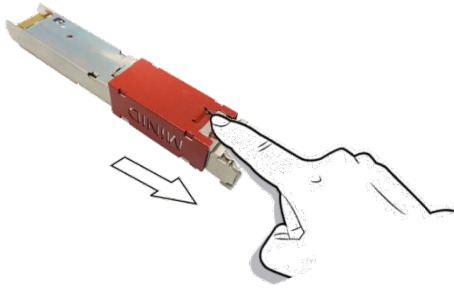
Note

The pictures below show the MiNID underside in order to illustrate the release latch.



Pushing Release Latch

3. While continuing to keep pressure on the latch, push MiNID gently towards the host device, and then pull MiNID gently out of the socket.



Removing MiNID

4. Release the latch after you have completely removed MiNID from the socket.



Warning

MiNID can become very hot while in operation. Use caution when extracting it.

Inserting MiNID SFP into a Host Unit

This section explains how to connect/eject MiNID to/from a host device.

You need to insert MiNID into the host device in order to configure it via the host, or in order to operate it. Refer to [Service Provisioning](#) for procedures to configure MiNID via SFP-CA.2 or host device (switch).

Note

You do not have to switch off the host unit when inserting or extracting MiNID.

➤ To insert MiNID:

1. Set the DIP switches to the required operation mode as specified in [Setting the DIP Switches](#).
2. Insert MiNID into a free SFP (MSA-compatible) socket in the host device.
3. Press MiNID firmly into the SFP socket until MiNID clicks into place and lift the latch to lock MiNID into place.

MiNID SFP is ready to operate.

➤ **To eject MiNID:**

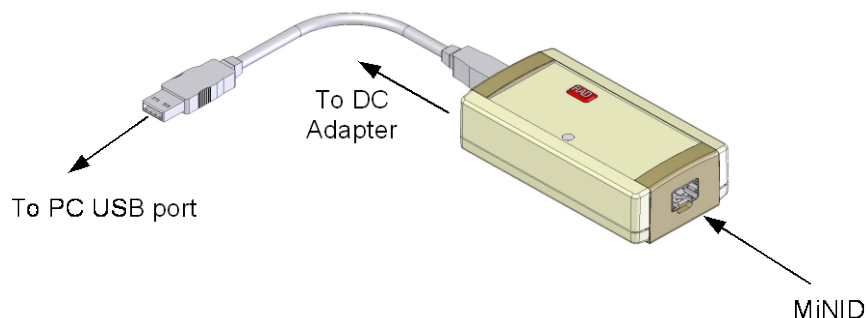
1. Disconnect any cables attached to MiNID.
2. To extract the MiNID SFP from the host device, open the latch and remove MiNID from the socket.

Inserting MiNID into an SFP-CA.2

➤ **To connect MiNID and SFP-CA.2 to the PC:**

1. Insert MiNID into the SFP socket of the SFP-CA.2 module.
2. Connect SFP-CA.2 to your PC via USB 2.0 port.

The **Ready** LED on SFP-CA.2 turns on.



SFP-CA.2 Configuration Module

➤ **To disconnect MiNID and SFP-CA.2 from the PC:**

1. Close all relevant management applications.
2. Eject MiNID from SFP-CA.2 in the same manner as from a host unit (see [Inserting MiNID SFP Sleeve into a Host Unit](#) and [Inserting MiNID SFP into a Host Unit](#)).
3. Disconnect SFP-CA.2 from the PC and from the power.

➤ **To configure MiNID via SFP-CA.2:**

1. Configure the PC parameters if necessary.
2. Insert any compatible SFP into MiNID, while it is not connected to SFP-CA.2 or a host device.
3. Set MiNID to Configuration mode, using its DIP switches (refer to [Setting the DIP Switches](#)).

Note *While MiNID is in Configuration mode, it responds to the default IP address 192.168.205.1, with no host VLAN, even if a different IP address was configured.*

4. Insert MiNID into the SFP socket of the SFP-CA.2 module.
5. Connect SFP-CA.2 to your PC via USB 2.0 port.
The **Ready** LED on SFP-CA.2 turns on.
6. Using either your browser or Telnet, connect to **192.168.205.1** and log in with **su** credentials (see **User Access**).

Note *If no fiber is connected to the MSA port, ensure that all fault propagation actions are disabled (this is the default setting).*

2.7 Basic Connectivity Test

Check connectivity between MiNID and the PC by sending a ping from the PC to the unit.

➤ To ping MiNID:

1. In the Windows Run prompt, type ping and enter the MiNID IP address (IPv4 or IPv6) as the destination IP address of the ping.
2. Press <Enter> to start sending pings.
3. A reply from MiNID indicates a proper communication link.
4. If the ping request times out, check the link between MiNID and the PC (physical path, configuration parameters, such as static route, etc.).

3 Operation

3.1 Turning On the Unit

MINID SFP sleeve and MINID SFP receive power from its host device, therefore they start to function when inserted into SFP-CA.2 or a host unit, or when the host device is powered on.

3.2 Indicators

MINID SFP sleeve and MINID SFP do not have LED indicators.

3.3 Startup

Configuration and Software Files

The following system files contain configuration settings or application software:

- **factory-default-config** – Contains the manufacturer default settings. At startup, factory-default-config is loaded if startup-config and user-default-config are missing or invalid.
- **running-config** – Contains the current configuration that the device is running
- **startup-config** – Contains saved non-default user configuration. You can use the save or copy command to create startup-config, or click <Save Configuration> in the Web interface. At startup, startup-config is loaded if it exists and is valid.
- **user-default-config** – Contains default user configuration. This file is not automatically created. You can use the copy command or Utilities > File > Copy screen to create it. At startup, user-default-config is loaded if startup-config is missing or invalid

- **sw-pack-1** or **sw-pack-2** – Contains application software. One of the software packs is designated as active.

Loading Sequence

At startup, the device attempts to load configuration files in the following sequence until a valid one is found:

- startup-config
- user-default-config
- factory-default-config

If an error is encountered while loading a file, MiNID stops loading that file, sends an event to indicate the error, and then looks for the next file in the sequence.

To display MiNID parameter values after startup, use the **info** command.

3.4 Working with Custom Configuration Files

In large deployments, often a central network administrator sends configuration files to the remote locations and all that remains for the local technician to do is to replace the IP address in the file or other similar minor changes, and then download the file to the device. Alternatively, the technician can download the file as is to the device, log in to the device and make the required changes, then save the configuration.

To download the configuration file, use the **copy** command (refer to [Administration](#)). After downloading the configuration file, the unit must be reset in order to execute the file. After the unit completes its startup, the custom configuration is complete.

If you have an appropriate existing MiNID configuration file, you can use Zero Touch provisioning (see [On-net Zero Touch Provisioning](#)) to have the configuration file automatically fetched by all your MiNID devices. Otherwise, you need to manually configure all your MiNID devices with any additional non-default settings.

There is no startup sequence other than Zero Touch, if enabled (see [On-net Zero Touch Provisioning](#) for further details).

On-net Zero Touch Provisioning

Zero Touch provisioning allows MiNID to acquire an IP address from a DHCP server and fetch configuration files from a TFTP server, eliminating the need to manually log into MiNID to provision it.

Zero Touch provisioning is enabled by configuring the following:

- DHCP must be enabled (DHCP in menu Configuration > System > Management > Host IP; dhcp in level config>mngmt>host)
- If the user management traffic is via a specific VLAN, the management VLAN must be configured (Host VLAN, Host VLAN ID, and Host VLAN Priority in menu Configuration > System > Management > Host IP; vlan-id and vlan-pbits in level config>mngmt>host).

Note *Zero touch must be enabled (**Zero Touch** in menu **Configuration > System > Management > Zero Touch**; **zero-touch** in level **config>mngmt>host**), however this is the default setting.*

Prerequisites

- A DHCP server for providing the TFTP server address and configuration file information, in addition to the usual IP address, default gateway, etc.
- A TFTP server with a configuration file, consisting of valid CLI commands.

Sequence

At reboot, when MiNID obtains a DHCP lease from the DHCP server, the lease provides the TFTP server address, either via option 150, or as a string ('xxx.xxx.xxx.xxx') via option 66. The DHCP lease provides the path and file name of the configuration file via DHCP option 67, if zero touch is to be used.

1. If the lease specified a configuration file, MiNID loads the configuration file from the TFTP server. If no configuration file was specified, the zero touch process does not start.
2. If the TFTP download fails, MiNID does the following:
 - a. Send the trap systemDownloadEnd (with failed indication) to any configured network managers
 - b. Start a 10-minute timer.
 - c. When the timer expires, MiNID attempts again to download the configuration file.
3. If the configuration file is received successfully from the TFTP server, MiNID does the following:

- a. Send the trap systemDownloadEnd (with success indication) to any configured network managers
- b. Verify that the file contains valid CLI commands
- c. Execute the commands in the file.

Saving Configuration Changes

When you make changes to the MiNID configuration, you need to save the changes so that they are available after the next device reset. You can save your configuration as follows:

- Web: Click <Save Configuration>
CLI: Use the save command to save running-config as startup-config
- Web: Use the Utilities > File > Copy screen to copy running-config to startup-config or user-default-config
CLI: Use the copy command to copy running-config to startup-config or user-default-config.

3.5 Configuration and Management

MiNID provides the following management interfaces:

- Web interface over HTTP
- CLI interface over Telnet / secure Telnet (SSH)

Once MiNID is accessible via IP address, it can be configured via Web or Telnet/ secure Telnet (SSH).

By default, MiNID can transmit management data over both of its ports. The MiNID SFP sleeve and MiNID SFP ports are designated as follows:

- SFP –Provides socket where an SFP can be inserted into MiNID
- MSA – Enables inserting MiNID into a customer device

Management can be disabled on either port (see [Management Access Control](#)); it cannot be disabled for both ports simultaneously.

MiNID SFP sleeve and MiNID SFP can be configured via the SFP-CA.2 adaptor, or via a switch that acts as a host device. See [Configuration of MiNID SFP via SFP-CA.2 \(Telnet/SSH/Web\)](#) and [Configuration via Host \(Telnet/SSH and CLI/Web Interface/RADview/SNMP\)](#).

3.6 CLI-Based Configuration

Working with Telnet/SSH

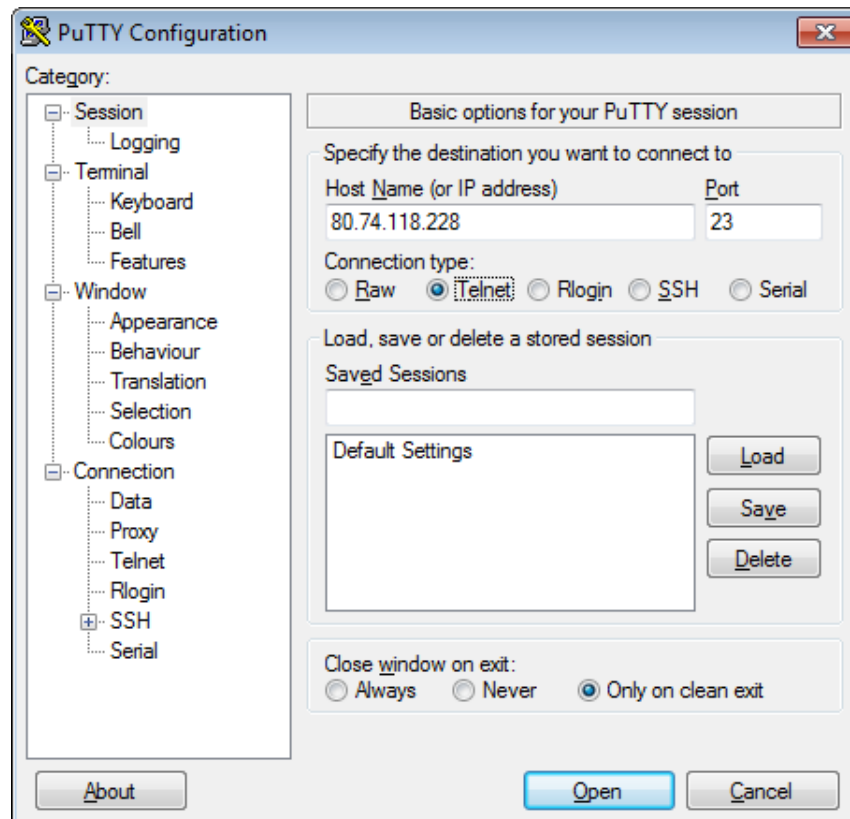
You can use any Telnet/SSH client to connect to the MiNID CLI interface.

Logging In

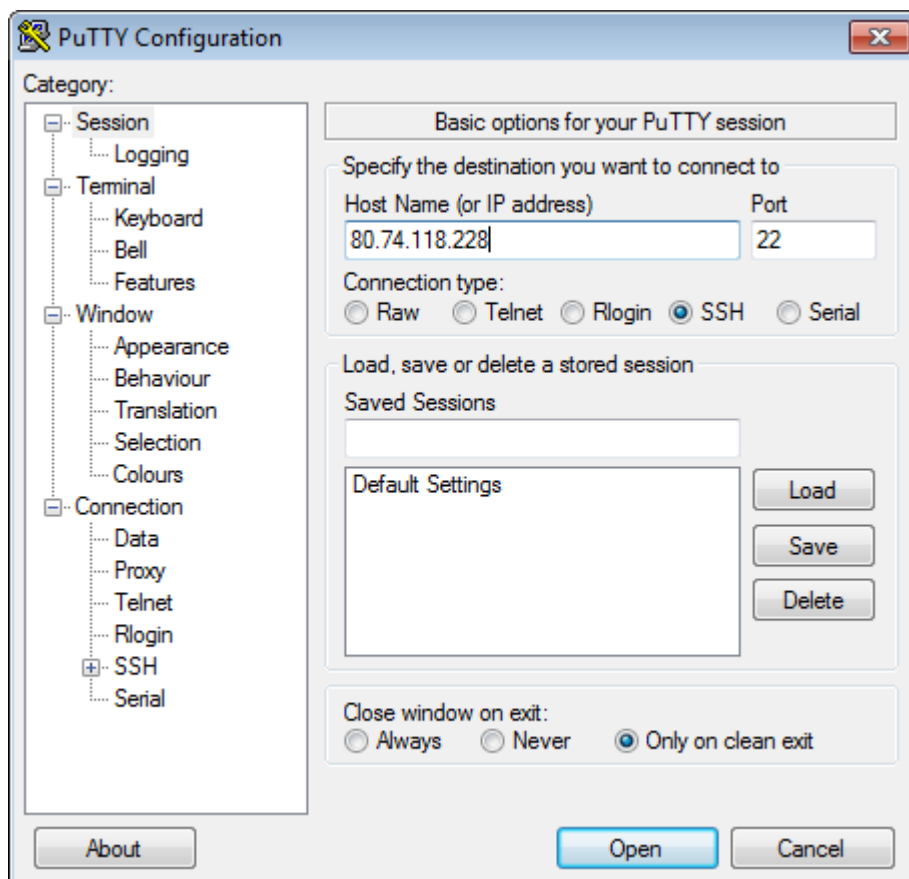
➤ **To connect to MiNID via Telnet/SSH:**

1. Point the Telnet/SSH client to MiNID's IP address. Depending on the client, this may be done with the command:
telnet <IP-address>

Or, you may need to type the IP address and select Telnet/SSH, and then select regular Telnet access or secure Telnet (SSH) access:

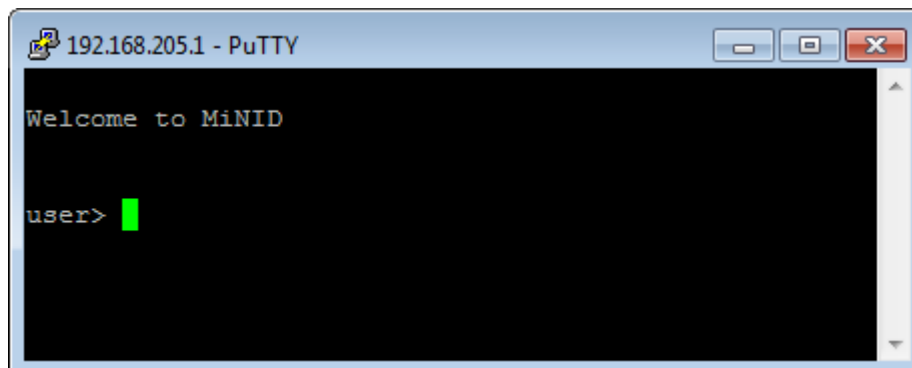


Regular Telnet Connection with the PuTTY Client

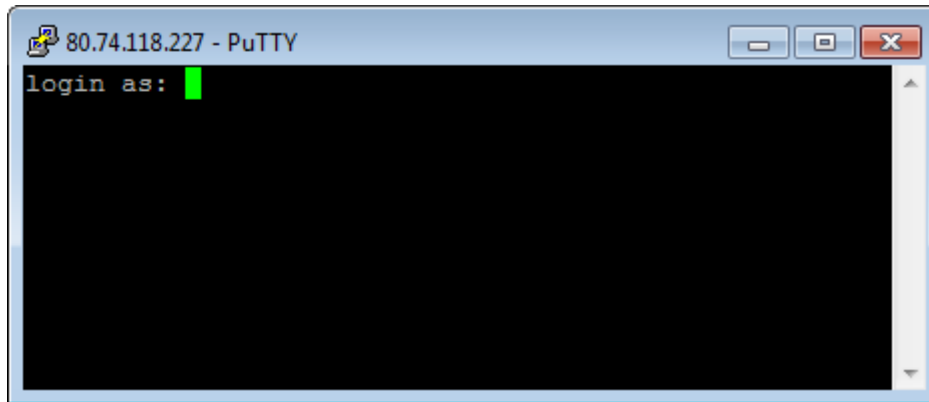


Secure Telnet Connection with the PuTTY Client

A login prompt appears.



CLI Login Prompt – Regular Telnet



CLI Login Prompt – Secure Telnet

2. Enter the relevant credentials (see User Access).

Login

To prevent unauthorized modification of the operating parameters, supports various access levels. Refer to [User Access](#) in the Management and Security chapter for more information on the access levels, as well as a list of the default users defined in the device.

► To log in to :

1. At the user prompt (user>), enter su and press <Enter>. The password prompt (<password>) appears.
2. Enter the password (default is 1234) and press <Enter>. The base prompt **MiNID#** appears.

If you have lost your superuser password, contact Technical Support via the [RADcare Online portal](#) or by [email](#).

Using the CLI

The CLI consists of commands organized in a tree structure of levels, starting at the base level. Each level (also referred to as context) can contain levels and commands (refer to the [Navigating](#) section for more information on the levels and commands available in MiNID). The level is indicated by the CLI prompt.

Note *Most commands are available only in their specific context. Global commands are available in any context. You can type ? at any level to display the available commands.*

CLI Prompt

The base level prompt contains the device name, which is MiNID by default (the device name can be configured in the system level; refer to the [Device Information](#) section in this manual). The prompt ends with #.

After you type a command at the CLI prompt and press <Enter>, MiNID responds according to the command entered.

Navigating

To navigate down the tree, type the name of the next level. The prompt then reflects the new location. To navigate up, use the global command **exit**. To navigate all the way up to the root, type **exit all** or press <CTRL> + 'Z'.

At the prompt, one or more level names separated by a space can be typed, followed (or not) by a command. If only level names are typed, navigation is performed and the prompt changes to reflect the current location in the tree. If the level names are followed by a command, the command is executed, but no navigation is performed and the prompt remains unchanged.

Note *To use show commands without navigating, type **show** followed by the level name(s) followed by the rest of the show command.*

In the following example, the levels and command were typed together and therefore no navigation was performed, so the prompt did not change.

```
MiNID# configure system date-and-time date-format yyyy-mm-dd
MiNID# show configure system system-date
2013-06-10 15:08:20 UTC +00:00
MiNID#
```

In the following example, the levels were typed separately and the navigation is reflected by the changing prompt.

```
MiNID# configure
MiNID>config# system
MiNID>config>system# date-and-time
MiNID>config>system>date-time# date-format yyyy-mm-dd
MiNID>config>system>date-time# exit
MiNID>config>system# show system-date
2013-06-10 15:13:23 UTC +00:00
```

```
MiNID>config>system#
```

Command Tree

The **tree** command displays a hierarchical list of all the commands in the CLI tree, starting from the current context.

➤ To view the entire CLI tree (commands only):

1. At the root level, type tree.

```
MiNID# tree
|
+---admin
|   |
|   +---factory-default-all
|   |
|   +---factory-default
|   |
|   +---reboot
|   |
|   +---software
|   |   |
|   |   +---install
|   |   |
|   |   +---software-confirm-required
|   |   |
|   |   +---undo-install
|   |   |
|   |   +---show status
|   |
|   +---startup-confirm-required
more..
```

2. Press <Enter> to see more or <CTRL-C> to return to the prompt.

Command Structure

CLI commands have the following basic format:

```
command [parameter]{ value1 | value2 | ... | valuen }  
[ optional-parameter <value> ]
```

where:

- | | |
|------------|---|
| { } | Indicates that one of the values must be selected |
| [] | Indicates an optional parameter |

<>	Indicates a value to be typed by the user according to parameter requirements
----	---

You can type only as many letters of the level, command, or parameter as required by the system to identify it. For example you can enter **config manag** to navigate to the **management** level.

Special Keys

The following keys are available at any time:

?	List all commands and levels available at the current level
<Tab>	Complete a string while typing commands and parameters
<Backspace>	Delete character before cursor
<Delete>	Delete character before cursor
Left/right arrow	Move cursor one character left/right
Up/down arrow	Command history for current level (keeps the last five commands per level)
<CTRL> + 'Z'	Exit all

Getting Help

You can get help in the following ways:

- Type ? to display general help (see General Help below)
- Type <command> ? to display information on a command and its parameters (see Command Help)
- Type ? to display the commands available in the level (see Level Help)
- Use <Tab> while typing commands and parameters, for string completion (see String Completion)
- Use ? after typing a command or parameter, for interactive help (see Interactive Help).

General Help

Type help or ? to display general CLI help, including:

- Commands and levels available at the current level
- Globally available commands.

Example of **help** or **?** command output from the root level:

```
MiNID# ?
admin
configure
inventory
file

-- Global Commands --
exit
info
logout
help
tree
ping
save
```

Command Help

- Enter **<command> ?** to display command and parameter information.

```
MiNID>config>mngmnt# timeout ?
- timeout <minutes>
```

Level Help

Enter **?** at the command prompt to display the commands available in the current level.

```
MiNID>config#?
access-control
management
router
oam
qos
flows
port
reporting
service
system
test
```

String Completion

MiNID automatically completes levels, commands, and parameters when you press **<Tab>** immediately after a string.

If the string can be completed in more than one way, MiNID appends the characters that are common to all the possibilities.

If the string can be completed in only one way, MiNID completes it and appends a space.

If the string is already a complete level/command/parameter or cannot be completed to a level/command/parameter, no completion is done.

Pressing **<Tab>** a second time displays any available command parameters.

Interactive Help

► To get interactive help:

- type **?**.

In general, typing a **?** directly after a string performs string completion, while typing **<space>** and then a **?** executes the command.

When a **<CR>** appears in a **?** list, the string you entered is itself a valid command needing no further additions. Pressing **<Enter>** executes the command or navigates to the indicated level.

Typing **?** immediately after a command or partial command with no **space** before the **?**, tells MiNID to display all possibilities for completing the string. Help output is always followed by the string you typed with the cursor at the end of the string waiting for input.

```
MiNID>config>oam>cfm>md(1)>ma(1)>mep(1)>service(1)# loss?  
loss-threshold  
loss-dei-bit-color
```

Typing **<?>** after a **space** between a command or level name and the **?** tells MiNID to try to execute the command. The **space** tells the CLI that you are finished typing and to try to match the string to an appropriate command. The string does not have to be a complete command.

If there is only one possible command starting with that string, pressing **<Enter>** will execute the command.

```
MiNID# con  
MiNID>config#
```

If there is more than one command that starts with the string, the CLI displays a list of possible completions.

```
MiNID>config>oam>cfm>md(1)>ma(1)>mep(1)>service(1)# loss ?  
loss-threshold  
loss-dei-bit-color
```

A string that is a complete command name followed by a **space ?** displays all possible command parameters.

```
MiNID>config>oam>cfm>md(1)>ma(1)>mep(1)>service(1)# loss-threshold ?  
loss-threshold
```

The next example shows a complete command to which a parameter could be appended. It also shows how a string that is a complete command is executed by pressing "?".

```
MiNID>config>oam>cfm>md(1)>ma(1)>mep(1)# vlan 100?  
- vlan <vlan> [p-bit <p-bit>] [tag-ether-type <ether-tag>]
```

Using Scripts

CLI commands can be gathered into text files. They may be created using a text editor, by recording the user commands or by saving the current configuration.

These files can be configuration files or scripts. Configuration files have specific names and contain CLI commands that MiNID can use to replace the current configuration, while scripts contain CLI commands that add to the current configuration. Configuration files can be imported from and exported to RAD devices via file transfer protocols.

For more information on configuration files, refer to the description in the [Operation](#) chapter.

In order to execute a CLI script, you have to copy/paste it to the CLI terminal, or send it to MiNID via the RADview Jobs mechanism, CLI script option.

3.7 Web-Based Configuration

Logging in

You can configure and manage MiNID locally or remotely using its web interface. Supported browsers are the following:

- Internet Explorer 10 or above
- Google Chrome

Note

To prevent configuration errors, flush the browser's cache whenever you return to the same screen.

➤ **If you have trouble with the web interface:**

- Enable scripts.

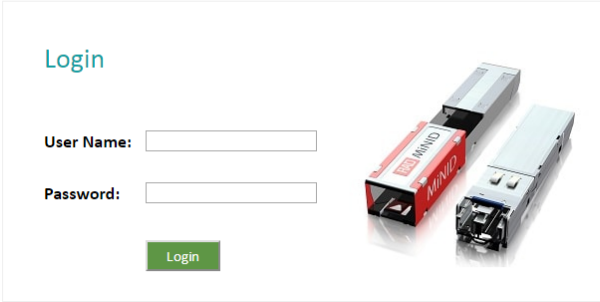
- Make sure that local and organizational firewalls allow access to the destination IP address.
- Disable pop-up blocking software, such as Google Popup Blocker. You may also have to configure spyware and adware protecting software to accept traffic from/to the destination IP address.

➤ **To log into the via the web interface:**

1. In the web browser, navigate to the IP address (see Router Interface for instructions on setting the IP address).

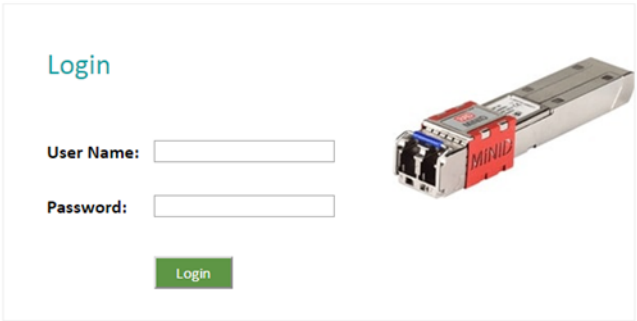
The MiNID login prompt appears.

RAD MiNID Configuration Server



The login interface is displayed within a white-bordered box. On the left, the word "Login" is written in teal. Below it are two input fields: "User Name:" and "Password:", each followed by a white text box. A green "Login" button is positioned below the password field. To the right of the input fields is a 3D rendering of a red and silver MiNID SFP sleeve and a silver SFP module.

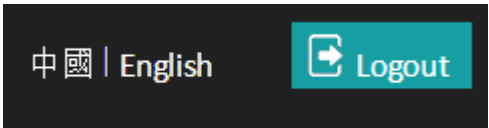
MiNID SFP Sleeve Login



MiNID SFP Login

➤ To switch languages:

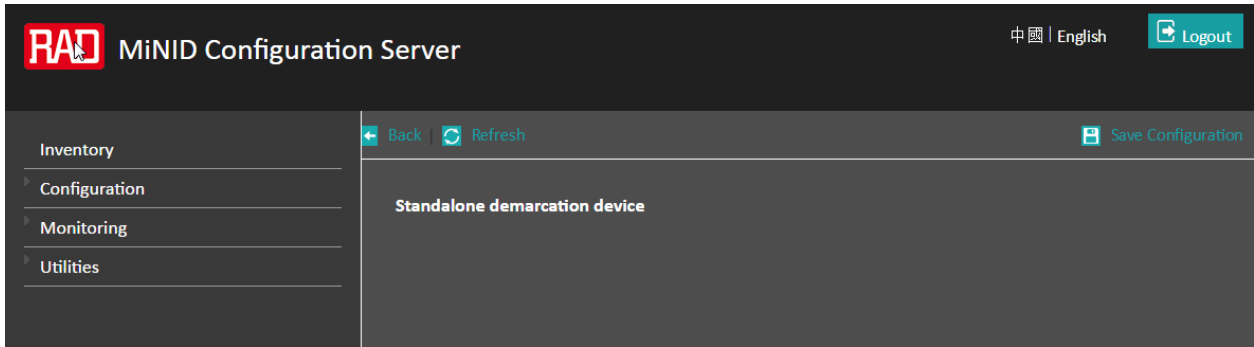
- 1. Click the desired language in the top right corner (next to Logout) in the web interface.



Switching the Language of the Web Interface

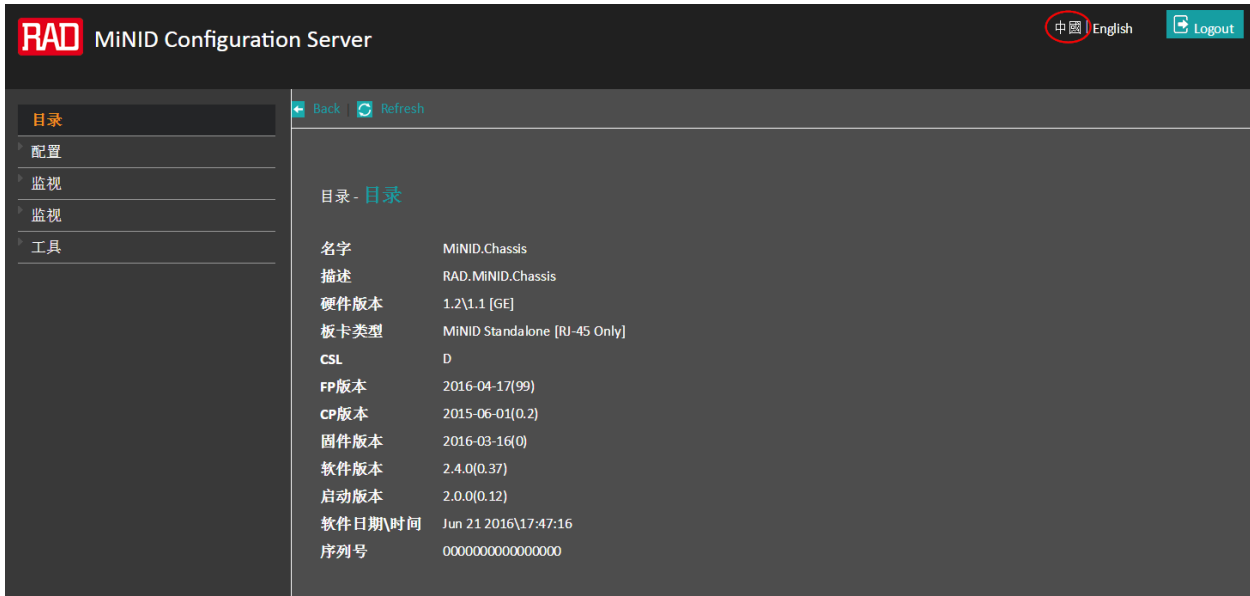
Using the Web Interface

A navigation tree is displayed on the left, as shown below. You can **Logout** of the interface by clicking the <Logout> icon on the top right.



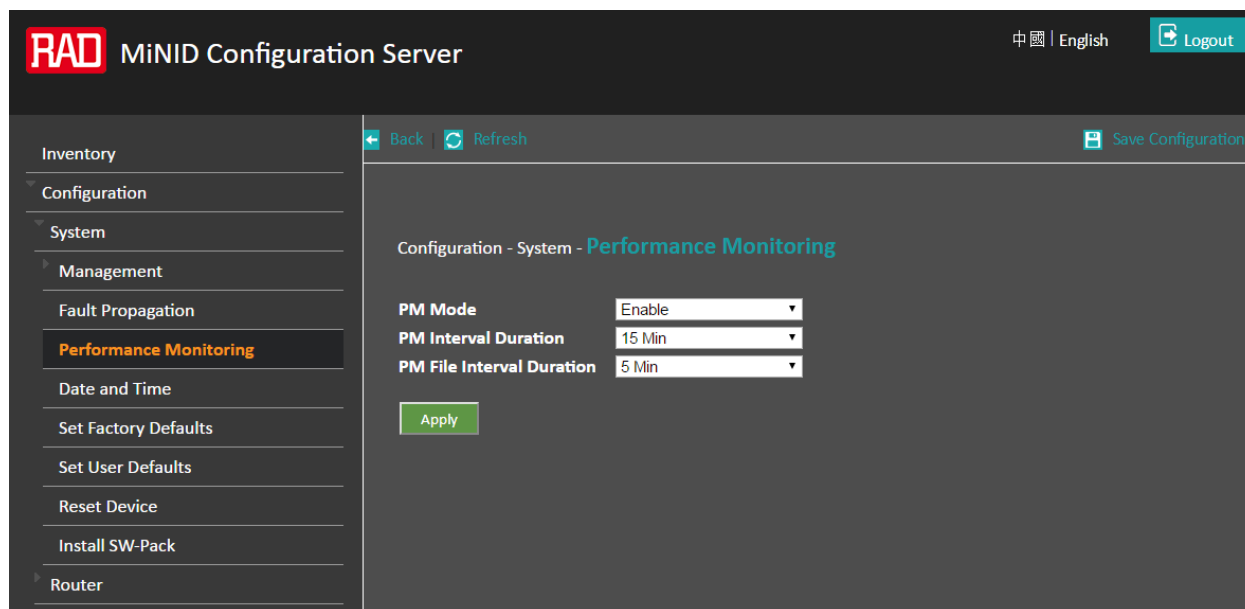
MiNID Main Menu

You can display the interface in Chinese by clicking the Chinese language  icon on the top right, next to the **<Logout>** icon. When you click this icon, you are logged out of the current interface, and the login page is displayed. Log in again to start working in the chosen interface.



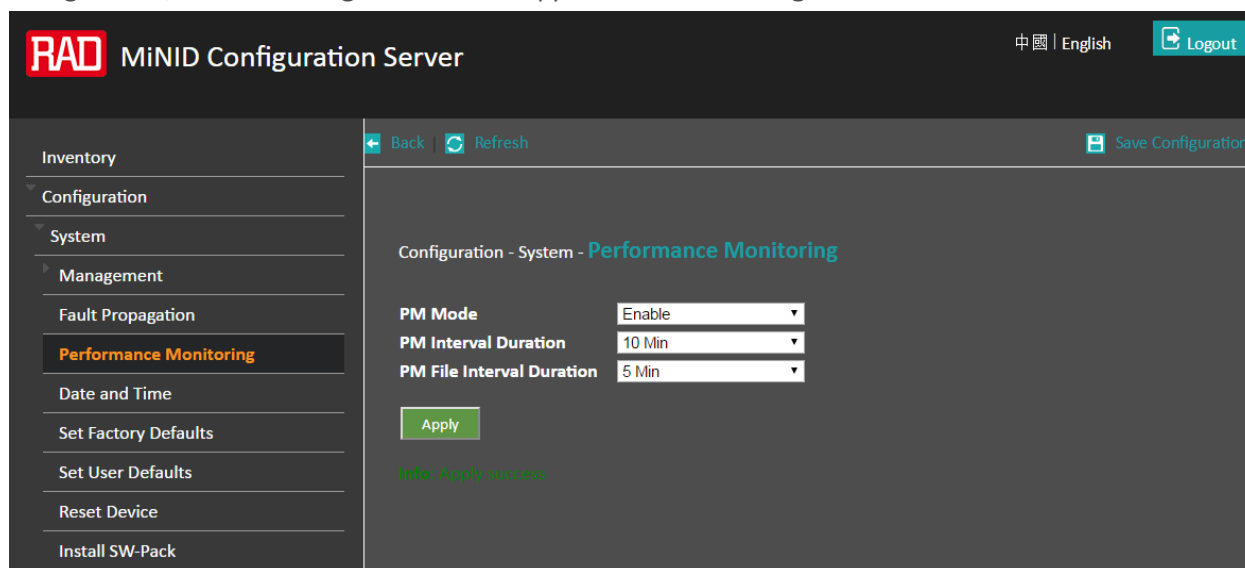
Web Interface in Chinese

Use the tree on the left to navigate to the various menus. Click the right arrow (▶) to expand a subtree, or downward arrow (▼) to collapse the subtree. You can also click a tree entry to toggle between expanding/collapsing the submenu. Tree entries without a ▶ or ▼ represent menus that can be opened by clicking the entry. The individual menus display information or provide configuration options. For example:



MiNID Web Interface

If you change a parameter setting and then click **<Apply>** to save the change in the device, the change is saved only in the device running configuration. Therefore any time that you change the device running configuration, a **<Save Configuration>** icon appears under the **<Logout>** icon.



When you click **<Save Configuration>**, the device running configuration is copied to the startup configuration, and the **<Save Configuration>** icon disappears.

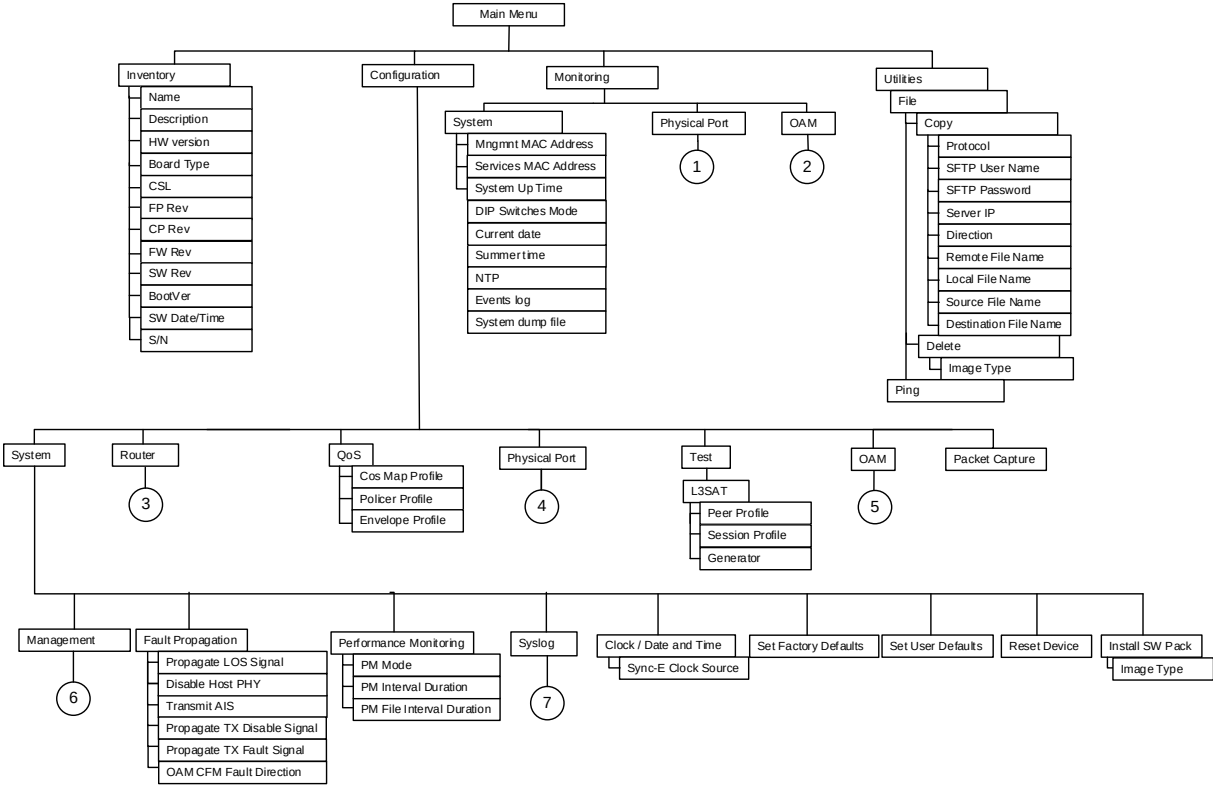
Menu Map

The main menu options are:

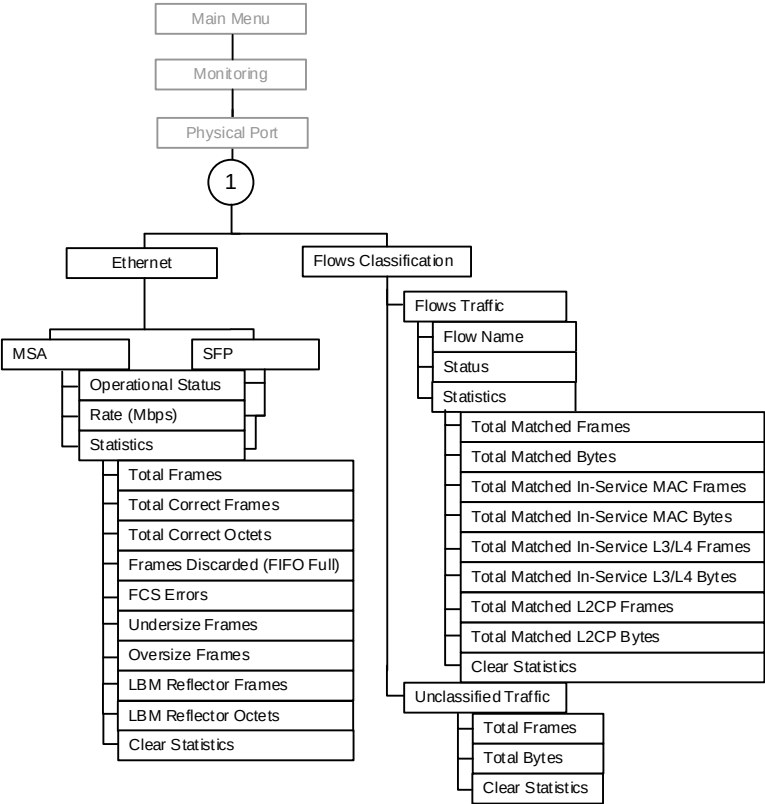
Inventory:	View hardware and software information
Configuration:	Set MiNID operational parameters
Monitoring:	View statuses, logs, and traffic statistics
Utilities:	Copy files Delete files Ping

The following figures detail the available display and configuration items.

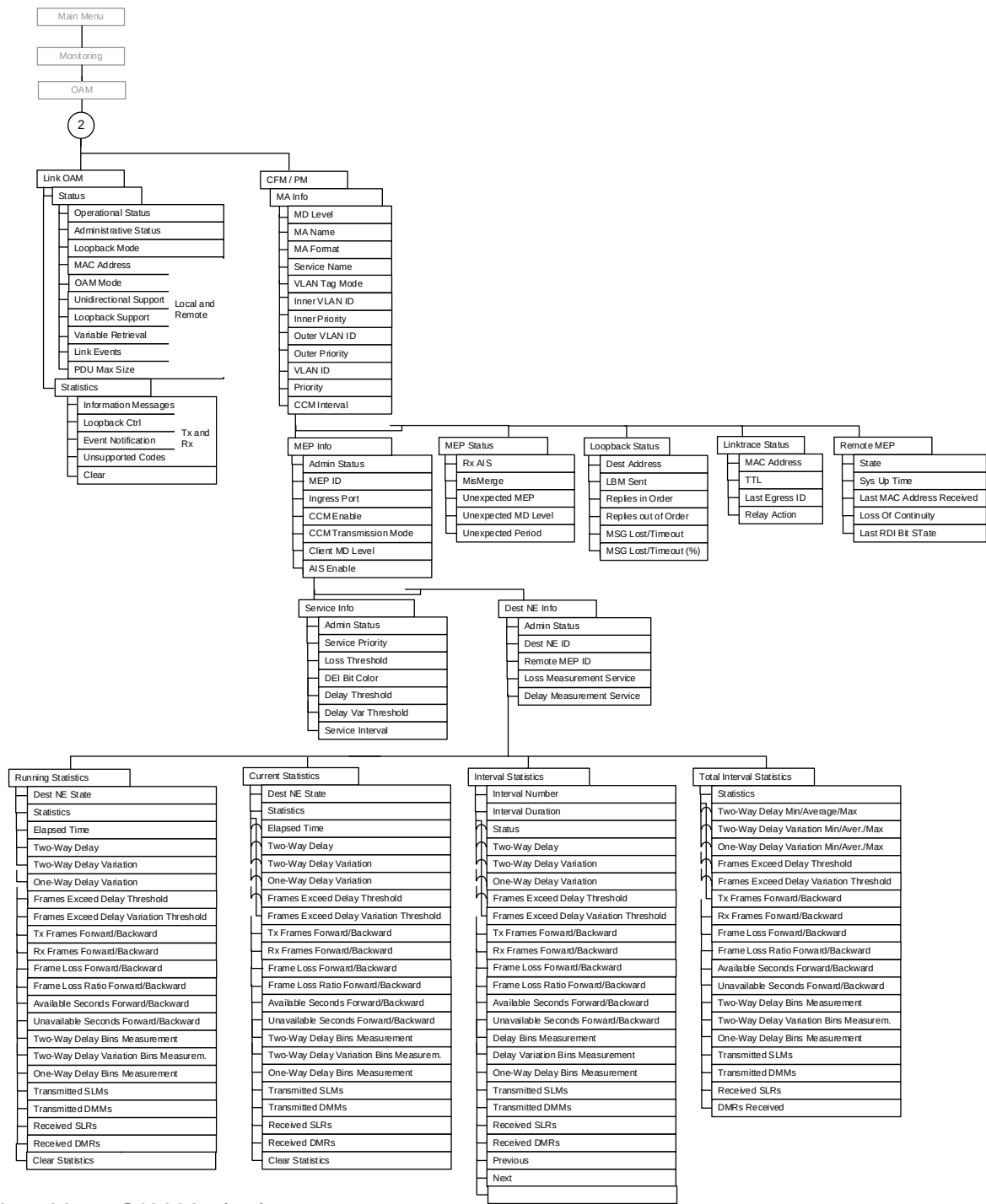
Note	<i>Some menus or menu items may not be available, depending on configuration.</i>
-------------	---



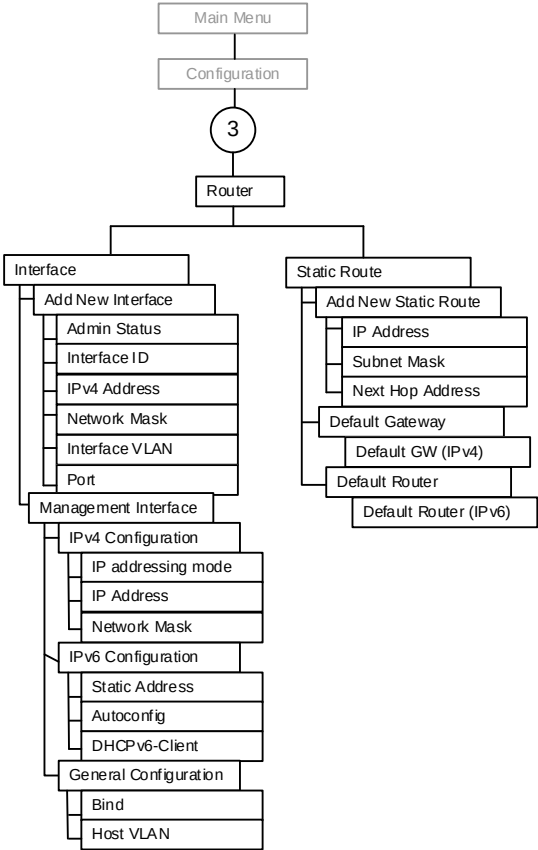
Menu Map (Except Physical Port Monitoring, OAM Monitoring, Router, Physical Port Configuration, OAM Configuration, Management, and Syslog)



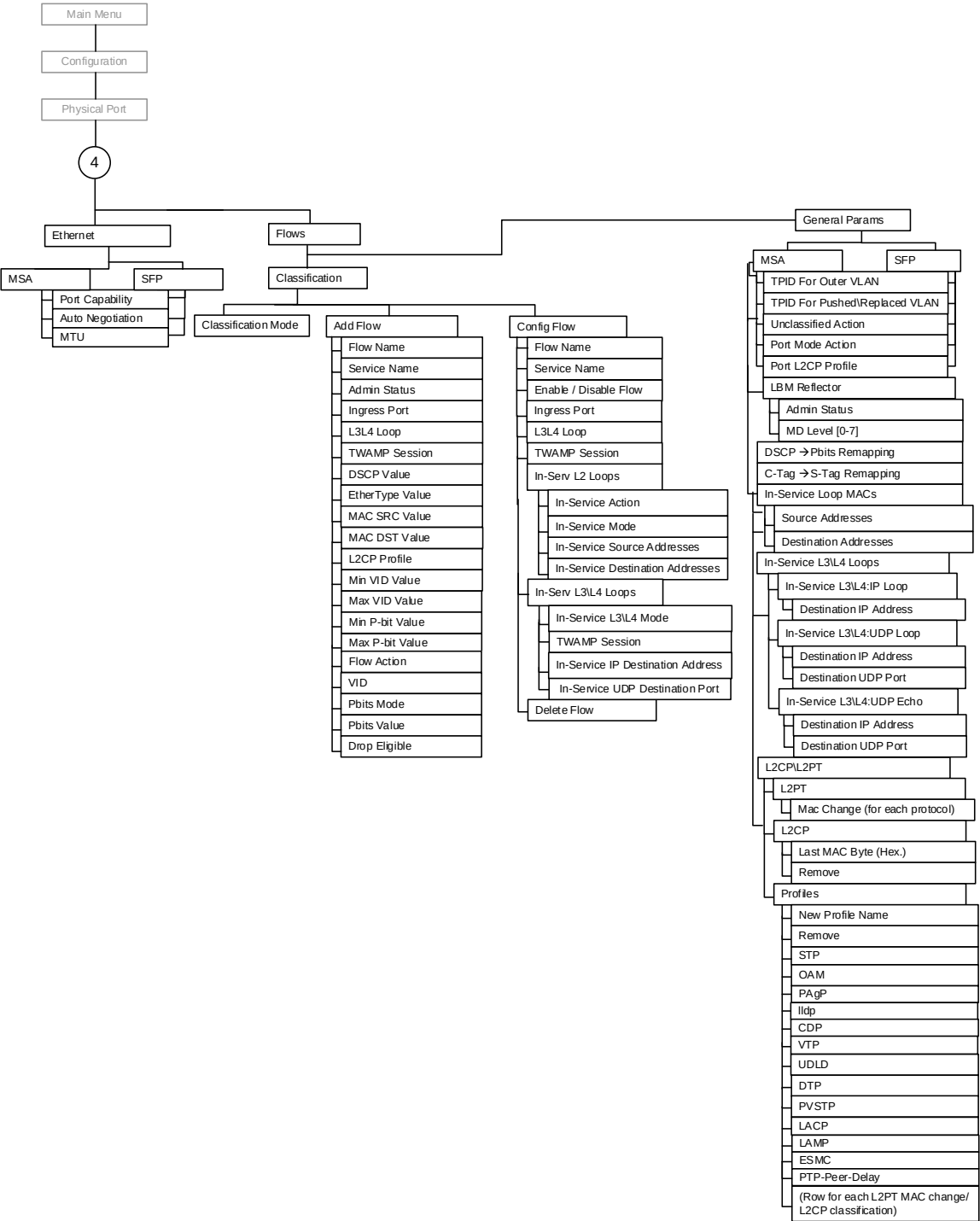
Menu Map – Physical Port Monitoring



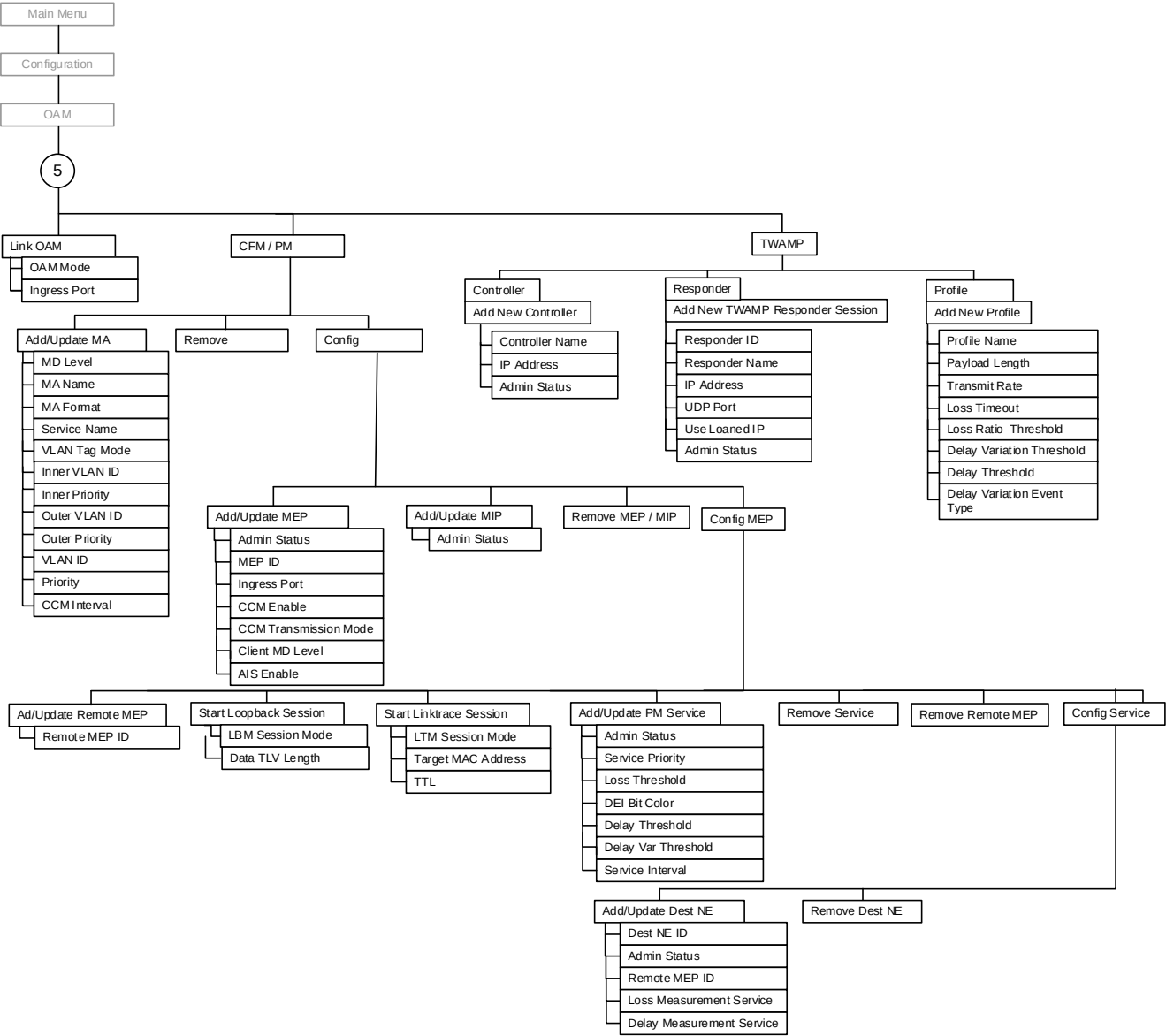
Menu Map – OAM Monitoring



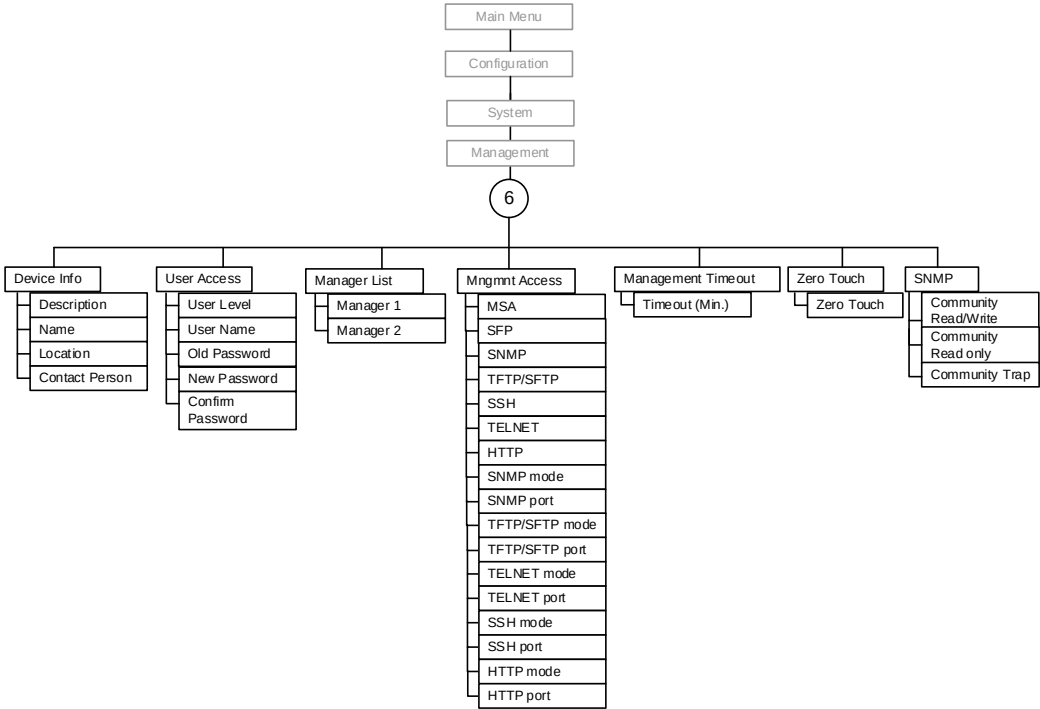
Menu Map – Router



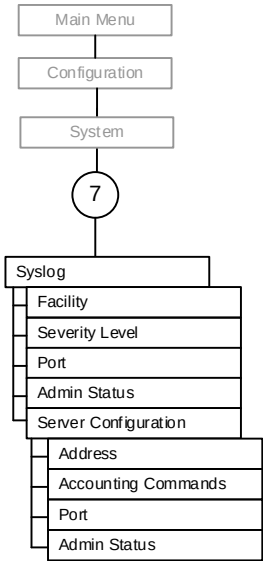
Menu Map – Physical Port Configuration



Menu Map – OAM Configuration



Menu Map – System Management



Menu Map – System Syslog

3.8 SNMP-Based Network Management

Note

*Make sure that SNMP is enabled and not set in the “read-only” mode (see **Management Access Control**) before starting to work with RADview or other SNMP-based NMSs.*

Working with RADview

RADview is a user-friendly and powerful SNMP-based element management system, used for planning, provisioning and managing heterogeneous networks. RADview provides a dedicated graphical user interface (GUI) for monitoring RAD products via their SNMP agents. RADview for MiNID is bundled in the RADview package for Windows or UNIX.

RADview is a Windows-based modular, client-server, scalable management system that can be used in a distributed network topology or a single-station configuration. RADview consists of the system (**EMS**) and the following optional modules:

- **Service Manager (SM)** – end-to-end Carrier Ethernet service provisioning for Ethernet Access products. This module includes the Service Center (SC) module, which is an end-to-end Carrier Ethernet and TDM service provisioning for CI (Critical infrastructure) products.
- **Performance Monitor (PM)** – portal for service SLA monitoring for both carriers and their customers.



Warning

To prevent conflicts and ensure a stable and consistent network setup, provision services using either RADview Service Manager or CLI, but not both simultaneously.

For more details about this network management software, and for detailed instructions on how to install, set up, and use RADview, contact your local RAD partner. See **Working with Other SNMP-Based NMSs** for details on the SNMP MIBs used by MiNID.

RADview for MiNID supports autodiscovery and alarms, as well as configuration via SNMP. You can double-click the MiNID icon in RADview to activate a connection to the MiNID web interface, or right-click the MiNID icon in RADview and select the option to open a Telnet session.

Working with Other SNMP-Based NMSs

MiNID can be integrated into third-party network management systems (NMS) with basic autodiscovery and alarm functionality, by having MiNID send basic status SNMP traps to the NMS. MiNID must be configured to define the NMS as an SNMP Manager.

SNMP (Simple Network Management Protocol) is an application-layer protocol that provides a message format for communication between managers and agents. MiNID acts as an SNMP agent, sending SNMP traps to up to two configured SNMP Managers. MiNID supports SNMPv1 and SNMPv2.

The supported SNMP features are based on the following standards:

Reference	Title
RFC 3418	Management Information Base (MIB) for the Simple Network Management Protocol (SNMP)
RFC 2863	The Interfaces Group MIB

The MiNID OIDs are:

- For SPF sleeve – 1.3.6.1.4.1.164.6.1.6.36
- For SFP – 1.3.6.1.4.1.164.6.1.6.54

3.9 Configuration of MiNID SFP via SFP-CA.2 (Telnet/SSH/Web)

You can use RAD’s SFP-CA.2 module for the following:

- Configure MiNID SFP and MiNID SFP Sleeve by using SFP-CA.2 as a hosting device (possible only with a PC that has Windows XP SP2 installed)
- Access the boot menu in order to upgrade MiNID software (possible only with a PC that has Windows XP SP2, Windows 7, or Windows 10 installed). Refer to Chapter 12 for more information on upgrading

Note *All SFP-CA.2 modules have the MAC address 00-00-E8-00-00-01.*

Preconfiguration for SFP-CA.2

When you connect SFP-CA.2 to a specific PC for the first time, if you wish to configure MiNID via SFP-CA.2 then you have to install the SFP-CA.2 driver on that PC, and configure the PC network parameters for communication with SFP-CA.2. Refer to the SFP-CA.2 supplement for instructions on how to download and install the driver.

➤ To configure the PC network parameters for the SFP-CA.2 connection to MiNID:

1. Connect the SFP-CA.2 configuration unit to a USB port on your PC.
The New Hardware is Detected notice appears.
2. Right-click **My Network Places**.
A new local area network connection appears in the list of network connections.
3. Right-click the new local area connection and rename it **SFP-CA**.
4. Right-click **Properties**, click **Configure**, and select the Advanced tab
The Network Connection Properties dialog box appears.
5. Choose **Select Media** and under **Value**, choose **Home LAN**, and then click **OK**.
The dialog box closes and your settings are applied.
6. Right-click the SFP-CA connection and click **Properties**.
The Local Area Connection Properties dialog box appears.
7. Select Internet Protocol (TCP/IP) and click **Properties**.
The Internet Protocol (TCP/IP) dialog box appears.
8. To enable entering TCP/IP settings, select **Use the following IP Address**.
The IP Address and Subnet Mask fields become available.
9. Enter the following TCP/IP settings and then click **OK**:
 - **IP Address:** 192.168.205.20
 - **Subnet Mask:** 255.255.255.0
10. Close **My Network Places**.
The PC is ready to communicate with MiNID.

Connecting SFP-CA.2

For instructions on how to connect SFP-CA.2, see the [Inserting into an SFP-CA.2](#) section in Chapter 2.

3.10 Configuration via Host (Telnet/SSH and CLI/Web Interface/RADview/SNMP)

A similar procedure to configuring via SFP-CA.2 can be used to configure MiNID SFP sleeve and MiNID SFP when it is inserted into the SFP port of a host device, as long as connectivity is established to an available Ethernet port.

► **To configure MiNID via a host:**

1. Disconnect MiNID from any device.
2. If you use MiNID SFP sleeve, insert any compatible SFP into it.
3. Set MiNID to Configuration mode, using its DIP switches (refer to Setting the DIP Switches in Chapter 2).

Note

While MiNID is in Configuration mode, it responds to the default IP address 192.168.205.1, with no host VLAN, even if a different IP address was configured.

4. Insert MiNID SFP sleeve or MiNID SFP into an available SFP socket of the switch.
5. Connect the available switch port to which connectivity has been established, to your PC, using an IP address in the subnet 192.168.205.XX.
6. Using either your browser or Telnet, connect to 192.168.205.1 and log in with su credentials (see User Access).

3.11 Turning Off the Unit

SFP sleeve and SFP receive power from their host device, therefore they stop functioning when removed from SFP-CA.2 or a host unit, or when the host device is powered off.

4 Service Provisioning

4.1 Ethernet Service

To create an Ethernet service via MiNID, you need to define how MiNID should handle each type of traffic by configuring service flows. See [Traffic Processing](#).

4.2 Service Summary

You can display the associations between service names and their associated flows/MAs.

Benefits

Viewing the associated entities of service names is useful for service administration, and to ensure correct discovery of service-related entities by network management systems.

Functional Description

If you have defined service names for flows and MAs (see [Service Demarcation](#) and [Configuring MAs](#), respectively), you can display the flows/MAs associated with the service names.

Viewing Service Summary

In the Web interface, you can view a summary per service name of the associated flows/MAs; in the CLI interface, you can view a summary of all service names and their associated flows/MAs.

➤ To view the service summary:

Do one of the following:

- In the web interface, go to **Configuration > Physical Port > Flows Classification > Config Flow** or **Configuration > OAM > CFM** (see **Monitoring and Diagnostics**), and click the service name for which you would like to view the associated flows/MAs.

MinID

Configuration Physical Port>Flows>Flow(11)>Service Info

Service ID Name

Flow

Name:

Status:

Classifier:

Ingress Port:

Gold

22

Enable

match vlan-min 0 pbit-min 5 vlan-max 4095 pbit-max 5

SFP/port 1

Name:

Status:

Classifier:

Ingress Port:

11

Enable

match vlan-min 0 pbit-min 5 vlan-max 4095 pbit-max 5

MSA/port 2

MA

MD Level:

MA Name:

2

MA4

Service Summary

- In the CLI, go to the **configure service** context, and enter:
show summary

For example:

```
# show summary
service name:      Gold
=====
Flow              |
-----
```

: 22
: 11

MA
:MA5

5 Ports

The MiNID ports are as follows, according to the ordering option:

- SFP sleeve – MiNID has two Ethernet ports, one at each end, designated as:
 - SFP – SFP slot, for provider-side SFP plug
 - MSA – SFP plug side, for plugging into a customer device
- MiNID SFP – MiNID has two Ethernet ports, one at each end, designated as:
 - SFP – SFP (5DH / 6DH according to the ordering option)
 - MSA – SFP plug side, for plugging into a customer device

5.1 Physical Ethernet Ports

You can configure autonegotiation behavior, and set the port MTU to a value from 2000–12000.

Note

- *In MiNID /FE, the physical link is set to 100 Mbps full duplex, and autonegotiation is disabled, therefore configuring autonegotiation is relevant only for MiNID /GbE.*
- *In MiNID /GbE, autonegotiation can be set for automatic discovery of the autonegotiation status between MiNID and the port with which it needs to communicate.*

Applicability and Scaling

Ethernet ports exist in all MiNID devices, according to the ordering options as stated above.

Standards Compliance

Autonegotiation is defined in [IEEE 802.3ab](#).

Benefits

Autonegotiation enables optimal communications between communicating ports of different devices, without error-prone manual configuration.

Factory Defaults

By default:

- Autonegotiation is disabled on both ports for MiNID/FE
- Autonegotiation is set for automatic discovery on both ports for MiNID/GbE
- MTU is set to 12000

Configuring Port Parameters

➤ To configure Ethernet port parameters:

Do one of the following:

- In the web interface:
 - a. Go to Configuration > Physical Port > Ethernet > SFP/Port 1 or Ethernet > MSA/Port 2.
 - b. Set Auto Negotiation:
 - Discovery – Configure automatic discovery of autonegotiation status
 - Enable – Enable autonegotiation
 - Disable – Disable autonegotiation
 - c. Set Force Direct I2C to Disable (valid for MSA port of MiNID SFP and SFP sleeve devices only).

To enable the Force Direct I2C mode, select **Enable**, save the configuration and reboot MiNID.

- a. Set Force Speed (for SFP sleeve only) to Disable.
- b. To enable the Force Speed mode, select Enable, save the configuration and reboot MiNID.

- c. Set MTU to the maximum data packet size that the port transmits (2000–12000).
- d. Click <Apply> to implement the changes, and then click <Save Configuration>.

MiNID

Configuration>Physical Port>Ethernet>SFP/Port 1

Port Capability	1000 Mbps, Full Duplex
Auto Negotiation	Discovery
Force Direct I2C	Disable
Force Speed	1000
MTU (Bytes)	12000

Port Parameters – SFP/Port 1

- In the CLI:
 - a. Go to the configure port ethernet sfp/1 or configure port ethernet msa/2 context. The prompt **config>port>eth(sfp/1)#** or **config>port>eth(msa/2)#** is displayed.
 - b. Set Auto Negotiation:
 - To configure automatic discovery of autonegotiation status, enter:
auto-negotiation discovery
 - To enable autonegotiation, enter:
auto-negotiation
 - To disable autonegotiation, enter:
no auto-negotiation
 - c. Force Direct I2C (for MiNID SFP and SFP sleeve devices only): By default, MiNID serves as a proxy between the host device and the SFP.
 - d. If force direct I2C is enabled, the proxy is disabled and MiNID is transparent between the host device and the SFP.
 - e. If force direct I2C is disabled, MiNID remains in proxy mode only when one of the following cases occurs:
 - Plugged SFP is DDM.

- Force Speed mode is enabled.
- f. If none of the conditions above are met, MiNID automatically forces the direct I2C connection and moves to transparent mode.

To enable the Force Direct I2C mode, go to the **configure port ethernet msa** context, enter **force-direct-i2c**, save the configuration, and reboot MiNID.

Note *Force Direct I2C is valid only for MSA port parameters.*

- a. Force Speed (for SFP sleeve only): Force the speed of the MSA port to 1 GbE. Save the configuration and remove the MiNID from the socket. Once you plug it back in, the new configuration is accepted. When working with 1GbE on the MSA side and FE on the SFP port, the traffic received from the MSA side must be shaped to 100 M.

Force Speed can be enabled only when the Force Direct I2C mode is disabled, i.e. when the host device is not connected directly to the SFP via I2C.

- a. To set MTU to the maximum packet size that the port transmits (2000–12000), enter:
mtu <mtu-size>

Viewing Port Status

You can view the operational status (Up/Down) and current traffic rate (100/1000 Mbps).

➤ To view port status:

Do one of the following:

- In the web interface, go to Monitoring > Physical Port > Ethernet > SFP/Port 1 or MSA/Port 2:

MiNID

Monitoring>Physical Port>Ethernet>MSA/Port 2

Operational status	UP
Rate (Mbps)	1000
Active Media	Fiber
Statistics	
SFP Details	

Port Status – MSA/Port 2

- In the CLI, go to the configure port context, and enter:
ethernet { sfp/1 | msa/2 }
show status

For example:

```
MiNID>config>port# ethernet msa
MiNID>config>port>eth(msa/2) # show status
Operational Status -      UP
Rate (Mbps)        -      1000
Active Media        -      Copper
```

Viewing Port Statistics

You can view per-port traffic statistics.

➤ To view port statistics:

Do one of the following:

- In the web interface, go to Monitoring > Physical Port > Ethernet > SFP/Port 1 or MSA/Port 2, and click Statistics:



Monitoring>Physical Port>Ethernet> MSA/Port 2 >MSA/Port 2 Statistics

	---RX---	---TX---
Total Frames	19519	1514
Total Correct Frames	19519	1514
Total Correct Octets	1346023	201966
Frames Discarded (FIFO Full)	0	0
FCS Errors	0	
Undersize Frames	0	
Oversize Frames	0	
LBM Reflector Frames	0	
LBM Reflector Octets	0	
Clear Statistics		

Port Statistics – MSA/Port 2

- In the CLI, go to the configure port context, and enter:

ethernet { sfp/1 | msa/2 }

show statistics

For example:

```
MiNID>config>port# ethernet msa/2
MiNID>config>port>eth(msa/2)# show statistics
```

	--Rx--	--Tx--
Total Frames	329	293
Correct Frames	329	293
Total Correct Octets	29074	24332
Frames Discarded (FIFO Full)	0	0
FCS Errors	0	--
Undersize Frames	0	--

Overflow Frames	0	--
LBM Reflector Frames	0	--
LBM Reflector Octets	0	--

For each of Tx (Transmit) and Rx (Receive), the displayed statistics include the following parameters:

Ethernet Statistics Counters

Parameter	Description
Total Frames	All traffic, in number of frames
Correct Frames	All traffic not included in FCS Errors and not included in Undersize/Oversize Errors, in number of frames
Total Correct Octets	Same as Correct Frames, in bytes
Frames Discarded (FIFO Full)	Buffer overflow
FCS Errors	Frames not matching their Frame Check Sequence checksums
Undersize Frames	Frames under 64 bytes
Oversize Frames	Frames over the MTU size (default = 12000 bytes)
LBM Reflector Frames	LBM frames received by reflector
LBM Reflector Octets	LBM frame octets received by reflector

You can clear the stored traffic statistics for each port, resetting all values to 0.

➤ To clear statistics for a port:

Do one of the following:

- In the web interface, go to Monitoring > Physical Port > Ethernet > SFP/Port 1 or MSA/Port 2 > Statistics, and click Clear Statistics.
- In the CLI, go to the configure port context, and enter:
ethernet { sfp/1 | msa/2 }
clear statistics

5.2 SFP and DDM

MINID can display SFP and DDM details.

Applicability and Scaling

SFP and DDM details can be displayed in all MiNID devices.

Benefits

DDM details are displayed via the MiNID.

Displaying SFP and DDM Details Using the Web Interface

If the SFP inserted into the MINID SFP sleeve is SGMII, DDM details are not displayed.

➤ To display SFP and DDM details:

- Navigate to MiNID > Monitoring > Physical Port >Ethernet > SFP.
SFP and DDM information is displayed.

MiNID

Monitoring>Physical Port>Ethernet>SFP

Operational Status	Down
Rate Mbps	Not Available
Active Media	None
Ethernet Compliance Codes	1000BASE SX
Connector Type	LC
Manufacturer Name	WTD

Typical Maximum Range (Meter)	550
Wave Length (nm)	850
Fiber Type	MM
RX Power (dBm)	-31.5 dBm
TX Power (dBm)	-5.8 dBm
Laser Bias (mA)	0.9 mA
Laser Temperature (Celsius)	57.0 C
Power Supply	3.18 V
Statistics	

Displaying SFP and DDM Details Using the CLI

- To display SFP and DDM details:
- Navigate to configure port ethernet sfp and enter show status.
The SFP and DDM details are displayed.

6 Management and Security

6.1 Access Control List

Access control lists are used to flexibly filter incoming management traffic by the source IP address.

Applicability and Scaling

A single access group list can be configured per device.

Standards Compliance

Relevant sections of [RFC 1812](#).

Benefits

The service providers use the ACLs to maintain the network security by preventing the malicious traffic from entering the device. The ACLs can be used to save the network resources by dropping the unwanted packets.

Functional Description

Devices featuring ACLs can flexibly filter management traffic, by denying or permitting IP packets to enter or exit entities in the device, according to the packet's IP address and mask.

ACL entries are sequentially numbered rules containing statements (Deny or Permit).

Packets are permitted or denied access, based on the following conditions:

- IP source

- Mask

For the ACL to take effect, it must be enabled using the **configure access-control** command in CLI or via **Configuration>System> Management>Access Control** page in the web interface.

When ACL is enabled, by default, MiNID denies any management packet, unless there is a matching rule that permits the source IP address of the packet.

Note

- *Working with ACL and IPv6, you must create a permit rule for the Link Local IP in addition to the Global IP address.*
- *You have to add rules to permit IP addresses of any server that communicates with MiNID (RADview, NTP, etc.) for proper operation, otherwise the traffic from these servers will be blocked by ACL.*

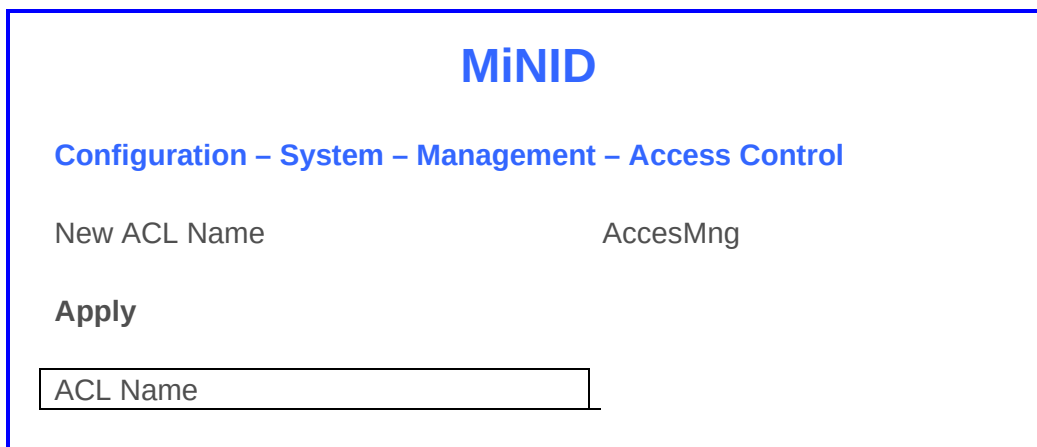
Factory Defaults

By default, no access list is defined on the MiNID.

Configuring ACL Using the Web Interface

➤ **To configure ACL:**

1. Navigate to Configuration > System > Management > Access Control.
2. In the **New ACL Name** field, type the ACL name.



MiNID

Configuration – System – Management – Access Control

New ACL Name AccesMng

Apply

ACL Name

3. Click <Apply>.

- 4. In the table, click the ACL name to configure the list.
The ACL Configuration page opens.

MinID

Configuration – System – Management – Access Control – Config
ACL AccessMng

Action

Permit

IP-net

0.0.0.0

/32

Sequence

10

Add

Access list records

Sequence #	Action	IP net
------------	--------	--------

Activate

Disable

Apply

- 5. Create the access list:
 - a. Select the desired action: Deny/Permit.
 - b. In the IP-net field, type the IP address and mask (the default mask is 32).
 - c. Click <Add>.The item is added to the **Access list records** table.
- 6. In the **Activate** field, select **Enable** to activate the ACL.
- 7. Click <Apply>.

Configuring ACL Using the CLI

The ACL configuration tasks are performed at the access control and management levels.

➤ To configure ACL:

- 1. Configure the ACL using the commands listed in the table below:

Task	Command	Comments
Create ACL name	<code>access-list <acl-name></code>	

Task	Command	Comments
Add statements to the list	permit ip <source-IP>/ <src-prefix-length> <sequence-number>	• In this step you can configure the IP address/mask and the sequence number. • A mask and sequence number are automatically assigned if none are configured. • You can either “permit” or “deny” a source IP • The sequence number is generated automatically if not configured.
Delete a statement from the ACL	delete <sequence-number>	
Attach access list to management	config>management>access>[no] access-group <acl-name>	

Example:

```
MiNID>config>access-control# access-list accessmng
MiNID>config>access-control# access-list (accessmng)# permit ip 168.15.168.5/32
Config>management>access> access-group <accessmng>

MiNID>config>access-control# access-list accessoki
MiNID>config>access-control# access-list (accessokig)# deny ip 168.15.169.5/32
Config>management>access> access-group <accessoki>
```

6.2 Authentication via TACACS+ Server

TACACS+ (Terminal Access Controller Access Control System Plus) is a security application that provides access control for routers, network access servers, and other networked computing devices via one or more centralized servers. TACACS+ provides separate authentication, authorization, and accounting services. It is used to communicate between MiNID and an authentication database. Because TACACS+ is based on TCP, implementations are typically resilient against packet loss.

Applicability and Scaling

Authentication with TACACS+ is applicable to all MiNID devices.

MiNID supports up to five accounting groups, with up to five TACACS+ servers. Each TACACS+ server can be bound to a single accounting group only.

Standards Compliance

TACACS+ Protocol Version 1.78 (IETF draft-grant-tacacs-02)

Benefits

The TACACS+ protocol allows centralized authentication and access control, avoiding the need to maintain a local user data base on each device on the network. The TACACS+ server encrypts the entire body of the packet but leaves a standard TACACS+ header.

Functional Description

TACACS+ is a protocol that provides access control for routers, network access servers and other networked computing devices via one or more centralized servers. TACACS+ is based on AAA model:

- Authentication – The action of determining identity of a user.
- Authorization – The action of determining what a user is allowed to do. It can be used to customize the service for the particular user.
- Accounting – The action of recording what a user is doing, and/or has done.

The TACACS+ client can be configured to use authentication/authorization functionality, accounting functionality, or both.

Components

The TACACS+ remote access environment has three major components: access client, TACACS+ client, and TACACS+ server.

- The access client is an entity which seeks the services offered by the network.

- TACACS+ client, running on MiNID, processes the requests from the access client and passes this data to TACACS+ server for authentication.
- The TACACS+ server authenticates the request, and authorizes services over the connection. The TACACS+ server does this by matching data from the TACACS+ client's request with entries in a trusted database.

TACACS+ server decides whether to accept or reject the user's authentication or authorization. Based on this response from the TACACS+ server, the TACACS+ client decides whether to establish the user's connection or terminate the user's connection attempt. The TACACS+ client also sends accounting data to the TACACS+ server to record in a trusted database.

TACACS+ uses TCP for its transport and encrypts the body of each packet. TACACS+ client and server can agree to use any port for authentication and accounting. TACACS+ supports authentication by using a user name and a fixed password.

Accounting

MiNID supports up to five accounting groups, with up to five TACACS+ servers. Each TACACS+ server can be bound to a single accounting group only.

A group can be defined with its own accounting level:

- Shell accounting, which logs the following events:
 - Successful logon
 - Logon failure
 - Logoff
 - Terminated management session
- System accounting, which records system events/alarms registered in local log file
- Command accounting, which logs the following events:
 - Any command entered by the user and successfully executed by MiNID

TACACS+ server log displays the exact commands as they were typed.

Factory Defaults

The following default values are defined:

Parameter	Default Value
privilege level	su = 12, user = 3
wait-to-restore-server	0

By default, no TACACS+ servers are defined. When the TACACS+ server is first defined, it is configured as shown below.

Parameter	Default Value
administrative status	shutdown
accounting port	49
authentication-port	49
key	empty string
retry	3
timeout	5 seconds

Note

When MiNID is set to the Config operation mode, authentication with TACACS+ server is disabled.

Configuring Authentication via TACACS+ Server

Configuring Access

To configure the authentication method, refer to the [Configuring Management Access](#) section.

Configuring Privilege Levels

You can define custom authorization levels by mapping the TACACS+ levels (from 0 to 15) to MiNID user levels (user, su).

TACACS+ Privilege Levels

Level	User	Allowed Actions	Description
3	user	Monitoring	Commands that do not affect services, traffic, or configuration
12	su	User management	Commands that manage users in the database

A user type can be mapped to multiple TACACS+ privilege levels, i.e. su can have level of 10, 12, and 15 simultaneously.

The TACACS+ privilege level should be mapped to only one user type.

➤ **To configure a custom privilege level in the web interface:**

1. Navigate to Configuration>System>Management>TACACS Plus.
2. In the levels table, select the boxes with the desired authorization levels.
3. Click <Apply>.

➤ **To delete a custom privilege level in the web interface:**

1. In the levels table, unselect the boxes that you do not need.
2. Click <Apply>.

➤ **To configure a custom privilege level in the CLI:**

1. At the **config>mngmnt>tacacsplus#** prompt, enter **privilege-level** <tacacs-privilege-level> {su | user}.

➤ **To delete a custom privilege level in the CLI:**

1. At the **config>mngmnt>tacacsplus#** prompt, enter **no privilege-level** <tacacs-privilege-level> {su | user}.

Note *At least one privilege level must be assigned to the **su** user level.*

Configuring TACACS+ Servers

MinID provides connectivity to up to five TACACS+ authentication servers. You must specify the associated server IP address, key, number of retries, etc.

Note *If you intend to use TACACS+ for authentication, verify that TACACS+ is selected as level-1 authentication method (refer to the **Management Access Control** section).*

- **To configure a TACACS+ server in the web interface:**
1. Navigate to Configuration>System>Management>TACACS Plus as shown below.

MiNID

Configuration>System>Management>Server

Wait To Restore Server 0

CLI Level/TACACS+ Level	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
SU													V			
USER				V												

Apply

Add New Group

Group Name

Add New Server

Server IP	Admin Status	
1.1.1.1	Active	Remove
10.10.10.10	Not Active	Remove
1.1.14.1	Active	Remove

TACACS+ Levels Configuration

1. For a new server, click **Add New Server**.
The TACACS+ server configuration page is displayed.

MiNID

Configuration>System>Management>TACACS Plus>New Server

IP Address	0.0.0.0
Accounting Port	49
Authentication Port	49
Group	None
Key	
Retry	3
Timeout[s]	5
Admin Status	Disable

Apply

- Enter the IP Address (IPv4 or IPv6) of the server.
- Enter the rest of the parameters according to the tasks listed in the table below.
- Click <Apply>.

➤ To configure a TACACS+ server in the CLI:

- At the **config>mngmnt>tacacsplus#** prompt, type **server** <ip-address> to specify the server IP address.
The **config>mngmnt>tacacsplus>server(<ip-address>)#** prompt is displayed.
- Enter the necessary commands according to the tasks listed below.

TACACS+ Server Configuration Parameters

Task	Command	Comments
Defining the TCP port to be used for accounting	accounting-port <port-number>	Possible values: 1–65535
Defining the TCP port to be used for authentication	authentication-port <port-number>	Possible values: 1–65535
Binding server to a group	[no] group <group-name>	

Task	Command	Comments
Defining a non-disclosed string (shared secret) used to encrypt the user password	key <string> [hash]	The shared secret is a secret key consisting of free text known to the client and the server for encryption. The hash keyword denotes that the string is hashed, rather than clear text; usually it is added by the device after hashing the clear text that the user enters, before saving it in the database. If you enter the password as a text string, do not use the hash parameter. Use it only if you are specifying the password as a hashed value (obtained by using the info command to display TACACS+ data). Note: In the web interface, you can only enter a text string. Any hashed string is interpreted as plain text.
Defining the number of recurrent authentication request attempts	retry <number-of-retries>	Possible values: 1–10
Defining the time period (in seconds) between two requests to the TACACS+ server	timeout <seconds>	Possible values: 1–255 Note: TACACS+ server reply is expected within 1 second.
Administratively enabling server	no shutdown	shutdown administratively disables the server
Displaying statistics	show statistics	
Clearing statistics	clear-statistics	

Sometimes, TACACS+ server doesn't reply, so another server should be connected. You can define how much time MiNID should wait before trying the non-replying server again.

➤ **To define the time to wait for the server restoration:**

1. At the **config>mngmnt>tacacsplus#** prompt, type **wait-to-restore** <seconds> to define the period of time in seconds.

Configuring Accounting Groups

➤ To configure accounting groups in the web interface:

1. Navigate to **Configuration>System>Management>TACACS Plus**.
2. For a new group, click **Add New Group**.

The group configuration page is displayed.

MiNID

Configuration>System>Management>TACACS Plus>New Group

Group Name	Group1
Accounting Shell	Disable
Accounting System	Disable
Accounting Commands	Disable

Apply

1. Set the Enable/Disable status for each accounting profile.
2. Click **<Apply>**.
3. Click **<Back>** to edit the server parameters.
4. Click the server name.
5. In the **Group** list, choose the desired group name to bind the group to the TACACS+ server.
6. Click **<Apply>**.

➤ To configure accounting groups in the CLI:

1. At the **config>mngmnt>tacacsplus#** prompt, type **group <group-name>** to configure an accounting group with the specified name.
The **config>mngmnt>tacacsplus>group(<group-name>)#** prompt is displayed.
2. To define the accounting for the group, enter:
accounting [shell] [system] [commands]

Note

- *You can enter any combination of the parameters shell, system, or commands, but you must enter at least one of them*
- *Type no accounting to disable TACACS+ accounting for the group.*

3. Type **exit** to return to the TACACS+ level.
The **config>mngmnt>tacacsplus#** prompt is displayed.
4. Type **server <ip-address>** to select the TACACS+ server to which to bind the group.
The **config>mngmnt>tacacsplus>server(<ip-address>)#** prompt is displayed.
5. At the **config>mngmnt>tacacsplus>server(<ip-address>)#** prompt, enter **group <group-name>** to bind the previously defined accounting group to the TACACS+ server.

Examples

Defining Server

The example below illustrates the procedure for defining a TACACS+ server.

- Server IP address: 175.18.172.150
- Key: TAC_server1

```
exit all
configure management tacacsplus
  server 175.18.172.150
    key TAC_server1
    no shutdown
exit all
save
```

➤ To display the configuration from the above example:

```
# configure management tacacsplus server 175.18.172.150
>config>mngmnt>tacacsplus>server(175.18.172.150)# info
  authentication-port 49
  key 244055BF667B8F89225048C6571135EF hash
  retry 3
  timeout 5
  no shutdown
```

Defining Accounting Group

The example below illustrates the procedure for defining an accounting group.

- Group name: TAC1
- Accounting: Shell, system, and commands
- Bound to server defined in Defining Server.

```
exit all
configure management tacacsplus
  group TAC1
    accounting shell system commands
  exit
  server 175.18.172.150
    group TAC1
  exit all
```

➤ To display the configuration from the above example:

```
# configure management tacacsplus server 175.18.172.150
>config>mngmnt>tacacsplus>server(175.18.172.150)# info
  key "244055BF667B8F89829AB8AB0FE50885" hash
  retry 1
  timeout 5
  authentication-port 49
  accounting-port 49
  group "TAC1"
  no shutdown
```

Viewing TACACS+ Statistics

Information presented by statistic counters refers only to authentication, not to authorization.

➤ To display TACACS+ statistics in the web interface:

1. Navigate to Configuration>System>Management>TACACS Plus.
2. Click the IP address of the server.

The TACACS+ server update page is displayed.

MiNID

Configuration>System>Management>TACACS Plus>Update Server

Statistics

IP Address	175.18.172.150
Authentication Port	49
Key	244055BF667B8F89225048C6571135EF
Retry	3
Timeout[s]	5
Admin Status	Enable

Apply

3. Click Statistics.
- The TACACS+ server statistics page is displayed. For statistics counters description, refer to the table below.

Configuration>System>Management>TACACS Plus>Statistics

Requests	5
Request Timeouts	60
Unexpected Responses	12
Server Error Responses	2
Incorrect Responses	3
Transaction Successes	256
Transaction Failures	5
Pending Requests	1

Clear Statistics

TACACS+ Server Statistics

➤ **To display TACACS+ statistics in the CLI:**

- At the `config>mngmnt>tacacsplus>server(<ip-address>)#` prompt, type:
`show statistics`.

The TACACS+ statistic counters are displayed.

```
>config>mngmnt>tacacsplus>server(175.18.172.150)# show statistics
Requests          5
Request Timeouts  60
Unexpected Responses 12
Server Error Responses 2
Incorrect Responses 3
Transaction Successes 256
Transaction Failures 5
Pending Requests  1
```

TACACS+ Statistic Counters

Counter	Description
Requests	Number of authentications performed toward a specific TACACS+ server
Request Timeouts	Number of transaction timeouts that occurred between the client and server
Unexpected Responses	Number of times the TACACS+ client receives a TACACS+ packet that is not expected at that time. Usually, this occurs due to a delayed response to a request that has already timed out
Server Error Responses	Number of errors received from the TACACS+ server
Incorrect Responses	Number of times the TACACS+ client: • Fails to decrypt the packet • Detects an invalid field in the TACACS+ packet • Receives a response that is not valid according to the initial request
Transaction Successes	Number of successful transactions between the client and TACACS+ server
Transaction Failures	Number of times the TACACS+ client's request is aborted by the TACACS+ server or the server fails to respond after maximum retry is exceeded
Pending Requests	Number of TACACS+ client's requests minus number of TACACS+ server responses or timeouts

➤ **To clear TACACS+ statistics in the web interface:**

1. Navigate to `Configuration>System>Management>TACACS Plus>Statistics` as shown on the above figure.
2. Click `Clear Statistics`.

TACACS+ statistic counters are set to 0.

► **To clear TACACS+ statistics in the CLI:**

1. At the config>mngmnt>tacacsplus>server(<ip-address>)# prompt, type:
clear-statistics.

TACACS+ statistic counters are set to 0.

6.3 Management Access Control

You can enable or disable management access via either port ({SFP/MSA}/{1/2}). You cannot disable access for both ports simultaneously. In addition, you can configure access via HTTP/SNMP/SSH/Telnet/TFTP/SFTP applications. By disabling Telnet, SSH, or SNMP, you can prevent unauthorized access to the system when the security of the MiNID IP address has been compromised. By denying specific IP addresses or subnets, you can block unauthorized access to the system.

Note *To allow management, enable at least one protocol.*

Applicability and Scaling

For loaned IP access, only four applications can be opened simultaneously out of six available applications (SNMP, TFTP/SFTP, SSH, Telnet, HTTP, and RPCAP).

Functional Description

Access via HTTP/SNMP/SSH/Telnet/TFTP/SFTP is configured according to the IP address mode configured in the router interface parameter settings:

- If the IP address is configured statically or via DHCP, then the access can be enabled or disabled per access type (HTTP/SNMP/SSH/Telnet/TFTP/SFTP). If a management packet contains the standard TCP/UDP port number of an enabled access type, then it is processed by MiNID, otherwise it is discarded.
- If the IP address is loaned from the host device, then in addition to specifying whether a particular access type is enabled, you also specify which TCP/UDP port number is used to indicate to MiNID whether to process the packet or forward it to the host device:

- Automatic – A predefined TCP/UDP port number is associated with the access type. The predefined TCP/UDP port numbers are:
 - SNMP port – 58888
 - TFTP/SFTP port – 58889
 - Telnet port – 0
 - SSH port – 58888
 - HTTP port – 0
- Port number configured by user – You configure the TCP/UDP port number associated with the access type.

If a management packet with a predefined/configured TCP/UDP port number associated with an enabled access type is received by MiNID, then the packet is processed by MiNID, otherwise it is forwarded to the host device.

Access can be granted by authorization via a local or TACACS+ server.

Factory Defaults

By default, access to both ports and all applications (predefined TCP/UDP port in the case of loaned IP) is enabled.

By default, SNMP read-only mode is disabled, which means that SNMP messages are available for reading and writing.

For Loaned IP access, RPCAP mode is disabled by default.

By default, authorization is performed via a local server; authorization via a TACACS+ server is disabled.

Parameter	Default Value
auth-policy	local
wait-to-restore-auth-method	No wait-to-restore-auth-method

Configuring Management Access

➤ To configure management access:

Do one of the following:

- In the web interface, go to Configuration > System > Management > Management Access:

MiNID

Configuration – System – Management - Management Access

MSA/Port 1	_____	Enable
SFP/Port 2	_____	Enable
SNMP Read Only	_____	Disable
Static IP/DHCP Access:		
SNMP	_____	Enable
TFTP/SFTP	_____	Enable
SSH	_____	Enable
TELNET	_____	Enable
HTTP	_____	Enable
Loaned IP Access:		
SNMP mode	_____	Auto
SNMP port	_____	58888
TFTP/SFTP mode	_____	Auto
TFTP/SFTP port	_____	58889
SSH mode	_____	Auto
SSH port	_____	58888
TELNET mode	_____	Auto
TELNET port	_____	0
HTTP mode	_____	Auto
HTTP port	_____	0
RPCAP mode	_____	Disabled
Authentication Policy		
First Level	Local	Second Level none
Wait to restore auth method	_____	
Apply	_____	

- a. Enable or Disable either of the ports.
- b. Enable or Disable SNMP read-only mode:
 - Disable – SNMP is in the read-and-write mode
 - Enable – SNMP is in the read-only mode
- c. According to the IP addressing mode configured in the host settings:
 - Static IP or DHCP – Enable or disable each type of access
 - Loaned IP – For each type of access, set the mode (and port if applicable) as follows:
 - Disabled – Access is not allowed via this access type
 - Auto – If a management packet is received by MiNID with a TCP/UDP port number that matches the predefined port for this access type, then the packet is processed by MiNID (see Functional Description for a list of the predefined TCP/UDP port numbers), otherwise the packet is forwarded to the host device.
 - Port – If a management packet is received by MiNID with a TCP/UDP port number that matches the configured port number for this access type, then the packet is processed by MiNID, otherwise the packet is forwarded to the host device.

Note *RPCAP port can be configured only using Packet Capture configuration. Refer to the **Packet Capture** section in **Chapter 11** for more details.*

- d. Authentication Policy:
 - First Level – authentication method that is applied first
 - Second Level – authentication method that is applied after the first method
- e. Click <Apply> to implement the changes, and then click <Save Configuration>.
- In the CLI:
 - a. Go to the config>mngmnt>access level.
 - b. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Setting the authentication policy	auth-policy 1st-level {local tacacs+} [2nd-level {local tacacs+ none}]	By default, only local authentication is set. See Authentication via TACACS+ Server
Setting the time period for the previously failed method to be restored	wait-to-restore-auth-method <seconds>	
Enabling or disabling MSA port	msa	Use no msa to disable the port
Enabling or disabling SFP port	sfp	Use no sfp to disable the port

Task	Command	Comments
Enabling or disabling SNMP read-only mode	[no] snmp-read-only	
Commands in level config>mngmnt>access>static-ip-access , to be used when loaned IP is not enabled in the host settings		
Enabling access for the various access types	snmp tftp ssh telnet http	The tftp command configures access for TFTP/SFTP
Disabling access for the various access types	no snmp no tftp no ssh no telnet no http	The tftp command configures access for TFTP/SFTP

Task	Command	Comments
Commands in level config>mngmnt>access>loaned-ip-access , to be used when loaned IP is enabled in the management router interface settings		
Enabling access for the various access types and specifying predefined or configured TCP/UDP port number	snmp loaned-ip {auto port} <port-num> tftp loaned-ip {auto port} <port-num> ssh loaned-ip {auto port} <port-num> telnet loaned-ip {auto port} <port-num> http loaned-ip {auto port} <port-num> rpcap loaned-ip {auto}	• auto – If a management packet is received by MiNID with a TCP/UDP port number that matches the predefined port for this access type, then the packet is processed by MiNID (see Functional Description for a list of the predefined TCP/UDP port numbers), otherwise the packet is forwarded to the host device. In this case, you do not specify the <port-num> parameter • port – If a management packet is received by MiNID with a TCP/UDP port number that matches <port-num>, then the packet is processed by MiNID, otherwise the packet is forwarded to the host device. Note: The tftp command configures access for TFTP/SFTP.
Disabling access for the various access types	no snmp no tftp no ssh no telnet no http no rpcap	Note: The tftp command configures access for TFTP/SFTP.

6.4 Network Managers

You can define up to two network managers to receive alarms/events/traps from MiNID.

Factory Defaults

By default, no SNMP Managers are configured (both manager IP addresses are set to 0.0.0.0).

Configuring Network Managers

➤ **To define network managers:**

Do one of the following:

- In the web interface, go to Configuration > System > Management > Manager List, enter one or two IP addresses of network manager servers, and click <Apply> to implement the changes, and then click <Save Configuration>:

MiNID

Configuration – System- Management - Manager List

Manager1	_____	192.168.42.248
Manager2	_____	10.10.36.61

Network Managers

- In the CLI, go to the config>mngmt level, and enter manager <IP address>.
You can do this twice to have two network managers. To remove a specific manager, enter **no manager** <IP address>.

6.5 Router Interface

MiNID supports up to five router interfaces.

Management traffic is transmitted via the router interface number 1 based on the configured IP address, static route and mask.

The router interface can be set with VLAN parameters, which will be added as a VLAN tag to the traffic transmitted via this interface.

Note *If VLAN is not configured, traffic is transmitted untagged.*

Functional Description

You can define the IP address as:

- DHCP – MiNID receives its IP address from a DHCP server for management router interface

Note *If DHCP is enabled, option 61 (client identifier) is supported by sending the device MAC address to the DHCP server.*

- Static – A specific IP address is configured for MiNID that is used for all types of access
- Loaned IP – MiNID uses the IP address of the host device. When packets arrive at MiNID, the packets are forwarded to the host device or are processed by MiNID, depending on the packet TCP/UDP port number (see Management Access Control for details).

The loaned IP feature allows a network management system such as RADview to communicate with MiNID and the MiNID host device by sending management packets to the same IP address but with different TCP/UDP port values, according to which packets should be processed by MiNID and which should be forwarded to its host device.

Note *If loaned IP address is enabled, TWAMP, Packet Capture and L3SAT configurations based on Router Interface # 1 cannot be enabled. However, TWAMP, Packet Capture and L3SAT configurations based on another router interface can be successfully enabled.*

Note *For loaned IP configuration having EtherType other than 0x8100, the EtherType should be defined specifically by **config flows tpid-outer-vlan** command.*

Factory Defaults

IP address is statically configured to 192.168.205.1.

No management VLAN is configured.

By default, only the router interface #1, which includes additional functionality for host management, is defined. No additional router interface is defined.

By default, IPv6 address is disabled.

Configuring Router Interface

➤ To configure a router interface using the web interface:

1. Navigate to Configuration > Router > Interface.

Configuration – Router - Interface

Add new Interface

ID	Admin Status	Interface IP	
1	Enabled	192.168.205.1/24	Delete Interface

Configuring a Router Interface

1. To add a new router interface:
 - Click Add new Interface.

Configuration – Router – Interface - New Interface

Admin Status	Disable
Interface ID	2
IPv4 Address	0.0.0.0
Network Mask	0.0.0.0
Interface Vlan	Untagged
Port	Port 1

Apply

New Router Interface

- Fill in the parameters as specified in the following table.
- Click <Apply> to create the router interface.

The **Configuration>Router>Interface** screen (see above) appears, with an entry containing the new router interface.

1. To delete a router interface, click **Delete Interface** in the corresponding row.
The router interface is deleted.
2. Click <Save Configuration>.

➤ To configure a router interface using the CLI:

1. Navigate to configure router interface <number>.
The **config>router>interface<number>#** prompt is displayed.
2. Enter all necessary commands according to the tasks listed below.

Router Interface Parameters

Task	Command	Comments
Assigning an IP address to the router interface	address <ip-address/mask>	The controller should have the same IP address as in the router interface
Associating router interface with VLAN	[no] vlan <vlan-id> pbit <p-bit> [inner-bit <vlan-id> inner-pbit <p-bit>]	

Task	Command	Comments
Binding router to Ethernet port	bind ethernet <1 2> or <SFP MSA>	For Router Interfaces 2-5, you can bind the router to either SFP or MSA port. For Router Interface 1, you can bind the router to both.
Administratively enabling or disabling the router	[no] shutdown	Enter shutdown to administratively disable the router

➤ **To disable a router interface:**

1. Go to configure router interface <number>.
2. Type shutdown.

Configuring Router Interface for Management

➤ **To configure the router interface in the web interface:**

1. Navigate to the Configuration > Router > Interface.
2. Click **1** in the row of the default router interface to change its parameters.

Configuration – Router – Interface - Update Management Interface

IPv4 Configuration:

IP addressing mode	Static IP
IP Address	192.168.205.1
Network Mask	255.255.255.0

IPv6 Configuration:

Static Address	Disable
Autoconfig	Disable
DHCPv6-Client	Disable

General Configuration:

Bind	Both
Host VLAN	Untagged

Apply

Management Router Interface Configuration

1. Enter all necessary commands according to the tasks listed in the table below.
2. Click **<Apply>**.

➤ To configure the router interface for management in the CLI:

Do one of the following:

1. Use a unified command:
 - a. Go to the config>router level.
 - b. Enter management configuration parameters:
management-cfg {router-interface} <ri-number> {address} <ri-ip-address-mask> [vlan <value> pbit <value> [inner-vlan <value> inner-pbit <value>]] {static-route} <dest-subnet> {address} <next-hop ip address>
2. Use a set of commands according to the tasks listed below.

Router Interface for Management

Task	Command	Comments
Configuring the IP address, if DHCP is not enabled	[no] address <ip-address/mask>	Specify the mask in CIDR notation, which means that it is a decimal number that specifies the number of high-order consecutive bits set to 1 in the mask. For example, the CIDR notation for mask 255.255.255.0, is 24. For router interface #1, the IP address can have IPv4 (e.g. 10.10.10.1) or IPv6 format (e.g. 10:10:10:10:10:10:10:10). • Note: Only IPv6 address can be deleted by entering no address in the router interface # 1.
Binding router to Ethernet port	bind ethernet {1} {2}	
Enabling or disabling DHCP	[no] dhcp	Use the no form to disable the mode (for management interface) You cannot enable DHCP (for IPv4) in the following cases: • Router interface is bound to a PPP port. • IPv4 address is configured. • Router interface is not unnumbered. DHCPv6 is enabled.
Enabling or disabling DHCPv6 client for the router interface	[no] dhcpv6-client	You can enable DHCPv6 client provided that the following conditions exist: • Router entity is Router # 1. • There is no other DHCPv6 client defined in the device. DHCPv4 is not enabled.
Enabling or disabling IPv6 autoconfiguration on the router interface	[no] ipv6-autoconfig	Enter no ipv6-autoconfig to disable IPv6 autoconfiguration.
Enabling or disabling loaned IP address mode	[no] loaned-ip	Use the no form to disable the mode. If loaned IP address is enabled, then DHCP cannot be enabled.

Task	Command	Comments
Displaying router interface status	show status	See Viewing Router Interface Status
Creating a management traffic VLAN by configuring the management VLAN ID and specifying its priority by setting the VLAN p-bits	[no] vlan <vlan-id> pbit <p-bit> [inner-vlan <vlan-id> inner-pbit <p-bit>]	Possible values for pbit: 0–7

Note IPv6 autoconfiguration must be enabled before you start configuring static address and DHCPv6 client.

Note Router Interface parameters cannot be changed if the interface is used for configuring TWAMP, Packet Capture, or L3SAT parameters.

Configuring Static Route

➤ **To configure the static route in the web interface:**

1. Navigate to Configuration > Router > Static Route.
2. Click Add new Static Route.

Configuration – Router - Static Route - New Static Route

IP Address	0.0.0.0
Subnet Mask	0.0.0.0
Next Hop Address	0.0.0.0
Apply	

1. Set the parameters and click <**Apply**>.

➤ **To configure the static route in the CLI:**

1. Go to the config>router level.

- 2. Type **static-route** <ip-address/mask> **address** <next-hop-ip-address>. You can define two separate entities for IPv4 and IPv6 default gateway.

Configuring Default Gateway

The default gateway can be configured, if DHCP is not enabled. You can set two separate entries: one for IPv4 address and another for IPv6 address.

➤ **To configure the default gateway in the web interface:**

- 1. Navigate to Configuration > Router > Static Route.

MinID

Configuration – Router - Static Route

Add new Static Route

Static Route IP	Network Mask	Next Hop IP	
0.0.0.0/0	0	0.0.0.0	Delete Static Route
::/0	0	::	Delete Static Route

- 1. In the **Next Hop IP** column, click the **0.0.0.0** entry to edit the IPv4 default gateway, or the **::** entry, to edit the IPv6 default router.
- 2. Enter the default gateway address and click **Apply**.

➤ **To configure the default gateway in the CLI:**

- 1. Go to the **config>router** level.
- 2. For IPv4 default gateway, type **static-route 0.0.0.0/0 address** <next-hop-ip-address>. For IPv6 default router, type **static-route 0::0/0 address** <next-hop-ip-address>.

Viewing the Configured Router Interface Parameters

➤ To view the configured router interface parameters in the web interface:

1. Navigate to Configuration > Router > Interface.
2. In the row of the router interface 1, click **1**.
The router interface parameters are displayed.

➤ To view the configured router interface parameters in the CLI:

1. Go to the **config>router** level.
2. Type **info**.

```
interface 1
  ipv6-autoconfig
  address 192.168.205.11/24
  no vlan
  bind ethernet sfp msa
  no dhcp
  no loaned-ip
  no dhcpv6-client
exit
static-route 0.0.0.0/0 address 192.168.205.11
static-route 0::0/0 address ::
```

Viewing Router Interface Status

You can view the router interface status by using the **show status** command:
config>router>interface(<interface-num>)>show status.

➤ To display the router interface status in the web interface:

1. Navigate to **Configuration > Router > Interface** and click the interface row in the table.

➤ To display the router interface status in the CLI:

```
MiNID>config>router>interface(1)# show status
Admin: up          Oper: Up
IP Addresses:
  101.101.101.1/24      (manual)          (preferred)
  1234:1234:1234:1234:1234:1234:1234:1234/126 (auto-configuration)(preferred)
  FE80::220:D2FF:FE12:3456/64      (link layer)      (preferred)

IPv4 Default Router:  1.1.1.254
IPv6 Default Router:  FE80::2AC7:CEFF:FEF4:68C4
```

The above fields are:

Router Interface Status Parameters

Field	Description
Admin	Administrative status: • up – ready to pass packets • down
Oper	Operational status: • up – ready to pass packets • down
IP Addresses	
IP Address/prefix length	IPv4 or IPv6 address and prefix length <i>Note: Supported for DHCPv6</i>
origin	Origin of the IP address. Possible origins are: • other • manual • DHCP • link layer • random
status	Status of the IP address. Available statuses (from the IPv6 Stateless Address Autoconfiguration protocol) are: • preferred (default) • deprecated • invalid • inaccessible • unknown • tentative • duplicate • optimistic
IPv4 Default Router	IP address of the IPv4 default router
IPv6 Default Router	IP address of the IPv6 default router

6.6 Session Timeout

Both management interfaces (web and CLI) disconnect the user session after a configurable time period without user action. The timeout value is a whole number of minutes in the range 1–60.

Factory Default

The default timeout is 5 minutes.

Configuring the Session Timeout

➤ **To configure the session timeout:**

Do one of the following:

- In the web interface, go to Configuration > System > Management > Management Timeout, set the Timeout and click <Apply> to implement the changes, and then click <Save Configuration>:



The screenshot shows the MiNID web interface. At the top, the title "MiNID" is displayed in blue. Below it, the navigation path "Configuration>System>Management>Management Timeout" is shown in blue. The main content area has a light gray background. On the left, the label "Timeout (Min.)" is followed by a text input field containing the value "5". A dashed line is visible below the input field.

- In the CLI, go to the config>mngmnt level, and enter timeout <minutes>.

6.7 SNMP Communities

You can configure the names for the read-only, read/write, and trap communities used for SNMPv1/v2 access.

➤ **To configure the SNMP communities:**

Do one of the following:

- In the web interface, go to Configuration > System > Management > SNMP

MiNID
Configuration>System>Management>SNMP

Community Read/Write	_____	private
Community Read only	_____	public
Community Trap	_____	public

SNMP Communities

- Set Community Read/Write to the name of a community with read/write authorization.
 - Set Community Read only to the name of a community with read-only authorization.
 - Set Community Trap to the name of a community for sending traps.
 - Click <Apply> to implement the changes, and then click <Save Configuration>.
- In the CLI, go to the `config>mngmnt>snmp` level, and enter the following command:
snmp community {rw | ronly | trap} <name>

Where:

- `rw` – indicates configuring read/write community to <name>
- `ronly` – indicates configuring read-only community to <name>
- `trap` – indicates configuring trap community to <name>.

6.8 User Access

Two user accounts exist for both of the two configuration interfaces (web and CLI). The usernames are fixed:

- `su`: Superuser account with permission for all configuration.
- `user`: Read-only user. Can view all current configuration.

The initial password for both accounts is: **1234**. For better security, change the password as explained below.

Up to four user sessions are accepted by the device at any one time. It is recommended not to exceed one web session at a time, which means that you can have one web session and three more CLI sessions simultaneously.

As the superuser (su), you can modify the username and password for both user accounts (**su** and **user**).

As the read-only user, you cannot configure your password. The following procedure is for the superuser to change credentials.

➤ **To configure the user accounts:**

Do one of the following:

- Use the web interface:
 - a. In the web interface, go to Configuration> System > Management > User Access:
The User access screen appears. If you are logged in as the superuser, you can change your password or the read-only user password. If you're logged in as the read-only user, you cannot modify any parameters.

MiNID

Configuration>System>Management>User Access

User Level	_____	Super User
User Name	_____	su
Old Password	_____	****
New Password	_____	
Confirm New Password	_____	

User Access – Logged In as Superuser

MiNID

Configuration>System>Management>User Access

User Level
User Name

User
user

User Access – Logged In as Read-Only User

- b. In User Level, select Super User or User, according to which user credentials you wish to configure.
- c. Define the User Name.
- d. Type the Old Password, type the New Password and Confirm it.
- e. Click <Apply> to implement the changes, and then click <Save Configuration>.
- Use the CLI:
 - a. Log into the CLI as the superuser, and go to the `config>mngmnt` level.
 - b. To define the username and password of the superuser, enter:
user <new username> **level su password** <new password>
 - c. To define the username and password of the regular user, enter:
user <new username> **level user password** <new password>
If you're logged in as the read-only user (not the superuser), the command **user** is not available.

Note

Make sure not to assign the same username to the super user and the read-only user.

You can view the current usernames by going to the **config>mngmnt** level and entering **show users**.

7 Resiliency (Fault Propagation)

Fault propagation enables you to specify what actions to take when failure indication is received by one of the MiNID ports, e.g. whether to pass on the failure indication to the other MiNID port or block it.

Note *If no fiber is connected to the MSA (SFP sleeve option) port, ensure that all fault propagation actions are disabled (this is the default setting).*

7.1 Standards Compliance

Loss of Signal (LOS), Tx fault, and Tx disable are defined in the **CFP Multi-Source Agreement**.

7.2 Functional Description

MiNID handles the following failure indications:

- LOS received by the SFP port –indicates loss of signal, for instance it can be caused by disconnecting the fiber from the SFP that is inserted in MiNID.

You can configure MiNID to do any of the following upon receiving LOS from the provider-side SFP port, depending on the ordering option:

- SFP sleeve:
 - Propagate LOS to the customer-side MSA port
 - Disable the physical interface (PHY) towards the host
 - Send OAM CFM AIS to the MSA port (in this case it is recommended to configure a MEP with ingress port MSA, in order to transfer the AIS to the remote MEP)

Note

- *You can enable sending AIS only if propagating LOS and disabling host PHY are both disabled, and you must disable sending AIS in order to enable propagating LOS or disabling host PHY*
- *If you wish to block the LOS, you should disable all three actions.*

- (SFP sleeve only) TX disable received by the MSA port –transmitted by host to request shutting down the transmitter optical output. You can configure whether MiNID propagates the TX disable signal from the MSA port to the SFP port.
- (SFP sleeve only) TX fault received by the SFP port –indicates that a fault has occurred in the SFP. You can configure whether MiNID propagates the TX fault signal from the SFP port to the MSA port.

Note

For the SFP sleeve option, if you want the same fault indication behavior as if the SFP in MiNID were inserted directly into the host device, you should enable fault propagation for the TX-fault and TX-disable signals.

- Disable – When LOC or AIS failure is observed on any MEP, it is not propagated.
- Interface Status TLV:
 - Tx handling –The CCM to the peer MEP will carry “Interface Status TLV” with value of “isUp”. When the other side port is down, an “Interface Status TLV” will be sent in the CCM message to the peer MEP carrying value of “isDown”. The switching from “isUp” to “isDown” is triggered by interface deactivation, and vice versa.
 - Rx handling –The receiving MEP will be able to deactivate the other side port (LOS signal, Tx Disable signal).

7.3 Factory Defaults

By default, all fault propagation actions are disabled, so LOS is blocked.

7.4 Configuring Fault Propagation

► To configure fault propagation actions:

Do one of the following:

- In the web interface, go to Configuration > System > Fault Propagation, enable or disable the relevant propagation actions (see Functional Description), and click <Apply> to implement the changes, and then click <Save Configuration>:

Note

The following appear only for the SFP sleeve option:

- Disable Host PHY
- Propagate TX Disable Signal

Propagate TX Fault Signal.

MiNID

Configuration>System>Fault Propagation

Propagate LOS Signal

Disable

Disable Host PHY

Disable

Transmit AIS

Disable

Propagate TX Disable Signal

Disable

Propagate TX Fault Signal

Disable

CFM Fault : IF Deactivation

Disable

IF Status TLV: IF Deactivation

Disable

Fault Propagation

- In the CLI:
 - Go to the configure system context, and enter fault-propagation, to go into the config>system>fault context.
 - To enable LOS propagation from SFP/port 1 to MSA/port 2, enter: los-propagation
 - To specify a direction for propagating OAM CFM faults (LOC, Rx AIS), enter:
[no] cfm-fault {sfp/Port 1-msa/Port 2 | msa/Port 2-sfp/Port 1}
 - For the MEP to deactivate the other side port, if-status-tlv {SFP|Port 1-MSA|Port 2/MSA|Port 2-SFP|Port 1}.

Note

The following are relevant only for the SFP sleeve option:

- e. To disable the physical interface (PHY) towards the host in the event that LOS is received, enter: phy-disable
- f. If los-propagation and phy-disable are both disabled, you can enable sending an OAM CFM AIS when LOS is received, by entering: ais
- g. To enable propagating the TX disable signal from the MSA port to the SFP port, enter: tx-disable
- h. To enable propagating the TX fault signal from the SFP port to the MSA port, enter: tx-fault

8 Traffic Processing

8.1 Flows

Traffic is classified into flows, which are unidirectional entities that have an action on traffic. MiNID supports the following port-level classification mechanisms:

- Port Classification (see [Port Mode](#))
- Flow Classification (see [Flow Classification](#))

Applicability and Scaling

This feature is applicable to all MiNID options.

Standards Compliance

[IEEE 802.1q](#)

[IEEE 802.1ad](#)

[RFC 862](#) – Echo Protocol

Benefits

Flows enable traffic differentiation for different customers and for different traffic types, thus creating separate Ethernet services. For example, a VLAN can be added to MSA-to-SFP/ Port 2-to-Port 1 traffic and removed from SFP-to-MSA/ Port 1-to-Port 2 traffic to function as an S-tag to create an EVC. VLAN p-bits can be applied according to C-tag p-bits or according to DSCP to maintain a desired SLA.

Port mode is appropriate when other services are handled on other ports, for clearer service separation and simpler SLA measurement. On the other hand, flow-based mode enables handling different service

types on device, providing a cheaper, more scalable solution. Per-flow statistics enable per-service SLA measurements.

The layer-3/4 loopback feature allows you to define loopback tests according to IP/UDP parameters, enabling MiNID to support diagnostics tests for Layer-3 networks.

Functional Description

MiNID can work in port or flow-based mode.

Port Mode

In port mode (all-in-one bundling), MiNID processes customer and network traffic data frames consistently for all traffic coming into each of its two ports (SFP/Port 1 and MSA/Port 2); see [Physical Ethernet Ports](#).

When switching from flow-based mode to port mode, configured flows are disabled but retained. From port mode you can only change to the last configured flow classification mode, upon which the configured flows are again available.

Flow Classification

In flow-based mode, MiNID classifies frames by any of the following:

- Customer VLAN ID range and P-bit range

Note

- *VLAN ID ranges of different flows may not overlap.*
- *Double VLAN classification is supported.*
- *One flow can be marked as Untagged.*

- DSCP
- MAC addresses:
 - MAC source address
 - MAC destination address
- Ethertype

In flow-based mode, up to 64 flows can be configured (not including management traffic; includes disabled flows. Each flow applies to only one direction (SFP/Port 1 to MSA/Port 2 or MSA/Port 2 to SFP/Port 1); up to 32 flows per direction can be configured). All configured flows classify traffic according to the same criterion type.

Changing from one flow classification mode to another deletes all configured flows.

Flow Actions

For each configured flow, and for the unclassified (non-matched) incoming traffic of each port, any of the following actions can be defined:

Drop	Drop the frame
Pass Through	Pass the traffic through the device without changes
VLAN	Push (add) for single and double VLANs, pop (remove) for single and double VLANs, or replace a VLAN tag, as specified. Pushing adds a VLAN tag before the existing tag, if there is one. Popping removes the first tag, if there is at least one. Replacing overwrites the first tag, if there is at least one. When pushing or replacing a VLAN tag, the new tag's P-bit can be configured to be set in any of the following ways, per flow: • Fixed Value for the flow • C-Tag Copy: The new tag's P-bit is copied from the originating frame's P-bit • DSCP Marking: The new tag's P-bit is set according to a separately configured mapping of originating frames' DSCP values to P-bit values. All flows configured for DSCP-to-P-bit mapping use the same mapping. • C-Tag Marking: The new tag's P-bit is set according to a separately configured mapping of originating frames' P-bit values to new P-bit values. All flows configured for P-bit-to-P-bit mapping (not copying) use the same mapping.
Loopback	The traffic is echoed back through the same port, with or without swapping the source and destination addresses (MAC, and where relevant also IP), as specified. The looped traffic is passed through to the other port (intrusive loopback).

Note

If the configured flow action is not supported by the arriving frame (e.g. non-IP frame for DSCP remarking or untagged frame for C-tag remarking), the new tag's P-bit is set to 0.

The Drop Eligible bit can be set to fixed value (Yes / No) or defined automatically by Policer if it is enabled.

L2CP/L2PT

MiNID can discard, pass through, tunnel, or peer packets that are classified per flow or per port as corresponding to a particular L2CP/L2PT protocol.

L2PT protocol classification is based on the packet destination MAC address, Ethertype, and Subtype as specified in the following table.

L2PT Protocol Classification

Protocol	MAC Address	Ethertype	Subtype
STP	01-80-C2-00-00-00	All	0x42
OAM	01-80-C2-00-00-02	0x8809	0x03
PAgP	01-00-0C-CC-CC-CC	0x0104	0xAA
LLDP	01-80-C2-00-00-0E	0x88CC	All
CDP	01-00-0C-CC-CC-CC	0x2000	0xAA
VTP	01-00-0C-CC-CC-CC	0x2003	0xAA
UDLD	01-00-0C-CC-CC-CC	0x0111	0xAA
DTP	01-00-0C-CC-CC-CC	0x2004	0xAA
PVSTP	01-00-0C-CC-CC-CD	0x010B	All
LACP	01-80-C2-00-00-02	0x8809	0x01
LAMP	01-80-C2-00-00-02	0x8809	0x02
ESMC	01-80-C2-00-00-02	0x8809	0x0A
PTP-Peer-Delay	01-80-C2-00-00-0E	0x88F7	All

L2CP protocol classification is user-configurable and based on the destination MAC address 01-80-C2-00-00-XX, where XX is 00–2F (excluding values used in the table above).

For each L2PT protocol, you can define changing the destination MAC address to a user-configured value.

If you configure a new L2CP protocol classification, or a change to an L2PT MAC address, that counts as an L2CP/L2PT entry. MiNID supports up to 31 L2CP/L2PT entries; 13 entries are already occupied by the predefined L2PT classifications, therefore you can configure up to 18 additional entries.

The L2CP/L2PT handling is done by creating L2CP profiles and associating them with a port or a flow. In an L2CP profile, you can configure an action for each protocol:

- Discard – Packet is dropped
- Pass through – Packet is transmitted unchanged to the egress port; MAC address is not changed
- Peer – Packet is delivered to host treatment (OAM protocol only)
- Tunnel – Packet is classified to a flow and handled accordingly; MAC address is changed if a new MAC address has been defined

Note *The action for OAM protocol must be set to peer in order for link OAM to function. However, when packets such as ESSM packets are received with MAC address corresponding to OAM protocol (01-80-C2-00-00-02), and OAM action is set to peer, they are discarded. If the OAM action is set to pass through, then the ESSM packets are not discarded, but link OAM does not function.*

When you add an L2CP/L2PT entry by configuring an L2CP protocol classification or an L2PT MAC address change, the entry is added to any existing L2CP/L2PT profile, with the action set to tunnel.

Note

- *You can create up to eight L2CP profiles.*
- *An L2CP profile can be associated with a flow only if the flow classification mode is VID+Pbits Range.*

Loopbacks

MiNID provides the following types of loopbacks:

- Layer-2 loopbacks (see [Layer-2 Loopbacks](#))
- Layer-3/4 loopbacks (see [Layer-3/4 Loopbacks](#))

Layer-2 Loopbacks

To enable testing flow connectivity without interrupting traffic, you can configure MiNID with dedicated source and/or destination MAC addresses, representing testing devices whose frames should be looped back.

Up to four source MAC addresses and up to four destination MAC addresses for MiNID as a whole can be configured for layer-2 loopback (referred to in the Web and CLI as in-service loopback). Frames with these MAC addresses can be looped back, according to service flow configuration. Flow configuration defines the following:

- The mode for matching flow-matching traffic to MAC addresses. One of:
 - **SA:** According to only source MAC address
 - **DA:** According to only destination MAC address

- **SA+DA:** Only when matching both source and destination MAC addresses
- Which of the four (or eight, in **SA+DA** mode) MAC addresses are enabled and which disabled for layer-2 loopback. Traffic is looped if it matches a flow for which its MAC address is enabled for layer-2 loopback.

Each flow's layer-2 loopback (for all the flow's enabled MAC addresses) can be configured to include source and destination swap. Source and destination swap is of MAC addresses, and, when the frame includes an IP header, of IP addresses.

Layer-3/4 Loopbacks

To enable testing flow connectivity without interrupting traffic, you can configure MiNID with a dedicated IP address and UDP port, representing testing devices whose frames should be looped back.

You can configure a dedicated IP address (and UDP port if applicable) for each layer-3/4 loopback mode, to be used as criteria whether to perform layer-3/4 loopbacks, and configure the layer-3/4 loopback mode per flow or per port (if port mode is enabled). The loopback modes are as follows:

- IP loop – Loopback packets whose destination IP address matches the dedicated IP address, with MAC + IP swap
- UDP loop – Loopback packets whose destination IP address matches the dedicated IP address, and UDP destination port matches the dedicated UDP port, with MAC + IP + UDP swap
- UDP echo – Perform UDP echo for packets whose destination IP address matches the dedicated IP address, and UDP destination port matches the dedicated UDP port.

The UDP echo is a mechanism where the source sends packets, and the responder sends the packets back to the source. If you want to work according to RFC 862 standard, configure the dedicated UDP port to 7.

IP Agnostic Loops

IP agnostic loops enable you to run UDP loop\UDP Echo or TWAMP tests without configuring the IP address. Only the UDP port is configured; the test runs for any IP address.

TWAMP Light Responders

Flows can be bound to TWAMP (Two-Way Active Measurement Protocol) responders, to be used to reflect TWAMP test packets (see **OAM (TWAMP)**). Flows are bound to TWAMP responders by specifying the flow layer-3/4 loopback type as TWAMP responder, and then selecting the TWAMP responder.

Note *Each flow can be bound to a single TWAMP responder.*

Factory Defaults

By default, no flows are configured.

General Flow Parameters

The following general (not per flow) classification and flow parameters can be configured in:

- The CLI **config>flows** context
- The Web interface, in and below Configuration > Physical Port > Flows Classification port name.

Flow General Parameter Defaults

Parameter	Description	Default Value
classification mode per port	How incoming frames should be classified: uniformly (port mode) or by a specified criterion	vlan-pbits-range
dscp-marking	Mapping of DSCP values to P-bit values, for enabled flows configured to push or replace VLAN tags with P-bits using this mapping	Each eight DSCP values (0-7, 8-15, ..., 56-63) are mapped sequentially to the eight P-bit values (0-7)
fallback-action	Action for frames from a specified ingress port that don't match any enabled flows, if the classification mode is not port mode	pass-through (for both ports)
fallback-l2cp	Set L2CP profile for the unclassified flow of the relevant port (for flow mode)	no fallback-l2cp
fallback-pm-collection	Set PM collection for the unclassified flow of the relevant port (for flow mode)	no fallback-pm-collection
fallback-policer	Set Policer profile for the unclassified flow of the relevant port (for flow mode)	
in-service-mac	MAC addresses (four source and four destination) to be made available to per flow loopback	in-service-mac src-addr <n> and in-service-mac dst-addr <n>, where n=1-4, all set to 00-00-00-00-00-00
inner-vlan-marking	Mapping of originating frame P-bit values to new tag P-bit values, for flows configured to push or replace VLAN tags with P-bits using this mapping	Each P-bit value is mapped to the same value
l2cp-mac	L2CP protocol classification	None

Parameter	Description	Default Value
l2cp-protocol	L2PT protocol MAC address changes	None (see L2PT Protocol Classification table for a list of the supported L2PT protocols)
l2cp-profile	Configuring L2CP/L2PT profile	None
port-l2cp	Defining L2CP/L2PT profile to use if classification mode is port mode	None
port-mode-action	Action for frames from a specified ingress port if the classification mode is port mode	pass-through (for both ports)
tpid-outer-vlan	TPID value used in flow packets and as the outer TPID of double-tagged OAM packets	0x88a8 (for both ports)
tpid-inner-vlan	TPID value for enabled flows configured to push or replace VLAN tags, for a specified traffic direction	0x8100 (for both ports)

Per-Flow Parameters

The following per-flow parameters can be configured in:

- The CLI **config>flows>flow(<flow-name>)** context
- The Web interface, in:
 - For new flows: Configuration > Physical Port > Flows > Flows Classification Port name>Add flow.
 - For existing flows: Configuration > Physical Port > Flows Classification port name> Config Flow > flow name
 - For flow layer-2 loopback parameters: Configuration > Physical Port > Flows > General Params > In-Serv Loops

Per-Flow Parameter Defaults

Parameter	Description	Default Value
classifier match	Frame values considered a match for this flow	Default values by Classification Mode are:

Parameter	Description	Default Value	
		Source MAC address	src-mac 00-00-00-00-00-00
		Destination MAC address	dst-mac 00-00-00-00-00-00
		VID range and P-bit range	vlan-min 0 pbit-min 0 vlan-max 0 pbit-max 0
flow-action	The ingress port of frames to which this flow applies, and the flow action	ingress-port sfp/1 action drop	
in-service-l3-l4-loops	Layer-3/4 loop mode	Disabled	
l2cp	L2CP profile assigned to flow	None	
service-name	Service name assigned to flow	None	
shutdown	Whether the flow is enabled (no shutdown) or disabled (shutdown)	shutdown	
Configured in in-service-loops context:			
in-serv-action	Layer-2 loopback with or without source-destination address swap	loop-mac-ip-swap	
in-serv-mode	Mode for matching flow-matching traffic to MAC addresses (see Layer-2 Loopbacks)	src-dst	
in-serv-dst-mac 1 to in-serv-dst-mac 4	Indicates whether this MAC destination address is enabled for layer-2 loopback for this flow	disable	
in-serv-src-mac 1 to in-serv-src -mac 4	Indicates whether this MAC source address is enabled for layer-2 loopback for this flow	disable	

Flow Configuration Tasks

General Tasks

The following tasks can be performed in the CLI **config>flows** context (unless otherwise specified), or in the Web interface.

General Configuration Tasks

Task	CLI Command	Web Interface Location	Comments
Define how incoming frames should be classified: uniformly (port mode) or by a specified criterion	classification mode <mode>	Configuration > Physical Port > Flows > Classification Mode	<mode> is one of the following: • port • dscp • ethertype • src-mac (source MAC address) • dst-mac (destination MAC address) • vlan-pbits-range (VLAN ID range and P-bit range)
Map a DSCP value to a P-bit value, for enabled flows configured to push or replace VLAN tags with P-bits using DSCP marking	dscp-marking dscp <DSCP value> s-pbit <new P-bit>	Configuration > Physical Port > Flows > General Params > DSCP->S-Tag Mapping	
Configure action for frames from a specified ingress port that don't match any enabled flows, if not in port mode	fallback-action ingress-port <sfp/1 msa/2> action <action> [<VLAN parameters>]	Configuration > Physical Port > Flows > General Params > General Params [SFP/Port 1 MSA/ Port 2]> Unclassified Action	<sfp/1 msa/2>defines the traffic direction that this flow applies to, by defining the ingress port as either SFP/port 1 or MSA/ port 2. For details of the <action> <VLAN parameters> parameters, see Action Parameters table

Task	CLI Command	Web Interface Location	Comments
Bind the unclassified flow to the specified L2CP profile	fallback-l2cp profile <profile-name> ingress-port <sfp/1 msa/2>	Configuration > Physical Port > Flows > General Params > General Params [SFP/Port 1 MSA/ Port 2]> Fallback L2CP Profile	
Enable PM collection and set the PM interval for the unclassified flow	fallback-pm-collection interval <interval-time>	Configuration > Physical Port > Flows > General Params > General Params [SFP/Port 1 MSA/ Port 2]> Fallback PM Collection	
Enable policer profile and bind it to the unclassified flow	fallback-policer ingress-port {1 2} policer {profile regular-accounting-only} <policer-profile-name>	Configuration > Physical Port > Flows > General Params > General Params [SFP/Port 1 MSA/ Port 2]> Fallback Policer Type	
Configure MAC destination or source address to be available for layer-2 loopback	in-service-mac dst-addr <index> <address> or in-service-mac src-addr <index> <address>	Configuration > Physical Port > Flows > General Params > In-Service Loop MACs	
Specify the IP address to match to packet destination IP addresses for IP loop layer-3/4 loopback	ip-loop ip-address <ip-dest-addr>	Configuration > Physical Port > Flows > General Params > In-Service L3\L4> IP Loop	Note: This must be a different IP address than the MiNID host IP address, or any IP address specified for a different loopback type or TWAMP responder.
Specify the IP address to match to packet destination IP addresses for UDP loop layer-3/4 loopback	udp-loop ip-address <ip-dest-addr>	Configuration > Physical Port > Flows > General Params > In-Service L3\L4> UDP Loop	Note: The combination of the IP address and UDP port specified for UDP loop must be different than specified for UDP echo
Specify the UDP port to match to packet destination IP addresses for UDP loop layer-3/4 loopback	udp-loop udp-port <udp-dest-port>	Configuration > Physical Port > Flows > General Params > In-Service L3\L4> UDP Loop	UDP port range 1–65535 See above note

Task	CLI Command	Web Interface Location	Comments
Specify the IP address to match to packet destination IP addresses for UDP echo layer-3/4 loopback	udp-echo ip-address <ip-dest-addr>	Configuration > Physical Port > Flows > General Params > In Service L3\L4> UDP Echo	Note: The combination of the IP address and UDP port specified for UDP echo must be different than specified for UDP loop
Specify the UDP port to match to packet destination IP addresses for UDP echo layer-3/4 loopback	udp-echo udp-port <udp-dest-port>	Configuration > Physical Port > Flows > General Params > In Service L3\L4> UDP Echo	UDP port range 1–65535 See above note
Configure port mode action for frames from a specified ingress port	port-mode-action ingress-port <sfp/1 msa/2> action <action> [<VLAN parameters>]	Configuration > Physical Port > Flows > General Params > General Params [SFP/ Port 1 MSA/ Port 2] > Port Mode Action	See Configuring Port Mode . <sfp/1 msa/2> defines the traffic direction that this action applies to, by defining the ingress port as either SFP/port 1 or MSA/ port 2. For details of the <action> <VLAN parameters> parameters, see Action Parameters table
Define L2CP/L2PT profile to use for frames from specified ingress port, when in port mode	port-l2cp profile <profile-name> ingress-port {sfp/1 msa/2}	Configuration > Physical Ports > Flows > General Params > [SFP/ Port 1 MSA/ Port 2]> Port L2CP Profile	
Create new named flow and (in CLI) enter its command context	[no] flow <name>	Configuration > Physical Port > Flows > Add Flow > Flow Name	<name> is the flow's (new) name
Configure existing flow / in CLI: enter flow's command context	flow <name>	Configuration > Physical Port > Flows > Config Flow > flow name	
Map an originating frame P-bit value to a new tag P-bit value, for flows configured to push or replace VLAN tags with P-bits using C-tag marking	inner-vlan-marking c-pbit <original P-bit> s-pbit <new P-bit>	Configuration > Physical Port > Flows > General Params > C-Tag->S-Tag Mapping	

Task	CLI Command	Web Interface Location	Comments
Configure changing destination MAC address to user-specified value, for a particular L2PT protocol	l2cp-protocol {stp oam lldp pagp udld cdp vtp dtp pvstp} mac-change <mac-addr>	Configuration > Physical Port > Flows > General Params > L2CP\L2PT > L2PT > Mac Change	
Configure L2CP protocol classification	l2cp-mac <last-mac-byte>	Configuration > Physical Port > Flows > General Params > L2CP\L2PT > L2CP > Last MAC Byte (Hex.)	
Create a new L2CP/L2PT profile and (in CLI) enter its command context	[no] l2cp-profile <profile-name>	Configuration > Physical Port > Flows > General Params > L2CP\L2PT > Profiles > New Profile Name	<profile-name> is the profile's (new) name.
Configure existing L2CP/L2PT profile / in CLI: enter profile's command context	l2cp-profile <profile-name>	Configuration > Physical Port > Flows > General Params > L2CP\L2PT > Profiles > profile name	
Configure L2CP protocol action for L2CP/L2PT profile	mac <last-mac-byte> { tunnel pass-through peer discard }	Configuration > Physical Port > Flows > General Params > L2CP\L2PT > Profiles > profile name	Command is in the profile context config>flows>profile(<profile-name>)
Configure L2CP/L2PT profile action for L2PT protocol	protocol {stp oam lldp pagp udld cdp vtp dtp pvstp} [l2pt] { tunnel pass-through peer discard }	Configuration > Physical Port > Flows > General Params > L2CP\L2PT > Profiles > profile name	Command is in the profile context config>flows>profile(<profile-name>)
Define outer TPID value for flow packets and double-tagged OAM packets	tpid-outer-vlan <ethertype> ingress-port {sfp/1 msa/2}	Configuration > Physical Port > Flows > General Params > General Params [SFP/ Port 1 MSA/ Port 2] > TPID For Outer VLAN	
Define TPID value for new VLAN tags of enabled flows configured to push or replace VLAN tags, for specified ingress port	tpid-pushed-replaced-vlan <ethertype> ingress-port {sfp/1 msa/2}	Configuration > Physical Port > Flows > General Params > General Params [SFP/ Port 1 MSA/ Port 2] > Tag Value TPID For Inner VLAN	

Task	CLI Command	Web Interface Location	Comments
View a convenient summary of the classification mode and configured flows	show summary	--	
View traffic handling statistics for unclassified (fallback) traffic	show unclassified-statistics	Monitoring > Physical Port > Classification > Unclassified Traffic > Unclassified [SFP/ Port 1 MSA/ Port 2]	
Clear traffic handling statistics for unclassified (fallback) traffic	clear-statistics		

Per-Flow Tasks

If you want MiNID to treat all traffic in each direction uniformly, configure flows as in [Configuring Port Mode](#). Otherwise, configure flows as in [Configuring Flows](#) or in [Configuring Flows Using the CLI](#).

The following tasks are performed in the CLI **config>flow(<name>)** context, or in the Web interface.

Per-Flow Configuration Tasks

Task	CLI Command	Web Interface Location	Comments
Define frame values to be considered a match for this flow	classifier match <criteria>	Configuration > Physical Port > Flows > [SFP/ Port 1 MSA/ Port 2] or Configuration > Physical Port > Flows > Config Flow > flow name	.
Define the ingress port of frames to which this flow applies, and the flow action	flow-action ingress-port <sfp/1 msa/2> action <action> [<VLAN parameters>]		
Configure flow mode for frame matching for layer-2 loopback	in-service-loops > in-serv-mode <mode>	Configuration > Physical Port > Flows > Config Flow > In-Serv Loops	
Enable/disable addresses that should be matched for layer-2 loopback	in-service-loops > in-serv-dst-mac <index> {enable disable} or in-service-loops > in-serv-src-mac <index> {enable disable}		<index> is 1–4

Task	CLI Command	Web Interface Location	Comments
Define whether this flow's layer-2 loopbacks should just loop or also swap addresses	in-service-loops > in-serv-action {loop loop-mac-ip-swap}		
Assign Layer-3/4 loopback or TWAMP responder	in-service-l3-l4-loops > in-serv-l3-l4-mode	Configuration > Physical Port > Flows > Add Flow or Configuration > Physical Port > Flows > Config Flow > flow name or Configuration > Physical Port > Flows > General Params > In Service L3\L4 Loops	
Assign L2CP profile to flow that has classification VID+Pbits Range	l2cp profile <profile-name>	Configuration > Physical Port > Flows > Add Flow or Configuration > Physical Port > Flows > Config Flow > flow name	
Assign service name to flow, to enable network management systems to locate entities related to specific services	service-name <service-name>		
Enable or disable the flow	[no] shutdown		To disable: shutdown. To enable: no shutdown
View traffic handling statistics for the fflow	show statistics	Monitoring > Physical Port > Classification >[SFP/ Port 1 MSA/ Port 2]> Statistics	
Clear traffic handling statistics for the flow	clear-statistics		

The CLI commands for flows (**flow-action**), unclassified traffic (**fallback-action**), and port mode (**port-mode-action**) include the <action> [<VLAN parameters>] parameters. The following table gives more details on the parameters.

Action Parameters

Parameter	Description								
<action>	One of the following (see Flow Actions): • drop • pass-through • push-vlan (single \ double) • pop-vlan (single \ double) • replace-vlan-tag • loop • loop-mac-ip-swap								
<VLAN parameters>	Use only if the <action> is push-vlan or replace-vlan-tag , to define the properties of the new VLAN tag. <VLAN parameters> is: vlan-id <VID> priority <P-bit mode> drop-eligible <yes no> The above parameters are: <table><tr><th>Parameter</th><th>Description</th></tr><tr><td><VID></td><td>The new tag's VLAN ID (0-4095)</td></tr><tr><td><P-bit mode></td><td>One of the following (See Flow Actions, under VLAN): • A fixed P-bit value in range 0-7 • For C-tag Marking: inner-vlan-marking • For C-tag Copy: inner-vlan-copy • For DSCP Marking: dscp-marking </td></tr><tr><td><yes no by-policer></td><td>The new tag's Drop Eligible Indicator (DEI) bit value (yes for 1) The by-policer option is available only for outer VLAN.</td></tr></table>	Parameter	Description	<VID>	The new tag's VLAN ID (0-4095)	<P-bit mode>	One of the following (See Flow Actions , under VLAN): • A fixed P-bit value in range 0-7 • For C-tag Marking: inner-vlan-marking • For C-tag Copy: inner-vlan-copy • For DSCP Marking: dscp-marking	<yes no by-policer>	The new tag's Drop Eligible Indicator (DEI) bit value (yes for 1) The by-policer option is available only for outer VLAN.
Parameter	Description								
<VID>	The new tag's VLAN ID (0-4095)								
<P-bit mode>	One of the following (See Flow Actions , under VLAN): • A fixed P-bit value in range 0-7 • For C-tag Marking: inner-vlan-marking • For C-tag Copy: inner-vlan-copy • For DSCP Marking: dscp-marking								
<yes no by-policer>	The new tag's Drop Eligible Indicator (DEI) bit value (yes for 1) The by-policer option is available only for outer VLAN.								

Configuring Port Mode

For a description of port mode, see [Port Mode](#).

➤ To configure port mode:

Do one of the following:

- Use the CLI:
 - i. Go to the configure flows context, and enter:
classification mode port
 - j. Define the action for each traffic direction by entering, for each of sfp/Port 1 and msa/Port 2:
port-mode-action ingress-port <sfp/1 | msa/2> action <action> [<VLAN parameters>]
 <sfp/1 | msa/2> defines the traffic direction that the action applies to, by defining the ingress port as either **sfp/Port 1** or **msa/Port 2**. For details of the <action> <VLAN parameters> parameters, see **Action Parameters** table.
- Use the web interface:
 - k. Go to Configuration > Physical Port > Flows.
 - l. By Classification Mode, select Port Mode, and click <Apply> to implement the changes, and then click <Save Configuration>:



Port Mode

- m. Go to Configuration > Physical Port > Flows > General Params > SFP/Port 1.

MiNID

Configuration>Physical Ports>Flows>General Params>SFP/Port 1

TPID For Outer VLAN	_____	0x88a8
TPID For Inner VLAN	_____	0x8100
	_____	0x8100
	_____	0x8100
Port Mode Policer Type	_____	Disabled
Port Mode PM Collection	_____	Disable
Port Mode Action	_____	Pass Through
L2CP Profile	_____	kl2cp
IP Agnostic loop		Disable (option available In "port mode" only)
In-Service L2 Loops (Port Mode)		
LBM reflector		
In-Service L3\L4 Loops (Port Mode)		

General Params [SFP/Port 1] – Port Mode

- n. Set the Port Mode Action (see Flow Actions) for SFP-to-MSA/ port 1–to–port 2 traffic. If you select Push VLAN or Replace VLAN, the additional following VLAN settings appear:
 - VID: The new tag's VLAN ID (0–4095)
 - Pbits Mode: See Flow Actions (under VLAN)
 - Pbits Value: The new tag's p-bit value if Pbits Mode is set to Fixed Value (0–7)
 - Drop Eligible: The new tag's Drop Eligible Indicator (DEI) bit value (Yes for 1, No for 0 and "by policer" for outer VLAN only, takes the DEI bit from the policer)
- o. Port L2CP Profile: Select an L2CP profile if necessary for your application
- p. If you want to configure LBM reflectors, click LBM reflector (see Configuring LBM Reflector).
- q. Click <Apply> to implement the changes, and then click <Save Configuration>
- r. Go to Configuration > Physical Port > Flows > General Params> MSA/Port 2, set the same parameters in the same way for MSA-to-SFP/ port 2–to–port 1 traffic, and click <Apply> to implement the changes, and then click <Save Configuration>.

Configuring Flows Using the Web Interface

Configuring General Parameters

➤ To configure general flow parameters:

1. To configure flow classification mode, go to **Configuration > Physical Port > FlowsPort Name**, and select the **Classification Mode** from one of the following, to set the criterion by which all traffic in both directions will be classified (see **Flow Classification**):
 - DSCP
 - EtherType
 - MAC SRC
 - MAC DST
 - Port Mode
 - VID + Pbits Range

Note If you change the flow classification mode, all configured flows are deleted.

MiNID

Configuration>Physical Ports>Flows>Flows SFP/Port 1 or MSA/Port 2

Classification Mode

VID + Pbits Range

Add Flow

Config Flow

Classification Mode

1. Click **<Apply>** to apply the changed flow classification mode, and then click **<Save Configuration>**.
2. Go to **Configuration > Physical Port > Flows > General Params > SFP/Port 1**, set the following fields for SFP-to-MSA/ port 1-to-port 2 traffic, and click **<Apply>** to implement the changes, and then click **<Save Configuration>**; then, go into **General Params > MSA/Port 2** and set the same fields for MSA-to-SFP/ port 2-to-port 1 traffic:

- **TPID For Outer VLAN** – Applied to flow packets and as the outer TPID of double-tagged OAM packets
- **TPID For Inner VLAN** –Applied to packets for this traffic direction if VLAN tags are pushed or replaced
- **Fallback L2CP Profile** – Binds the L2CP profile to the unclassified flow of the port
- **Unclassified Action** – Applied to all traffic in this direction that is not matched by any enabled flow. Available actions are the same as for flows.

If you want to configure LBM reflectors, click **LBM reflector** (see **Configuring LBM Reflector** for details).

MiNID

Configuration>Physical Ports>Flows>General Params>SFP/Port 1

TPID For Outer VLAN	_____	0x88a8
TPID For Inner VLAN	_____	0x8100
	_____	0x8100
	_____	0x8100
Fallback Policer Type	_____	Disabled
Fallback PM Collection	_____	Disable
Fallback L2CP Profile	_____	
Unclassified Action	_____	Pass Through
Port Mode Action	_____	Pass Through

LBM reflector

General Parameters [SFP/Port 1] – Flow Mode

1. If you have any flows with Push or Replace VLAN actions with a P-bits Mode of **C-Tag Marking**, go to **Configuration > Physical Port > Flows Classification > General Params > C-Tag->S-Tag Mapping**, configure the P-bits-to-P-bits mapping and click **<Apply>** to implement the changes, and then click **<Save Configuration>**:

MinID

Configuration>Physical Ports>Flows>General Params>CTag Remap

00	0	01	0	02	3	03	3
04	3	05	3	06	7	07	0

P-bit-to-P-bit Mapping

1. If you have any flows with Push or Replace VLAN actions with a P-bits Mode of **DSCP Marking**, go to **Configuration > Physical Port > Flows Classification > General Params > DSCP->S-Tag Mapping**, configure the DSCP-to-P-bits mapping and click **<Apply>** to implement the changes, and then click **<Save Configuration>**:

MiNID

Configuration>Physical Ports>Flows>General Params>DSCP Remap

0							
0	01	02	03	04	05	06	07
0	1	2	3	4	5	6	7
0							
8	09	10	11	12	13	14	15
0	1	2	3	4	5	6	7
0							
1							
6	17	18	19	20	21	22	23
0	1	2	3	4	5	6	7
0							
2							
4	25	26	27	28	29	30	31
0	1	2	3	4	5	6	7
0							
3							
2	33	34	35	36	37	38	39
0	1	2	3	4	5	6	7
0							
4							
0	41	42	43	44	45	46	47
0	1	2	3	4	5	6	7
0							
4							
8	49	50	51	52	53	54	55
0	1	2	3	4	5	6	7
0							
5							
6	57	58	59	60	61	62	63
0	1	2	3	4	5	6	7

DSCP-to-P-bit Mapping

1. To configure layer-2 loopback general parameters, go to **Configuration > Physical Port > Flows > General Params > In Service Loop MACs**, and define the **Source** and **Destination** MAC addresses to be available for flow layer-2 loopback:

MiNID

Configuration>Physical Port>Flows>General Params>In Service Loop MACs

Source Addresses	Destination Addresses
12-34-56-78-90-ab	cd-ef-12-34-56-78
00-00-00-00-00-00	00-00-00-00-00-00
00-00-00-00-00-00	00-00-00-00-00-00
00-00-00-00-00-00	00-00-00-00-00-00

In-Service Loopback MACs

1. Click **<Apply>** to implement the changes to the layer-2 loopback MAC addresses, and then click **<Save Configuration>**.
2. To configure layer-3/4 loopback general parameters, go to **Configuration > Physical Port > Flows > General Params > In Service L3\L4 Loops**.

MiNID

Configuration > Physical Port > Flows > General Params > In-Serv L3\L4 Loops

In-Service L3\L4:IP Loop
In-Service L3\L4:UDP Loop
In-Service L3\L4:UDP Echo

In-Service Layer-3/4 Loopback General Parameters

1. Select the loop type for which you wish to configure the general parameters:
 - IP Loop – Click **<In-Service L3\L4:IP Loop>**
 - UDP Loop – Click **<In-Service L3\L4: UDP Loop>**
 - UDP Echo – Click **<In-Service L3\L4: UDP Echo>**.

MiNID

Configuration > Physical Port > Flows > General Params > In-Serv L3\L4 Loops > IP Loop

Destination IP Address

192.168.205.100

In-Service Layer-3/4 Loopback General Parameters – IP Loop

MiNID

Configuration > Physical Port > Flows > General Params > In-Serv L3\L4 Loops > UDP Loop

Destination IP Address

192.168.205.101

Destination UDP Port

153

In-Service Layer-3/4 Loopback General Parameters – UDP Loop

MiNID

Configuration > Physical Port > Flows > General Params > In-Serv L3\L4 Loops > UDP ECHO

Destination IP Address

192.168.205.102

Destination UDP Port

222

In-Service Layer-3/4 Loopback General Parameters – UDP Echo

- 1. Set the general parameters (see **General Configuration Tasks** table).

Note

When IP agnostic loop is enabled, the IP loop option under In-Serv L3\L4 Loops is not available. In addition, configuration of an IP address for UDP loop, UDP Echo and TWAMP responder is not relevant (the option is blocked).

Configuring Flows

➤ To add or configure flows:

- To add a new flow, go to Configuration > Physical Port > Flows > SFP/Port 1 or MSA/Port 2, and click <Add Flow>.

MiNID

Configuration>Physical Port>Flows> Add Flow

Flow Name	_____	Flow 11
Service Name	_____	Gold
Flow Enable	_____	Enable
Ingress Port	_____	SFP/Port 1
L2CP Profile	_____	kl2cp
L3L4 Loop	_____	Disabled
PM Collection	_____	Disable
Policer Type	_____	Disabled
Classification Tag Mode	_____	Single
Min VID Value	_____	0
Max VID Value	_____	4095
Min Pbit Value	_____	5
Max Pbit Value	_____	5
Flow Action	_____	Push VLAN

Apply

Add Flow with Single Tag VLAN

MiNID

Configuration>Physical Port>Flows>Add Flow

Flow Name	_____	Flow 11
Service Name	_____	Gold
Admin Status	_____	Enable
Ingress Port	_____	SFP/Port 1
L2CP Profile	_____	kl2cp
L3L4 Loop	_____	Disabled
PM Collection	_____	Disable
Policer Type	_____	Disabled
Classification Tag Mode	_____	Double
Outer VID Value	_____	
Outer Min Pbit Value:	_____	
Outer Max Pbit Value	_____	
Inner Min VID Value	_____	
Inner Max VID Value	_____	
Inner Min P-bit Value	_____	
Inner Max P-bit Value	_____	
Flow Action	_____	Push VLAN

Outer VLAN Information:			
VID	Pbits Mode	Pbits Value (Fixed Value Only)	Drop Eligible
2222	Fixed Value	3	No

Add Flow with Double Tag VLAN

MinID

Configuration>Physical Ports>Flows>Add Flow

Flow Name	_____	Flow 11	
Service Name	_____	Gold	
Admin Status	_____	Enable	
Ingress Port	_____	SFP/Port 1	
L2CP Profile	_____	kl2cp	
L3L4 Loop	_____	Disabled	
PM Collection	_____	Disable	
Policer Type	_____	Envelope	
Envelope Profile	_____		
Cos Map Profile	_____		
Classification Tag Mode	_____	Single	
Min VID Value	_____		
Max VID Value	_____		
Min Pbit Value	_____		
Max Pbit Value	_____		
Flow Action	_____	Push VLAN	
Outer VLAN Information:			
VID	Pbits Mode	Pbits Value (Fixed Value Only)	Drop Eligible
2222	Fixed Value	3	No

Add Flow with Policer

1. To configure an existing flow:
- s. Go to Configuration > Physical Port > Flows > Classification SFP/MSA, and click <Config Flow>.

MinID

Configuration>Physical Port>Flows>Config Flow

Flow Name	Service Name					
22	Gold	Disable Flow	In-Serv Loops	In-Serv L3\L4 Loops	Test	Delete Flow
11	Gold	Disable Flow	In-Serv Loops	In-Serv L3\L4 Loops	Test	Delete Flow

Configure Flows

- u. If necessary, click Disable Flow in the row corresponding to the flow, as the flow must be administratively disabled in order to edit it.
- v. Click the flow name to display the Flow Info screen.

MinID

Configuration>Physical Ports>Flows>Flow (Flow name) > Flow Info

Flow Name	Flow 11
Service Name	Gold
Flow Enable	Disable
Ingress Port	SFP/Port 1
L2CP Profile	kl2cp
L3L4 Loop	TWAMP Responder
TWAMP Session	tw1
Min VID Value:	0
Max VID Value:	4095
Min P-bit Value:	5
Max P-bit Value:	5
Flow Action	Push VLAN

VLAN Header Information:

VID	Pbits Mode	Pbits Value (Fixed Value Only)	Drop Eligible
2222	Fixed Value	3	No

Flow Info

1. In the Add Flow or Flow Info screens, configure the settings:
- **Flow Name** (if adding the flow)
 - **Service Name** – The service name enables network management systems to locate entities related to specific services
 - **Admin Status** (if adding the flow) – A disabled flow retains its configuration for future use, and is included in the count of flows in connection to the maximum of 16 configured flows.
 - **Ingress Port** – **MSA/Port 2** for customer-to-provider/Ethernet port 2 traffic or **SFP/Port 1** for provider-to-customer /Ethernet port 1 traffic
 - **L2CP Profile** – Select an L2CP profile if necessary for your application (this parameter appears only if classification mode is VID+Pbits Range)

- **L3L4 Loop** – Set the flow layer-3/4 loopback mode to one of the following (see [Layer-3/4 Loopbacks](#) for more information on the layer-3/4 loopback types):
 - Disabled
 - IP Loop – Layer 3/4 loopback, type IP loop
 - UDP Loop – Layer 3/4 loopback, type UDP loop
 - UDP Echo – Layer 3/4 loopback, type UDP echo

Note *Before setting layer-3/4 loopback mode to IP Loop, UDP Loop, or UDP Echo, you need to configure the general layer-3/4 loopback parameters as described in [Configuring General Parameters](#).*

- TWAMP responder – associate the flow with a TWAMP responder
If this option is selected, the **TWAMP Session** parameter appears in the screen, and you can select from a dropdown list of the configured TWAMP responders. See [OAM \(TWAMP\)](#) for more information on TWAMP responders.

Note *The flow layer-3/4 loopback mode can also be set via the In Serv L3\L4 Loops screen (see [Configuring Layer-3/4 Loopbacks](#)).*

- **PM Collection** – The PM Collection enables setting a PM Interval for monitoring microbursts in network traffic
- **PM Interval** – Set the duration of the time period for measuring the microbursts
- **Policer Type** – Standardized according to MEF 10.2
- **Test** – This command allows the user to change the flow action to either of the following:
 - Loop With MAC (IP) Swap
 - Discard

without changing the flow action by configuration. This command cannot be saved in a database, so if the unit is reset, the flow action returns to its configured action. This command can be used for temporary change of the flow action.
- **Flow criteria** – These appear differently depending on the current classification mode.
If the classification mode is **VID+Pbits Range** and you want to define a flow by just VLAN ID, configure it for the full range of possible P-bits (0-7) and specify the desired VLAN ID range. Similarly, if you want to define the flow by just priority, configure it for the full range of possible VLAN IDs (0-4095) and specify the desired P-bit range.
If the Classification Tag Mode is “Double”, configure the following parameters:
 - Outer VID Value: The VID of the outer VLAN
 - Outer Min P-bit Value: Minimum P-bit value for outer VLAN
 - Outer Max P-bit Value: Maximum P-bit value for outer VLAN
 - Inner Min VID Value: Minimum VLAN ID value for inner VLAN
 - Inner Max VID Value: Maximum VLAN ID value for inner VLAN

- Inner Min P-bit Value: Minimum P-bit value for inner VLAN
 - Inner Max P-bit Value: Maximum P-bit value for inner VLAN
 - Single flow under this classification can be set as “Untagged”.
 - Flow Action – See **Flow Actions**.
If the Flow Action is **Push VLAN** or **Replace VLAN**, the additional following VLAN settings appear:
 - VID: The new tag’s VLAN ID (0-4095)
 - Pbits Mode: See Flow Actions (under VLAN)
 - Pbits Value: The new tag’s P-bit value if Pbits Mode is set to Fixed Value (0-7)
 - Tag Ether Type: The VLAN Ethertype
 - Drop Eligible: The new tag’s Drop Eligible Indicator (DEI) bit value (Yes for 1)
2. Click **<Apply>** to implement the changes, and then click **<Save Configuration>**.

Configuring Layer-2 Loopbacks

This procedure describes how to configure flow/port layer-2 loopbacks. Before configuring layer-2 loopbacks, configure the general layer-2 loopback parameters as described in **Configuring General Parameters**.

➤ To configure layer-2 loopbacks:

1. If flow classification is not per port:
 - w. Go to Configuration > Physical Port > Classification > Config Flow.
 - x. In the row corresponding to the flow, click In-Serv Loops.

MiNID

Configuration>Physical Ports>Flows>Flow (cust-traf)>In-Serv Loops

In-Service Action

In-Service Mode

Loop With MAC (IP) Swap

SA+DA

In-Service Source Addresses:

00-00-00-00-00-00

00-00-00-00-00-00

00-00-00-00-00-00

00-00-00-00-00-00

Disable

Disable

Disable

Disable

In-Service Destination Addresses:

00-00-00-00-00-00

00-00-00-00-00-00

00-00-00-00-00-00

00-00-00-00-00-00

Disable

Disable

Disable

Disable

Layer-2 Loop Configuration – Flow

- If flow classification is per port:
 - Go to Configuration > Physical Port > Flows> General Params [SFP/Port 1] to configure layer-2 parameters for SFP/Port 1
 - In the General Params[SFP/Port 1] screen, click In-Service L2 Loops (Port Mode)

MinID

Configuration>Physical Ports>Flows>SFP/Port 1> In-Service L2 Loops (Port Mode)

In-Service Action	Loop With MAC (IP) Swap
In-Service Mode	SA+DA

In-Service Source Addresses:

00-00-00-00-00-00

00-00-00-00-00-00

00-00-00-00-00-00

00-00-00-00-00-00

Disable

Disable

Disable

Disable

In-Service Destination Addresses:

00-00-00-00-00-00

00-00-00-00-00-00

00-00-00-00-00-00

00-00-00-00-00-00

Disable

Disable

Disable

Disable

Layer-2 Loop Configuration – SFP/Port 1

1. By In-Service Action, define whether the flow/port layer-2 loopbacks should just Loop or Loop with MAC (IP) Swap (See **Layer-2 Loopbacks**).
2. By In-Service Mode, set the flow/port mode for frame matching (See **Layer-2 Loopbacks**).
3. **Enable** each MAC address that should be matched for layer-2 loopback for the flow/port.
4. Save.
5. Click **<Apply>** to implement the changes, and then click **<Save Configuration>**.
6. If flow classification is per port, go to **Configuration > Physical Port > Flows > General Params [MSA/Port 2]**, set the same parameters in the same way for MSA/port 2, and click **<Apply>** to implement the changes, and then click **<Save Configuration>**.

Configuring Layer-3/4 Loopbacks

This procedure describes how to configure flow/port layer-3/4 loopbacks, or bind a TWAMP responder to a flow/port. Before configuring layer-3/4 loopback, configure the general layer-3/4 loopback parameters as described in **Configuring General Parameters**.

➤ **To configure layer-3/4 loopbacks:**

- 1. If flow classification is not per port:
 - a. Go to Configuration > Physical Port > Classification > Config Flow.
 - b. In the row corresponding to the flow, click In-Serv L3\L4 Loops.

Note *The flow layer-3/4 loopback parameters can also be configured by clicking the flow name to edit the flow parameters (see **Configuring Flows**).*

MiNID

Configuration>Physical Ports>Flows>Flow (data1)>In-Serv L3\L4 Loops

In-Service L3\L4 Mode

UDP Loop

In-Service IP Destination Address:

192.168.205.101

In-Service UDP Destination Port

153

Layer-3/4 Loop Mode Configuration – Flow

- 1. If flow classification is per port:
 - a. Go to Configuration > Physical Port > Flows > General Params > SFP/Port 1 to configure layer-3/4 loopback mode for SFP/Port 1.
 - b. In the General Params [SFP/Port 1] screen, click In-Service L3\L4 Loops (Port Mode).

MinID

Configuration > Physical Ports > Flows > SFP/Port 1 > In-Service L3\L4 Loops (Port Mode)

In-Service L3\L4 Mode	UDP Loop
In-Service IP Destination Address:	192.168.205.101
In-Service UDP Destination Port	153

Layer-3/4 Loopback Mode Configuration – SFP/Port 1

- By **In-Service L3\L4 Mode**, set the flow/port layer-3/4 loopback mode to one of the following (see **Layer-3/4 Loopbacks** for more information on the layer-3/4 loopback types):
 - Disabled
 - IP Loop – Layer 3/4 loopback, type IP loop
 - UDP Loop –Layer 3/4 loopback, type UDP loop
 - UDP Echo – Layer 3/4 loopback, type UDP echo

Note

*If the IP address (and UDP port if applicable) have not been set for a particular layer-3/4 loopback type, it does not appear in the dropdown list for **In-Service L3\L4 Mode***

- TWAMP responder – associate the flow/port with a TWAMP responder.
If this option is selected, the **TWAMP Session** parameter appears in the screen, and you can select from a dropdown list of the configured TWAMP responders. See **OAM (TWAMP)** for more information on TWAMP responders.
- Click **<Apply>** to implement the changes, and then click **<Save Configuration>**.
 - If flow classification is per port, go to **Configuration > Physical Port > Flows > General Params> MSA/Port 2**, set the same parameters in the same way for MSA/port 2, and click **<Apply>** to implement the changes, and then click **<Save Configuration>**.

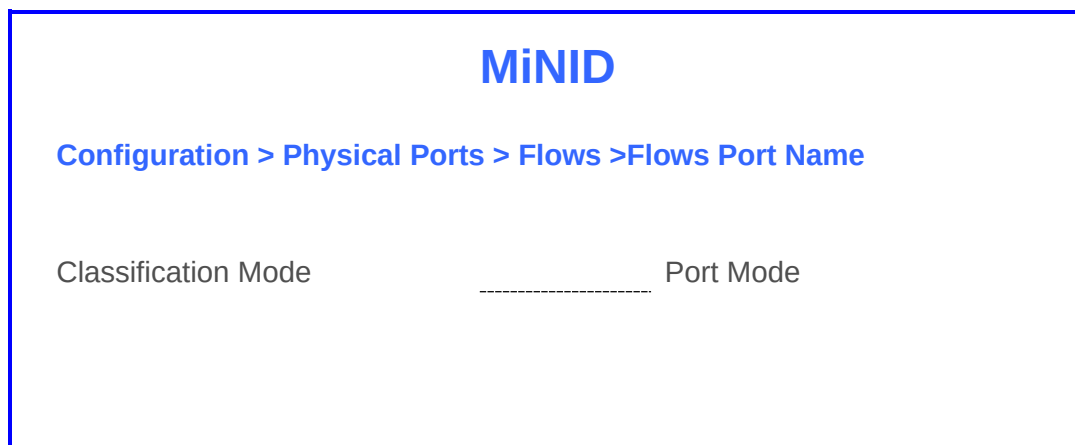
Configuring IP-Agnostic Loops

Note When IP agnostic loop is enabled, the IP loop option under In-Serv L3\L4 Loops is not available. In addition, configuration of an IP address for UDP loop, UDP Echo and TWAMP responder is not relevant (the option is blocked).

Note IP-agnostic loops can be configured in “Port Mode” only.

➤ To configure IP-agnostic loops:

1. Go to Configuration>Physical Ports>Flows>Port Name and select Port Mode in Classification Mode.



1. Go to Configuration> Physical Ports> Flows> General Params> SFP/Port 1/ MSA/Port 2 and enable the IP-agnostic loop.

MiNID

Configuration > Physical Ports > Flows > General Params > SFP/Port 1/ MSA/Port 2

TPID For Outer VLAN	0×88a8
TPID For Inner VLAN	0×8100
	0×8100
	0×8100
Port Mode Policer Type	Disabled
Port Mode PM Collection	Disable
L2CP Profile	
Port Mode Action	Pass Through
IP Agnostic Loop	Enable

IP-Agnostic Loop Configuration

1. Click <Apply> to implement the changes, and then click <Save Configuration>.

Configuring Flows Using the CLI

Configuring General Parameters

➤ To configure general flow parameters:

1. Go to the **configure flows** context.
2. Set the classification mode with the following command. This is the criterion by which all traffic in both directions will be classified (see **Flow Classification** for mode descriptions):

classification mode <mode>

where <mode> is one of:

- dscp
- ethertype
- **src-mac** (source MAC address)
- **dst-mac** (destination MAC address)

- **vlan-pbits-range** (VLAN ID range and P-bit range)

Note *If you change the flow classification mode, all configured flows are deleted.*

3. By default, traffic that does not match any enabled flow is passed through. To change the action for unmatched traffic in either direction, enter:

fallback-action ingress-port {sfp/1 | msa/2} action <action> [<VLAN parameters>]

For details of the <action> <VLAN parameters> parameters, see [Action Parameters](#) table.

4. The outer TPID value for flow packets and double-tagged OAM packets is specified by **tpid-outer-vlan**. To change the value for all flows of one direction, in the **configure flows** context, enter:

tpid-outer-vlan <ethertype> **ingress-port {sfp/1 | msa/2}**

where <ethertype> is the hexadecimal TPID value to apply, and {sfp/1 | msa/2} is either **sfp/1** (for SFP-to-MSA/ port 1-to-port 2 traffic) or **msa/2** (for MSA-to-SFP/ port 2-to-port 1 traffic).

The inner TPID value for flow packets and double-tagged OAM packets is **tpid-inner-vlan**. To add or change the value in the **configure flows** context, enter:

tpid-inner-vlan <ethertype> **ingress-port {sfp/1 | msa/2}**.

You can configure up to 3 inner TPIDs. To delete one, type [no] before one of the entries.

5. All flows in each direction that push or replace VLAN tags apply the TPID value specified by **tpid-inner-vlan**. To change the value for all flows of one direction, in the **configure flows** context, enter:

tpid-inner-vlan <ethertype> **ingress-port {sfp/1 | msa/2}**

where <ethertype> is the hexadecimal TPID value to apply, and {sfp/1 | msa/2} is either **sfp/1** (for SFP-to-MSA/port 1-to-port 2 traffic) or **msa/2** (for MSA-to-SFP/ port 2-to-port 1 traffic).

6. If you have any flows with Push or Replace VLAN actions configured with P-bits mode of C-tag marking, configure the P-bits-to-P-bits mapping. For each P-bit value of the originating frame that needs to be mapped to a non-default value, enter:

inner-vlan-marking c-pbit <original P-bit> **s-pbit** <new P-bit>

7. If you have any flows with Push or Replace VLAN actions configured with P-bits Mode of DSCP Marking, configure the DSCP-to-P-bits mapping. For each DSCP value of the originating frame's IP payload that needs to be mapped to a non-default value, enter:

dscp-marking dscp <DSCP value> **s-pbit** <new P-bit>

8. To configure up to four source MAC addresses and up to four destination MAC addresses for flow layer-2 loopback, enter:

in-service-mac {src | dst}-addr <index> <address>

where the above parameters are:

Parameters for in-service-mac

Parameter	Description
<src dst>	Whether configuring a source address or a destination address
<index>	A number in range 1–4 identifying this source or destination address, for reference when enabling/disabling loopback for this address per-flow
<address>	The MAC address, in format xx-xx-xx-xx-xx-xx , where xx is a valid hexadecimal number in two digits

9. To configure layer-3/4 loopback general parameters:

- a. Enter in-service-l3-l4, in order to go into the config>flows>in-serv-l3-l4 context.
- b. Enter all necessary commands according to the tasks listed below (see **General Configuration Tasks** table for more information on the parameters).

10. To configure IP-agnostic loops:

- a. Enter classification mode port to enable IP agnostic loop.
- b. Enter all necessary commands according to the tasks listed below (see **General Configuration Tasks** table for more information on the parameters).

Task	Command	Comments
Specify IP agnostic loop	ip-agnostic-loop {sfp/1 msa/2}	SFP sleeve option – Port mode must be configured to enable IP agnostic loop
Specify the IP address to match to packet destination IP addresses for IP loop layer-3/4 loopback	ip-loop ip-address <ip-dest-addr>	<ip-dest-addr> – IP address in format aaa.bbb.ccc.ddd
Specify the IP address to match to packet destination IP addresses for UDP loop layer-3/4 loopback	udp-loop ip-address <ip-dest-addr>	<ip-dest-addr> – IP address in format aaa.bbb.ccc.ddd
Specify the UDP port to match to packet destination UDP ports for UDP loop layer-3/4 loopback	udp-loop udp-port <udp-dest-port>	<udp-dest-port> – UDP port number
Specify the IP address to match to packet destination IP addresses for UDP echo layer-3/4 loopback	udp-echo ip-address <ip-dest-addr>	<ip-dest-addr> – IP address in format aaa.bbb.ccc.ddd

Task	Command	Comments
Specify the UDP port to match to packet destination UDP ports for UDP echo layer-3/4 loopback	udp-echo udp-port <udp-dest-port>	<udp-dest-port> – UDP port number

Configuring Flows

➤ To configure flows:

- Go to the **configure flows** context, and enter:
flow <flow-name>:
where <name> is the flow name.
The CLI enters the flow context.
- Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Setting the flow classification	classifier match <criteria>	The parameter <criteria> depends on the current classification mode (see Classifier Criteria table)
Associating the flow with a classifier profile	classifier <classifier-profile-name>	If the flow is multi-CoS, the classification must be one of the following: - VLAN, VLAN + inner VLAN, VLAN range - Match all - Untagged
Associating the flow with a CoS mapping profile	cos-mapping profile <cos-mapping-profile-name>	See CoS Map Profiles in Chapter 11 . Enter no cos-mapping to disable the profile
Specifying flow action	flow-action ingress-port {sfp/1 msa/2} action <action> <VLAN parameters>	See Flow Actions . The parameter { sfp/1 msa/2 } defines the traffic direction that this flow applies to, by defining the ingress port as SFP/port 1 or MSA/ port 2. For details of the <action> <VLAN parameters> parameters, see Action Parameters table.
Configuring layer-2 loopbacks	in-service-loops	See Configuring Layer-2 Loopbacks

Task	Command	Comments
Configuring layer-3/4 loopbacks	in-service-l3-l4-loops	See Configuring Layer-3/4 Loopbacks
Associating an L2CP profile with the flow	l2cp profile <profile-name>	Relevant only if classification mode is vlan-pbits-range
Enabling PM collection and setting the duration of the collection in seconds	pm-collection interval <seconds>	See Microburst Monitoring in Chapter 11 Enter no pm-collection to disable the collection
Setting the policer type used for the flow	policer [profile regular-accounting-only envelope]	The following policer types can be defined: • profile – Regular policer profile (See Policer Profiles in Chapter 11) • regular-accounting-only – Profile used to only count the statistics, without limiting the traffic (See • envelope – Envelope profile (See Envelope Bandwidth Profiles in Chapter 11). Enter no policer to disable the policer
Associating a service name to with the flow, to enable network management systems to locate entities related to specific services	service-name <service-name>	
Test	test <loop-mac-ip-swap>/<discard>	Enter no test to remove test
Administratively enabling or disabling flow	shutdown	Enter no shutdown to enable the flow
Clearing flow statistics	clear-statistics	
Viewing flow statistics	show statistics	See Viewing Flow Statistics

Classifier Criteria

Classification Mode	<criteria>	Example
DSCP	dscp <value>	classifier match dscp 63
Ethertype	ether-type <hexadecimal value>	classifier match ether-type 0x8100
Source MAC	src-mac <MAC address>	classifier match src-mac 12-34-56-78-90-ab

Classification Mode	<criteria>	Example
Destination MAC	dst-mac <MAC address>	classifier match dst-mac cd-ef-12-34-56-78
Source and destination MACs	src-mac <source MAC> dst-mac <destination MAC>	classifier match src-mac 12-34-56-78-90-ab dst-mac cd-ef-12-34-56-78
VLAN ID and P-Bit	vlan-min <VID minimum> pbit-min <P-bit minimum> vlan-max <VID maximum> pbit-max <P-bit maximum>	classifier match vlan-min 0 pbit-min 4 vlan-max 4095 pbit-max 4

Configuring Layer-2 Loopbacks

This procedure describes how to configure flow/port layer-2 loopbacks. Before configuring layer-2 loopbacks, configure the general layer-2 loopback parameters as described in [Configuring General Parameters](#).

➤ To configure layer-2 loopback:

1. Go to the **configure flows flow**(<flow-name>) context.
2. According to the flow classification:
 - Any classification other than per port – Enter **in-service-loops** to go into the **config>flows>flow**(<flow-name>)>**in-serv** context
 - Classification per port – Enter one of the following, according to which port you want to configure:
 - **in-service-loops sfp/1** to go into the context **config>flows>in-serv(sfp/1)**
 - **in-service-loops msa/2** to go into the context **config>flows>in-serv(msa/2)**
3. Configure the flow/port mode for layer-2 loopback frame matching, by entering: **in-serv-mode** <mode>
where <mode> is one of (see [Layer-2 Loopbacks](#)):
 - **src**
 - **dst**
 - **src-dst**
4. For each MAC address that should be matched for layer-2 loopback for the flow/port, enter: **[no] in-serv-<src|dst>-mac** <index>
where the above parameters are:

in-serv Parameters

Parameter	Description
<src dst>	Whether enabling a source address or a destination address
<index>	A number in range 1–4 identifying this source or destination address, referring to the addresses configured in step 3 above

- 5. Define whether the flow/port layer-2 loopbacks should just loop or also swap addresses (see **Layer-2 Loopbacks**). Enter:
in-serv-action <loop|loop-mac-ip-swap>

Configuring Layer-3/4 Loopbacks

This procedure describes how to configure flow/port layer-3/4 loopbacks, or bind a TWAMP responder to a flow/port. Before configuring layer-3/4 loopbacks, configure the general layer-3/4 loopback parameters as described in **Configuring General Parameters**.

➤ **To configure layer-3/4 loopbacks:**

- 1. Go to the **configure flows flow(<flow-name>)** context.
- 2. According to the flow classification:
 - Any classification other than per port – Enter **in-service-l3-l4-loops** to go into the **config>flows>flow(<flow-name>)>in-serv -l3-l4** context
 - Classification per port – Enter one of the following, according to which port you want to configure:
 - in-service-l3-l4-loops sfp/1 to go into the context config>flows>in-serv-l3-l4(sfp/1)
 - in-service-l3-l4-loops msa/2 to go into the context config>flows>in-serv-l3-l4(msa/2)

- 3. Configure the layer-3/4 loopback mode by entering the following command:
[no] in-serv-l3-l4-mode <{ip-loop | udp-loop | udp-echo-loop | twamp}

Note Before setting layer-3/4 loopback mode to IP Loop, UDP Loop, or UDP Echo, you need to configure the general layer-3/4 loopback parameters as described in **Configuring General Parameters**.

- 4. If you specified **twamp** as the loopback mode, enter the following command to bind a TWAMP responder to the flow (see **OAM (TWAMP)** for more information on TWAMP responders):
responder <responder-name>

Configuring IP-Agnostic Loops

Note IP-agnostic loops can be configured in “Port Mode” only.

► To configure IP-agnostic loops:

1. Configure classification mode “Port”.
2. Enable IP-agnostic loop for SFP port 1 MSA/port 2.

```
MiNID>config>flows# classification mode port
MiNID>config>flows# ip-agnostic-loop {sfp/1 | msa/2}
```

3. Save the configuration.

Configuring LBM Reflector

You can configure a generic LBM reflector that enables replying with LBR to any LBM packet received with any VLAN and selected MD levels, per SFP/port 1 or MSA/port 2. By default, the LBM reflector is disabled.

When the LBM reflector is enabled, LBRs are sent in reply to any received LBM with an MD level for which the LBM reflector is enabled.

You can view counters for each port that show how many LBM packets have been received (see [Viewing Port Statistics](#)).

Configuring LBM Reflector Using the Web Interface

► To configure LBM reflector parameters:

1. Navigate to the General Parameters screen for the SFP/port 1 or MSA/port 2, according to which port’s parameters you want to configure.
2. Click LBM reflector.

Note The figure below illustrates the screen for **Configuration>Physical Ports>Flows>General Params>SFP/Port 1>LBM Reflector**, which is the same screen as **Configuration>Physical Ports>Flows>General Params>MSA/Port 2>LBM Reflector** except for the menu path at the top of the screen.

MiNID

Configuration>Physical Ports>Flows>General Params>SFP/Port 1>LBM Reflector

Admin Status

Disable

01234567

MD Level:

☐☐☐☐☐☐☐☐

Apply

LBM Reflector – SFP/Port 1 (Disabled)

3. Set **Admin status** according to whether you want to enable or disable the LBM reflector:
 - If you enable the LBM reflector, all MD levels are automatically selected. You can select and deselect MD levels as needed, by clicking the corresponding levels.
 - If you disable the LBM reflector, all MD levels are automatically deselected.

MiNID

Configuration>Physical Ports>Flows>General Params>SFP/Port 1>LBM Reflector

Admin status

Enable

01234567

MD Level:

☒☒☒☒☒☒☒☒

LBM Reflector – SFP/Port 1 (Enabled)

4. Click **<Apply>** to implement your changes, and then click **<Save Configuration>**.

Configuring LBM Reflector Using the CLI

➤ To configure LBM reflector parameters:

1. Navigate to **configure flows lbm-reflector sfp/1** or **configure flows lbm-reflecto msa/2**, according to whether you want to configure the parameters for SFP/port 1 or MSA/ port 2.

The `config>flows>lbm-reflector(sfp/1)#` or `config>flows>lbm-reflector(msa/2)#` prompt is displayed.

2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Specifying MD level for which to send LBR in response to LBM received with the specified MD level	<code>md-level <md-level></code>	Enter no md-level <md-level> in order to stop replying to LBMs received with the specified MD level MD level range: 0–7 To specify multiple MD levels, use the md-level command for each MD level
Administratively enabling LBM reflector	<code>no shutdown</code>	Enter shutdown to disable the LBM reflector. When you disable, all MDs are deselected (same as typing no md-level <md-level> for all MD levels). <i>Note: At least one MD level must be selected via command md-level before you can enable the LBM reflector.</i>

Example

- To configure MiNID to answer LBMs with MD level 2, 4, or 7, received from SFP/port 1:

```
exit all
configure flows lbm-reflector sfp/1
  shutdown
  md-level 2
  md-level 4
  md-level 7
  no shutdown
```

Configuring L2CP/L2PT

You can configure the following:

- MAC address change for packets classified to L2PT protocols (see [Configuring L2PT Protocol MAC Change Using the Web Interface](#) and [Configuring L2PT Protocol MAC Change Using the CLI](#))

- L2CP protocol classification (see **Configuring L2CP Protocol Classification Using the Web Interface** and **Configuring L2CP Protocol Classification Using the CLI**), based on the destination MAC address 01-80-C2-00-00-XX, where XX is 00–2F (excluding values used in **Flow General Parameter Defaults** table)
- L2CP/L2PT profiles (see **Configuring L2CP/L2PT Profiles Using the Web Interface** and **Configuring L2CP/L2PT Profiles Using the CLI**). Any configured L2CP protocol classifications or L2PT MAC address changes are added to existing profiles, and appear in the list of protocols when configuring profiles.

Configuring L2PT Protocol MAC Change Using the Web Interface

➤ To configure MAC address change:

1. In the web interface, go to Configuration > Physical Port > Flows Classification > General Params>L2CP\L2PT > L2PT.

MiNID

Configuration>Physical Ports>Flows>General Params>L2PT

Protocol Name	MAC Change	
STP	00-00-00-00-00-00	Mac Change
OAM	00-00-00-00-00-00	Mac Change
PAgP	00-00-AA-CC-00-BB	Mac Change
LLDP	00-00-00-00-00-00	Mac Change
CDP	00-00-00-00-00-00	Mac Change
VTP	00-00-00-00-00-00	Mac Change
UDLD	00-00-00-00-00-00	Mac Change
DTP	00-DD-00-02-00-04	Mac Change
PVSTP	00-00-00-00-00-00	Mac Change
LACP	00-00-00-00-00-00	Mac Change
LAMP	00-00-00-00-00-00	Mac Change
ESMC	00-00-00-00-00-00	Mac Change
PTP-Peer-Delay	00-00-00-00-00-00	Mac Change

L2PT Protocol MAC Changes

Note

The value **00-00-00-00-00-00** in the **MAC Change** field indicates that no MAC change is done.

- To configure a change of destination MAC address for a particular protocol, click <Mac Change> in the corresponding entry.

MINID

Configuration>Physical Ports>Flows>General>L2PT>Mac Change

Mac Change _____ 00-00-00-00-00-00

MAC Change

3. In the **MAC Change** field, enter the MAC address to which the destination MAC address should be changed, and then click **<Apply>** to apply the change, then click **<Save Configuration>**.

Configuring L2PT Protocol MAC Change Using the CLI

➤ To configure MAC address change:

1. Navigate to the configure flows context, and enter:
`l2cp-protocol {stp | oam | lldp | pagp | udld | cdp | vtp | dtp | pvstp} mac-change <mac-addr>`
 where `<mac-addr>` specifies the MAC address to which the destination MAC address should be changed for the particular protocol.

For example, to specify changing MAC address to 00-00-AA-CC-00-BB for LLDP protocol:

```
MiNID>config>flows# l2cp-protocol lldp mac-change 00-00-AA-CC-00-BB
```

2. You can display the configured L2PT protocol MAC changes by typing **info**, and searching the output for lines beginning with **l2cp-protocol**:

```
MiNID>config>flows# info
l2cp-protocol stp mac-change 00-00-00-00-00-00
l2cp-protocol oam mac-change 00-00-00-00-00-00
l2cp-protocol pagp mac-change 00-00-00-00-00-00
l2cp-protocol lldp mac-change 00-00-AA-CC-00-BB
l2cp-protocol cdp mac-change 00-00-00-00-00-00
l2cp-protocol vtp mac-change 00-00-00-00-00-00
l2cp-protocol udld mac-change 00-00-00-00-00-00
l2cp-protocol dtp mac-change 00-DD-00-02-00-04
l2cp-protocol pvstp mac-change 00-00-00-00-00-00
```

Configuring L2CP Protocol Classification Using the Web Interface

- To configure L2CP protocol classification:
1. In the web interface, go to Configuration > Physical Port > Flows Classification > General Params>L2CP\L2PT > L2CP.

MiNID

Configuration>Physical Ports>Flows>General>L2CP

Last MAC Byte
(Hex.)

01-80-C2-00-00-01

Remove

01-80-C2-00-00-0F

Remove

L2CP Protocol Last MAC Byte

Note A list of configured L2CP protocol classifications, if any exist, is displayed on the screen.

2. Enter the value for the last byte (e.g. to classify packets with destination MAC address 01-80-C2-00-00-07, enter **07**), and then click <Apply> to apply the change, then click <Save Configuration>.

Configuring L2CP Protocol Classification Using the CLI

- To configure L2CP protocol classification:
1. Navigate to the **configure flows** context, and enter:
l2cp-mac <last-mac-byte>
where <last-mac-byte> is a value from 0x00–0x2F that specifies the last byte of the MAC address 01-80-C2-00-00-XX. You cannot use values that are already used in **Flow General Parameter Defaults** table or have already been used to configure an L2CP profile classification.

For example, to specify classifying packets with destination MAC address 01-80-C2-00-00-0F:

```
MiNID>config>flows# l2cp-mac 0x0f
```

2. You can display the configured L2CP classifications by typing **info**, and searching the output for lines beginning with **l2cp-mac**:

```
MiNID>config>flows# info
l2cp-mac 0x01
l2cp-mac 0x0f
```

Configuring L2CP/L2PT Profiles Using the Web Interface

➤ **To create a new L2CP/L2PT profile:**

1. In the web interface, go to Configuration > Physical Port > Flows Classification > General Params>L2CP\L2PT > Profiles.

MiNID

[Configuration>Physical Ports>Flows>L2CP Profiles](#)

New Profile Name _____ rl2cp

Profile Name	
kl2cp	Remove
ml2cp	Remove

L2CP Profiles

2. Enter the name of the profile in **New Profile Name**, and then click **<Apply>**, to create the profile and add it to the list of profiles, and then click **<Save Configuration>**.

➤ **To configure an existing L2CP/L2PT profile:**

1. In the web interface, go to Configuration > Physical Port > Flows Classification > General Params>L2CP\L2PT > Profiles (see the figure above).
2. Click the name of the profile that you want to configure.

The profile screen displays an entry for each L2PT protocol specified in **Flow General Parameter Defaults** table, as well as entry for each configured L2PT protocol MAC change and each configured L2CP protocol classification.

MiNID

Configuration>Physical Ports>Flows>L2CP Profile>Cfg Profile

Profile Name

STP

Tunnel

OAM

Peer

PAgP

Tunnel

LLDP

Tunnel

CDP

Tunnel

VTP

Tunnel

UDLD

Tunnel

DTP

Tunnel

PVSTP

Pass-Through

LACP

Tunnel

LAMP

Tunnel

ESMC

Tunnel

PTP-Peer-Delay

Tunnel

PagpL2PT

Tunnel

L2CP Profile

- Configure the protocol actions (**Discard**, **Pass-Through**, **Tunnel**, or **Peer** for OAM protocol), and click **<Apply>** to apply your changes, then click **<Save Configuration>**.

Configuring L2CP/L2PT Profiles Using the CLI

➤ To add or configure an L2CP/L2PT profile:

- Navigate to **configure flows l2cp-profile** <profile-name>, where <profile-name> is the name of the profile.
The **config>flows>profile** (<profile-name>)# prompt is displayed.
- Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Specifying the action for L2CP protocol	<code>mac <last-mac-byte> {tunnel pass-through peer discard}</code>	<last-mac-byte> is a value from 0x00–0x2F that specifies the last byte of the MAC address 01-80-C2-00-00-XX. You cannot use values that are already used in Flow General Parameter Defaults table or have already been used to configure an L2CP profile classification tunnel – Frames are classified to flow and handled according to flow pass-through –Frames are transmitted unchanged to egress port peer – Frames are delivered to host treatment (OAM only) discard –Frames are dropped
Specifying the action for L2PT protocol	<code>protocol {stp oam lldp pagp udld cdp vtp dtp pvstp lacp lamp esmc ptp-peer-delay} [l2pt] {tunnel pass-through peer discard}</code>	See above for an explanation of the actions

Examples

Loopback in Port Mode

The following example assumes an initial connectivity testing phase, during which all traffic should be looped back. To achieve this, MiNID is configured for port mode on each port, with a loopback action with source-destination address swap.

```

configure
flows
classification mode port sfp/1
classification mode port msa/2
port-mode-action ingress-port sfp/1 action loop-mac-ip-swap
port-mode-action ingress-port msa/2 action loop-mac-ip-swap

```

Filtering Traffic by VID and Applying an S-Tag

In the following example, only customer network traffic with VIDs in the 100-200 range are to be accepted by the provider network, and an S-tag should be applied with VID 1000 and the same P-bit as in the C-tag. In the opposite direction, only provider traffic with VID 1000 should be passed through to the customer, with the S-tag removed.

```
configure
flows
classification mode vlan-pbits-range sfp/1
flow cust
classifier match vlan-min 100 pbit-min 0 vlan-max 200 pbit-max 7
copies P-bit from C-tag
flow-action ingress-port sfp/1 action push-vlan vlan-id 1000 priority inner-vlan-copy
tag-ether-type 0x8100 drop-eligible no
no shutdown
exit

flow net
classifier match vlan-min 1000 pbit-min 0 vlan-max 1000 pbit-max 7
flow-action ingress-port sfp action pop-vlan
no shutdown
exit
fallback-action ingress-port sfp/1 action drop
fallback-action ingress-port msa/2 action drop
Tpid-outer-vlan 0x88a8 ingress-port sfp/1
```

Layer-2 Loopback

In the following example, there is a testing device on each side of MiNID, and traffic should be looped back only when one is communicating with the other, in the single flow that is configured for each direction. The loopback should include address swap.

The address of the testing device on the customer side is: **12-34-56-78-90-ab**.

The address of the testing device on the provider side is: **cd-ef-12-34-56-78**

```
configure
flows
in-service-mac src-addr 1 12-34-56-78-90-ab
in-service-mac dst-addr 1 cd-ef-12-34-56-78
in-service-mac src-addr 2 cd-ef-12-34-56-78
in-service-mac dst-addr 2 12-34-56-78-90-ab
flow cust-traf
in-service-loops
in-serv-mode src-dst
in-serv-src-mac 1
in-serv-dst-mac 1
in-serv-action loop-mac-ip-swap
exit
exit
```

```
flow net-traf
in-service-loops
in-serv-mode src-dst
in-serv-src-mac 2
in-serv-dst-mac 2
in-serv-action loop-mac-ip-swap
exit all
```

Configuration Errors

The following messages may be generated by MinID when a configuration error is detected.

Flow Configuration Errors

Message	Cause	Corrective Action
A Parameter Is Missing	You entered a command in invalid syntax	Try again with valid syntax
P-Bit Value Is Out Of Range (0-7)	You tried to define a flow classification with an out-of-range P-bit	Try again with valid P-bit values
DSCP Value Is Out Of Range (0-63)	You tried to define a flow classification with an out-of-range DSCP value	Try again with a valid DSCP value
Vlan Id Is Out Of Range (0-4095)	You tried to define a flow classification with an out-of-range VID	Try again with valid VIDs
Min Vlan Id Is Bigger Then Max Vlan Id	In VID+P-bit classification mode you tried to define a flow classification with a vlan-min greater than the vlan-max	Try again with valid values
Vlan-Id\Pbits Can Not Be Overlapped	In VID+P-bit classification mode, you tried to define a flow classification with values already defined for another flow	Design your flows differently
MAC Address Format Is Invalid	You entered a command with a MAC address parameter in an invalid format	Try again with format XX-XX-XX-XX-XX-XX (each XX a valid hexadecimal number)

Message	Cause	Corrective Action
The Requested Flow Classification Already Exist	You tried to define a flow classification with a value for which a flow already exists	Delete or reconfigure the other flow
The Requested Flow Name Already Exist	In the web interface, you tried to Add a flow with a name already used for another flow	Use a different name
The Requested Flow Does Not Exist		
Flow Name Can't Be Empty	In the web interface, you tried to save a new flow without a name	Enter a name and try again
Flow Name Too Long, Max Name Length Is 10 Chars	You tried to create a flow with a name longer than ten characters	Shorten the name
Can't Enable a Flow Without a Classification Value	You tried to Enable (or, in CLI: no shutdown) a flow without a classification criterion	Configure the classification criterion and try again
Can't Enable a Flow Without an Action	You tried to Enable (or, in CLI: no shutdown) a flow without a flow action	Configure the flow action and try again
Can't Change Flow Configuration While Flow Is Enable	In CLI, you tried to reconfigure an enabled flow	Disable the flow and try again
Can't Add A New Flow, Resources Unavailable	You tried to create a new flow, but there are already the maximum number of flows configured (including disabled)	Delete a flow and try again

Viewing Flow Summary

In the web interface, you can view a list of existing flows in **Configuration > Physical Port > Flows Classification > Config Flow**. From the list, you can **Enable / Disable** or **Delete** each flow. Clicking a flow’s name displays its details, and if the flow is disabled you can then edit the details. You can also view the list and details (but not make any changes) in **Monitoring > Physical Port > Flows Classification > Flows Traffic**.

In the CLI, you can view flow details with the **info** command, for an individual flow in its context or for all flows in the **config>flows** context. You can also view a convenient (but less detailed) summary of the classification mode and configured flows by going to the **config>flows** context and entering:

show summary

For example:

```
MiNID>config>flows# show summary
Classification Mode:    VID+Pbits Range

Name:                  cust
Status:                Enable
Action:                push-vlan
Ingress-Port:          SFP/1
Match Value:           VID Range [100-200]+ P-bit [0-7]

Name:                  net
Status:                Enable
Action:                pop-vlan
Ingress-Port:          MSA/2
Match Value:           VID Range [1000-1000]+ P-bit [0-7]

MiNID>config>flows#
```

Viewing Flow Statistics

You can view traffic handling statistics for each flow and for unclassified (fallback) traffic. The statistics for each of these can be manually cleared, and changing the classification mode clears all of them.

Viewing Statistics for Unclassified (Fallback) Traffic

Statistics include the number of handled frames and the number of handled bytes, for both ports.

➤ **To view or clear statistics for unclassified (fallback) traffic:**

Do one of the following:

- In the CLI **configure flows** context:
 - View the statistics by entering:
show unclassified-statistics

For example:

```
MiNID# configure flows
MiNID>config>flows# show unclassified-statistics
Unclassified Port 1:
```

```
Total Unclassified Frames:      18678103
Total Unclassified Bytes:       1771890591
Total Unclassified In-Service MAC Frames:      0
Total Unclassified In-Service MAC Bytes:      0
Total Unclassified In-Service L3\L4 Frames:    0
Total Unclassified Matched In-Service L3\L4 Bytes: 0
Total Unclassified L2CP Frames:  341207
Total Unclassified L2CP Bytes:   21837248
Unclassified Port 2:
Total Unclassified Frames:      18678103
Total Unclassified Bytes:       1771890591
Total Unclassified In-Service MAC Frames:      0
Total Unclassified In-Service MAC Bytes:      0
Total Unclassified In-Service L3\L4 Frames:    0
Total Unclassified Matched In-Service L3\L4 Bytes: 0
Total Unclassified L2CP Frames:  341207
Total Unclassified L2CP Bytes:   21837248
```

- Clear the statistics by entering:
clear-statistics
- In the web interface, go to Monitoring > Physical Port > Classification > Unclassified traffic>Unclassified Port <Name>.

MinID

Monitoring>Physical Ports>Classification>Unclassified Port 1 Statistics

Total Matched Frames	18682494
Total Matched Bytes	1772301596
Total Matched In-Service MAC Frames	0
Total Matched In-Service MAC Bytes	0
Total Matched In-Service L3\L4 Frames	0
Total Matched In-Service L3\L4 Bytes	0
Total Matched L2CP Frames	341286
Total Matched L2CP Bytes	21842304

	Green Bytes	Yellow Bytes	Red Bytes
Total Matched Cos 0	0	0	0
Total Matched Cos 1	0	0	0
Total Matched Cos 2	0	0	0
Total Matched Cos 3	0	0	0
Total Matched Cos 4	0	0	0
Total Matched Cos 5	0	0	0
Total Matched Cos 6	0	0	0
Total Matched Cos 7	0	0	0
Total Matched Green Bytes	0	0	0
Total Matched Yellow Bytes	0	0	0
Total Matched Red Bytes	0	0	0

Statistics for Unclassified Traffic

You can click **Back** and clear these statistics by clicking **Clear Statistics**.

Note The above statistics are shared between both ports.

Viewing Statistics for a Specified Flow

Statistics include the number of handled frames and the number of handled bytes. Layer-2 (in-service) and layer-3/4 loopback statistics are also displayed. The layer-3/4 statistics include any TWAMP looped packets and bytes if the flow is bound to a TWAMP responder.

➤ **To view or clear statistics for a specified flow:**

Do one of the following:

- In the flow’s CLI **config>flows>flow(name)** context:
 - View the statistics by entering:
show statistics
- For example:

```
MiNID>config>flows# flow net
MiNID>config>flows>flow(net)# show statistics
Total Matched Frames      :      17962
Total Matched Bytes       :     1235401
Total Matched In-Service MAC Frames :      0
Total Matched In-Service MAC Bytes  :      0
Total Matched In-Service L3\L4 Frames:      0
Total Matched In-Service L3\L4 Bytes :      0
Total Matched L2CP Frames   :      0
Total Matched L2CP Bytes    :      0
MiNID>config>flows>flow(net)#
```

- To view statistics of all flows, use the **show summary** command.
 - Clear the statistics by entering:
clear-statistics
- In the web interface, go to **Monitoring > Physical Port > Classification > Port name**, and in the relevant row, click **Statistics**:

MiNID

Monitoring>Physical Ports>Classification>Flow Traffic port name

Flow Name	Status	
cust	Disable	Statistics
net	Enable	Statistics

Statistics for Flow Traffic

The statistics are displayed:

MiNID

Monitoring>Physical Ports>Classification>Statistics

Total Matched Frames	0
Total Matched Bytes	0
Total Matched In-Service MAC Frames	0
Total Matched In-Service MAC Bytes	0
Total Matched In-Service L3\L4 Frames	0
Total Matched In-Service L3\L4 Bytes	0
Total Matched L2CP Frames	0
Total Matched L2CP Bytes	0

Clear Statistics

Displayed Flow Statistics

You can **Clear** these statistics.

Viewing Statistics for a Flow with Envelope Policer

➤ To view or clear statistics for a specified flow:

Do one of the following:

- In the flow’s CLI **config>flows>flow(name)** context:
 - View the statistics by entering:
show statistics
- For example:

```
MiNID>config>flows>flow(Silver)# show statistics
Total Matched Frames      :      0
Total Matched Bytes       :      0
Total Matched In-Service MAC Frames :      0
Total Matched In-Service MAC Bytes  :      0
Total Matched In-Service L3\L4 Frames :      0
Total Matched In-Service L3\L4 Bytes :      0
Total Matched L2CP Frames :      0
Total Matched L2CP Bytes  :      0
Cos Number :      4
```

Total Matched Green Bytes	:	0
Total Matched Yellow Bytes	:	0
Total Matched Red Bytes	:	0
Cos Number :	5	

Total Matched Green Bytes	:	0
Total Matched Yellow Bytes	:	0
Total Matched Red Bytes	:	0
Cos Number :	7	

Total Matched Green Bytes	:	0
Total Matched Yellow Bytes	:	0
Total Matched Red Bytes	:	0

- To view statistics of all flows, use the **show summary** command.
- Clear the statistics by entering:
clear-statistics
- In the web interface, go to **Monitoring > Physical Port > Classification > Flows Traffic port name**, and in the relevant row, click **Statistics**:

The statistics are displayed:

MiNID

Monitoring>Physical Port>Classification>Statistics

Total Matched Frames	0		
Total Matched Bytes	0		
Total Matched In-Service MAC Frames	0		
Total Matched In-Service MAC Bytes	0		
Total Matched In-Service L3\L4 Frames	0		
Total Matched In-Service L3\L4 Bytes	0		
Total Matched L2CP Frames	0		
Total Matched L2CP Bytes	0		
	Green Bytes	Yellow Bytes	Red Bytes
Total Matched Cos 0	0	0	0
Total Matched Cos 1	0	0	0
Total Matched Cos 2	0	0	0
Total Matched Cos 3	0	0	0
Total Matched Cos 4	0	0	0
Total Matched Cos 5	0	0	0
Total Matched Cos 6	0	0	0
			0
Clear Statistics			

Displayed Statistics for Envelope Policer Flow

You can **Clear** these statistics.

Testing Flow Configuration

- To test flow configuration:
1. Send various types of traffic via MiNID, according to each flow, in both directions.
 2. Check that the traffic arrives properly at its destination.

3. View MiNID statistics (see [Viewing Flow Statistics](#)) and check that no traffic has been lost.

8.2 Quality of Service (QoS)

The MiNID Quality of Service (QoS) parameters include the following profiles:

- CoS map profiles
- Policer profiles
- Envelope profiles

Applicability and Scaling

This feature is applicable to all MiNID options.

Standards Compliance

MEF 10.3

Benefits

- QoS allows you to optimize bandwidth, avoiding the need to allocate excessive bandwidth to facilitate the necessary bandwidth for traffic at different requirements of speed and quality.

CoS Map Profiles

Class of Service (CoS) mapping profiles map the following user priorities to internal CoS values:

- **p-bit** — when ingress traffic is prioritized according to 802.1p requirements

Factory Defaults

By default, there are no CoS mapping profiles. When you create a CoS mapping profile, it is configured as follows:

- Classification p-bit
- Mappings:
 - Map 0 to CoS 7
 - Map 1 to CoS 6
 - Map 2 to CoS 5
 - Map 3 to CoS 4
 - Map 4 to CoS 3
 - Map 5 to CoS 2
 - Map 6 to CoS 1
 - Map 7 to CoS 0

Configuring CoS Mapping Profiles

➤ To define a CoS profile:

1. Navigate to the qos context (**config>qos**).
2. Define a CoS profile and assign a classification to it:
cos-map-profile <cos-mapping-profile-name>
3. Map the user priority to a CoS value (user priority values 0–7 for p-bit CoS values 0–7):
map <0-7> to <0-7>

Examples

➤ To create and configure a CoS mapping profile:

- Profile name: my-p-bit
- Map priority 6–7 to CoS 0
- Map priority 3–5 to CoS 1
- Map priority 0–2 to CoS 2

```
exit all
configure qos cos-map-profile my-p-bit
  map 6..7 to-cos 0
  map 3..5 to-cos 1
  map 0..2 to-cos 2
exit all
```

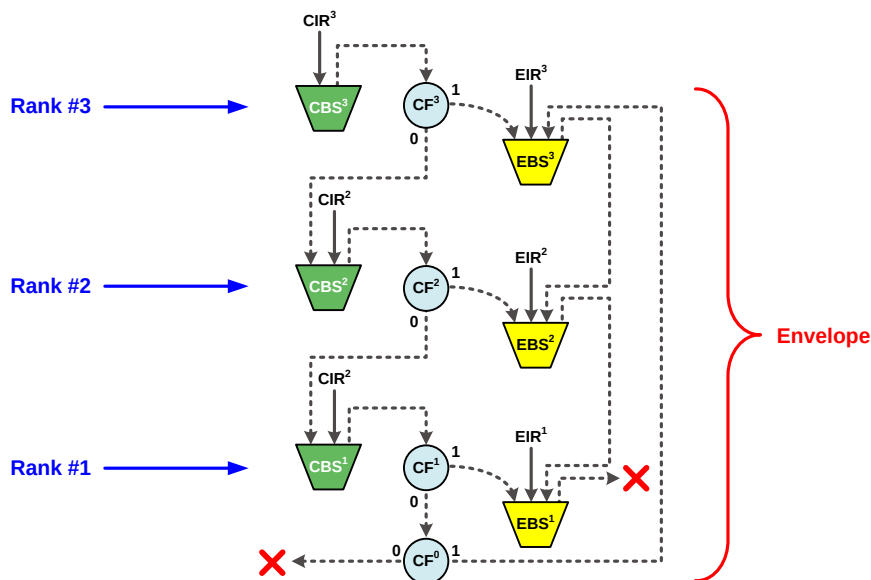
➤ To create and configure a CoS mapping profile for a multi-CoS flow:

- Profile name: p-bit-multi
- Map priority 0 to CoS 7
- Map priority 7 to CoS 0
- Map untagged traffic to CoS 0

```
exit all
configure qos cos-map-profile p-bit-multi classification p-bit
untagged-map to-cos 0
exit all
```

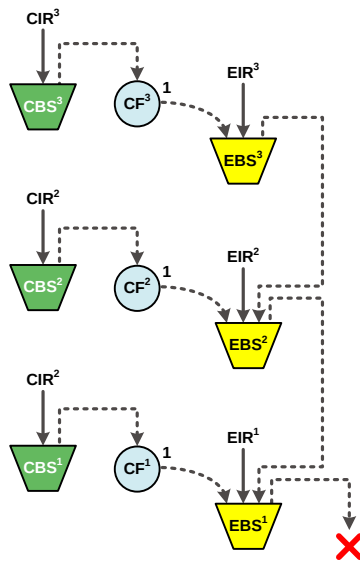
Envelope Bandwidth Profiles

An envelope profile as defined in MEF 10.3 contains a set of bandwidth profiles, each of which has been assigned a unique rank from 1 (lowest) to 4 (highest). Excess bandwidth from a higher rank can overflow to a lower rank to be shared, either to the committed or to the excess bucket. In MiNID, each profile corresponds to a separate CoS. The figure below illustrates an envelope profile with three CoSs. The coupling flags specify the path of overflow bandwidth. For the CoS coupling flags (CFi), 0=committed token bucket of the next lower rank, and 1=excess token bucket of the same rank. For coupling flag 0, 0=discard, and 1=excess token bucket of the highest rank.

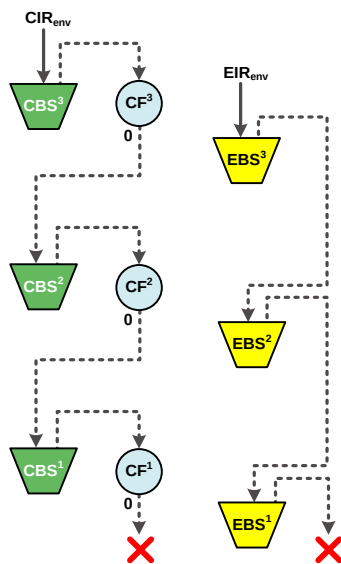


MEF 10.3 Bandwidth Profiles

When the envelope profile is assigned to a multi-CoS flow, it enables the flow to share excess bandwidth. The bandwidth sharing can be overflowed to the excess bucket or independent from the excess bucket.



Sharing Excess Bandwidth



Uncoupled from EIR/EBS

Configuring Envelope Profiles Using the Web Interface

➤ To configure an envelope profile:

1. Navigate to Configuration > QoS > Envelope Profile.

1. **MiNID**

2.

Configuration>QoS> Envelope Profiles

New Profile Name

Apply

Profile Name

2. In the **New Profile Name** field, type the envelope profile name.
3. Click <Apply>.

The new envelope profile is added to the list.

MiNID

Configuration>QoS> Envelope Profile

New Profile Name

Apply

Profile Name

1 Remove

4. Click the desired profile to configure profile parameters.

MiNID

Configuration>QoS> Envelope Profile>Cfg Profile 1

CF Policy	sharing-excess-bw
Color Aware	Disable
Compensation	0
Cos	0
CIR	0
CIR Max	1000000
CBS	0
EIR	1000000
EIR Max	1000000
EBS	32767
Coupling Flag	

Apply

5. Configure the profile:

- **CIR:** Defines the Committed Information Rate (CIR) for the current profile. The CIR specifies a bandwidth with committed service guarantee (“green bucket” rate).
- **CBS:** Defines the Committed Burst Size (CBS) for the current profile. The CBS specifies the maximum guaranteed burst size (“green bucket” size).
- **EIR:** Defines the Excess Information Rate (EIR). The EIR specifies an extra bandwidth with no service guarantee (“yellow bucket” rate).
- **EBS:** Defines the Excess Burst Size (EBS). The EBS specifies the extra burst with no service guarantee (“yellow bucket” size).
- **Compensation:** You can specify the amount of bytes that the policer can compensate for the layer 1 overhead (preamble and IFG) and the overhead for the added VLAN header in case of stacking.
- **Color Aware:** The policer profile can be specified as color aware.

6. Click <Apply>.

Configuring Envelope Profiles Using the CLI

This section explains how to configure envelope profiles, to apply to multi-Cos flows per MEF 10.3.

Adding Envelope Policer Profiles

- 1. Navigate to **configure qos**.
The **config>qos#** prompt is displayed.
- 2. Type:
envelope-profile <envelope-profile-name>
An envelope profile with the specified name is created and the following prompt is displayed:
config>qos>envelope-profile(<envelope-profile-name>)\$
The new envelope profile parameters are configured by default.
- 3. Configure the envelope profile as described in [Configuring Envelope Profile Parameters](#).

Configuring Envelope Profile Parameters

- 1. Navigate to **configure qos envelope-profile <envelope-profile-name>** to select the envelope profile to configure.
The **config>qos>policer-profile(<envelope-profile-name>)#** prompt is displayed.
- 2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Defining policy for excess bandwidth sharing	cf-policy {sharing-excess-bw uncoupled-bw-sharing}	sharing-excess-bw –Excess bandwidth is shared to excess token bucket (see Sharing Excess Bandwidth figure). Selecting this parameter automatically sets coupling-flag-0 to 0, and sets each CoS coupling flag to 1. uncoupled-bw-sharing – Excess bandwidth is shared independently from EIR/EBS (see Uncoupled from EIR/EBS figure). Selecting this parameter automatically sets coupling-flag-0 to 0, and sets each CoS coupling flag to 0. If you enter no cf-policy, then coupling-flag-0 and each CoS coupling flag determine the bandwidth sharing.
Specifying if the envelope profile is color aware	color-aware	
Specifying the compensation (bytes)	compensation <value>	<value> is -(-32)-31

Task	Command	Comments
Specifying the CIR (Kbps), CBS (bytes), EIR (Kbps), and EBS (bytes) bandwidth limits, for a particular CoS	<code>cos <value> bandwidth [cir <cir-kbit-sec> [cir-max <cir-max-kbit-sec> [cbs <cbs-bytes> [eir <eir-kbit-sec> [eir-max <eir-max-kbit-sec> [ebs <ebs-bytes> [coupling-flag <coupling-fla g>]</code>	Range for cos value is 0–7; you can define up to four or eight cos values in an envelope profile. Range for <cir-kbit-sec> , <cir-max-kbit-sec> , <eir-kbit-sec> , and <eir-max-kbit-sec> : 0–1000000 (0–1 Gbps) Range for <cbs-bytes> , <ebs-bytes> : 0–1000000 (0–1Mbytes) <cir-max-kbit-sec> must be greater than or equal to <cir-kbit-sec> . <eir-max-kbit-sec> must be greater than or equal to <eir-kbit-sec> . coupling-flag controls the path of overflow tokens: 0=overflow to committed token bucket, 1=overflow to excess token bucket.
Specifying path of overflow bandwidth (see CF0 in MEF 10.3 Bandwidth Profiles figure)	<code>coupling-flag-0 <value></code>	<value> is 0–1: 0=discard, and 1=excess token bucket of the highest rank

Policer Profiles

This section explains how to configure non-envelope policer profiles to limit traffic. The profiles can be applied to flows.

Configuring Policer Profiles Using the Web Interface

- To configure a policer profile:
1. Navigate to **Configuration > Qos > Policer Profile**.

MiNID

Configuration>QoS> Policer Profiles

New Profile Name

Apply

Profile Name

- 2. In the **New Profile Name** field, type the profile name.
 - 3. Click <Apply>.
- The new policer profile is added to the list.

MiNID

Configuration>QoS> Policer Profiles

New Profile Name

Apply

Profile Name

1Remove

- 4. Click the desired profile to configure profile parameters.

MiNID

Configuration>QoS>Policer Profile>Cfg Profile

CIR [Kbps]	0
CBS [Bytes]	0
EIR [Kbps]	1000000
EBS [Bytes]	32767
Coupling Flag	Disable
Color Aware	Disable
Compensation	0

Apply

- Use the table below to fill in and select the desired parameters.
- Click <Apply>.

Configuring Policer Profiles Using the CLI

➤ To add a policer profile:

- Navigate to **configure qos**.
The **config>qos#** prompt is displayed.
- Type:
policer-profile <policer-profile-name>
A policer profile with the specified name is created and the following prompt is displayed:
config>qos>policer-profile(<policer-profile-name>)\$
The new policer profile parameters (except for name) are configured by default as described in the table below.
- Configure the policer profile as described in **Configuring Policer Profile Parameters**.

➤ To configure policer profile parameters:

- Navigate to **configure qos policer-profile <policer-profile-name>** to select the policer profile to configure.
The **config>qos>policer-profile(<policer-profile-name>)#** prompt is displayed.

2. Enter all necessary commands according to the tasks listed below.

Policer Profile Parameters

Task	Command	Comments
Specifying the CIR (Kbps), CBS (bytes), EIR (Kbps), and EBS (bytes) bandwidth limits	bandwidth [cir <cir-kbit-sec>] [cbs <cbs-bytes>] [eir <eir-kbit-sec>] [ebs <ebs-bytes>]	Notes: • <i>CIR & EIR allowed values: 0–1000000</i> • <i>CBS & EBS allowed values: 0–1000000</i> • <i>CIR can be set to zero only if CBS is set to zero and vice versa</i> • <i>EIR can be set to zero only if EBS is set to zero</i> • <i>CBS should be greater than the maximum frame size</i>
Specifying if the policer profile is color aware	color-aware	
Specifying the compensation (bytes)	compensation <-32-31>	
Specifying whether to check CIR+EIR when determining packet color	coupling-flag	Relevant only for color-aware policer profile in

Examples

➤ **To create and configure a policer profile named Policer4:**

- CIR = 50,000 Kbps
- CBS = 28,000 bytes
- EIR = 30,000 Kbps
- EBS = 20,000 bytes
- Compensation = 30

```
exit all
configure qos policer-profile Policer4
  bandwidth cir 50000 cbs 28000 eir 30000 ebs 20000
  compensation 30
```

```
exit all
```

- To display the configuration information for policer profile Policer4:

```
MiNID# configure qos policer-profile Policer4
MiNID>config>qos>policer-profile(Policer4)# info
    bandwidth cir  49984 cbs  28000 eir  29952 ebs  20000
    compensation   30
```


9 Timing and Synchronization

9.1 Network Time Protocol (NTP)

Network Time Protocol (NTP) is a networking protocol for clock synchronization between computer systems over packet-switched, variable-latency data networks.

Factory Defaults

The default configuration for NTP parameters:

- No NTP servers defined
- Poll interval set “fast mode”
- Date-format “yyyy-mm-dd”
- No “summer time”

Configuring NTP

➤ **To configure NTP in the web interface:**

1. Navigate to Configuration > System > Date and Time.

Configuration>System>Date and Time**NTP**

Date Format	yyyy-mm-dd
Date	2016-3-29
Time	12:02:46
Zone (utc)	00:00
Summer Time	No summer time
Type	
Apply	

1. Fill in the fields according to the tasks listed below:
 - Date Format: Available formats (yyyy-mm-dd | dd-mm-yyyy | mm-dd-yyyy | yyyy-dd-mm)
 - Date: Current date
 - Time: Current time (hh:mm[:ss])
 - Zone: Desired time zone (Allowed range of values: -12:00 to +12:00, in 30-minute increments)
 - Summer-Time Type:
 - No summer time – No summer time differences
 - Recurring – Define a scheme for the range of summer time dates, including the Offset in minutes
 - Date – Define a specific date and time, including the Offset in minutes
2. Click <Apply> to save changes.
3. Click NTP.

Configuration>System>Date and Time>Config NTP

Add New NTP Server

Poll-Interval	Fast-Mode
Apply	
NTP Server	

- 1. Select the desired poll interval and click <Apply>.
- 2. Click Add New NTP Server.

Configuration>System>Date and Time>NTP> Add NTP Server

Server Index	
Admin Status	Enable/Disable
Server IP Address	
UDP Port Number	
Apply	

- 3. Fill in the fields according to the tasks listed below:
 - Server Index: The index is automatically selected
 - Admin Status: Enable/Disable
 - Server IP Address: NTP server’s IP address
 - UDP Port Number: The port used by the NTP server
- 4. Click <Apply> to save the configuration.

➤ To display NTP Server Information:

- Navigate to Monitoring > System > NTP.

Monitoring>System> NTP						
NTP Server	Address	Type	UDP Port Received	TStamp	DateTime (UTC)	Stratum
						Query-Server

All existing NTP servers and information are displayed.

Setting the Date and Time

➤ To set the system date and time in the CLI:

1. Navigate to configure system date-and-time.
The **config>system>date-time#** prompt is displayed.
2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Specifying the desired date format	date-format { yyyy-mm-dd dd-mm-yyyy mm-dd-yyyy yyyy-dd-mm }	
Defining the date	date <date>	Date is according to the configured date format
Defining the time zone relative to Coordinated Universal Time (UTC)	zone utc [<[{+ -}]<hh>[:<mm>]>]	Allowed range of values: -12:00 to +13:00, in 1-hour increments
Defining the time	time <hh:mm[:ss]>	

Displaying the Date and Time

➤ To display the date and time in the CLI:

- From the system context (config>system), enter:
show time

Setting NTP Parameters

➤ To configure NTP parameters in the CLI:

1. Navigate to configure system date-and-time ntp.
The **config>system>date-time>ntp#** prompt is displayed.
2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Setting polling interval (in minutes) for NTP requests	poll-interval < minutes >	Allowed range is 8–131072
Defining and configuring NTP servers	server < server-id >	
Displaying NTP status	show status	

Defining NTP Servers

➤ To define an NTP server in the CLI:

1. Navigate to config system date-and-time ntp.
The **config>system>date-time>ntp#** prompt is displayed.
2. Type server <server-id> to define an NTP server with ID <server-id>.
The following prompt is displayed: **config>system>date-time>ntp>server(<server-id>)\$. .**
3. Configure the NTP server parameters as needed, as described in Setting NTP Server Parameters.

Setting NTP Server Parameters

➤ To configure NTP server parameters in the CLI:

1. Navigate to config system date-and-time ntp.

The **config>system>date-time>ntp#** prompt is displayed.

- 2. Type server <server-id> to select the NTP server to configure.

The following prompt is displayed: **config>system>date-time>ntp>server(<server-id>)#**

- 3. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Setting the IP address of the server	address <IP-address >	
Setting UDP port for NTP requests, to a specific UDP port or to default UDP port (123)	udp port <udp-port >	• Allowed value 123 or • Allowed range of 1024–65535
Enabling and defining summer time parameters	summer time <recurring/date >	• Recurring- Define dates for a recurring summer time • Date- Manually define summer time dates
Administratively enabling server	no shutdown	Using shutdown disables the server
Sending query to server and displaying result	query-server	

Examples

➤ **To set the date and time:**

- Format = mm-dd-yyyy
- Date = May 17, 2016
- Time = 5:40pm
- Zone = GMT –4 hours and 30 minutes.

```
exit all
configure system date-and-time
  date-format mm-dd-yyyy
  date 05-17-2016
  time 17:40
  zone utc -04:00
```

➤ To define NTP server:

- Server ID = 1
- IP address = 192.1.1.1
- Administratively enabled.

```
exit all
configure system date-and-time ntp
server 1
address 192.1.1.1
no shutdown
```

➤ To display server information:

```
# configure system date-and-time ntp server 1
>config>system>date-time>ntp>server(1)# query-server
Query Server Reply
```

```
-----
Server   : 192.1.1.1      UDP    : 123
Date     : 00-00-0       Time   : 00:00:00
Stratum  : 0
```

```
>config>system>date-time>ntp>server(1)# exit
```

```
>config>system>date-time>ntp# show status
```

```
System Uptime : 006 Days 116:19:55
```

```
System Time   : 2016-05-04          13:01:09
```

NTP Server	Type	UDP Port	Tstamp	Date	Time(UTC)	Stratum	Received
192.1.1.1	Enable	123	00-00-0000	00:00:00	1	000 Days 00:00:25	

9.2 Sync-E Clock Source

SFP sleeve and SFP unit can transparently transfer the Sync-E clock and ESSM packets between its ports. You can define whether to transfer from the MSA port or the SFP port.

Applicability and Scaling

Copper SFPs do not support Sync-E transfer, except for SFP-ToC, which supports only the direction SFP to MSA.

Standards Compliance

ITU-T G.8262

Benefits

You have the flexibility to define which port transfers the Sync-E clock.

Functional Description

You can configure the Sync-E clock source as the MSA port (to transfer the Sync-E clock from the MSA port to the SFP port), or the SFP port (to transfer the Sync-E clock from the SFP port to the MSA port).

If a physical fault (LOS) occurs on the clock source port, automatically sets the clock source to the system internal oscillator.

When the port fault is cleared, the clock source returns to the user-configured clock source.

Note

Port status polling rate is 400 ms, therefore it can take up to 400 ms for the clock source to change back to the configured source.

Factory Defaults

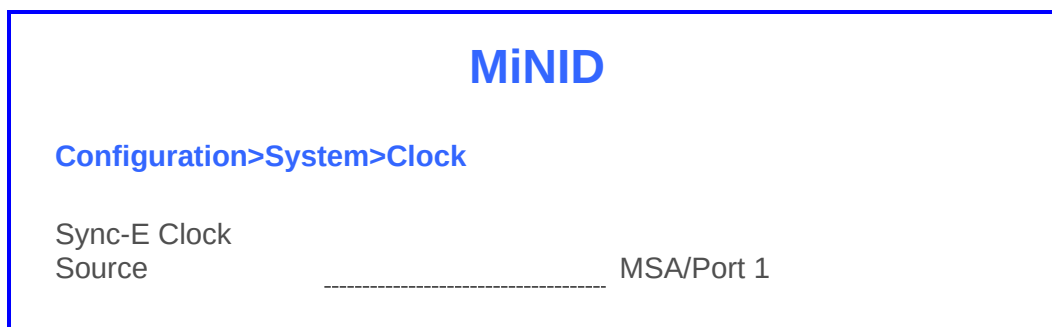
By default, transfers the Sync-E clock from the MSA port to the SFP port.

Configuring the Sync-E Clock Source

➤ To define the Sync-E clock source:

Do one of the following:

- In the web interface, go to Configuration > System > Clock, select SFP/Port 2 to transfer timing packets from the SFP/Port 2 port to the MSA/Port 1 port; or MSA/Port 1 to transfer timing packets from the MSA/Port 1 port to the SFP/Port 2 port; then click <**Apply**> to implement the changes, and then click <**Save Configuration**>.



- In the CLI, navigate to the configure system clock context, and do one of the following:
 - To transfer timing packets from the SFP/Port 2 port to the MSA/Port 1 port, enter:
sync-e-src sfp
 - To transfer timing packets from the MSA/Port 1 port to the SFP/Port 2 port, enter:
sync-e-src msa

10 Administration

10.1 Device Information

To distinguish MiNID from other devices in the system, you can configure it with descriptive information, including a description, a name and location, and a contact person.

Factory Defaults

The **device-name** is **MiNID** by default. The parameters **contact** and **location** are empty (no value).

Configuring Device Information

➤ To enter device information:

Do one of the following:

- In the web interface, go to Configuration > System > Management > Device Info, enter the information and click <Apply> to implement the changes, and then click <Save Configuration>.

MiNID

Configuration>System>Management>Device Info

Description	Ethernet Demarcation SFP
Name	MiNID
Location	RAD
Contact Person	RAD

- In the CLI, go to the configure system context, and configure any of the following (to empty the parameter, use no):
 - [no] contact <contact name>
 - [no] device-name <device name>
 - [no] location <location>

Note *The device information that you enter should not contain spaces, in both the web interface and CLI.*

10.2 File Operations

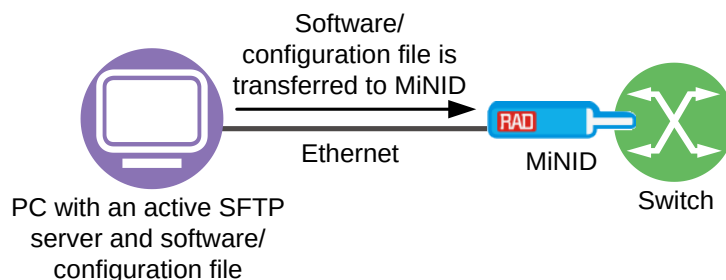
You can perform the following file operations:

- Copy application software, boot software, or configuration files, either locally in MiNID or between MiNID and a remote PC via SFTP or TFTP
- Display information on the configuration files and software files (software packs)
- Delete files

SFTP Application

The SFTP protocol is used to provide secure file transfers via the product's Ethernet interface. SFTP is a version of FTP that encrypts commands and data transfers, keeping your data secure and your session private. For SFTP file transfers, an SFTP server application must be installed on the local or remote computer.

A variety of third-party applications offer SFTP server software. For more information, refer to the documentation of these applications.



Downloading a Software or Configuration File via SFTP

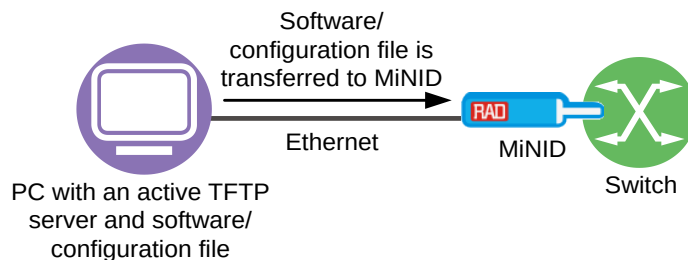
If you use a local laptop and SFTP is the preferred transfer method, an SFTP server application must be installed on it. SFTP file transfers are carried out through TCP port 22. You should check that the firewall you are using on the server computer allows communication through this port.

TFTP Application

The TFTP protocol is typically used for remote IP-to-IP file transfers via the product's Ethernet interface. It can be used, however, for local file transfer as well, as the transfer rate of the Ethernet interface is much faster than using a terminal emulation program with the SFP-CA.2 adaptor for the SFP sleeve.

For TFTP file transfers, a TFTP server application must be installed on the local or remote computer. As it runs in the background, the TFTP server waits for any TFTP file transfer request originating from the product, and carries out the received request automatically.

A variety of third-party TFTP applications are available that allow the instant creation of a TFTP server on a client computer. For more information, refer to the documentation of these applications.



Downloading a Software or Configuration File via TFTP

If you use a local laptop and TFTP is the preferred transfer method, a TFTP server application must be installed on it. TFTP file transfers are carried out through port 69. You should check that the firewall you are using on the server computer allows communication through this port.

Copying Files

You can use the Web interface or CLI to copy files within the MiNID unit, or download/upload files to the MiNID unit via SFTP/TFTP.

After a file copying operation completes, an event is written in the event log that indicates the result of the file copy.

If a configuration file is downloaded to MiNID, it is executed after the next device reset if applicable according to the MiNID startup sequence (refer to [Operation](#)).

The following files can be uploaded or downloaded:

- boot – Boot software (can be downloaded only)
- log-file – Event log file (can be uploaded only)
- running-config – Configuration file containing current configuration that the device is running (can be uploaded only)
- startup-config – Configuration file containing saved non-default user configuration
- sw-pack-1, sw-pack-2 – Application software files (can be downloaded only)
- user-default-config – Configuration file containing default user configuration

The following files can be copied within the MiNID unit:

- Source file can be startup-config, running-config or user-default-config
- Destination file can be startup-config or user-default-config

Note

- *A file cannot be copied to itself*
- *A configuration file cannot be copied to the log file, and the log file cannot be copied to a configuration file.*

Copying Files Using the Web Interface

➤ **To copy files via the web interface:**

1. Go to Utilities > File > Copy.

MinID

Utilities>File>Copy

Protocol

Server IP

Direction

Remote File Name

Local File Name

TFTP

0.0.0.0

Remote to Local

sw-pack-1

sw-pack-1

2. Fill in the parameters according to the table below.
3. Click <Apply> to initiate the file copy.
You are asked to confirm the copying.
4. Click <Yes> to confirm.
The file copying is performed.

Copy File Web Parameters

Parameter	Description
Protocol	Specifies protocol for copying: - TFTP – File transfer via TFTP server - SFTP – File transfer via SFTP server - Local – Copy files within the MinIDunit
SFTP User Name	User name for SFTP server Note: Appears only if protocol is SFTP
SFTP Password	Password for SFTP server Note: Appears only if protocol is SFTP
Server IP	IP address of the PC where the SFTP/TFTP server is located Note: Appears only if protocol is SFTP or TFTP

Parameter	Description
Direction	• Remote to Local – File transfer from SFTP/TFTP server to MiNID • Local to Remote – File transfer from MiNID to SFTP/TFTP server Note: Appears only if protocol is SFTP or TFTP
Remote File Name	Name of file on remote PC Note: Appears only if protocol is SFTP or TFTP
Direction	• Remote to Local – File transfer from SFTP/TFTP server to MiNID • Local to Remote – File transfer from MiNID to SFTP/TFTP server Note: Appears only if protocol is SFTP or TFTP
Remote File Name	Name of file on remote PC Note: Appears only if protocol is SFTP or TFTP
Local File Name	Name of local file, can be one of the following: • startup-config • user-default-config • running-config • log-file Note: Appears only if protocol is SFTP or TFTP
Source File Name	Source file for local copy, must be one of the following: • startup-config • user-default-config • running-config Note: Appears only if protocol is Local
Destination File Name	Destination file for local copy, must be one of the following: • startup-config • user-default-config Note: Appears only if protocol is Local

Copying Files Using the CLI

➤ To copy files:

- 1. Navigate to the **file** level.
- 2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Copying file from remote PC via SFTP/TFTP server to MinID	copy sftp://<username>:<password>@<ipv4-address>/ <remote-file-name> { sw-pack-1 sw-pack-2 boot startup-config user-default-config copy tftp://<ipv4-address>/<remote-file-name> { sw-pack-1 sw-pack-2 boot startup-config user-default-config}	<username> – User name for SFTP server <password> – Password for SFTP server <ipv4-address> – IP address of the PC where the SFTP/TFTP server is located <remote-file-name> – Name of file on remote PC
Copying file from MinID to remote PC via SFTP/TFTP server	copy { startup-config running-config user-default-config log-file } sftp://<username>:<password>@<ipv4-address>/ <remote-file-name> copy { startup-config running-config user-default-config log-file } tftp://<ipv4-address>/<remote-file-name>	See above for parameter descriptions
Copying file within MinID	copy {startup-config running-config user-default-config } {startup-config user-default-config }	

Deleting Files

You can delete the following files:

- startup-config
- user-default-config

➤ **To delete a file:**

Do one of the following:

- In the web interface:
 - a. Go to Utilities > File > Delete.

MiNID

Utilities>File>Delete

Image Type

startup-config

- b. Select the file to delete, and click <**Apply**>.

Note

*The **Image Type** dropdown list contains only the files that can be deleted, and exist in the device.*

You are prompted to confirm the deletion.

- c. Confirm the deletion.

The file is deleted.

- In the CLI:
 - a. At the file# prompt, enter:
delete <file-name>
You are prompted to confirm the deletion.
 - b. Confirm the deletion.
The file is deleted.

Example

```
>file# delete user-default-config
Are You Sure? [y/n] y
>file#
*****user-default-config Deleted Successful*****
```

Displaying File Information

You can display the following:

- Information on the software files, including the status and contents of the active software version and the software packs, as well as the date the software packs were downloaded if applicable. For information on installing a software pack as active, refer to Chapter 12 (Upgrade) for details.
- Contents of startup and user configuration files (CLI only).

Displaying File Information Using the Web Interface

➤ To display information on the software files:

- Navigate to Configuration > System > Install SW-Pack .

MiNID

[Configuration](#) > [System](#) > [Install](#)

Image Type _____ sw-pack-1

Apply

Name	Status	Version	Date
Active	Exist	2.4.0 (0.37)	
sw-pack-1	Exist	2.4.0 (0.37)	21-06-2016
sw-pack-2	Exist	2.1.0 (0.53)	17-05-2015

Display Software File Information

Displaying File Information Using the CLI

➤ To display information on the software files:

- At the file# prompt, enter:
show sw-pack

```
MiNID# file
MiNID>file# show sw-pack
```

Active	Status: Exist	Version: 2.4.0(0.37)
SW-PACK 1	Status: Exist	Version: 2.4.0(0.37) Date: 21-06-2016

SW-PACK 2	Status: Exist	Version: 2.1.0(0.53) Date: 17-05-2015
-----------	---------------	---------------------------------------

➤ **To display the contents of the startup or user configuration files:**

- At the file# prompt, enter one of the following:
 - show startup-config
 - show user-default-config

The contents of the specified configuration file are displayed.

10.3 Inventory

You can view the MiNID inventory table, displaying the unit's description, hardware and software revisions, as well as the device serial number (S\N).

Standards Compliance

RFC 4133: Entity MIB

Benefits

You can monitor the installed components and hardware and software revisions.

Displaying Inventory Information

➤ **To display the inventory table:**

Do one of the following:

- In the web interface, go to Inventory:

MiNID

Inventory

Name	MiNID.Chassis
Description	RAD.MiNID.Chassis
Hw version	1.2\1.1 [GE]
Board Type	MiNID SFP / MiNID Sleeve /
CSL	D
FP Rev	2016-04-17(99)
CP Rev	2015-06-01(0.2)
FW Rev	2016-03-16(0)
SW Rev	2.4.1(0.03) [PLATINUM \ DEMARC]
BootVer	2.0.0(0.12)
SW Date\Time	Sep 18 2016\15:51:56
S\N	1234567890ABCDEF

- In the CLI, in the root context enter:
inventory
For example:

```
MiNID# inventory
                        Inventory
[-----]
Name:      MiNID.Chassis
Description: RAD.MiNID.Chassis
HW Rev:    1.2\1.1 [GE]
CSL:       D
Board Type: MiNID Sleeve
FP Rev :   2016-06-26(101)
CP Rev :   2016-09-05(0.4)
FW Rev :   2016-06-14(0)
SW Rev:    2.6.0(0.07) [PLATINUM]
BootVer:   2.0.0(0.12)
SW Date\Time: Sep 18 2018\15:51:56
S\N:       1234567890ABCDEF
[-----]
```

10.4 Reset

You can reset MiNID in the following ways:

- Reset to factory defaults. All settings return to factory defaults, except for host IP and VLAN parameters. To reset these, use the DIP switches (refer to [Setting the DIP Switches](#)).
- Reset to user defaults (from user-default-config)
- Overall reset (restart) of the device

Resetting to Factory Defaults

➤ **To reset MiNID to factory defaults:**

Do one of the following:

- In the web interface, go to Configuration > System > Set Factory Defaults.
- In the CLI, in the admin context, enter:
factory-default

In both interfaces, you are prompted to confirm.

Resetting to User Defaults

➤ **To reset MiNID to user defaults:**

Do one of the following:

- In the web interface, go to Configuration > System > Set User Defaults.
- In the CLI, in the admin context, enter:
user-default

In both interfaces, you are prompted to confirm.

Restarting the Unit

► To restart MiNID:

Do one of the following:

- In the web interface, go to Configuration > System > Reset Device.
- In the CLI, in the admin context, enter:
reboot

In both interfaces, you are prompted to confirm the reboot.

11 Monitoring and Diagnostics

11.1 Layer-3 Service Activation Test

The Layer-3 Service Activation Test (L3 SAT) provides an out-of-service (intrusive) IP/UDP test to assess the proper configuration and performance of an IP transport service prior to customer notification and delivery.

Applicability and Scaling

This feature is applicable to all MiNID options.

Standards Compliance

ITU-T Y.1564

Benefits

The Y.1564 testing methodology allows service providers to have a standard way of measuring the performance of IP transport services. The tests are performed per multiple traffic streams simultaneously, confirming policing per EVC or EVC.CoS.

Functional Description

L3 SAT testing has the following objectives:

- Validate that the IP transport service is correctly configured
- Validate the quality of the services as delivered to the end user

L3 SAT tests can be performed over Layer-3 networks.

L3 SAT supports service performance test.

Performance Test

The performance test validates the quality of the services over a user-configurable period of time, as follows:

- Traffic is generated for all services at the configured bandwidth level.
- For all the test sessions, test packets are sent simultaneously at 100% of the bandwidth configured per test session
- Per test session, the duration of the performance test is evenly divided between the different packet sizes, e.g. per test session, each packet size is transmitted for an equal amount of time

The performance test is declared successful if the results are within SAC limits.

Test Elements

L3 SAT includes the following elements:

- Generator – initiates multiple test sessions for multiple responders, sends out the test and OAM frames, receives responses from the responder, processes the resulting measurements, and displays test reports. Generator can support two responder types: TWAMP Light and IP Loop.
- Peer – is used to run TWAMP test sessions. Only one peer can be configured per generator with IP address corresponding to responder
- Test session – Only one test session can be configured per peer
- Responders – receive test and OAM frames from generator and transmit responses to generator. Responders can be of the following types:
 - TWAMP Light – receives a test packet, reflects the test packet after it, adds an Rx stamp and a Tx stamp
 - IP loop – filters incoming traffic by destination IP address, and loops it back while performing MAC address swap and IP address swap

L3 SAT generator supports a special packet format (short-packet mode) configured at the generator level. This mode can be enabled only with ip-loop responder.

Factory Defaults

By default, there are no L3 SAT entities configured in MiNID.

When a peer profile is created, it has the following default configuration:

Parameter	Default	Remarks
performance-duration	2h	
udp-port	53248	
responder-type	twamp-light	

When a session profile is created, it has the following default configuration.

Parameter	Default	Remarks
delay-threshold	200000	
delay-variation-threshold	100000	
ip-size	256	
loss-ratio-threshold	100	

When a peer is created, there is no default configuration.

When a test session is created, it has the following default configuration.

Parameter	Default	Remarks
<name>	<Not applicable>	<name> must be specified when the test session is created.
session-profile	<Not applicable>	<profile-name> must be specified when the test session is created.
bandwidth	100	<kbps> must be specified when the test session is created.
dscp	0	

When a generator is created, it has the following default configuration.

Parameter	Default	Remarks
<name>	<Not applicable>	<name> must be specified when the test session is created.

Parameter	Default	Remarks
session-profile	<Not applicable>	<profile-name> must be specified when the test session is created.

Parameter	Default	Remarks
peer	<not applicable>	This parameter has no default configuration.
local-ip-address	<not applicable>	This parameter has no default configuration.

Configuring L3 SAT Entities

To configure L3 SAT, perform the following steps:

1. In the responder device:
 - a. Configure relevant router interface and flow.
 - b. Configure and activate L3 SAT responder.
2. In the generator device:
 - a. Configure relevant router interface and flow.
 - b. Configure L3 SAT peer and session profile.
 - c. Configure and activate L3 SAT generator and relevant peer and test session.

Note*Only one peer profile and one session profile could be defined.*

Configuring a Generator

- **To configure an L3 SAT generator using the web interface:**
1. Navigate to Configuration > Test > L3SAT > Generator.
 2. Type in the name in the New Generator Name field.
 3. Click <Apply>.
- The new generator appears as a row in the generators table.

MiNID

Configuration>Test>L3SAT>Generator

New Generator

Name

Apply

Generator Name	Router Interface	Router Interface oper status	Generator Status	
Gen1	N/A	N/A	Not Ready	Remove

1. Click the generator name in the table.
The generator profile page opens.
2. Set the generator Local IP Address and Admin Status.
3. Click <Apply>.

MiNID

Configuration>Test>L3SAT>Generator> Generator Profiles (Gen1)

Local IP Address

Router Interface 1

Add New Peer

Admin Status

Disable

Short Packet Mode

Disable

Apply

Peer	Short Packet Mode	Last Connectivity Sub-test	Responder Type	Test Name	UDP Port	Status	Summary Report	Remove
------	-------------------	----------------------------	----------------	-----------	----------	--------	----------------	--------

➤ **To configure an L3 SAT generator using the CLI:**

1. Navigate to configure test l3sat.
The **config>test>l3sat#** prompt is displayed.
2. Enter:
generator <name>
The **config>test>l3sat>generator<name>#** prompt is displayed.
3. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Assigning a router interface to the L3SAT generator	local-ip-address <router-interface-1 router-interface-2 router-interface-3 router-interface-4 router-interface-5>	If Loaned IP is enabled on router interface #1, generator IP address can be configured on any other router interface
Defining peer entity (corresponding to generator) to run L3SAT session.	peer <ip-address>	
Configuring the L3SAT test mode and the packet format	[no] short-packet-mode	short-packet-mode: 64–2028 bytes frame no short-packet-mode: 95–10240 bytes frame
Administratively enabling or disabling the generator	no shutdown	Enter shutdown to administratively disable the generator
Viewing generator status	show status	

Configuring a Peer

➤ **To configure an L3 SAT peer using the web interface:**

1. On the Generator Profile page, click Add New Peer.
2. On the Peer Profile page, set the parameters as detailed in the table below.
3. Click <Apply>.

MinID

Configuration>Test>L3SAT>Generator> Profiles Config>Add Peer

Activate

Peer IP Address

Peer Profile

Apply

Disable

0.0.0.0

peer_profile

Test session Name	Session Profile	Bandwidth	DSCP	Report	Remove
-------------------	-----------------	-----------	------	--------	--------

➤ To configure an L3 SAT peer using the CLI:

- 1. Navigate to configure test l3sat generator <name>.
The **config>test>l3sat>generator(<name>)#** prompt is displayed.
- 2. Enter:
peer <ip-address>
The prompt **config>test>l3sat>generator(<name>)> peer(<ip-address>)#** is displayed.
- 3. Enter all necessary commands according to the tasks listed below.

L3SAT Peer Parameters

Task	Command	Comments
Activating or deactivating the peer test sessions	activate no activate	You can activate a peer only if at least one test session has been configured
Assigning a peer profile to use for the peer parameters	peer-profile <profile-name>	
Assigning a test session	test-session <name> session-profile <session-profile> bw <bw-kbps> [dscp <dscp-number>]	bw –Rate of the test session traffic in Kbps. Range 4-1000000 kbps dscp - Priority value for the test session traffic. Range: : 0-63 (default: 0)

Task	Command	Comments
Displaying results and measurements for a specific test	show report	Available only if peer was activated
Displaying summary of test results and measurements	show summary-report	Available only if peer was activated
Displaying the peer status	show status	

Configuring a Peer Profile

You can configure only one peer profile to associate with the peer.

➤ To configure an L3 SAT peer profile using the web interface:

1. Navigate to Configuration > Test > L3SAT > Peer Profile.
2. In the New Peer Profile Name field, fill in the name.
3. Click <Apply>.

The peer profile appears in the table.

MiNID

Configuration>Test>L3SAT>Peer Profile

New Peer Profile Name

Apply

Peer Profile Name

peer_prof

Remove

4. Click the peer profile name.
The Peer Profile configuration page opens.
5. Edit the peer profile parameters according to the tasks in the table below.

MiNID

Configuration>Test>L3SAT>Peer Profile>Peer Profiles Cfg (peer_prof)

Performance Duration

2h

UDP Port

53248

Responder Type

TWAMP Light

Apply

➤ To configure an L3 SAT peer profile using the CLI:

- 1. Navigate to configure test l3sat.
The **config>test>l3sat>#** prompt is displayed.
- 2. Enter:
peer-profile <name>
The **config>test>l3sat>peer-profile<name>#** prompt is displayed.
- 3. Enter all necessary commands according to the tasks listed below.

L3SAT Peer Profile Parameters

Task	Command	Comments
Configuring the duration of the session	performance-duration <15m 2h 24h custom>	Custom duration is measured in minutes.
Specifying the type of the responder	responder-type {twamp-light ip-loop}	
Specifying UDP port number	udp-port <port-number>	Range is <1-65535>

Configuring a Session Profile

You can configure only one L3 SAT session profile to associate with the peer.

➤ To configure an L3 SAT session profile using the web interface:

- 1. Navigate to Configuration > Test > L3SAT > Session Profile.
- 2. In the New Session Profile Name field, fill in the name.

- 3. Click **<Apply>**.
The session profile appears in the table.

MiNID

Configuration>Test>L3SAT>Session Profile

New Session Profile Name _____

Apply

Peer Profile Name	
session_prof	Remove

- 4. Click the session profile name.
The Session Profile configuration page opens.
- 5. Edit the session profile parameters according to the tasks in the table below.
- 6. Click **<Apply>**.

MiNID

Configuration>Test>L3SAT>Session Profile>Session Profiles Cfg
(session_prof)

Loss Ratio Threshold

100

Delay Threshold

200000

Delay Variation Threshold

100000

Packet Size

128

256

512

1024

1280

1500

Custom

0

☐

☒

☐

☐

☐

☐

☐

L1 Rate

Disable

Apply

➤ To configure an L3 SAT session profile using the CLI:

- 1. Navigate to configure test l3sat.
The **config>test>l3sat>#** prompt is displayed.
- 2. Enter:
session-profile <name>
The **config>test>l3sat>session-profile<name>#** prompt is displayed.
- 3. Enter all necessary commands according to the tasks listed below.

L3 SAT Session Profile Parameters

Task	Command	Comments
Defining delay threshold in microseconds	delay-threshold	Range 0–1000000
Defining delay variation threshold in microseconds	delay-variation-threshold	Range 0–1000000

Task	Command	Comments
Defining if the test session bandwidth parameter is a Layer 1 or Layer 2 rate	l1-rate	Default: no l1-rate (Layer 2 rate)
Defining allowable test packet loss in ppm (parts per million)	loss-ratio-threshold	
Defining test packet size in bytes	packet-size {128 256 512 1024 1280 1500} } [custom <custom-size>]	- You can specify up to four packet sizes - Range for custom size: 95–2028

Configuring Router Interface and Flow

To configure a router interface, see [Configuring Router Interface](#).

To configure a flow, see [Configuring Flows Using the Web Interface](#) or [Configuring Flows Using the CLI](#).

Viewing L3 SAT Status

You can view the status of the test as it is running:

- Generator status
 - in the web interface - the Generator row in the table as on [Configuring a Generator](#).
 - in the CLI - **show status** in the config test l3sat generator level

```
MinID>config>test>l3sat>generator(gen)# show status
Router Interface           : 1
Router Interface oper status : UP
Generator Status           : Ready
```

- Peer test session status
 - in the Web interface – the Peer row in the table as on [Configuring a Peer](#)
 - in the CLI – show status in level config test l3sat generator peer.

```
MinID-Generator>config>test>l3sat>generator(1)>peer(11.11.11.3)# show status
Last Connectivity Sub-test : Pass
Short Packet Mode          : Enable
Responder Type              : IP Loop

Test Name      UDP Port      status
-----
```

```
MinID_SA                               862                               Pass
MinID-Generator>config>test>l3sat>generator(1)>peer(11.11.11.3)#
```

Viewing L3 SAT Test Reports

The generator calculates performance parameters according to the received test packets, for the peer and its active test session. Some performance parameters are recalculated on the fly (duration, delay, PDV, etc.); other parameters are recalculated at the end of the session (lost packets, loss ratio) The performance parameters are presented in test reports that can be viewed per peer and test session.

➤ To display detailed test results in the CLI:

```
MinID-Generator>config>test>l3sat>generator(1)>peer(11.11.11.3)# show report
End Points
Generator Address      : 11.11.11.1
Responder Address      : 11.11.11.3
Short Packet Mode      : Enable
Responder Type         : IP Loop
UDP Port               : 862
Test Rate Mode         : L1

Test
Scope                  : Performance
Peer Profile Name      : peer_profile
BW [Kbps] - L1         : 4
DSCP                   : 0
Frame Size             : 128

Session Profile Name   : l3sat_session
Start Date & Time      : 00:39:50 01-01-1970
End Date & Time        : 00:40:53 01-01-1970
Total Duration         : 00:01:03
Overall Result         : Pass

Performance Phase
-----
Duration [min]         : 1:03
Performance Result     : Pass

Parameter              Frame Size  Threshold
#1
128 bytes
-----
Duration [min]         1:00
Tx Rate [Kbps]         4
IR - mean [Kbps]       4.990
PL - count             0
PLR                    0.000%      1.00E-01
PTD - min [ms]         0.020
```

```
PTD - mean [ms]          0.020      200000
PTD - max [ms]           0.021
PDV - mean [ms]          0.000
PDV - max [ms]           0.000      100000
-----
Result                   Pass
MinID-Generator>config>test>l3sat>generator(1)>peer(11.11.11.3)#
```

L3 SAT Report Parameters

Counter	Description
Information rate (IR)	Number of received times test packet Ethernet frame length (in bits), divided by the elapsed time (in seconds)
Frame rate (frms/sec)	Shown on web only. Frame rate takes the total frame size into consideration (frame size plus preamble and inter-frame gap)
Packet loss (PL)	Number of lost test packets. A test packet is considered lost if it was not received back at the generator, or was received with a round-trip delay of over two seconds.
Packet loss ratio (PLR)	Number of lost packets divided by the number of transmitted packets
One-way inter-packet delay variation (IPDV-Fwd)), forward – mean	Average forward IPDV. IPDV is calculated according to RFC 3393, from the variations of the delays between valid packets.
One-way inter-packet delay variation (IPDV-Fwd), forward – max	Maximum forward IPDV
One-way inter-packet delay variation (IPDV-Bck), backward – mean	Average backward IPDV. IPDV is calculated according to RFC 3393, from the variations of the delays between valid packets.
One-way inter-packet delay variation (IPDV-Bck), backward - max	Maximum backward IPDV
Round-trip packet transfer delay (PTD) –min	Minimum round-trip PTD. The round-trip PTD is calculated from the test packet embedded timestamps. A round-trip PTD over two seconds is ignored, as the packet is considered lost.
Round-trip PTD – mean	Average round-trip PTD
Round-trip PTD – max	Maximum round-trip PTD
Round-trip delay variation (PDV) – mean	Average round-trip PDV. The round-trip PDV is calculated according to ITU-T Y.1540, by subtracting the minimum PTD from the 99.9% percentile of the PTD values

Counter	Description
Round-trip delay variation (PDV) – max	Maximum round-trip PDV

➤ To view a summary report in the web interface:

1. Navigate to generator profile (Configuration > Test > L3SAT > Generator) and click the generator name in the table.
The generator profile configuration screen opens.
2. In the peer row, click **summary report**.

➤ To view a summary report in the CLI:

```
MiNID-Generator>config>test>l3sat>generator(1)>peer(11.11.11.3)# show summary-report
End Points
Generator Address      : 11.11.11.1
Responder Address      : 11.11.11.3
Short Packet Mode      : Enable
Responder Type         : IP Loop
Test Rate Mode         : L1

Test
Scope                  : Performance
Peer Profile Name      : peer_profile
Start Date & Time      : 00:39:50 01-01-1970
End Date & Time        : 00:40:53 01-01-1970
Total Duration         : 1:03
Overall Result         : Pass

Test Name              BW      DSCP      Performance Result
                       [Kbps]
-----
MiNID_SA               4        0        Pass

MiNID-Generator>config>test>l3sat>generator(1)>peer(11.11.11.3)#
```

11.2 Microburst Monitoring

Granular bandwidth utilization and Microburst monitoring is a Platinum feature available with the Platinum installation package. This feature enables you to monitor network traffic, on per-flow or

multiple flows basis, and detect unexpected bursts of data in very short time intervals measured in microseconds.

Applicability and Scaling

This feature is applicable to all MiNID options.

Benefits

Microburst monitoring provides clear indication of traffic and bursts passed, dropped or exceeding thresholds based on CIR/CBS and EIR/EBS. In addition, it helps facilitate bandwidth on demand.

Functional Description

Similar to OAM services statistics, microbursts statistics are collected for a certain PM interval, while this interval is defined from 1 to 900 seconds.

When a Regular policer profile is used, excess bandwidth is shared or discarded, and the corresponding statistics are collected. When a Regular Accounting Only profile is used, only the statistics are collected, but the network traffic remains intact.

Configuring Microburst Monitoring

➤ To configure microburst monitoring, perform the following steps:

1. Configure a policer profile (refer to [Policer Profiles](#)).
2. Configure a flow.

➤ To configure a flow with the web interface:

1. Navigate to **Configuration>Physical Ports>Flows>** [SFP/ Port 1 | MSA/ Port 2] and click **Add Flow**.
2. Set the PM collection parameters:
 - a. Set PM Collection to Enable.

- b. Set PM Interval to the desired number of seconds to monitor traffic behavior during the interval.
 - c. Set Policer type to Regular or Regular Accounting Only.
3. Click **Apply**.
4. Set the rest of the parameters as described in **Per-Flow Tasks**.

MiNID

Configuration>Physical Ports>Flows>Add Flow

Flow Name	_____	Flow 1
Service Name	_____	Gold
Admin Status	_____	Enable
Ingress Port	_____	SFP/Port 1
L2CP Profile	_____	
L3L4 Loop	_____	Disabled
PM Collection	_____	Enable
PM Interval	_____	1
Policer Type	_____	Regular Accounting Only
Policer Profile	_____	mburst
Classification Tag Mode	_____	Single
Min VID Value	_____	0
Max VID Value	_____	4095
Min P-bit Value	_____	5
Max P-bit Value	_____	5
Flow Action	_____	Push VLAN

Apply

Add Flow with PM Collection

5. Navigate to **Monitoring > Physical Port > Ethernet > Port** [SFP/ Port 1 | MSA/ Port 2]. Click **Statistics**.
Port statistics is displayed for the specified PM Interval.

➤ **To configure a flow with the CLI:**

1. Go to the **configure flows** context, and enter:
flow <name>:
 where <name> is the flow name
 The CLI enters the flow context.
2. Type **pm-collection interval** <seconds> to enable the mode of collecting PM data and set the duration of the collection in seconds.
3. Type **policer regular-accounting-only** <profile-name> or **policer profile** <profile-name> to associate the defined profile with the flow.
4. Type **show statistics** to display the flow statistics.

Viewing Flow Statistics

➤ **To display the flow statistics, do one of the following:**

- In the CLI, go to **config flows flow** <flow-id> and type **show statistics**.

```
MiNID>config>flows>flow(1)# show statistics
Total Matched Frames           :      10000
Total Matched Bytes            :      64000
Total Matched In-Service MAC Frames :      0
Total Matched In-Service MAC Bytes :      0
Total Matched In-Service L3\L4 Frames :      0
Total Matched In-Service L3\L4 Bytes :      0
Total Matched L2CP Frames      :      0
Total Matched L2CP Bytes       :      0
Total Matched Green Bytes      :      50000
Total Matched Yellow Bytes     :      10000
Total Matched Red Bytes        :      4000
```

- In the web interface, navigate to **Monitoring > Physical Port > Classification > Port** and click **Statistics** in the corresponding Flow row.

11.3 OAM (CFM)

Ethernet Connectivity Fault Management (CFM) is a service-level OAM protocol that provides tools for monitoring and troubleshooting end-to-end Ethernet services. This includes proactive connectivity monitoring, fault verification, fault isolation, and performance monitoring. CFM uses standard Ethernet frames and can be run on any physical media that can transport Ethernet service frames.

Applicability and Scaling

This feature is applicable to all MiNID options.

Standards Compliance

IEEE 802.1ag-D8

ITU-T Y.1731

Benefits

OAM functionality ensures that network operators comply with QoS guarantees, detect anomalies before they escalate, and isolate and bypass network defects. As a result, operators can offer binding service-level agreements.

Functional Description

MiNID provides the OAM (CFM) functions listed below in packet-switched networks:

- End-to-end Connectivity Fault Management (CFM) per IEEE 802.1ag:
 - Continuity check (CC)
 - Non-intrusive loopback, used to detect loss of bidirectional continuity
 - Link Trace for fault localization
- End-to-end performance monitoring per ITU-T Y.1731:
 - Delay measurement
 - Loss measurement
 - Unavailability/Availability

Frames must contain at least one VLAN to be considered as OAM (CFM) frames. You can configure maintenance associations to work in the following modes:

- Untagged mode – OAM frames that run without a VLAN.
- Single-tag mode – Only frames with one VLAN are classified as OAM frames.
- Double-inner-tag mode – Only frames with two VLANs (inner and outer) are classified as OAM frames. The inner VLAN is used for handling incoming OAM frames

- Double-outer-tag mode – Only frames with two VLANs (inner and outer) are classified as OAM frames. The outer VLAN is used for handling incoming OAM frames.

OAM Elements

The Ethernet OAM mechanism is implemented by the following elements:

- Maintenance Domains (MDs) – MDs designate the part of the network for which connectivity faults can be managed. Each maintenance domain has a unique MD level that designates the scope of its monitoring, from 0–7.
- Maintenance Associations (MAs) – Every MA belongs to a maintenance domain (MD), and inherits its level from the MD to which it belongs. An MD can contain up to eight MAs, which inherit the MD level and must have unique names. An MA can contain up to two maintenance end points (MEPs), one for each port, or one maintenance intermediate point (MIP).
- Maintenance End Points (MEPs) – MEPs are active OAM CFM entities that provide MA endpoints. MEPs are identified based on the corresponding MD level, MA name, and MEP ID (is unique in the corresponding MA). Additionally, each MEP is associated with multiple remote MEPs, a VLAN, and a port (SFP/port 1 or MSA/port 2) through which to transmit frames corresponding to the MEP. All OAM frames generated by a particular MEP use the MEP VLAN properties (VLAN ID and p-bit information).

MEPs generate and receive CFM PDUs and track responses, as follows:

- Validating received OAM frames. Incoming frames are processed by the MEP if the frame VLAN properties match the MEP VLAN properties.
- Forwarding OAM frames with MD level higher than the MEP MD level
- Discarding OAM frames with a lower MD level than the MEP MD level
- Transmitting periodic continuity check messages (CCMs), if CCM transmission is enabled, to the configured remote MEP(s)
- Performing OAM loopback (LBM):
 - Transmitting LBM frames when LBM is initiated
 - Receiving OAM loopback response (LBR) frames corresponding to previously transmitted LBMs
 - Responding to LBM frames with LBR frames.
- Performing OAM link trace (LTM):
 - Transmitting LTM frames when LTM is initiated
 - Receiving OAM loopback response (LTR) frames corresponding to previously transmitted LTMs
 - Responding to LTM frames with LTR frames.
- Performing delay measurement on PM services belonging to the MEP:

- Transmitting delay measurement message (DMM) frames periodically when accordingly configured
 - Receiving delay measurement response (DMR) frames corresponding to previously transmitted DMMs
 - Responding to DMM frames with DMR frames.
 - Performing loss measurement on PM services belonging to the MEP:
 - Transmitting loss measurement message (LMM/SLM) frames periodically when accordingly configured
 - Receiving loss measurement response (LMR/SLR) frames corresponding to previously transmitted LMMs
 - Responding to LMM/SLM frames with LMR/SLR frames.
 - Reporting defects when detected.
- PM Services – Provide loss and delay measurements for the relevant MEP. Up to 32 PM services can be created per device; 8 PM services per MEP.
 - Destination Network Elements (Dest NEs) –Provide performance monitoring for the relevant service
 - Maintenance Intermediate Points (MIPs) – Unlike MEPs, MIPs are passive OAM entities that respond to OAM frames but do not initiate actions. MIPs are identified based on the corresponding MD level and MA name. Additionally, each MIP is associated with a VLAN. All OAM frames generated by a particular MIP use the MIP VLAN properties (VLAN ID and p-bit information).
- MIPs perform the following:
- Validating received LTM and LBM frames. Incoming frames are processed by the MIP if the frame VLAN properties match the MIP VLAN properties
 - Responding to LBMs with LBRs
 - Responding to LTMs with LTRs, and forwarding the received LTM frames.

Note

MIPs ignore CCM frames.

OAM Messages

The Ethernet service OAM mechanism uses cyclic messages for availability verification, fault detection, and performance data collection. The main message types are described in the following sections.

Continuity Check Messages (CCMs)

CCMs are sent periodically (if enabled), in order to detect connectivity loss, referred to as loss of continuity (LOC), between each MEP and its remote MEPs in an MA. The CCM functionality also allows detection of unintended connectivity. The CCM interval is user-configurable at the MA level to 100 ms, 1 sec, 1 min, or 10 min.

If an error is detected by the MEP, CCM frames are transmitted to the appropriate remote MEP with the RDI bit set. When the MEP receives notification that the error was cleared, it clears the RDI bit.

CCMs are transmitted as unicast or multicast:

- Unicast – CCMs are transmitted to a configurable MAC address
- Multicast – CCMs are transmitted to the MAC address corresponding to the MEP MD level, as shown below.

Multicast MAC Address

01-80-C2-00-00-3y	
MD Level	Four Address Bits “y”
7	7
6	6
5	5
4	4
3	3
2	2
1	1
0	0

Alarm Indication Signal (AIS)

If a MEP detects a connectivity failure, it propagates AIS frames in the direction away from the detected failure, with the level set to the client MD level configured for the MEP (the default client MD level is the MEP MD level incremented by 1). The AIS is sent for the following trigger events:

- LOS signal from network side (the MEP stops transmitting AIS frames when it receives indication that the defect is cleared)

- LOC detected (the MEP stops transmitting AIS frames when it receives indication that the defect is cleared)
- CCM received with Interface Status TLV with status set to down (the MEP stops transmitting AIS frames when it receives indication that the defect is cleared)
- AIS received (the MEP stops transmitting AIS frames when no AIS frames are received within an interval of 3.5 seconds).

AIS frames are transmitted periodically at 1-second intervals. The frames are multicast AIS frames with VLAN parameters (VLAN ID, priority) as configured in the MA.

- Note**
 - *AIS transmittal can be enabled or disabled per MEP. If disabled, the MEP does not transmit AIS frames and ignores any received AIS frames.*
 - *If the MEP MD level is 7 (the highest MD level) it does not transmit AIS frames.*
 - *MIPs do not transmit AIS frames.*

When the original defect is cleared, the MEP clears the AIS defect condition.

Loopback Messages (LBMs)

The OAM loopback functionality (ETH-LB) is used to verify the bidirectional connectivity of a MEP with a MIP or remote MEP, and is initiated upon user request. The OAM loopback is a ping-like request/reply function, where an LBM is sent by a MEP to a MIP or remote MEP, and a loopback response (LBR) is expected in response to the LBM.

LBM transmission mode is one of the following:

- Multicast – One LBMs is transmitted to the MAC address corresponding to the MEP MD level, as shown in the table in **Continuity Check Messages (CCMs)**.
- Unicast – LBMs are transmitted to a user-configurable MAC address
- Remote MEP –MEP LBMs are transmitted in unicast mode to the remote MEP MAC address.

LBM supports user-configurable data TLV length to provide the capability of testing the loopback with different frame sizes, with a data pattern of all zeros.

- Note**

An LBM session cannot be started if one is already in progress.

LBMs are sent at the rate of approximately 40 frames per second. LBRs are expected to arrive within 5 seconds after the last LBM frame transmitted. If an LBR frame arrives after this time interval, it is considered as invalid and is discarded. LBMs are sent with transaction identifiers that must match the received LBRs, therefore the MEP also detects and reports out-of-order LBRs.

When a MEP/MIP receives an LBM, it replies with an LBR if the LBM is valid. The LBM is considered valid if the following are true:

- The LBM MD level is equal to the MEP/MIP MD level
- According to the frame transmission mode of multicast or unicast:
 - Unicast –The LBM MAC address is equal to the MEP/MIP MAC address
 - Multicast – The LBM MAC address is equal to the multicast MAC address corresponding to the MEP/MIP MD level (see the table in [Continuity Check Messages \(CCMs\)](#)).

Note *In addition to LBM/LBR sent and replied to by MEPs, has an LBM reflector feature that enables replying with LBR to any LBM packet received with any VLAN and any MD level, per port SFP/port 1 or MSA/port 2. See [Configuring LBM Reflector](#) in Chapter 8.*

Link Trace Messages (LTMs)

The Ethernet Link Trace function (ETH-LT) traces the path to a target MEP/MIP by retrieving adjacency relationships between a MEP and a MIP or remote MEP, therefore it can be used for fault localization, as the retrieved sequence of MIPs/MEPs is likely to be different if a fault exists. The ETH-LT function is initiated upon user request.

The result of running an ETH-LT is a sequence of MIPs/MEPs from the source MEP to the target MIP/MEP. Each MIP /MEP is identified by its MAC address.

When ETH-LT is initiated, the MEP sends a multicast LTM frame containing the TTL value, and the target device MAC address that terminates the link trace. The LTM frame is sent to the multicast MAC address corresponding to the MEP MD level (see the table in [Continuity Check Messages \(CCMs\)](#)).

When a MEP/MIP receives an LTR, it verifies if the transaction identifier value is expected by the receiving MEP/MIP. If the LTR frame contains an unknown transaction identifier, it is considered as invalid and is discarded. LTRs are expected to arrive within 5 seconds after the last LTM frame transmitted. If an LTR frame arrives after this time interval, it is considered as invalid and is discarded.

When a MEP/MIP receives an LTM, it generates a unicast LTR to the initiating MEP if the LTR is valid. It also forwards the LTM to the target destination MAC address, if it is not the target. When the frame is forwarded, the TTL is decremented by 1. The LTM is considered valid if the following are true:

- The LTM MD level is equal to the MEP/MIP MD level
- The LTM TTL is greater than 0

Performance Monitoring

PM services provide loss and two-way delay measurements for MEPs, based on ITU-T Y.1731. The measurements are between the corresponding local MEP and its configured remote MEP. Performance monitoring frames are unicast frames sent periodically to the remote MEP MAC address, at user-configurable intervals (100 ms, 1 sec, or 10 sec).

The following performance parameters are measured:

- Loss measurement (LM):
 - Frame Loss Ratio (FLR) –Number of service frames not delivered, divided by the number of service frames received. supports two synthetic loss measurement methods and a real traffic loss measurement method:
 - SLM Synthetic – Synthetic SLM/SLR frames are used to calculate the loss ratio
 - LMM Synthetic – Synthetic LMM/LMR frames are used to calculate the loss ratio
 - User Data – Real traffic LMM/LMR frames with OAM type CCM, APS, or CSF are used to calculate the loss ratio. With this method, you can configure whether to count green, yellow, or both green and yellow frames, according to the frame DEI bit.
- Delay measurement (DM):
 - Frame Delay (FD) –Round trip delay for a frame, defined as the time elapsed since the start of transmission of the first bit of the frame by a source node, until the reception of the last bit of the loopbacked frame by the same source node, when the loopback is performed at the frame's destination node
 - Frame Delay Variation (FDV) –Difference between two subsequent frame delay measurements.

When a MEP receives an LM or DM frame, it is considered valid if it has a valid MD level, and destination MAC address equal to the receiving MEP. If valid, the MEP responds with an LMR/SLR or DMR frame. When a MEP receives an LMR/SLR or DMR frame, it calculates the loss or delay measurements.

Note

Loss measurement and delay measurement can be enabled or disabled per PM service.

Capacity

OAM frame size is 64–2048 bytes.

MiNID supports the following per device:

- Up to eight MDs

- Up to 32 MAs
- Up to 32 MEPs or MIPs (one MIP per MA)
- Up to 32 remote MEPs
- Up to 32 PM services
- Up to 32 destination NEs (up to 32 destination NEs per PM service)
- Up to 32 simultaneous LM or DM sessions.

Factory Defaults

By default, there are no configured OAM CFM entities.

Configuring OAM (CFM/PM)

The Ethernet OAM (CFM) configuration procedure includes the following steps, detailed in this section:

1. Configure a Maintenance Domain (MD).
2. Configure Maintenance Associations (MA).
3. For each Maintenance Association, configure a MIP or configure MEPs.
4. For each MEP, configure remote MEPs.
5. For each MEP, configure PM services.
6. For each PM service, configure destination NEs.

Configuring MDs

The following parameters are configured in the MD.

MD Parameters

Parameter		Description
Web	CLI	
MD Index	<md index>	

Parameter		Description
Web	CLI	
MD level	<mdid>	Maintenance domain level of MD to which MA belongs; used as maintenance domain index in the CLI
MD name format	<string mac-and-uint dns>	Maintenance domain name format: • String-Alphanumerical value • DNS • MAC and UINT • No MD Name Present
MD name	<name>	Maintenance domain name

- **To add an MD in the web interface:**
1. Navigate to Configuration>OAM>CFM\PM.
 2. Select **Legacy** or **SOAM** Alarm type.
 3. Click Add New MD.
 4. Fill in the parameters specified in the table above.
 5. Click <**Apply**>.
- A message is displayed that the MD was successfully created.

- **To configure the MD in the web interface:**
1. Navigate to Configuration>OAM>CFM\PM.
 2. Click **config** in the MD index
- A new MA can be created in this screen.

- **To add an MD in the CLI:**
- Navigate to configure>oam cfm and configure the parameters listed in the table above.

```
MinID# configure oam cfm maintenance-domain 1
MinID>config>oam>cfm.md<1># md-level 1
MinID>config>oam>cfm.md<1># name string one
```

Configuring MAs

The following parameters are configured in MAs.

MA Parameters

Parameter		Description
Web	CLI	
[Not visible]	<maid>	Maintenance association index (1–8); must be unique in the device. When an MA is created via the Web, the MA index is automatically generated by MinID.
MA Name	name	Maintenance association name, according to MA name format; must be unique in the corresponding MD
MA Format	format	MA name format: • Character string – Alphanumeric value (up to 43 characters) • 2-octet integer – Unsigned integer decimal number (0–65535) • ICC based –Name is specified as the ITU carrier code that is assigned to the relevant network operator/service provider. The codes are maintained by ITU-T as defined in ITU-T Rec. M.1400
Vlan Tag Mode	tag-mode	Indicates whether MA works in single-tag mode, double-tag mode with inner VLAN used to handle incoming OAM frames, double-tag mode with outer VLAN used to handle incoming OAM frames or untagged mode
CCM Interval	ccm-interval	Specifies the interval for periodic transmission of CCMs (see Continuity Check Messages (CCMs)). Allowed values are: • 100ms • 1s • 10s • 1min • 10min
Interface Status TLV	interface-status-tlv	Allowed values are: • Disable • Enable

➤ To add an MA in the web interface:

1. Navigate to Configuration > OAM > CFM\PM.

MiNID

Configuration>OAM>CFM\PM>Config MD (index)

Add new MA

MA Number	MA Name	MD Level	Vlan Tag Mode	Service Name		
1	MA1	3	Single	Gold	remove	config
2	MA2	3	Double-Inner	Silver	remove	config
3	MA3	2	Double-Outer	Gold	remove	config
4	MA4	1	Double-Inner	Silver	remove	config
5	MA5	4	Single	Gold	remove	config
6	MA6	3	Untagged	Gold	remove	config

Configuring OAM (CFM)

- 2. Click Add new MA.

MiNID

Configuration>OAM>CFM\PM>Add MA (MD index)

MA Number

3

MA Name

MA6

MA Format

Character String

Vlan Tag Mode

Single

CCM Interval

1s

Interface Status

Disable

TLV

Apply

New MA

MiNID

Configuration>OAM>CFM\PM>Add MA (MD index)

MA Number	3
MA Name	MA6
MA Format	Character String
Vlan Tag Mode	Double-Inner
CCM Interval	1s
Interface Status TLV	Disable

Apply

New MA After Changing VLAN Tag Mode to Double-Inner

1. Fill in the parameters as specified in the MA Parameters table.
2. Click **<Apply>** to create the MA.
A message is displayed that the MA was successfully created.
3. Click **<OK>** to close the message.
4. Click **<Save Configuration>**.

➤ **To configure a maintenance association in the web interface:**

1. Navigate to Configuration>OAM>CFM.
2. Click **config** in the relevant MD row.
3. Click the MA name.

MiNID

Configuration>OAM>CFM

MA Name	MA4
MA Format	Character String
Vlan Tag Mode	Single
CCM Interval	1s
Interface Status TLV	Disable

Updating MA Parameters – Single Tag Mode

MiNID

Configuration>OAM>CFM

MA Name	MA5
MA Format	Character String
Vlan Tag Mode	Double-Inner
CCM Interval	1s
Interface Status TLV	Disable

Updating MA Parameters – Double Tag Mode

4. Edit the parameters as specified in the MA Parameters table.

Note You can change **Vlan Tag Mode** only if there are no MEPs in the MA.

5. Click <Apply> to apply your changes, and then click <Save Configuration>.

Maintenance associations can be configured with VLAN “untagged” mode. Only one MA under an MD can be configured as “untagged”, and only two MEPs can be configured per untagged MA (one per port).

➤ **To configure an MA with VLAN untagged mode (web interface):**

1. Navigate to Configuration > OAM > CFM.
2. Click the MA name.

MiNID

Configuration>OAM>CFM

MD Level	2
MA Name	MA4
Vlan Tag Mode	Untagged
Service Name	Gold

3. Edit the parameters as specified in the MA Parameters table.

➤ **To delete a maintenance association (web interface):**

1. Navigate to Configuration>OAM>CFM.
2. In the row corresponding to the MA, click **remove**.
You are prompted for confirmation.
3. Click <Yes> to delete the MA.

Note

- Before creating a maintenance association, use **info** at the **configure oam cfm** level to display all configured OAM CFM entries, in order to find an unused MA index
- If you want to configure an MA that you created via the web interface, you need to navigate to the corresponding maintenance domain CLI level and use **info** in order to find the MA ID, as the web interface shows the MA name but does not show its index.

➤ **To add or configure a maintenance association (with the CLI):**

1. Navigate to **configure oam cfm maintenance-domain** <mdid> **maintenance-association** <maid>, where <maid> is an unused MA ID if you are creating an MA.
The **config>oam>cfm>md(<mdid>)>ma(<maid>)#** prompt is displayed.
2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Specifying the interval between continuity check messages	ccm-interval { 100ms 1s 10s 1min 10min }	
Configuring MEP for the MA	mep <mepid>	See Configuring MEPs
Configuring MIP for the MA	mip <mipid>	See Configuring MIPs
Specifying the name and name format of the maintenance association	name <ma-name> format { icc uint string }	<ma-name> – MA name, according to the specified format: • Format string – Specify <ma-name> as an alphanumeric string • Format uint – Specify <ma-name> as an unsigned integer decimal number (0–65535) • Format icc – Specify <ma-name> as the ITU carrier code that is assigned to the relevant network operator/service provider. The codes are maintained by ITU-T as defined in ITU-T Rec. M.1400.
Assigning service name to the MA	service-name <service-name>	
Specifying whether MA works in single, double tag mode or untagged mode	tag-mode { single-vlan double-inner-vlan double-outer-vlan untagged }	single-vlan – MA works in single-tag mode double-inner-vlan – MA works in double-tag mode and uses inner VLAN to handle incoming OAM frames double-outer-vlan – MA works in double-tag mode and uses outer VLAN to handle incoming OAM frames untagged – MA works in an untagged mode
Enabling or disabling Interface Status Tlv	[no] interface-status-tlv	Allowed values: • Enable • Disable

➤ To delete a maintenance association (CLI):

- At the config>oam>cfm>md(<mdid>)# prompt enter:
 no maintenance-association <maid>
 The maintenance association is deleted.

Configuring MEPs

Maintenance endpoints (MEPs) reside at the edge of a maintenance domain. They initiate and respond to CCMs, linktrace requests, and loopbacks to detect, localize, and diagnose connectivity problems.

Up to 32 MEPs can be created in an MA, as long as a MIP has not been created in the MA. The following parameters are configured in MEPs.

MEP Parameters

Parameter		Description
Web	CLI	
MEP Enable	[no] shutdown	Administratively enables or disable the MEP
MEP ID	<mepid>	MEP identification (1–8191); must be unique in the MA
Remote MEP ID	remote-mep	Remote MEP identification (1–8191) ; must be unique in the MA. Multiple remote MEP IDs can be configured. OAM messages such as CCM, LBM, LTM, etc. are sent to all remote MEPs
Ingress Port	bind	Port for transmitting and receiving frames: • SFP/Port 1 • MSA/Port 2
Inner Vlan Id	vlan	Inner VLAN ID (0–4095) assigned to MEP; must be unique in the MEP <i>Note: Appears in Web menu only if VLAN tag mode is double-inner or double-outer</i>
Inner Priority	priority	Inner VLAN P-bit (0–7) assigned to MEP <i>Note: Appears in Web menu only if VLAN tag mode is double-inner or double-outer</i>
Outer Vlan Id	outer-vlan	Outer VLAN ID (0–4095) assigned to MEP; must be unique in the MEP <i>Note: Appears in Web menu only if VLAN tag mode is double-inner or double-outer</i>

Parameter		Description
Web	CLI	
Outer Priority	outer-priority	Outer VLAN P-bit (0–7) assigned to MEP <i>Note: Appears in Web menu only if VLAN tag mode is double-inner or double-outer</i>
Vlan Id	vlan	VLAN ID (0–4095) assigned to MEP; must be unique in the MEP <i>Note: Appears in Web menu only if VLAN tag mode is single</i>
Priority	priority	P-bit (0–7) assigned to MEP <i>Note: Appears in Web menu only if VLAN tag mode is single</i>
Service Name	service-name	Service name associated with MA, to enable network management systems to locate entities related to specific services
CCM Enable	ccm-initiate	Enables or disables transmission of CCMs (see Continuity Check Messages (CCMs)) to any configured remote MEPs
CCM Transmission Mode	dest-addr-type	CCMs are transmitted as unicast or multicast frames: • Unicast – CCMs are transmitted to the MAC address specified by Remote MEP MAC Address. In this mode, only one remote MEP can be configured for the MEP. • Multicast – CCMs are transmitted to the MAC address corresponding to the MEP MD level, as shown in the Multicast MAC Address table. In this mode, multiple remote MEPs can be configured for the MEP.
Remote MEP MAC Address	dest-mac-addr	Specifies the MAC address to which to send unicast CCMs <i>Note: Appears in Web interface/available in the CLI only if CCM transmission mode is set to unicast.</i>
AIS Enable	ais	Enable or disable transmission of AIS frames (see Alarm Indication Signal (AIS))
Client MD Level	client-md-level	Transmitted in AIS frames (default client MD level is the MD level corresponding to the MEP, incremented by 1)

➤ **To add a MEP with the web interface:**

1. Navigate to Configuration>OAM>CFM\PM.
2. In the row corresponding to the MD for which to add the MEP, click **config**.
3. In the row corresponding to the MA for which to add the MEP, click **config**.

MiNID

Configuration>OAM>CFM\PM>Config MA (ID: MA1)

Add new MEP

Add new MIP

MEP/MIP		
MEP (ID: 16)	Remove	Config MEP

Configuring MA

4. Click Add new MEP.

MiNID

Configuration>OAM>CFM\PM>Config MA>Add MEP (MA MA1)

Admin Status	-----	Disable
MEP ID [1..8191]	-----	1
Ingress Port	-----	SFP/Port 1
Vlan Id [1..4094]	-----	1
Classification Priority	-----	0
Tag-Ether-Type	-----	0x8100
CCM	-----	Disable
CCM Priority	-----	0
CCM Transmission Mode	-----	Multicast
Client MD Level	-----	5
AIS	-----	Disable

New MEP for Single Tag OAM

MiNID

Configuration>OAM>CFM\PM>Config MA>Add MEP (MA MA1)

Admin Status	Disable
MEP ID [1..8191]	1
Ingress Port	SFP/Port 1
Inner Vlan Id [1..4094]	1
Classification Inner Priority	Any
Tag-Ether-Type Inner	0x8100
Outer Vlan Id [1..4094]	1
Outer Priority	0
Tag-Ether-Type Outer	0x88A8
CCM	Disable
CCM Priority	0
CCM Transmission Mode	Multicast
Client MD Level	6
AIS	Disable

New MEP for Double Tag OAM

5. Fill in the parameters as specified in the MEP Parameters table.
6. Click **<Apply>** to create the MEP.
A message is displayed that the MEP was successfully created.
7. Click **<OK>** to close the message.
8. Click **<Save Configuration>**.

► **To configure MEP parameters in the web interface:**

1. Navigate to Configuration>OAM>CFM.
2. In the row corresponding to the MD to which the MEP belongs, click **config** to display the **Configuration>OAM>CFM>Config MA** screen.

3. In the row corresponding to the MA to which the MEP belongs, click config to display the Configuration>OAM>CFM>Config MEP screen.
4. In the row corresponding to the MEP, click the MEP name.

MiNID

Configuration>OAM>CFM\PM>Config MA>Update MEP (MA MA6)

Admin Status	_____	Disable
MEP ID [1..8191]	_____	2
Vlan Id [1..4095]	_____	1
Priority	_____	0
Ingress Port	_____	MSA/Port 2
CCM Enable	_____	Enable
CCM Transmission Mode	_____	Multicast
Client MD Level	_____	5
AIS Enable	_____	Enable

Updating MEP Parameters

5. Edit the parameters as specified in the MEP Parameters table.
6. Click <**Apply**> to apply your changes, and then click <**Save Configuration**>.

➤ To add remote MEP IDs for the MEP (web):

Note *You can add remote MEPs only if the MEP is administratively enabled.*

1. Navigate to Configuration>OAM>CFM.
2. In the row corresponding to the MA to which the MEP belongs for which you wish to add the service, click **config** to display the **Configuration>OAM>CFM>Config MA** screen.
3. In the row corresponding to the MEP, click Config MEP to display the Configuration>OAM>CFM>Config MEP screen.
4. Click Add Remote MEP.

MiNID

Configuration>OAM>CFM

Remote MEP ID		
[1..8191]		1

Adding Remote MEP

5. Fill in the remote MEP ID.
6. Click **<Apply>** to save the remote MEP ID.
7. Repeat the above steps to add more remote MEP IDs if necessary.
8. Click **<Save Configuration>**.

► **To delete remote MEPs (web):**

1. Navigate to Configuration>OAM>CFM.
2. In the row corresponding to the MA to which the MEP belongs for which you wish to add the service, click **config** to display the **Configuration>OAM>CFM>Config MA** screen.
3. In the row corresponding to the MEP, click Config MEP to display the Configuration>OAM>CFM>Config MEP screen.
4. Click **Remove** in the row corresponding to the remote MEP that you wish to remove.
The remote MEP ID is removed from the MEP configuration.
5. Click **<Save Configuration>**.

► **To delete a MEP (web):**

1. Navigate to Configuration>OAM>CFM.
2. In the row corresponding to the MA to which the MEP belongs, click config to display the Configuration>OAM>CFM>Config MA screen.
3. In the row corresponding to the MEP, click **Remove**.
You are prompted for confirmation.
4. Click **<Yes>** to delete the MEP.
5. Click **<Save Configuration>**.

➤ To add or configure a maintenance endpoint (with the CLI):

1. Navigate to configure oam cfm maintenance-domain <mdid> maintenance-association <maid> mep <mepid>, where <mepid> is 1–8191.
The prompt **config>oam>cfm>md(<mdid>)>ma(<maid>)>mep(<mepid>)#** is displayed.
2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Enabling AIS transmission	ais	To disable AIS transmission, enter no ais
Binding the MEP to an Ethernet port	bind ingress-port {sfp/1 msa/2}	To unbind the MEP, enter no bind
Enabling CCM transmission	ccm-initiate	To disable CCM transmission, enter no ccm-initiate
Defining the MD level transmitted in AIS frames	client-md-level	Default client MD level is the MD level corresponding to the MEP, incremented by 1
Defining the MAC address type sent in CCMs	dest-addr-type ccm {unicast multicast}	• Unicast – CCMs are transmitted to the MAC address specified by Remote MEP MAC Address • Multicast – CCMs are transmitted to the MAC address corresponding to the MEP MD level, as shown in the Multicast MAC Address table.
Defining a unicast MAC address if you defined unicast MAC address type for CCM messages via the dest-addr-type command	dest-mac-addr <mac-addr>	MAC address is in format xx-xx-xx-xx-xx-xx
Initiating OAM loopback	lbn	See Performing OAM Loopback
Initiating OAM linktrace	linktrace	See Performing OAM Link Trace
Specifying inner VLAN or single VLAN p-bit assigned to MEP	priority	Allowed values: 0–7

Task	Command	Comments
Defining remote MEP(s) with which the MEP communicates	remote-mep <remote-mep-id>	Allowed range for remote MEP ID(s) is: 1–8191 Type no remote-mep <remote-mep-id> to delete a remote MEP ID
Configuring service for the MEP	service <serviceid>	See Configuring PM Services
Displaying MEP status	show status	
Administratively enabling MEP	no shutdown	To deactivate the MEP, enter shutdown
Associating the MEP with a single/double tagged VLAN	vlan <vlan-id>[p-bit <p-bit>] [tag-ether-type <ether-tag>] vlan <vlan-id>[p-bit <p-bit>] [tag-ether-type <ether-tag>] [inner-vlan <inner-vlan> [p-bit <p-bit>]] inner-tag-ether-type <inner-ether-tag>]]	For single-tagged VLAN For double-tagged VLAN Allowed values for VLAN ID: 1–4094

➤ **To delete a maintenance endpoint (CLI):**

- At the **config>oam>cfm>md**(<mdid>)>**ma**(<maid>)# prompt, enter:
no mep <mepid>

The maintenance endpoint is deleted.

Configuring MIPs

Maintenance intermediate points reside in the middle of maintenance associations. They respond to linktrace requests and loopbacks; they do not initiate messages.

One MIP can be created in an MA, as long as no MEPs have been created in the MA.

➤ **To add and configure a MIP in the web interface:**

- Navigate to Configuration > OAM > CFM.
- In the row corresponding to the MA for which to add the MIP, click config to display the Configuration > OAM > CFM > Config MA screen.

- 3. Click Add new MIP.

MiNID

Configuration>OAM>CFM\PM> Config MA>Add MIP (MA MA1)

Admin Status	-----	Disable
Vlan ID (1...4094)	-----	1
Priority	-----	0
Tag-Ether-Type	-----	0x8100

New MIP for Single-Tag VLAN

MiNID

Configuration>OAM>CFM\PM> Config MA>Add MIP (MA MA3)

Admin Status	-----	Disable
Inner Vlan ID (1...4094)	-----	1
Inner Priority	-----	0
Tag-Ether-Type Inner	-----	0x8100
Outer Vlan ID (1...4094)	-----	1
Outer Priority	-----	0
Tag-Ether-Type Outer	-----	0x88A8
Apply	-----	

New MIP for Double-Tag VLAN

- 4. Set **Admin Status** to enable or disable the MIP.
- 5. Type in the VLAN ID and priority in the corresponding fields.
- 6. Click **<Apply>** to create the MIP.
A message is displayed that the MIP was successfully created.

7. Click **<OK>** to close the message.
8. Click **<Save Configuration>**.

➤ **To delete a MIP in the web interface:**

1. Navigate to Configuration>OAM>CFM.
2. In the row corresponding to the MA to which the MIP belongs, click config to display the Configuration>OAM>CFM>Config MA screen.
3. Click **Remove**.
You are prompted for confirmation.
4. Click **<Yes>** to delete the MIP.

➤ **To add or configure a MIP in the CLI:**

1. Navigate to configure oam cfm maintenance-domain <mdid> maintenance-association <maid> mip.
The **config>oam>cfm>md(<mdid>)>ma(<maid>)>mip#** prompt is displayed.
2. Enter **no shutdown** to administratively enable the MIP, or **shutdown** to administratively disable the MIP.

➤ **To delete a MIP in the CLI:**

- At the **config>oam>cfm>md(<mdid>)>ma(<maid>)#** prompt enter:
no mip <mipid>
The MIP is deleted.

Configuring PM Services

Services define delay and delay variation thresholds. If the thresholds are exceeded, the service is declared as degraded. You can also define priority of OAM messages originating from this service.

You can configure up to 32 services on a MEP, corresponding to each p-bit (32 services are supported for the entire device). If a MEP is administratively disabled, its services are also disabled. If a MEP is deleted, its services are also deleted.

The following parameters are configured in PM services.

PM Service Parameters

Parameter		Description
Web	CLI	
Admin Status	[no] shutdown	Administratively enables or disables the service. Loss/delay measurement is performed only if the service is enabled.
Service Priority	priority	P-bit assigned to service (0–7)
Loss Threshold	loss-threshold	If loss measurement rises above this threshold, corresponding event and trap are sent; if it falls below the threshold, corresponding event and trap are sent Allowed range: 1–40 [%]
DEI Bit Color	loss-dei-bit-color	If loss measurement is set to User Data, you can filter packets according to the DEI (Drop Eligible Indicator) bit: - Both – Use green and yellow packets for loss measurement - Green – Use only green packets for loss measurement - Yellow – Use only yellow packets for loss measurement
Delay Threshold	delay-threshold	If frame delay measurement rises above this threshold, corresponding event and trap are sent; if it falls below the threshold, corresponding event and trap are sent Allowed range: 1–5000000 [μs]
Delay Var Threshold	delay-var-threshold	If frame delay variation measurement rises above this threshold, corresponding event and trap are sent; if it falls below the threshold, corresponding event and trap are sent Allowed range: 1–5000000 [μs]
Service Lmm Interval	lmm-interval	Service interval for loss measurement messages
Service Dmm Interval	dmm-interval	Service interval for delay measurement messages transmission of DM frames

➤ To add PM services in the web interface:

Note You can add PM services only if the corresponding MEP is administratively enabled.

1. Navigate to Configuration > OAM > CFM\PM.
2. In the row corresponding to the MA to which the MEP belongs for which you wish to add the service, click **config** to display the **Configuration > OAM > CFM\PM>Config MA** screen.
3. In the row corresponding to the MEP, click **Config MEP**.

MiNID

Configuration>OAM>CFM\PM>Config MA>Config MEP (ID: 1 on MA MA 6)

Add Remote MEP

Start Loopback Session

Start Linktrace Session

Add PM Service

Service (P-bit: 1)	Configure	Remove
Remote MEP (ID:8)	Remove	

Configuring MEP

4. Click Add PM Service.

MiNID

Configuration>OAM>CFM>Config MA>PM Service

Admin Status

Service Priority

Loss Threshold ([1..40]%)

DEI Bit Color

Delay Threshold ([1..4000000]μs)

Delay Var Threshold ([1..4000000]μs)

Service Lmm Interval

Service Dmm Interval

Disable

0

1

Both

1000

1000

100 ms

100 ms

PM Service

5. Fill in the parameters as specified in the PM Service Parameters tablei.
6. Click <Apply> to create the PM service, and then click <Save Configuration>.

➤ **To configure PM service parameters in the web interface:**

- 1. Navigate to Configuration>OAM>CFM.
- 2. In the row corresponding to the MA to which the MEP belongs for which you wish to configure the service, click **config** to display the **Configuration>OAM>CFM\PM>Config MA** screen.
- 3. In the row corresponding to the MEP, click Config MEP to display the Configuration>OAM>CFM>Config MEP screen.
- 4. In the row corresponding to the service, click the service name to display the Configuration>OAM>CFM>PM Service screen.
- 5. Edit the parameters as specified in.
- 6. Click **<Apply>** to apply your changes, and then click **<Save Configuration>**.

➤ **To delete a PM service (web):**

- 1. Navigate to Configuration > OAM > CFM.
- 2. In the row corresponding to the MA to which the MEP belongs for which you wish to delete the service, click **config** to display the **Configuration>OAM>CFM>Config MA** screen.
- 3. In the row corresponding to the MEP, click Config MEP to display the Configuration>OAM>CFM>Config MEP screen.
- 4. In the row corresponding to the service, click **Remove**.
The service is deleted.

➤ **To add or configure a MEP service in the CLI:**

- 1. Navigate to **configure oam cfm maintenance-domain <mdid> maintena?nce-association <maid> mep <mepid> service <serviceid>**, where <serviceid> is 1–8.
The prompt **config>oam>cfm>md(<mdid>)>ma(<maid>)>mep(<mepid>)>service(<serviceid>)#** is displayed.
- 2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Clearing statistics	clear-statistics	
Specifying delay threshold in microseconds	delay-threshold <delay-thresh>	Allowed range: 1–5,000,000
Specifying delay variation threshold in microseconds	delay-var-threshold <delay-var-thresh>	Allowed range: 1–5,000,000

Task	Command	Comments
Specifying the interval for delay measurement messages	dmm-interval {100ms 1s 10s}	
Specifying the interval for loss measurement messages	lmm-interval {100ms 1s 10s}	
Specifying packet filtering according to DEI if loss measurement is set to user-data	loss-dei-bit-color {both green yellow}	• both – Use green and yellow packets for loss measurement • green – Use only green packets for loss measurement • yellow – Use only yellow packets for loss measurement
Specifying loss threshold	loss-threshold <loss-threshold>	Allowed range (1-40); Entered value is used as percentage for threshold
Associating this service with a priority	priority <p-bit>	The allowed range is 0–7 Note: Only one service can be defined on each p-bit.
Activating the service	no shutdown	To disable the service, enter shutdown You can activate a service only if the corresponding MEP is active

➤ **To delete a PM service in the CLI:**

- At the **config>oam>cfm>md(<mdid>)>ma(<maid>)>mep(<mepid>)#** prompt, enter:
no service <mepid>

The PM service is deleted.

Configuring Destination NEs

You can define multiple destination NEs for a PM service, to enable separation between the different remote MEPs that are defined for the MEP. Independent loss measurement and delay measurement can be configured for each destination NE.

Note

If the destination NE loss measurement method is user data, then only one destination NE can be configured for the PM service.

The following parameters are configured in destination NEs.

Destination NE Parameters

Parameter		Description
Web	CLI	
Dest NE ID	dest-ne	ID assigned to destination NE. This is automatically set by the Web interface to 1 for the first destination NE, and then incremented for each subsequent destination NE.
Admin Status	[no] shutdown	Administratively enables or disables the destination NE. Loss/delay measurement is performed only if the destination NE is enabled.
Remote MEP ID	remote-mep	Remote MEP ID to use for the loss and delay measurement; this can be set only to a remote MEP ID that is associated with the corresponding MEP
Loss Measurement Service	loss	Enables or disables loss measurements for the destination NE: • Disable – Loss measurement is not performed • LMM Synthetic – Loss measurement is performed using synthetic LMM/LMR frames • SLM Synthetic – Loss measurement is performed using synthetic SLM/SLR frames • User Data – Loss measurement is performed on user data (real traffic) using LMM/LMR frames
Delay Measurement Service	delay	Enables or disables delay measurements for the destination NE
Delay Measurement Bin Profile	delay-measurement-bin-profile	For Two way delay only
Delay Var Measurement Bin Profile	delay-var-measurement-bin-profile	For one and two way delay variation

➤ **To add destination NEs in the web interface:**

Note

You can add destination NEs only if the corresponding PM service is administratively enabled.

1. Navigate to Configuration>OAM>CFM.

- 2. In the row corresponding to the MA to which the MEP belongs for which you wish to add the destination NE, click **config** to display the **Configuration > OAM > CFM > Config MA** screen.
- 3. In the row corresponding to the MEP, click Config MEP to display the Configuration > OAM > CFM > Config MA > Config MEP screen.
- 4. In the row corresponding to the service, click **Configure**.

MiNID

Configuration>OAM>CFM\PM>Config MA>Config MEP>Config Service (P-bit: 0 on MEP2, MA MA1)

Add Destination Network Element

Dest NE (ID: 1)	Remove
-----------------	--------

Configuring PM Service

- 5. Click Add Destination Network Element.

MiNID

Configuration>OAM>CFM\PM>Config MA>Config MEP>Configure PM Service>Add Dest NE (MA: MA1, MEP-ID: 2, Service P-bit: 0)

Dest NE Number	1
Admin Status	Disable
Remote MEP ID	4
Loss Measurement Service	Disable
Delay Measurement Service	Disable
Delay Measurement Bin Profile	bin-profile-0
Delay Var Measurement Bin Profile	bin-profile-0

Destination NE

- 6. Fill in the parameters as specified in the Destination NE Parameters table.
- 7. Click **<Apply>** to create the destination NE.

➤ **To configure a destination NE in the web interface:**

1. Navigate to Configuration>OAM>CFM.
2. In the row corresponding to the MA to which the MEP belongs for which you wish to configure the destination NE, click **config** to display the **Configuration>OAM>CFM>Config MA** screen.
3. In the row corresponding to the MEP, click Config MEP to display the Configuration>OAM>CFM>Config MEP screen.
4. In the row corresponding to the service, click Configure to display the Configuration>OAM>CFM>Config PM Service screen.
5. In the row corresponding to the destination NE, click the destination NE ID.
6. Edit the parameters as specified in the Destination NE Parameters table.
7. Click **<Apply>** to save your changes, and then click **<Save Configuration>**.

➤ **To delete a destination NE (web):**

1. Navigate to Configuration>OAM>CFM.
2. In the row corresponding to the MA to which the MEP belongs for which you wish to delete the destination NE, click **config** to display the **Configuration > OAM > CFM > Config MA** screen.
3. In the row corresponding to the MEP, click Config MEP to display the Configuration > OAM > CFM > Config MEP screen.
4. In the row corresponding to the service, click Configure to display the Configuration > OAM > CFM > Config PM Service screen.
5. In the row corresponding to the destination NE, click **Remove**.
The destination NE is deleted.
6. Click **<Save Configuration>**.

➤ **To add or configure a destination NE in the CLI:**

1. Navigate to configure oam cfm maintenance-domain <mdid> maintenance-association <maid> mep <mepid> service <serviceid> dest-ne <dest-neid>.
The prompt **config>oam>cfm>md(<mdid>)>ma(<maid>)>mep(<mepid>)>service(<serviceid>)>dest-ne(<dest-neid>)#** is displayed.
2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Clearing statistics	clear-statistics	
Enabling delay measurement	delay	Enter no delay to disable delay measurement
Delay Measurement Bin Profile	delay-measurement-bin-profile	
Delay Var Measurement Bin Profile	delay-var-measurement-bin profile	
Enabling loss measurement	loss { slm Imm-synthetic user-data }	• slm – Loss measurement is performed using synthetic SLM/SLR frames • Imm-synthetic – Loss measurement is performed using synthetic LMM/LMR frames • user-data – Loss measurement is performed on user data (real traffic) using LMM/LMR frames (only one destination NE per PM service is allowed with this method) Enter no loss to disable loss measurement <i>Note: Only a single LM on real traffic is available per MinID port for untagged OAM</i>
Defining remote MEP to associate with the destination NE	remote-mep <remote-mep-id>	This must be a remote MEP ID that has been configured for the corresponding MEP, and has not been associated with any other destination NE corresponding to the PM service
Activating the destination NE	no shutdown	To disable the destination NE, enter shutdown You must define a valid remote MEP ID before you can activate the destination NE
Displaying statistics	show-statistics	See Viewing Destination NE Statistics

➤ To delete a destination NE:

- At the **config>oam>cfm>md(<mdid>)>ma(<maid>)>mep(<mepid>)>service(<serviceid>)#** prompt, enter:
no dest-ne <dest neid>
The destination NE is deleted.

Configuring Measurement Bin Profiles

➤ To add and configure a new measurement bin profile (web):

1. Navigate to Configuration > OAM > CFM/PM > Measurement Bin Profiles.

MiNID

Configuration>OAM>CFM/PM>Measurement Bin Profiles

Add New Measurement Bin Profile

Profile Name

2. Click Add New Measurement Bin Profile.

MiNID

Configuration>OAM>CFM/PM>New Bin Profile

Profile Name

Bin-profile-2

Number of Thresholds

1

Upper-Th-1

10

Apply

3. Type the name of the profile in the **Profile Name** field.
4. Select the number of thresholds from the drop down list.

- 5. In the **Upper-Th-x** field, type the upper threshold variable.
- 6. Click **<Apply>**.

MiNID

Configuration>OAM>CFM/PM>Measurement Bin Profiles

Add New Measurement Bin Profile

Profile Name

Bin-profile-1	remove
Bin-profile-2	remove
Bin-profile-3	remove

- 7. To edit bin profile configuration, click the profile name and reconfigure the fields.
- 8. To remove the bin profile, click **remove**.

➤ **To configure a measurement bin profile in the CLI:**

- Go to **MiNID>config>oam>cfm#** and configure the following:

Task	Command	Comments
Specifying the bin rprofile name	measurement-bin-profile	
Specifying the thresholds for the bin profile	thresholds	The command is followed by the series of chosen thresholds

Configuring Availability Parameters

➤ **To configure availability in the web interface:**

- 1. Navigate to **Configuring > OAM > CFM/PM > Availability Params**.
The default parameters are displayed.

MiNID

Configuration>OAM>CFM/PM>Availability Params

Delta-t	1sec
n	10
Forward Threshold	50
Backward Threshold	50

Apply

2. Edit the parameters according to the following:
 - Delta-t: Small interval time (in seconds)
 - n: Number of consecutive small time intervals over which to measure availability
 - Forward Threshold: Forward frame loss ratio threshold, for which unavailability occurs if exceeded
 - Backward Threshold: Backward frame loss ratio threshold, for which unavailability occurs if exceeded
3. Click **Apply**.

➤ **To configure availability in the CLI:**

- Navigate to **MiNID>config>oam>cfm#** and configure the following:

Task	Command	Comments
Defining parameters for availability calculations	availability [delta-t { 1 2 3 4 5 6 10 12 15 20 }] [n <n>] [forward-thr <forward-thr-percents>] [backward-thr <backward-thr-percents>]	These parameters define availability performance measurement, based on frame loss during a sequence of consecutive small time intervals: • delta-t – Time interval (in seconds) • n –Number of consecutive small time intervals over which to measure availability • forward-thr – Forward frame loss ratio threshold, for which unavailability occurs if exceeded • backward-thr – Backward frame loss ratio threshold, for which unavailability occurs if exceeded

Performing OAM Loopback

This diagnostic utility verifies OAM connectivity on Ethernet connections.

➤ To run an OAM loopback in the web interface:

1. Navigate to Configuration > OAM > CFM.
2. In the row corresponding to the MA to which the MEP belongs for which you wish to run OAM loopback, click **config** to display the **Configuration > OAM > CFM\PM > Config MA** screen.
3. In the row corresponding to the MEP, click Config MEP to display the Configuration > OAM > CFM\PM > Config MA > Config MEP screen.
4. Click Start Loopback Session.

MiNID

Configuration>OAM>CFM>Config MA>Config MEP>LBM Session

LBM Session Mode

Multicast

Data TLV Length [0..1450]

0

Starting LBM Session

5. Set **LBM Session Mode** to one of the following:

- **Multicast** –To send LBMs to the MAC address corresponding to the MEP MD level, as shown in the MEP Parameters table.
 - **Unicast** –To send LBMs to a specific MAC address
 - **Remote MEP** – To send LBMs to the remote MEP MAC address
6. If LBM Session Mode is set to Unicast, the parameters Destination MAC Address and Number Of LoopBack Messages appear in the screen.

MiNID

Configuration>OAM>CFM

LBM Session Mode	Unicast
Destination MAC Address	00-00-00-00-00-00
Number of LoopBack Messages [1..1000]	1
Data TLV Length [0..1450]	0

LBM Session – Unicast

1. If LBM Session Mode is set to Remote MEP, the parameters Remote MEP ID and Number of LoopBack Messages parameter appear.

MiNID

Configuration>OAM>CFM

LBM Session Mode	Remote MEP
Remote MEP ID	55
Number of LoopBack Messages [1..1000]	20
Data TLV Length [0..1450]	250

LBM Session – Remote MEP

1. Set the parameters as needed, and then click **Run** to start the loopback.

➤ To view the OAM loopback results in the web interface:

- 1. Navigate to Monitoring > OAM > CFM(see [Error! Reference source not found.](#)).
- 2. In the row for the corresponding MA, click <Info> to display the Monitoring>OAM>CFM>MA Info screen (see [Error! Reference source not found.](#)).
- 3. In the row for the corresponding MEP, click **LB Status**.

MiNID

Monitoring>OAM>CFM

Dest Address	00-20-D2-51-42-52
LBM Sent	20
Replies In Order	20
Replies Out Of Order	0
MSG Lost\Timeout	0
MSG Lost\Timeout (%)	0.00

OAM Loopback Status

➤ To run an OAM loopback in the CLI:

- At the **config>oam>cfm>md(<mdid>)>ma(<maid>)>mep(<mepid>)#** prompt, enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Sending LBMs to specific MAC address	lbm address <mac-address> [repeat <repeat-num>] [data-tlv-length <length>]	• MAC address is in the format <xx-xx-xx-xx-xx-xx> • Allowed range of repeat-num is 1–1000 • Allowed range of data-tlv-length is 0–1450
Sending LBMs to multicast MAC address	lbm multicast [data-tlv-length <length>]	Multicast MAC address corresponds to the MEP MD level, as shown in Error! Reference source not found.

Task	Command	Comments
Sending LBMs to remote MEP	lbm remote-mep <remote-mep-id> [repeat <repeat-num>] [data-tlv-length <length>]	
Checking OAM loopback results	show lbm-results	

Performing OAM Link Trace

This diagnostic utility traces the OAM route to the destination.

➤ To run an OAM link trace with the web interface:

1. Navigate to Configuration>OAM>CFM (see [Error! Reference source not found.](#)).
2. In the row corresponding to the MA to which the MEP belongs for which you wish to run OAM link trace, click **config** to display the **Configuration>OAM>CFM\PM>Config MA** screen .
3. In the row corresponding to the MEP, click Config MEP to display the Configuration>OAM>CFM>Config MA>Config MEP screen.
4. Click Start Linktrace Session.
The LTM Session screen appears.

MinID

Configuration>OAM>CFM>Config MA>Config MEP>LTM Session

LTM Session Mode

Target MAC Address

TTL [1..24]

Unicast

00-20-D2-51-42-52

24

Starting LTM Session – Unicast

1. Set **LTM Session Mode** to one of the following:
 - **Unicast** –To send LTMs to a specific MAC address, specified by **Target MAC Address**
 - **Remote MEP** – To send LTMs to a specific remote MEP, specified by **Remote MEP ID**

MiNID

Configuration>OAM>CFM>Config MA>Config MEP>LTM Session

LTM Session Mode	_____	Remote MEP
Remote MEP ID	_____	55
TTL [1..24]	_____	24

Starting LTM Session – Remote MEP

1. Set **TTL** (time-to-live) to the number of hops. Each unit in the link trace decrements the TTL until it reaches 0, which terminates the link trace.
2. Click Run to start the link trace.

➤ To view the OAM link trace results:

1. Navigate to Monitoring>OAM>CFM.
2. In the row for the corresponding MA, click <Info> to display the Monitoring>OAM>CFM>MA Info screen.
3. In the row for the corresponding MEP, click **LT Status**.

Monitoring>OAM>CFM

MAC Address	TTL	Last Egress ID	Relay Action
00-20-D2-51-42-52	23	(0010) 00-20-D2-51-7E-2D	RlyHit

OAM Link Trace Status

➤ To run an OAM link trace in the CLI:

- At the **config>oam>cfm>md(<mdid>)>ma(<maid>)>mep(<mepid>)#** prompt, enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Sending LTMs to specific MAC address	linktrace address <mac-address> [ttl <ttl>]	• MAC address is in the format <xx-xx-xx-xx-xx-xx> • Allowed range for <ttl> is 1–24. This parameter specifies number of hops. Each unit in the link trace decrements the TTL until it reaches 0, which terminates the link trace.
Sending LTMs to remote MEP	linktrace remote-mep <remote-mep-id> [ttl <ttl>]	
Checking the OAM link trace results	show linktrace-results	

Examples

This section contains CLI configuration examples.

Creating MA, MEP, and Service

This example illustrates creating the following:

- MD ID 4
- MA ID 5: Name MA5, VLAN 17
- MEP ID 16:
 - Bound to port: SFP/1
 - Enabled AIS and CCM transmission
 - Remote MEP IDs 32, 64
- Service 4: enabled loss (standard) and delay measurement

➤ **To create MEP and service:**

```
exit all
configure oam cfm maintenance-domain 4 maintenance-association 5
name MA5 format string
```

```

vlan 17

mep 16
  bind ingress-port sfp/1
  ais
  ccm-initiate
  remote-mep 32
  remote-mep 64
  no shutdown

  service 4
    delay
    loss slm
    no shutdown
    exit all

```

➤ To display the configured entities:

```

# configure oam cfm maintenance-domain 4 maintenance-association 5
>config>oam>cfm>md(4)>ma(5)# info
      ccm-interval 100ms
      mep 16
        ais
        bind ingress-port sfp/1
        ccm-initiate
        client-md-level 5
        dest-addr-type ccm multicast
        remote-mep 32
        service 4
          delay
          delay-threshold 1
          delay-var-threshold 1
          lmm-interval 1s
          dmm-interval 1s
          loss slm
          priority 0
          no shutdown
        exit
      exit
    no shutdown
  exit
  name MA5 format string
  priority 0
  vlan 17
exit

```

Creating MA and MIP

This example illustrates creating the following:

- MD ID 3
- MA ID 2: Name MA2, VLAN 999

- MIP

➤ To create MIP:

```
exit all
configure oam cfm maintenance-domain 3 maintenance-association 2
name MA2 format string
vlan 999
mip
no shutdown
exit all
```

Displaying MEP Status

The following illustrates displaying MEP status for the MEP configured in the above example, assuming that in another device in the network, MEP 32 has been configured with remote MEP 16 in MD ID 4, MA5.

```
>config>oam>cfm>md(4)>ma(5)>mep(16)# show status
MEP Defect:
-----
Rx AIS : OFF
Cross Connected CCM (Mismatch; Unexpected MD Level) : OFF
Invalid CCM (Unexpected MEP; Unexpected CCM Period) : OFF

Remote MEP      Remote MEP Address      Operational Status
-----
32              00-20-D2-51-42-52      OK
64              00-00-00-00-00-00      Fail

>config>oam>cfm>md(4)>ma(5)>mep(16)# show remote-mep remote-mep-id 32
Operational State      : RMEP_OK
SysUpTime              : 25199 sec.
Remote MEP Address     : 00-20-D2-51-42-52
Last RDI Bit           : False
Loss Of Continuity     : OFF
```

Performing OAM Loopback

This example illustrates performing OAM loopback on the previously configured MEP:

- Use remote MEP ID
- Send 20 LBMs
- LBM frame length 250

```
>config>oam>cfm>md(4)>ma(5)>mep(16)# lbm remote-mep 32 repeat 20 data-tlv-length 250
>config>oam>cfm>md(4)>ma(5)>mep(16)# show lbm-results
Destination MAC : 00-20-D2-51-42-52
```

```

LBM Sent           :      20
Replies In Order   :      20
Replies Out Of Order :      0
Messages Lost\Timeout :      0
Messages Lost\Timeout(%) :    0.00

```

Performing OAM Link Trace

This example illustrates performing OAM link trace on the previously configured MEP:

- Use remote MEP MAC address
- TTL=20

```

>config>oam>cfm>md(4)>ma(5)>mep(16)# linktrace address 00-20-D2-51-42-52 ttl 20
>config>oam>cfm>md(4)>ma(5)>mep(16)# show linktrace-results
MAC                               TTL      Last      Relay
Address                           Egress-Id Action
-----
00-20-D2-51-42-52                19      (0010) 00-20-D2-51-7E-2D  RlyHit

```

Displaying Running Statistics

```

# configure oam cfm maintenance-domain 4 maintenance-association 5 mep 16 service 4
dest-ne 1 show-statistics
>config>oam>cfm>md(4)>ma(5)>mep(16)>service(4)>dest-ne(1)>show-stats# running
Service State      : Unavailable
Statistics         : Running
Elapsed Time (Sec.) : 168517
-----
Two-Way Delay (ms) :      0.000 (ms)
Two-Way Delay Variation (ms) :    0.000 (ms)
One-Way Delay Variation (ms) :    0.000 (ms)
Frames Exceed Delay Threshold :      0
Frames Exceed Delay Variation Threshold :    0

Tx Frames      : 0
Rx Frames      : 0
Frame Loss     : 0
Frame Loss Ratio (%) : 0.000000
Unavailable Seconds : 168517
Available Seconds : 0

Forward      Backward
Tx Frames    : 0      0
Rx Frames    : 0      0
Frame Loss   : 0      0
Frame Loss Ratio (%) : 0.000000 0.000000
Unavailable Seconds : 168517 168517
Available Seconds : 0      0

Tx      Rx
LMs : 0  LMRs : 0
DMs : 0  DMRs : 0

```

Displaying Current Interval Statistics

```
# configure oam cfm maintenance-domain 4 maintenance-association 5 mep 16 service 4
dest-ne 1 show-statistics
>config>oam>cfm>md(4)>ma(5)>mep(16)>service(4)>dest-ne(1)>show-stats# current
Service State      : Unavailable
Statistics         : Current (Last 15 Min.)
Elapsed Time (Sec.) : 64
-----
```

	Min (ms)	Average (ms)	Max (ms)
Two-Way Delay	0.000	0.000	0.000
Two-Way Delay Variation	0.000	0.000	0.000
One-Way Delay Variation	0.000	0.000	0.000
Frames Exceed Delay Threshold			0
Frames Exceed Delay Variation Threshold			0

	Forward	Backward
Tx Frames	0	0
Rx Frames	0	0
Frame Loss	0	0
Frame Loss Ratio (%)	0.000000	0.000000
Unavailable Seconds	64	64
Available Seconds	0	0

	Tx	Rx
LMMS : 0		LMRs : 0
DMMS : 0		DMRs : 0

Displaying Selected Interval Statistics

```
# configure oam cfm maintenance-domain 4 maintenance-association 5 mep 16 service 4
dest-ne 1 show-statistics
>config>oam>cfm>md(4)>ma(5)>mep(16)>service(4)>dest-ne(1)>show-stats# interval 4
Statistics         : Interval
Interval Number    : 4
Interval Duration  : 08:46:49 29-03-2016 - 09:01:39 29-03-2016
-----
```

	Min (ms)	Average (ms)	Max (ms)
Two-Way Delay	0.000	0.000	0.000
Two-Way Delay Variation	0.000	0.000	0.000
Frames Exceed Delay Threshold			0
Frames Exceed Delay Variation Threshold			0

	Forward	Backward
Tx Frames	0	0
Rx Frames	0	0
Frame Loss	0	0
Frame Loss Ratio (%)	0.000000	0.000000
Unavailable Seconds	900	900

```
Available Seconds      : 0          0
```

```

      Tx      Rx
LMMS : 0      LMRs : 0
DMMS : 0      DMRs : 0
```

Displaying Total Interval Statistics

```
# configure oam cfm maintenance-domain 4 maintenance-association 5 mep 16 service 4
dest-ne 1 show-statistics
```

```
>config>oam>cfm>md(4)>ma(5)>mep(16)>service(4)>dest-ne(1)>show-stats# total-intervals
Statistics      :      Total Interval
```

```
-----
                        Min      Average Max
                        (ms)      (ms)      (ms)
Two-Way Delay          0.000    0.000    0.000
Two-Way Delay Variation 0.000    0.000    0.000
Frames Exceed Delay Threshold :      0
Frames Exceed Delay Variation Threshold :      0
```

```

                        Forward      Backward
Tx Frames      : 0          0
Rx Frames      : 0          0
Frame Loss      : 0          0
Frame Loss Ratio (%) : 0.000000    0.000000
Unavailable Seconds : 85500        85500
Available Seconds  : 0          0
```

```

      Tx      Rx
LMMS : 0      LMRs : 0
DMMS : 0      DMRs : 0
```

Displaying All Interval Statistics

This example illustrates displaying all intervals if there are two valid intervals.

```
# configure oam cfm maintenance-domain 4 maintenance-association 5 mep 16 service 4
dest-ne 1 show-statistics
```

```
>config>oam>cfm>md(4)>ma(5)>mep(16)>service(4)>dest-ne(1)>show-stats# all-intervals
```

```
Interval No.      :      1
Interval Duration  :      09:31:20 29-03-2016 - 09:46:10 29-03-2016
```

```
-----
                        Min      Average Max
                        (ms)      (ms)      (ms)
Two-Way Delay          0.000    0.000    0.000
Two-Way Delay Variation 0.000    0.000    0.000
Frames Exceed Delay Threshold :      0
Frames Exceed Delay Variation Threshold :      0
```

```

                        Forward      Backward
Tx Frames      : 0          0
```

Rx Frames	:	0	0
Frame Loss	:	0	0
Frame Loss Ratio (%)	:	0.000000	0.000000
Unavailable Seconds	:	900	900
Available Seconds	:	0	0
Tx		Rx	
LMMS : 0		LMRs : 0	
DMMS : 0		DMRs : 0	
Interval No.	:	2	
Interval Duration	:	09:16:30 29-03-2016	- 09:31:20 29-03-2016

		Min	Average Max
		(ms)	(ms) (ms)
Two-Way Delay		0.000	0.000 0.000
Two-Way Delay Variation		0.000	0.000 0.000
Frames Exceed Delay Threshold	:		0
Frames Exceed Delay Variation Threshold	:		0
		Forward	Backward
Tx Frames	:	0	0
Rx Frames	:	0	0
Frame Loss	:	0	0
Frame Loss Ratio (%)	:	0.000000	0.000000
Unavailable Seconds	:	900	900
Available Seconds	:	0	0
Tx		Rx	
LMMS : 0		LMRs : 0	
DMMS : 0		DMRs : 0	

Viewing OAM Status

Viewing MEP Status

- To view the MEP status in the web interface:
1. Navigate to Monitoring>OAM>CFM\PM.

MiNID

Monitoring>OAM>CFM\PM

MD Index	MD Level	MD Name	MD Name Format	
1	5	MD1	String	Info
2	3	MD2	String	Info
3	4	MD2	String	Info
4	2	MD4	String	Info

Summary

Monitoring OAM (CFM) – MD Level

2. In the row for the corresponding MD, click <Info>.

MiNID

Monitoring>OAM>CFM\PM>MD Info

MD Index

MD Level

MD Name Format

MD Name

1

5

String

MD1

MA Index	MA Name	MD Level	Vlan Tag Mode	Service Name	
1	MA1	5	Single	23	Info
2	MA3	5	Double-Inner	Gold	Info
3	MA6	5	Double-Outer		Info

Monitoring OAM (CFM) – MA Level

3. In the row for the corresponding MA, click <Info>.

MiNID

Monitoring>OAM>CFM\PM>MA Info

MD Level5
MA NameMA1
MA FormatCharacter String
Service Name23
Vlan Tag ModeSingle
CCM Interval100ms
Interface Status TLVDisable

MEP (ID: 2)	Info	MEP Status	LB Status	LT Status	Remote MEP – ID: 100
-------------	------	------------	-----------	-----------	----------------------

Monitoring OAM (CFM) – MA Info (Single-Tag Mode)

MiNID

Monitoring>OAM>CFM\PM>MA Info

MD Level	5
MA Name	MA6
MA Format	Character String
Service Name	Goldr
Vlan Tag Mode	Double-Inner
CCM Interval	100ms
Interface Status	
TLV	Disable

MEP (ID: 11)	Info	MEP Status	LB Status	LT Status
MEP (ID: 2)	Info	MEP Status	LB Status	LT Status

Monitoring OAM (CFM) – MA Info (Double-Tag Mode)

- In the row for the corresponding MEP, click <MEP Status>.

Note
If the MEP is administratively disabled, then the indication **Disabled** appears instead of the **Status** and **Statistics** links.

The MEP status is displayed.

MiNID

Monitoring>OAM>CFM>MA MEP Status

Rx AIS	OFF
Defects:	
MisMerge*	OFF
Unexpected MEP	OFF
Unexpected MD Level	OFF
Unexpected Period	OFF

Monitoring OAM (CFM) – MEP Status

*MisMerge denotes a state when a CCM frame with the same MD Level but with a MAID (MEGID) different than the receiving MEP’s own MAID (MEGID) is received.

- 5. To view status for a remote MEP, click <Back> to return to the MEP status screen, and then click the remote MEP ID for which you wish to view the status.

MiNID

Monitoring>OAM>CFM

State	RMEP_OK
Sys Up Time	207 sec.
Last MAC Address Received	00-20-D2-51-7E-2D
Loss of Continuity	OFF
Last RDI Bit State	False

Monitoring OAM (CFM) – Remote MEP Status

➤ **To view MEP status in the CLI:**

1. Navigate to configure oam cfm maintenance-domain <mdid> maintenance-association <maid> mep <mepid>.
The prompt **config>oam>cfm>md(<mdid>)>ma(<maid>)>mep(<mepid>)#** is displayed.
2. Enter:
show status
The MEP status is displayed. See [Displaying MEP Status](#) for an example.
3. For each configured remote MEP, enter :
show **remote-mep remote-mep-id** <remote-mep-id>
The remote MEP status is displayed. See [Displaying MEP Status](#) for an example.

Viewing OAM Statistics

Viewing Destination NE Statistics

You can view the following types of statistics for the OAM services destination NEs:

- Running –OAM statistics collected since the destination NE was activated.
- Interval – OAM statistics for the current interval or selected intervals
- Total intervals – Sums of the statistics of all valid intervals (with the exception that the Two-Way Delay and Two-Way Delay Variation counters are averages).

Note *In order to be able to view the statistics, the destination NE, corresponding MEP, and corresponding service must be administratively enabled, for at least the duration of the configured PM interval.*

Viewing Statistics Using the Web Interface

➤ **To display the OAM CFM statistics:**

1. Navigate to Monitoring>OAM>CFM (see [Error! Reference source not found.](#)).
2. In the row for the corresponding MA, click <Info> to display the Monitoring>OAM>CFM>MA Info screen (see [Error! Reference source not found.](#)).
3. In the row for the corresponding MEP, click <Info>.

MiNID

C

Monitoring >OAM>CFM

Admin Status	Enable
MEP ID [1..8191]	1
Ingress Port	SFP/Port 1
CCM Enable	Enable
CCM Transmission Mode	Multicast
Client MD Level	4
AIS Enable	Enable

Service (P-bit: 0)	Info	Dest NE – ID: 1 Dest NE – ID: 2
Service (P-bit: 5)	Info	Dest NE – ID: 7 Dest NE – ID: 8

Monitoring OAM (CFM) – MEP Info

4. Click the destination NE for which you wish to view statistics.

MinID

Monitoring>OAM>CFM

Admin Status

Enable

Dest NE ID

1

Remote MEP ID

55

Loss Measurement Service

LMM Synthetic

Delay Measurement Service

Enable

Running statistics

Current statistics

Interval statistics

Total Interval statistics

Monitoring OAM (CFM) – Dest NE Info

5. According to which type of statistics to display:
- To display the running statistics, click **Running Statistics**.
The statistics are displayed. To clear the statistics, click **<Clear Statistics>**.
 - To display the current interval statistics, click **Current**.
The statistics are displayed as illustrated in the figure below; for explanation, see **OAM Statistic Counters** and **OAM Loss and Delay Measurement Counters** tables. To clear the statistics, click **<Clear Statistics>**.
 - To display the statistics for selected intervals, click **Interval Statistics**.
The statistics are displayed for interval 1; see **OAM Statistic Counters** and **OAM Loss and Delay Measurement Counters** tables.

The duration and validity of the interval are indicated in the interval statistics. An interval is considered as not valid if it has not yet ended, or if statistics were cleared during the interval.

You can view selected intervals from the last 96 intervals (the amount of available intervals depends on how much time has passed since the destination NE was administratively enabled) by clicking **<Previous>** or **<Next>** to go to the previous or next interval, respectively, or by entering the interval number and then clicking **<Change>**. If you attempt to view an interval that is not available, a message to that effect is displayed.
 - To display the total interval statistics, click **Total Interval Statistics**.

The statistics are displayed as illustrated below; see **OAM Statistic Counters** and **OAM Loss and Delay Measurement Counters** tables.

MiNID

Monitoring>OAM>CFM

Dest NE State	Unavailable	
Statistics	Running	
Elapsed Time (Sec.)	414	
Two-Way Delay (ms)	0.000 (ms)	
Two-Way Delay Variation (ms)	0.000 (ms)	
One-Way Delay Variation (ms)	0.000 (ms)	
Frames Exceed Delay Threshold	0	
Frames Exceed Delay Variation Threshold	0	
	Forward	Backward
Tx Frames	0	
Rx Frames	0	
Frame Loss	0	
Frame Loss Ratio	0.000000%	0.000000%
Available Seconds	0	0
Unavailable Seconds	414	414
Two-Way Delay Bins Measurement - No Bin Profile Bound:		
Bin Number	Range [us]	Counter
1		
2		
3		
Two-Way Delay Variation Bins Measurement - No Bin Profile Bound:		
Bin Number	Range [us]	Counter
1		
2		
3		
One-Way Delay Bins Measurement - No Bin Profile Bound:		
Bin Number	Range [us]	Counter
1		
2		
3		
Transmitted	Received	
LMMs: 0	LMRs: 0	
DMMs: 0	DMRs: 0	
Clear Statistics		

MiNID

Monitoring>OAM>CFM

Dest NE State	Unavailable		
Statistics	Current (Last 15 minutes)		
Elapsed Time (Sec.)	532		
	Min	Average	Max
Two-Way Delay (ms)	0.000	0.000	0.000
Two-Way Delay Variation (ms)	0.000	0.000	0.000
One-Way Delay Variation (ms)	0.000	0.000	0.000
Frames Exceed Delay Threshold	0		
Frames Exceed Delay Variation Threshold	0		
	Forward	Backward	
Tx Frames	0		
Rx Frames	0		
Frame Loss	0		
Frame Loss Ratio	0.000000%	0.000000%	
Available Seconds	0	0	
Unavailable Seconds	414	414	
Two-Way Delay Bins Measurement - No Bin Profile Bound:			
Bin Number	Range [us]	Counter	
1			
2			
3			
Two-Way Delay Variation Bins Measurement - No Bin Profile Bound:			
Bin Number	Range [us]	Counter	
1			
2			
3			
One-Way Delay Bins Measurement - No Bin Profile Bound:			
Bin Number	Range [us]	Counter	
1			
2			
3			
Transmitted	Received		
LMMs: 0	LMRs: 0		
DMMs: 0	DMRs: 0		
Clear Statistics			

MiNID

Monitoring>OAM>CFM

Interval Number	1		
Interval Duration	23:11:13 05-01-2016 – 23:26:02 05-01-2016		
Status	Valid		
		Min	
Two-Way Delay (ms)		0.000	
Two-Way Delay Variation (ms)		0.000	
One-Way Delay Variation (ms)		0.000	
Frames Exceed Delay Threshold	0		
	0		
Frames Exceed Delay Variation Threshold			
	Forward	Backward	
Tx Frames	0	0	
Rx Frames	0	0	
Frame Loss	0	0	
Frame Loss Ratio	0.000000%	0.000000%	
Available Seconds	0	0	
Unavailable Seconds	900	900	
Delay Bins Measurement - No Bin Profile Bound:			
Bin Number	Range [us]	Counter	
1			
2			
3			
Delay Variation Bins Measurement - No Bin Profile Bound:			
Bin Number	Range [us]	Counter	
1			
2			
3			
One-Way Delay Variation Bins Measurement - No Bin Profile Bound:			
Bin Number	Range [us]	Counter	
1			
2			
3			
Transmitted		Received	
LMMs: 0		LMRs: 0	
DMMs: 0		DMRs: 0	

Previous
Interval(1-96)

Next
1
Change

Monitoring OAM (CFM) – Interval Statistics

Monitoring>OAM>CFM

Statistics		15 Min. Interval	
	Min	Average	Max
Two-Way Delay (ms)	0.000	0.000	0.000
Two-Way Delay Variation (ms)	0.000	0.000	0.000
One-Way Delay Variation (ms)	0.000	0.000	0.000
Frames Exceed Delay Threshold		0	
Frames Exceed Delay Variation Threshold		0	
	Forward	Backward	
Tx Frames	0	0	
Rx Frames	0	0	
Frame Loss	0	0	
Frame Loss Ratio	0.000000%	0.000000%	
Available Seconds	0	0	
Unavailable Seconds	5400	5400	
Two-Way Delay Bins Measurement - No Bin Profile Bound:			
Bin Number	Range [us]	Counter	
1			
2			
3			
Two-Way Delay Variation Bins Measurement - No Bin Profile Bound:			
Bin Number	Range [us]	Counter	
1			
2			
3			
One-Way Delay Bins Measurement - No Bin Profile Bound:			
Bin Number	Range [us]	Counter	
1			
2			
3			
Transmitted		Received	

LMMs: 0	LMRS: 0
DMMs: 0	DMRs: 0

Monitoring OAM (CFM) – Total Interval Statistics

Viewing Statistics Using the CLI

➤ To display the OAM CFM statistics:

- 1. Navigate to configure oam cfm maintenance-domain <mdid> maintenance-association <maid> mep <mepid> service <serviceid> dest-ne <dest-neid> show-statistics.

The following prompt is displayed:

```
config>oam>cfm>md(<mdid>)>ma(<maid>)>mep(<mepid>)>service(<serviceid>)>
dest-ne(<dest-neid>)>show-stats#
```

- 2. Enter all necessary commands according to the tasks listed below.

Note*If the destination NE is administratively disabled, a message stating that is displayed if you attempt to display the statistics.*

Task	Command	Comments
Viewing statistics for all intervals	all-intervals	If available, the statistics are displayed as shown in Displaying All Interval Statistics ; see the tables below
Viewing running statistics	running	If available, the statistics are displayed as shown in Displaying Running Statistics ; see the tables below
Viewing statistics for current interval	current	If available, the statistics are displayed as shown in Displaying Current Interval Statistics ; see the tables below

Task	Command	Comments
Viewing statistics for selected interval	interval <interval-number>	If available, the statistics are displayed as shown in Displaying Selected Interval Statistics ; see the tables below
Viewing statistics for total intervals	total-intervals	If available, the statistics are displayed as shown in Displaying Total Interval Statistics ; see the tables below

OAM Statistic Counters

Parameter	Description
Tx Frames Forward	Total number of frames transmitted from local service to remote service since the service was activated
Rx Frames Forward	Total number of frames received by remote service since the service was activated
Frame Loss Forward	Total number of frames lost from local service to remote service since the service was activated (Far End Tx Frames - Far End Rx Frames) (the type of frames counted is either user data or synthetic, according to the method configured by the loss single-ended command)
Unavailable Seconds Forward	Number of seconds the remote service is considered unavailable in direction local service to remote service
Tx Frames Backward	Total number of frames transmitted from remote service to local service since the service was activated
Rx Frames Backward	Total number of frames received by local service since the service was activated
Frame Loss Backward	Total number of frames lost from service to local service since the service was activated (Near End Tx Frames - Near End Rx Frames)
Unavailable Seconds Backward	Number of seconds the remote service is considered unavailable in direction remote service to local service
Available Seconds Forward	Number of seconds the remote service is considered available in direction local service to remote service

Parameter	Description
Available Seconds backward	Number of seconds the remote service is considered available in direction remote service to local service
Two-Way Delay (ms)	Minimal, average and maximal of all frame delay values
Two-Way Delay Variation (ms)	Minimal, average and maximal difference between the frame delay values
One-Way Delay Variation (ms)	Minimal, average and maximal difference between the frame delay values
Frames Exceed Delay Threshold	Number of frames whose delay value exceeded the delay threshold configured for the service
Frames Exceed Delay Variation Threshold	Number of frames whose delay variation exceeded the delay variation threshold configured for the service

OAM Loss and Delay Measurement Counters

Parameter	Description
Transmitted LMMs	Number of transmitted loss measurement messages Note: Relevant only for LMM synthetic and user data loss measurement.
Transmitted SLMs	Number of transmitted standard loss measurement messages Note: Relevant only for SLM synthetic loss measurement.
Transmitted DMMs	Number of transmitted delay measurement messages
Received LMRs	Number of received loss measurement replies Note: Relevant only for LMM synthetic and user data loss measurement.
Received SLRs	Number of received standard loss measurement replies Note: Relevant only for SLM synthetic loss measurement.
Received DMRs	Number of received delay measurement replies

Viewing OAM/CFM Summary

MiNID displays a summary of OAM/CFM statistics.

➤ **To view the summary using the web interface:**

1. Navigate to >Monitoring>OAM>CFP/PM.
2. Click **Summary**.

The summary is displayed.

► **To view the summary using the CLI:**

1. Go to configure oam cfm.
2. Type in show summary.

The summary is displayed.

11.4 OAM (EFM)

This section covers the monitoring of Ethernet links using link OAM (EFM) (OAM Ethernet at the First Mile). can act as the active or passive side in an IEEE 802.3-2005 application. The link OAM can be activated for one port at a time.

Applicability and Scaling

This feature is applicable to all MiNID options.

Standards Compliance

IEEE 802.3-2005

Benefits

Link OAM provides remote management and fault indication for the Ethernet links, as well as detection of remote link failure.

Functional Description

The OAM (EFM) discovery process allows a local data terminating entity (DTE) to detect Ethernet OAM capabilities on a remote DTE. Once Ethernet OAM support is detected, both ends of the link exchange state and configuration information, such as mode, PDU size, loopback support, etc. If both DTEs are

satisfied with the settings, OAM is enabled on the link. However, the loss of a link or a failure to receive OAMPDUs for five seconds causes the discovery process to restart.

DTEs may be in either active or passive mode. DTEs in active mode initiate the OAM (EFM) communications and can issue queries and commands to a remote device. DTEs in passive mode generally wait for the peer device to initiate OAM communications, and respond to commands and queries, but do not initiate them.

A flag in the OAMPDU allows an OAM entity to convey the failure condition Link Fault to its peer. Link Fault refers to the loss of signal detected by the receiver; a Link Fault report is sent once per second with the Information OAMPDU.

Factory Defaults

By default, OAM EFM is not enabled for the ports.

Configuring Link OAM

The following parameters are configured for link OAM.

Link OAM Parameters

Parameter		Description
Web	CLI	
OAM Mode	mode	Specifies active or passive mode
OAM Port	bind	Port for which link OAM is active: - SFP/1 - MSA/2.

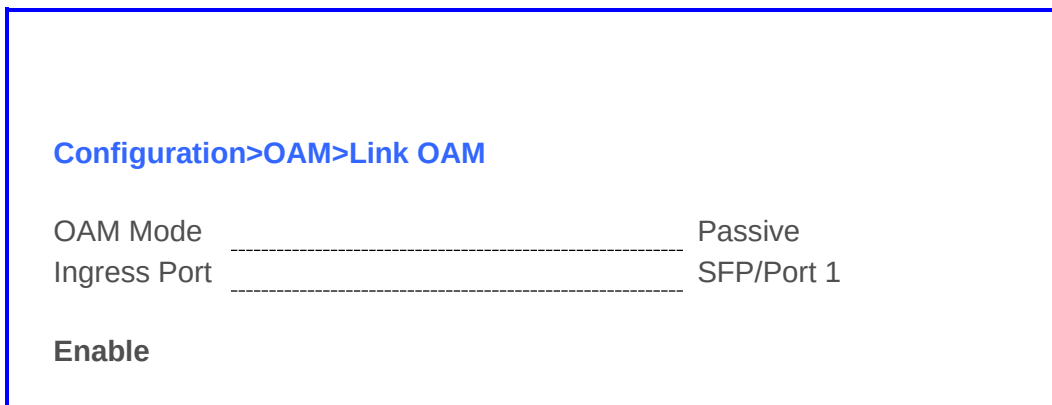
Note

In order for link OAM to function, an L2CP profile must be defined with the OAM protocol action set to peer, and flow classification mode must be port mode or VID+Pbits Range, to enable attaching the L2CP profile to the relevant port/flow.

Configuring Link OAM Using the Web Interface

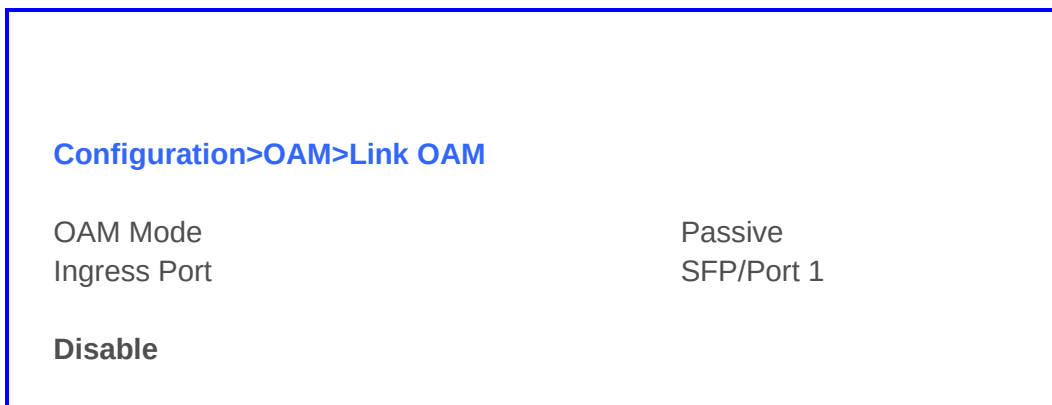
► To configure link OAM:

1. Navigate to Configuration > OAM > Link OAM.
2. If OAM (EFM) is disabled, the screen appears as shown in the following figures.



The screenshot shows the web interface for configuring Link OAM. At the top, the breadcrumb navigation 'Configuration>OAM>Link OAM' is displayed in blue. Below this, there are two rows of configuration options. The first row is 'OAM Mode' with a dropdown menu currently set to 'Passive'. The second row is 'Ingress Port' with a dropdown menu currently set to 'SFP/Port 1'. At the bottom of the form, there is a button labeled 'Enable'.

Link OAM Disabled



This screenshot is identical to the one above, showing the 'Configuration>OAM>Link OAM' page with 'OAM Mode' set to 'Passive' and 'Ingress Port' set to 'SFP/Port 1'. However, the button at the bottom is labeled 'Disable' instead of 'Enable'.

Link OAM Enabled

1. To enable, select the mode and port, and then click **Enable**.
2. To disable, click **Disable**.

Configuring Link OAM Using the CLI

➤ To configure link OAM:

- 1. Navigate to configure oam efm.
The **config>oam>efm#** prompt is displayed.
- 2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Specifying for which port to activate link OAM	bind ingress-port {sfp/1 msa/2}	
Enabling loopback	loopback	Type no loopback to disable loopback <i>Note: Available only in passive mode</i>
Specifying active or passive mode	mode {active passive}	
Enabling link OAM	no shutdown	Type shutdown to disable link OAM
Displaying link OAM (EFM) status	show status	
Clearing link OAM (EFM) statistics	clear-statistics	
Displaying link OAM (EFM) statistics	show statistics	

Examples

➤ To enable active link OAM for port SFP/1:

```
exit all
configure oam efm
  shutdown
  mode active
  bind ingress-port sfp/1
  loopback
  no shutdown
exit all
```

➤ To display the link OAM status:

```
>config>oam>efm# show status
Administrative Status : Enabled
```

Operational Status	: Send Local
Loopback Status	: REMOTE
OAM EFM Information	

	Local Remote
MAC Address	00-20-D2-51-42-52 --
OAM Mode	Active --
Unidirectional	Unsupported --
Var Retrival	Unsupported --
Link Events	Unsupported --
Loopback	Supported --
PDU Size	1518 --

➤ To display the link OAM statistics:

>config>oam>efm# show statistics		
OAM EFM Statistics		

	Rx	Tx
Information	0	0
Event Notification	0	0
Loopback Ctrl	0	0
Unsupported Codes	0	

11.5 OAM (TWAMP)

MiNID provides up to 32 TWAMP Light Controller sessions that interoperate with TWAMP responders on other devices, by generating TWAMP test packets. You can display the session measurements and results.

MiNID also provides up to 16 TWAMP Light responders that interoperate with TWAMP generators on other devices, by responding to TWAMP test packets with measurement packets.

In addition, MiNID supports one-way loss TWAMP statistics. In order to calculate one-way packet loss, the following are required:

- A responder that can generate an independent sequence number
- A controller that is aware of the responder’s independent sequence number

Applicability and Scaling

This feature is applicable to all MiNID options.

Standards Compliance

RFC 5357 – A Two-Way Active Measurement Protocol (TWAMP)

Benefits

You can measure the IP performance of your Layer-3 network at all locations, without the need for a special performance management system.

Functional Description

TWAMP Responders

The TWAMP Light mechanism provides for monitoring sessions where information is exchanged between TWAMP controllers and responders. The controller establishes the test session with the responder, and then the controller transmits test packets to the responder, which reflects the test packets to the controller. The controller processes the resulting measurements and calculates metrics that can be displayed in test reports.

The TWAMP responders work in Layer-2 probe mode, meaning they are bound to layer-2 interfaces (port/flow). A responder can be bound to more than one flow. The association between a TWAMP responder and flow/port is performed from the flow/port menus/CLI in the same way that flow/port layer-3/4 loopbacks are configured (see **Flows** for details). You can view statistics for TWAMP looped packets and bytes by viewing the associated flow statistics for layer-3/4 loopbacks.

Note

The TWAMP Light responders work in non-standard mode, i.e. the parameters for the TWAMP Light responder session are configured manually in via the web/CLI, rather than by TWAMP control protocol.

You can configure the following parameters for a TWAMP controller.

TWAMP Controller Parameters		
Parameter		Description
Web	CLI	
Controller Name	controller <controller-name>	Name of TWAMP controller in , up to 32 characters

Parameter		Description
Web	CLI	
IP Address	local-ip-address <router-interface-1 router-interface-2 router-interface-3 router-interface-4 router-interface-5>	Router interface associated with the controller, from which you can send TWAMP test packets and ARP requests.
Admin Status	shutdown	Administratively enable or disable the TWAMP controller

You can configure the following parameters for a TWAMP responder.

TWAMP Responder Parameters

Parameter		Description
Web	CLI	
Responder ID	Not applicable	Identification number assigned by MinID, not user-configurable, displayed only in Web
Responder Name	responder <responder-name>	Name of TWAMP responder in MinID, up to 32 characters
TWAMP Local IP	ip-address	IP address (IPv4 or IPv6) associated with the responder, to which to send TWAMP test packets, ARP requests, or pings
TWAMP UDP Port	udp-port	UDP port (range 1–65535) associated with responder <i>Note: The combination of IP address and UDP port must be unique for each TWAMP responder.</i>
Use Loaned IP	ip-loan	If enabled, responds only to TWAMP test packets sent to the TWAMP responder IP address, and does not respond to ARP/ping requests sent to the TWAMP responder IP address. This is useful if you have configured the responder with the same IP address as the host device (i.e. the IP address was loaned from the host), and you want the host to answer ARP/ping requests.
Admin Status	shutdown	Administratively enable or disable the TWAMP responder

MAC Allocation Logic

The following MAC addresses are available in the device:

- **Device Management MAC** – used for MiNID management and for TWAMP sessions 1 and 3
- **Router Int 2,4 MAC address** – used for TWAMP sessions 2 and 4
- **Device Services MAC address** – used for OAM/CFM only (not for TWAMP).

They can be seen in the CLI and in Web GUI as follows:

MiNID

Monitoring - System

Device Management MAC address

00-20-D2-51-7E-2D

Router Int2,4 MAC address

00-20-D2-51-7E-2F

Device Services MAC address

00-20-D2-51-7E-2E

System Up Time (hh:mm:ss)

05:35:48

DIP Switches Mode

Normal

Current date

10 May 2025 14:15:51+00:00

Summer time

not configured

NTP

Events Log

```

MiNID>config>system# show status
Device Management MAC address      :      00-20-D2-E2-37-93
Router Int 2,4 MAC address         :      00-20-D2-E2-37-94
Device Services MAC address        :      00-20-D2-E1-B3-C6
System UP Time (hh:mm:ss)          :      948:05:02
DIP Switches Mode                  :      NORMAL
    
```

Factory Defaults

By default, no TWAMP controllers or responders are configured.

Configuring TWAMP

- **To configure TWAMP, perform the following steps:**
1. In the responder device:
 - a. Configure relevant router interface and flows.
 - b. Configure and activate TWAMP responder and relevant test session(s).
 2. In the controller device:
 - a. Configure relevant router interface and flows.
 - b. Configure TWAMP profile(s).
 - c. Configure and activate TWAMP controller and relevant peers and test sessions.

Profile is a set of properties that a TWAMP session has. You have to define at least one profile in order to define a TWAMP controller.

Configuring TWAMP Controllers

- **To configure a TWAMP controller in the web interface:**
1. Navigate to Configuration > OAM > TWAMP > Controller.

Configuration>OAM>TWAMP>Controller

Add new Controller

Controller Name				
Controller1	Add Peer	Show Peers	Status	Delete Controller

Configuring TWAMP Controllers

2. To add a new TWAMP controller:
 - a. Click Add new Controller.

Configuration>OAM>TWAMP>Controller>TWAMP Controller>New Controller

Controller Name		
IP Address		Router Interface 1
Admin Status		Disable

Apply

New TWAMP Controller

- b. Fill in the parameters as specified in the above figure.
 - c. Click <Apply> to create the TWAMP controller.
- The **Configuration>OAM>TWAMP Controller** screen appears, with an entry containing the new TWAMP controller.

- 3. Click **Add Peer** to define remote responder parameters in a TWAMP session.

Configuration>OAM>TWAMP>Controller>Peer>TWAMP Peer> New Peer

Peer IP		0.0.0.0
Activation		OFF
Responder Sequence Num		OFF
Calculation Mode		Round-Trip

Apply

New TWAMP Peer

- a. Fill in the parameters as specified in the table below.
- b. Click <Apply> to create the peer.

➤ **To add or configure a TWAMP controller in the CLI:**

- 1. Navigate to **configure oam twamp controller** <controller-name> [<number>], where <controller-name> is the name of the controller and <number> is the controller ID number.
- The **config>oam>twamp>controller(<controller-name>)#** prompt is displayed.

2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Administratively enabling or disabling the controller	no shutdown	Enter shutdown to administratively disable the controller. You should enable the controller only after the responder has been configured and enabled.
Configuring controller IP address	local-ip-address <router-interface-1 router-interface-2 router-interface-3 router-interface-4 router-interface-5>	If Loaned IP is enabled on router interface #1, controller IP address can be configured on any other router interface
Assigning an IP address to the remote responder (peer).	peer <ip-address>	See Running Test Sessions via Controller Peers
Viewing controller status	show-status	

Running Test Sessions via Controller Peers

Do one of the following:

- In the web interface:
 - a. Navigate to Configuration > OAM > TWAMP > Controller.
 - b. In the row of the configured controller, click Show Peers.
 - c. In the row of the selected peer, click Add Test Session.

Configuration – OAM – TWAMP – Controller- Peer - Test Session - TWAMP Test Session - New Test Session

Test Session Name	_____	
TWAMP Profile	_____	<profile-name>
Display Num	_____	0
UDP Port	_____	0
DSCP	_____	0

Apply

Configuring New Test Session

- a. Fill in the parameters as specified in the table below.

- b. Click <Apply> to create the test session.
- In the CLI:
 - a. Navigate to configure oam twamp controller <name>peer <ip-address>. The **config>oam>twamp>controller (<name>)>peer(<ip-address>)#** prompt is displayed.
 - b. Enter all necessary commands according to the tasks listed below.
 - c. Enter all necessary commands according to the tasks listed below.

Peer Parameters

Task	Command	Comments
Activating all configured test sessions in one-time mode activate	activate duration <minutes>	The tests run for the specified amount of time. Type no activate to deactivate the one-time (non-continuous) command.
Activating all configured test sessions in continuous mode	activate continuous	The tests run until they are stopped. Type no activate to deactivate the continuous command.
Defining the type of calculation for the TWAMP metrics	calculation-mode <round-trip>	TWAMP controller calculates standard TWAMP metrics.
Specifying whether the responder transmits an independent sequence number	responder -seq-num	Notes: The controller calculates one-way loss and availability only if this is enabled. The corresponding responder must be configured to transmit the responder sequence number.
Configuring test session	test-session <number> name <name-string> udp-port <udp-port-number> test-profile <profile-name> dscp <dscp-number>	The UDP and DSCP can be used to distinguish between test sessions. UDP port number: 1-65535 Test profile name: Up to 32 characters DSCP number: 0-63 (default: 0)
Viewing test report	show-report <test-name> {interval current all} [<interval-num>]	See Viewing TWAMP Reports
Viewing summary of test reports	show summary-report	See Viewing TWAMP Reports
Viewing test status	show status	See Viewing TWAMP Status

- **To delete a TWAMP controller:**
- 1. Navigate to configure oam twamp.
 - 2. Type **no controller** <controller-name>.

Configuring TWAMP Profiles

- **To configure TWAMP profiles in the web interface:**
- 1. Navigate to Configuration>OAM>TWAMP>Profile.

Configuration>OAM>TWAMP>Profile

Add new Profile

Profile Name		
Profile 1	Delete Profile	

Configuring TWAMP Profiles

- 2. To add a new TWAMP profile:
 - c. Click Add new Profile.

Configuration>OAM>TWAMP>Profile>TWAMP Profile>New Profile

Profile Name	
Payload Length	256
Transmit Rate	10
Loss Timeout	2000000
Loss Ratio Threshold	1000
Delay Variation Threshold	1000
Delay Treshold	1000
Delay Variation Event Type	pdv

Apply

New TWAMP Profile

- d. Fill in the parameters as specified in the table below.
 - e. Click <Apply> to create the TWAMP profile.
- The **Configuration>OAM>TWAMP>Profile** screen appears, with an entry containing the new TWAMP profile.

- To delete a TWAMP profile, click **Delete Profile** in the corresponding row.
The TWAMP profile is deleted.
- Click <Save Configuration>.

➤ To add or configure a TWAMP profile in the CLI:

- Navigate to **configure oam twamp profile** <profile-name>, where <profile-name> is the name of the profile.
The **config>oam>twamp>profile(<profile-name>)#** prompt is displayed.
- Enter all necessary commands according to the tasks listed below.

TWAMP Profile Parameters

Task	Command	Comments
Defining delay threshold in microseconds	delay - threshold	

Task	Command	Comments
Configuring delay variation event type to define whether the PDV or IPDV metric is used for the delay threshold	delay-variation-event-type	pdv – Packet delay variation metric ipdv –Inter-packet delay variation metric
Defining delay variation threshold in microseconds	delay-variation-threshold	
Defining test packet loss timeout in microseconds	loss-timeout	
Defining test session loss ratio threshold in ppm (packet per million)	loss-ratio-threshold	Range is 1000 – 10000 ppm Default: 1000 ppm
Defining test packet payload length in bytes	payload-length	
Defining test profile packet transmit rate in PPS	transmit-rate <pps>	

➤ **To delete a TWAMP profile:**

1. Navigate to configure oam twamp.
2. Type **no profile** <profile-name>.
The TWAMP profile is deleted.

Configuring TWAMP Responders

➤ **To configure TWAMP responders in the web interface:**

1. Navigate to Configuration > OAM > TWAMP > Responder.

Configuration – OAM – TWAMP - TWAMP Responder

Add New TWAMP Responder Session

ID	Responder Name	Admin Status	TWAMP Local IP	TWAMP UDP Port	Tx Sequence Num	Use Loaned IP			
1	tw1	Disable Session	10.10.2.2	3	ENABLED	ENABLED	Status	Unbind All Flows	Delete Session
2	tw2	Enable Session	1.1.1.1	5555	DISABLED	DISABLED	Status	Unbind All Flows	Delete Session
3	tw3	Enable Session	10.10.2.2	4	DISABLED	DISABLED	Status	Unbind All Flows	Delete Session
4	tw4	Disable Session	2.2.2.2	888	ENABLED	ENABLED	Status	Unbind All Flows	Delete Session

Configuring TWAMP Responders

2. To add a new TWAMP responder:
- a. Click Add New TWAMP Responder Session.

Configuration>OAM>TWAMP>Responder>TWAMP Responder>New Responder Session

Responder ID5

Responder Name

IP Address

UDP Port0

Tx Sequence NumEnable

Use Loaned IPDisable

Admin StatusDisable

New TWAMP Responder

b. Fill in the parameters as specified in See [Peer Parameters](#) table.

c. Click <Apply> to create the TWAMP responder.

The **Configuration > OAM > TWAMP Responder** screen (see above) appears, with an entry containing the new TWAMP responder.

3. To enable or disable a TWAMP responder, click Enable Session/Disable Session in the corresponding row.
4. To unbind all flows associated with a TWAMP responder, click Unbind All Flows in the corresponding row.
5. To delete a TWAMP responder, click Delete Session in the corresponding row.
6. The TWAMP responder is deleted, if it is administratively disabled and is not bound to any flows or ports.
7. Click <Save Configuration>.

➤ **To add or configure a TWAMP responder with the CLI:**

1. Navigate to **configure oam twamp responder** <responder-name>, where <responder-name> is the name of the responder.

The **config>oam>twamp>responder(<responder-name>)#** prompt is displayed.

2. Enter all necessary commands according to the tasks listed below.

Task	Command	Comments
Assigning an IP address to the responder	ip-address <ip-address>	The address can be in either IPv4 or IPv6 format.
Specifying that the IP address assigned to the responder is loaned from the host device	ip-loan	Enter ip-loan for to answer only TWAMP test packets sent to the IP address Enter no ip-loan for to answer ARP/ping requests/TWAMP test packets sent to the IP address See Peer Parameters table for more details on IP loan
Generating an independent sequence number for the TWAMP responder packets	tx-sequence-num	
Administratively enabling or disabling the responder	no shutdown	Enter shutdown to administratively disable the responder

Task	Command	Comments
Assigning a UDP port to the responder	udp-port <udp-port>	1–65535
Unbinding any flows associated with the TWAMP responder	unbind-all-flows	

- **To delete a TWAMP responder:**
1. Navigate to configure OAM TWAMP.
 2. Type **no responder** <responder-name>.

Viewing TWAMP Status

You can view the following statuses using the CLI:

- Controller status – show status in level config oam twamp controller
- Responder status – show status in level config oam twamp responder
- Status of peer test sessions – show-status in level config oam twamp controller peer.
- The data is also available in the web interface:
- Controller status – Navigate to the Controller page and click **Status** in the row of the corresponding controller
- Responder status - Navigate to the Responder page and click **Status** in the row of the corresponding responder
- Status of peer test sessions – Navigate to the TWAMP Peers page and click **Status** in the row of the corresponding peer.

- **To view controller status:**

```
MinID>config>oam>twamp>controller(1)# show status
IPPM Type           : TWAMP Light
Router Interface     : 1
Controller Status    : Ready
```

- **To view responder status:**

```
MinID>config>oam>twamp>responder(tw1)# show status
IPPM Type           : TWAMP Light
Responder Status     : Ready
```

Responder Name	UDP	Tx-Packets	Rx-Packets
-----	-----	-----	-----
tw1	3	1107	1107

➤ To view peer test status:

MinID>config>oam>twamp>controller(1)>peer(33.33.116.6)# show-status

IPPM Type : TWAMP Light

Activation Mode : continuous

Calculation Mode : round-trip

Start Time : 00:00:54 01-01-1970

Controller	Test-Name	Peer	UDP	Status	Tx-Packets	Rx-Packets
-----	-----	-----	-----	-----	-----	-----
1	1	33.33.116.6	5001	In Progress	1770	1770
1	2	33.33.116.6	5002	In Progress	1770	1770
1	3	33.33.116.6	5003	In Progress	1770	1770
1	4	33.33.116.6	5004	In Progress	1770	1770
1	5	33.33.116.6	5005	In Progress	1770	1770
1	6	33.33.116.6	5006	In Progress	1770	1770
1	7	33.33.116.6	5007	In Progress	1770	1770
1	8	33.33.116.6	5008	In Progress	1771	1771

Viewing TWAMP Reports

TWAMP Metrics

The TWAMP controller calculates performance measurement metrics according to the received test packets for each peer and its active test sessions. The metrics are recalculated every minute. You can view the metrics for the current interval, selected interval, or all intervals by viewing the TWAMP report.

When the TWAMP responder receives a test packet, it reflects the test packet after it adds an Rx stamp and a Tx stamp.

When the TWAMP controller receives a test packet from the responder, it calculates the following:

- Delay = Responder Rx stamp - Controller Tx stamp + Controller Rx stamp –Responder Tx stamp
- Forward delay = Responder Rx stamp - Controller Tx stamp
- Backward delay = Controller Rx stamp - Responder Tx stamp
- Packet validity – determined by checking if the delay is less than the value configured by command loss-timeout. If so, the Rx valid count is incremented, otherwise the packet is dropped.

The following table lists the metrics that are displayed in the TWAMP reports, subject to the restrictions specified above for the calculation modes. See [Viewing the Metrics](#) for instructions on viewing the reports and for examples of TWAMP reports.

TWAMP Report Metrics

Counter	Description
Availability Count (sec)	Number of available seconds. A minute is declared as unavailable if it has more than 75% packet loss, therefore it is available if packet loss is 25% or less. When a minute is declared unavailable, the delay, delay variation, loss measurements, and their derived metrics are ignored for that minute.
Availability Count Fwd/Back	Number of available seconds. A minute is declared as unavailable if it has more than 75% packet loss, therefore it is available if packet loss is 25% or less. When a minute is declared unavailable, the delay, delay variation, loss measurements, and their derived metrics are ignored for that minute. These statistics are for the forward and backward directions. <i>Note: These statistics appear only when responder seq-num is enabled.</i>
Delay Average (ms)	Average of packet delay values
Delay Max (ms)	Maximum of packet delay values
Delay Min (ms)	Minimum of packet delay values
Delay Threshold Crossing Count	Number of packets with delay larger than the delay threshold configured for the corresponding test profile
Duplicate Packets Back	Number of duplicate packets in backward direction. A packet is considered duplicate (backward) if its responder Tx timestamp matches that of a previously received packet in backward direction.
Duplicate Packets Fwd	Number of duplicate packets in forward direction. A packet is considered duplicate (forward) if its controller sequence number or controller Tx timestamp matches that of a previously received packet in forward direction.
Duplicate Ratio Back	Duplicate Packets Back divided by Tx Packets Back, converted to a percentage
Duplicate Ratio Fwd	Duplicate Packets Fwd divided by Tx Packets Fwd, converted to a percentage
Fragmented Packets Back	Number of fragmented packets in backward direction. When the TWAMP controller recognizes a fragmented packet, it increments the Fragmented Packets Back counter.

Counter	Description
Fragmented Packets Fwd	Number of fragmented packets in forward direction. When the TWAMP responder receives a fragmented packet, when it reflects it to the controller, the responder sends indication of fragmentation, if tx-extended-info was enabled. When this indication is received, the controller increments the Fragmented Packets Fwd counter.
IPDV Average (ms)	Average of IPDV (Inter Packet Delay Variation) values in both directions
IPDV Max (ms)	Maximum of IPDV (Inter Packet Delay Variation) values in both directions. Inter Packet Delay Variation is calculated according to RFC 5481, from the variations of the delays between valid packets.
IPDV-Back Average (ms)	Average of IPDV (Inter Packet Delay Variation) values in backward direction
IPDV-Back Max (ms)	Maximum of IPDV (Inter Packet Delay Variation) values in backward direction
IPDV-Fwd Average (ms)	Average of IPDV (Inter Packet Delay Variation) values in forward direction
IPDV-Fwd Max (ms)	Maximum of IPDV (Inter Packet Delay Variation) values in forward direction
Loss Packets	Number of packets lost
Loss Packets Fwd/Back	Number of packets lost in each of the directions, calculated by Tx Packets –Rx valid count <i>Note: These statistics appear only when responder seq-num is enabled.</i>
Loss Ratio	Loss Packets divided by Tx Packets, converted to a percentage
Loss Ratio Fwd/Back	Loss Packets divided by Tx Packets, converted to a percentage in each of the directions <i>Note: These statistics appear only when responder seq-num is enabled.</i>
PDV Average (ms)	Average of PDV (Packet Delay Variation) values, calculated by subtracting Delay Min from Delay Average
PDV Max (ms)	Maximum of PDV (Packet Delay Variation) values. Packet Delay Variation is calculated according to ITU-T Y.1540, by subtracting the minimum delay from the 99.9% percentile of the delay values
Reordered Packets Back	Number of reordered packets in backward direction. A packet is considered reordered (backward) if its responder sequence number is smaller than that of a previously received packet in backward direction.
Reordered Packets Fwd	Number of reordered packets in forward direction. A packet is considered reordered (forward) if its controller sequence number or controller Tx timestamp is smaller than that of a previously received packet in forward direction.

Counter	Description
Reordered Ratio Back	Reordered Packets Back divided by Tx Packets converted to a percentage
Reordered Ratio Fwd	Reordered Packets Fwd divided by Tx Packets converted to a percentage
Tx Packets	Number of packets transmitted in forward direction (controller to responder)
Tx Packets Fwd/Back	Number of packets transmitted forwarded (from controller to responder) and back (from responder to controller)
<i>Note: These statistics appear only when responder seq-num is enabled.</i>	

Viewing the Metrics

You can view the following reports in level **config oam twamp controller peer**:

- Summary report of all peer test sessions – **show-summary-report**
- Test report of a particular session:
 - Test report of all intervals – **show-report <name> all**
 - Test report of current interval – **show-report <name> current**
 - Test report of selected interval – **show-report <name> interval <interval-num>**

➤ To view a summary report:

```

MiNID>config>oam>twamp>controller(1)>peer(33.33.116.6)# show-summary-report
IPPM Type           : TWAMP Light
Controller IP Address : 33.33.116.4
Responder IP Address  : 33.33.116.6
Activation Mode       : continuous
Calculation Mode      : round-trip

```

Controller	Test-Name	IP DSCP	Size (bytes)	Loss Ratio	Delay Max (ms)	PDV Max (ms)	IPDV Max (ms)	Result
1	1	1	256	0	1.891	0.246	1.790	NA
1	2	1	256	0	0.321	0.147	0.221	NA
1	3	1	256	0	0.390	0.173	0.290	NA
1	4	1	256	0	0.374	0.176	0.274	NA
1	5	1	256	0	0.301	0.144	0.201	NA
1	6	1	256	0	0.351	0.164	0.250	NA
1	7	1	256	0	1.889	0.092	1.789	NA
1	8	1	256	0	0.316	0.181	0.216	NA

➤ To view a test report:

```

MiNID>config>oam>twamp>controller(1)>peer(2.2.2.2)# show-report 1 all
Test Name                               : 1
IPPM Type                               : TWAMP Light
Controller IP Address                    : 33.33.116.4 / 1229
Responder IP Address                     : 33.33.116.6 / 5001
IP DSCP                                  : 11
Payload Length (bytes)                   : 256
Calculation Mode                         : round-trip
Test Session Start Time                  : 00:00:54 01-01-1970

Test Interval                           : Current
Elapsed Time                            (sec): 420

Tx Packets                              : 4200
Loss Packets                            : 0
Loss Ratio                              : 0
Availability Count                       (sec): 420
Tx Packets      Fwd / Back               : 0      0
Loss Packets    Fwd / Back               : 0      0
Loss Ratio      Fwd / Back               : 0      0
Availability Count Fwd / Back (sec): 0      0
Duplicate Packets Fwd / Back             : 0      0
Duplicate Ratio Fwd / Back               : 0      0
Reordered Packets Fwd / Back             : 0      0
Reordered Ratio Fwd / Back               : 0      0
Fragmented Packets Fwd / Back            : 0      0

Delay Threshold Crossing Count           : 1
Delay      Min / Max / Average (ms) : 0.099      1.891      0.101
PDV        Max / Average (ms)       : 0.182      0.002
IPDV       Max / Average (ms)       : 1.790      0.002
IPDV-Fwd   Max / Average (ms)       : 0.002      0.000
IPDV-Back  Max / Average (ms)       : 1.790      0.002

```

11.6 Packet Capture

MiNID sends packets to a PC running Wireshark or another packet analyzer supporting RPCAP. Then the traffic can be recorded and analyzed on the PC.

Performance was tested with Wireshark version 1.12 and 2.0.3.

Applicability and Scaling

This feature is applicable to all MiNID options.

Standards Compliance

RPCAP – Remote Packet Capture

Benefits

MiNID can replace an equipped PC running Wireshark or dumpcap (Wireshark's CLI) that is required at the viewing point. MiNID captures bidirectional traffic and sends it to a packet analyzer running on a remote PC over IP/UDP/RPCAP. By configuring the analyzer filters, you can troubleshoot your network, analyze traffic, and use the data for software and communications protocol development.

By applying relevant traffic filters, tunnels only selected packets, thus reducing the network workload and preventing the PC from collapsing when there is a large number of packets.

Functional Description

MiNID enables you to capture real-time traffic and to transport it to a remote viewing /recording station. The station running Wireshark initiates the RPCAP tunneling session from the MiNID.

MiNID can manipulate the captured traffic by editing or discarding frames. All flow actions are supported.

The following scenario and example illustrate how packet capturing is performed.

The user wishes to perform the flow action of push VLAN between the two ports of SFP and MSA):

- If the user has defined the packet capture on one of the ports (for example, the SFP port), and the traffic arrives from the SFP (in the direction of the MSA), the user sees the captured packets **before** the push VLAN flow action has been performed.
- If the traffic arrives from the MSA (in the direction of the SFP), the user sees the packets captured **after** the push VLAN flow action has been performed.

Note

*Some of the traffic processed by is not captured; only traffic passing **through** is captured. The feature of packet capture does not mirror traffic, such as OAM packets, which is addressed to itself.*

Note

Packet Capture applies to the IP address of the relevant router interface.

It is possible that timestamping of the captured traffic is altered due to synchronization with the NTP server clock. To prevent the time amendments during packet capturing, usage of NTP as a clock source can be disabled by choosing the Free Run option.

Factory Defaults

By default, packet capturing is disabled.

When packet capturing is enabled, it is configured as shown below.

Parameter	Default	Remarks
capture-dscp	0	DSCP value for the mirrored traffic
local-ip-address	router-interface-1	Dedicated IP interface for packet capture
rpcap-tcp-port	2002	TCP port number
inactive-timeout	5	Packet capture session inactivity timeout in minutes
timestamp-source	freerun	Source of timestamping for captured traffic

Configuring Packet Capture

Enabling Packet Capture

➤ To enable packet capture using the web interface:

1. Navigate to Configuration > Packet Capture.

Configuration>Packet Capture

Capture	Disable
Capture-DSCP	0
Local IP Address	Router Interface 1
RPCAP TCP Port	2002
Inactive Timeout	Enable
Inactive Timeout [Min.]	5
Timestamp Source	Free Run

Show Status

Apply

Enabling Packet Capture

- 1. Fill in the relevant parameters as described in the table below. Set the Capture status to Enable.
- 2. Click <Apply>.

➤ To enable packet capture using the CLI:

- 1. Navigate to configure>packet-capture.
The **config>packet-capture#** prompt is displayed.
- 2. Enter all necessary commands according to the tasks listed below.

Packet Capture Parameters

Task	Command	Comments
Enabling or disabling packet capturing	[no] capture	Type capture to administratively enable packet capturing
Setting the DSCP value for the mirrored traffic	capture-dscp	Range [0-63]
Defining packet capture session inactivity timeout in minutes	[no] inactive-timeout	When timeout period is over, packet capturing is disabled. Possible values: 1-65536 minutes

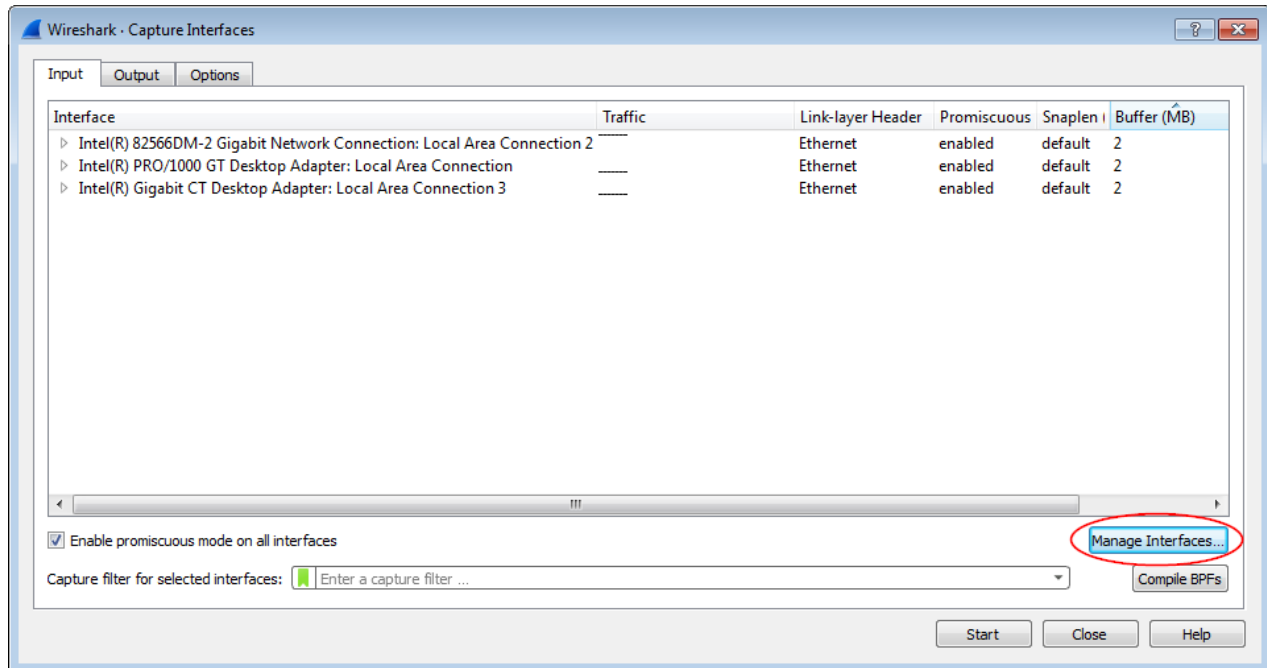
Task	Command	Comments
Setting dedicated IP interface for packet capture	local-ip-address <router-interface-1 router-interface-2 router-interface-3 router-interface-4 router-interface-5>	If Loaned IP is enabled on router interface #1, packet capture can be set on any other router interface
Setting the TCP port	rpcap-tcp-port	
Displaying the status of the current Packet Capture session	show status	See Viewing Statistics for Packet Capture
Setting the source of timestamping for captured traffic	timestamp source <freerun ntp>	freerun – internal clock is used as a timestamping source ntp – NTP server clock is used as a timestamping source

Configuring a Wireshark Session

You can have only one running Wireshark session. To open a new session, you have to close the previous one.

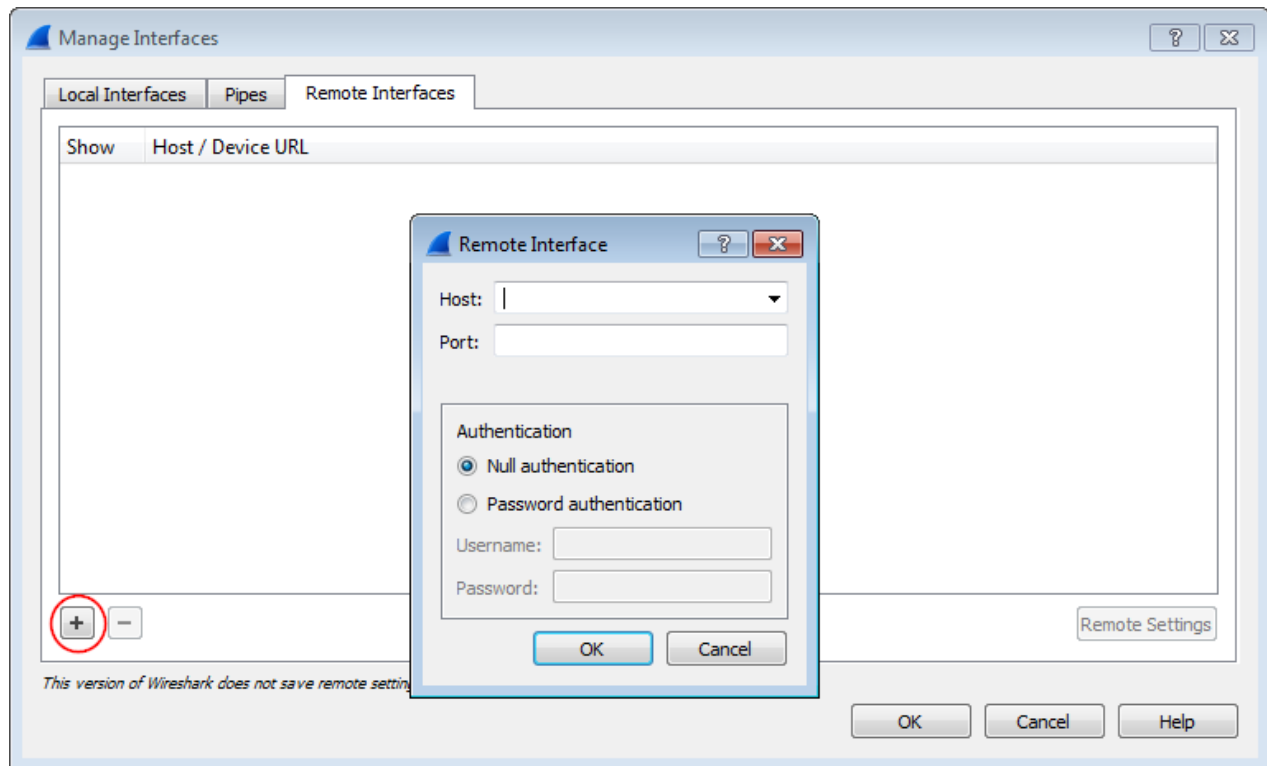
➤ **To define a new session:**

1. Open Wireshark. Click the **Capture options**  button on the Wireshark toolbar.
The **Capture Interfaces** window opens.



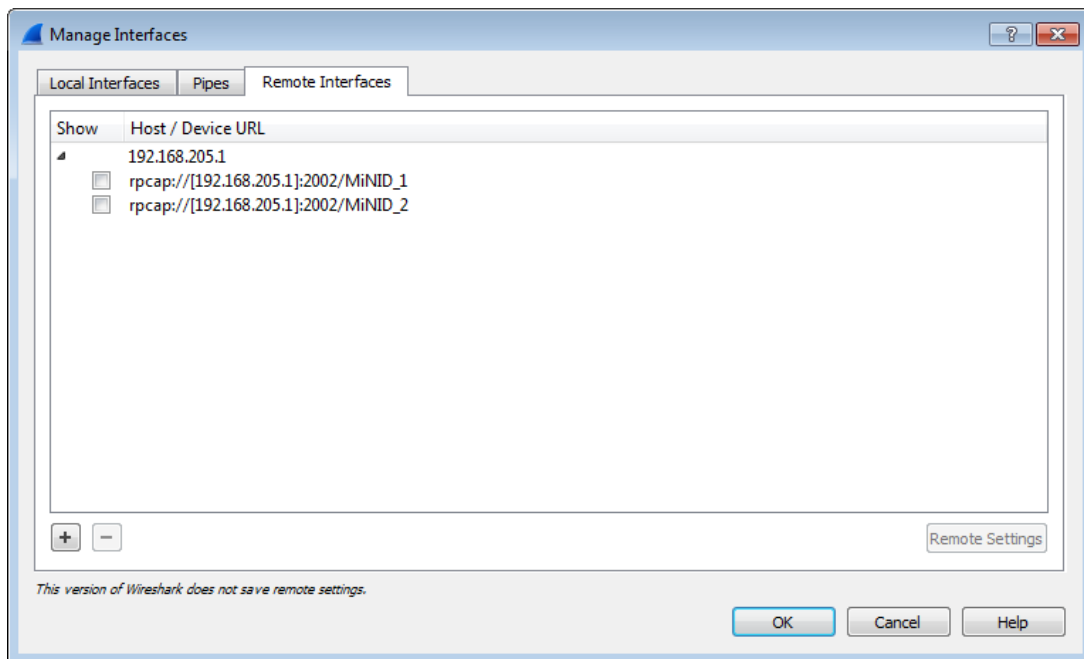
Capture Interfaces Window of Wireshark

2. Click the **Manage Interfaces** button.
The **Manage Interfaces** window opens. Switch to the **Remote Interfaces** tab.
3. Click the + button in the bottom left to add a new remote interface.
The **Remote Interface** window opens.



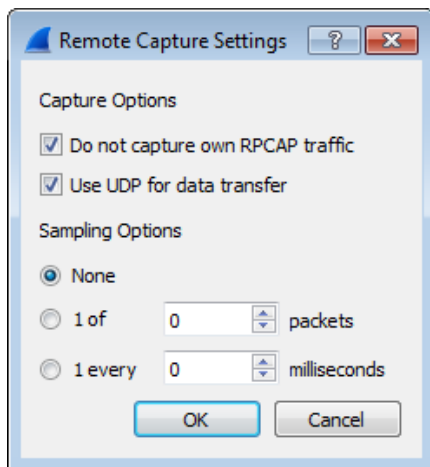
Remote Interfaces Window of Wireshark

4. Fill in the local IP address (by default, 192.168.205.1) and TCP port (by default, 2002). Click **OK**. Two new interfaces are added.



Adding Ports as Remote Interfaces

5. Select both ports. For each of the ports, click the **Remote Settings** button. The **Remote Capture Settings** window opens.
6. Under Capture Options, select Use UDP for data transfer. Click OK.

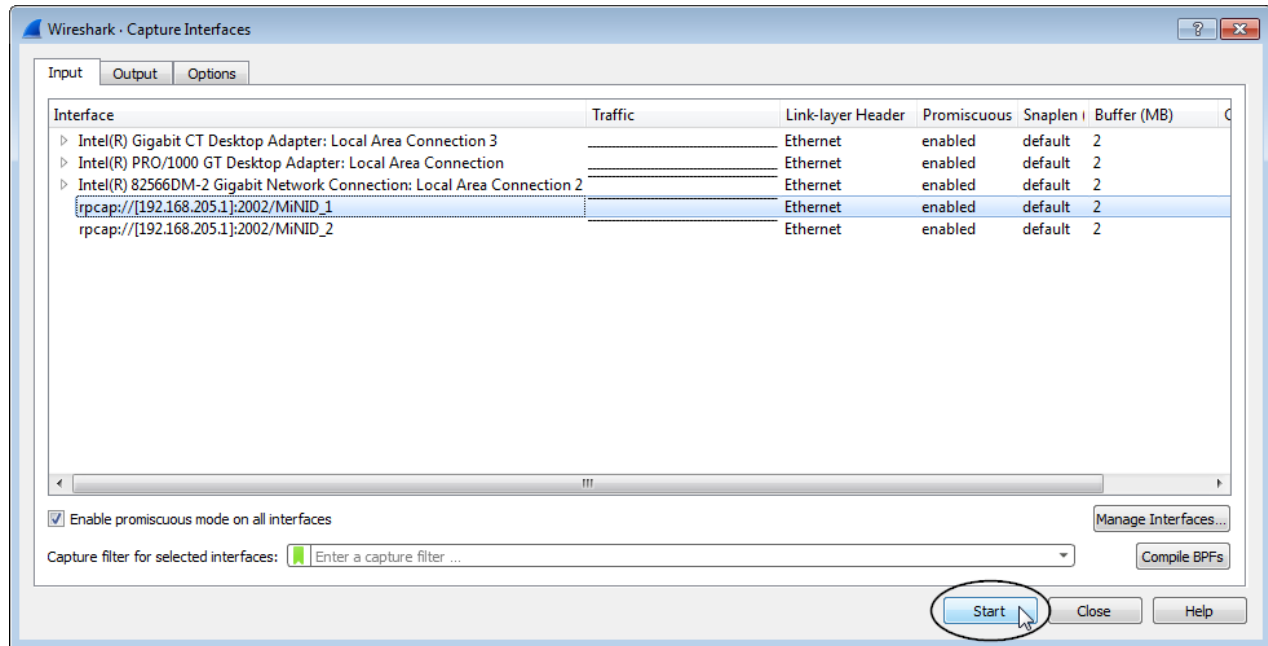


Remote Capture Settings

7. Close the **Manage Interfaces** window by clicking **OK**.

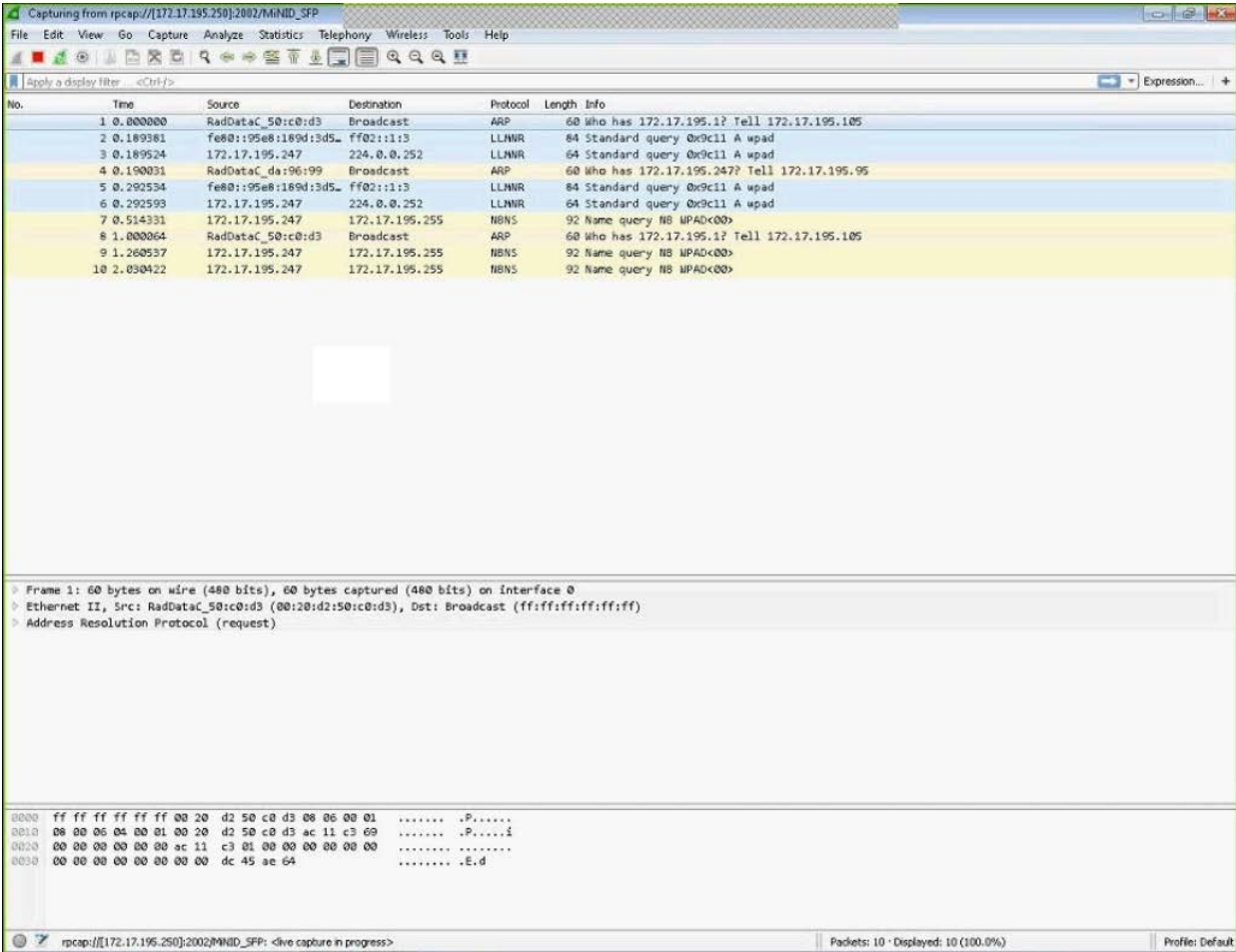
Both ports appear in the list as remote interfaces.

8. In the **Capture Interfaces** window, select one of the ports by clicking on its row.
9. Click the **Start** button to initiate packet capturing.



Starting a Session

Wireshark starts to display all traffic in the main window.



Captured Traffic in Wireshark Window

10. To end the session, click the **Stop capturing packets**  button on the Wireshark toolbar.

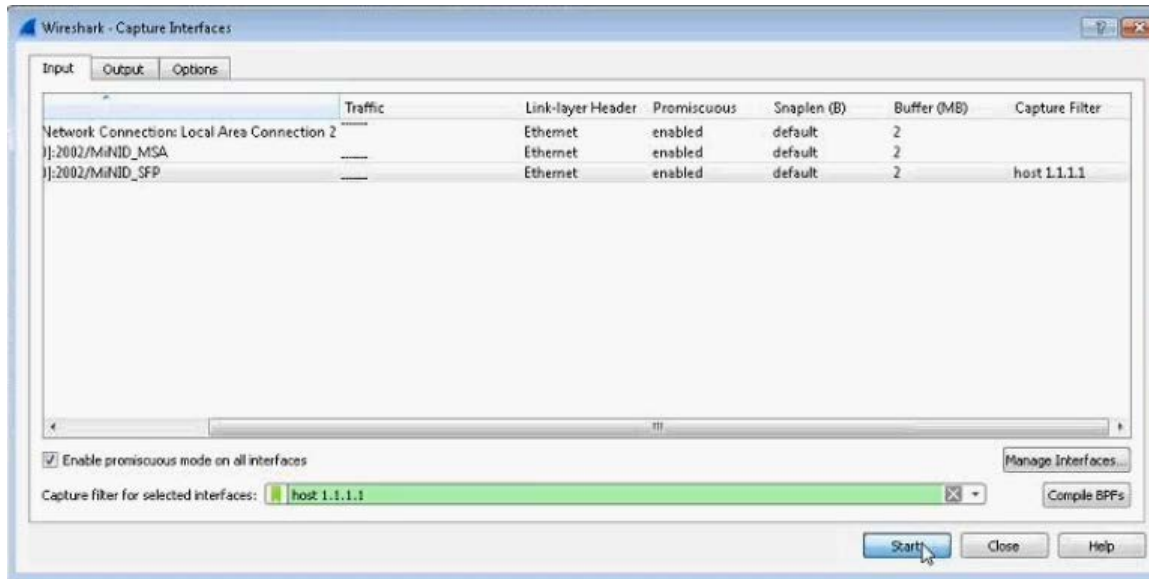
Configuring Display Filters in Wireshark

In addition to capturing the whole traffic, you can apply various filters to choose only specific type of traffic.

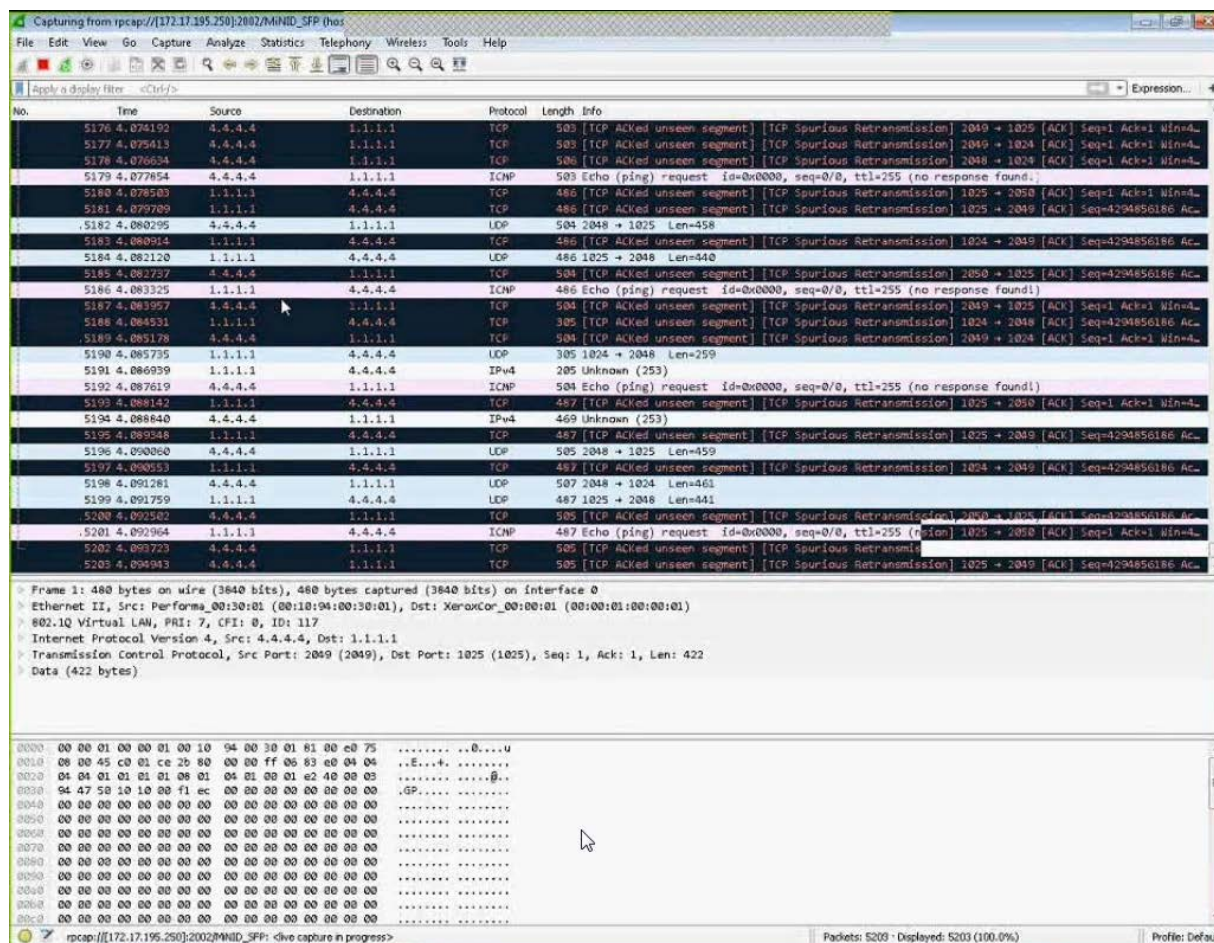
Note MinID supports filtering of IPv4 packets. IPv6 packets are ignored.

► To define a display filter:

1. Click the **Capture options**  button on the Wireshark toolbar.
The **Capture Interfaces** window opens.
2. In the **Capture filter for selected interfaces** field, type the filtering criteria. For displaying only one host, type **host <host-IP>**. Click **Start**.



Now Wireshark displays only traffic coming from and arriving to the host with the specified IP.



Captured Traffic with a Filter Applied

The following filtering criteria can be added:

Filters Applied to Captured Traffic

Criteria	Command	Comment
Traffic coming from and to a selected host	host <host-ip>	ARP and reverse ARP packets for the host are captured as well, when only IP address is specified. This parameter is required to get a correct filter expression
Traffic coming from a selected host	src host <host-ip>	

Criteria	Command	Comment
Traffic arriving to a selected host	dst host <host-ip>	
Applied protocol	<protocol name>	IP traffic includes only IPv4 packets, not IPv6 packets. Protocol name should be paired with ipv4 Example: tcp and ipv4
Port used in a source host	src port <port number>	
Port used in a destination host	dst port <port number>	

Note

*You can define up to two filter expressions concatenated by **or**. The parameters within a filter expression are joined by **and**.*
Example:
(src host 1.1.1.1 and dst host 4.4.4.4
and tcp src port 1025 and tcp dst port 2050)
or
(src host 4.4.4.4 and dst host 1.1.1.1
and tcp src port 2050 and tcp dst port 1025)

Configuring Packet Truncation

MiNID enables you to take only the most important data (start of a packet) and pass it to the viewing station. The truncated packet size should be specified in Wireshark. The software displays the full packet size if you specify the size of the truncated file more than 12,000 bytes. If the value is less than 32 bytes or more than 255 bytes, Wireshark displays an error message.

➤ **To display truncated packets:**

1. In Wireshark, open the **Capture Interfaces** window.
2. In the row of the selected port, double-click the **Snapplen (B)** field to open it editing.
3. Type the desired file size to be captured in the range of 32-255 bytes. Click **Start**.
Wireshark displays truncated packets.

Viewing Statistics for Packet Capture

➤ To view statistics in:

1. In the web interface, click **Show Status** in the Packet Capture page.
2. In the CLI, navigate to **config> packet-capture** and type **show status**.
Relevant statistic parameters are displayed.

```
Capture state           : In Progress
Destination address     : 192.168.205.55
Encapsulation protocol  : RPCAP
TCP port                : 2002
UDP port                : 59093

-----
Capture port            : 1
Capture packet size     : First 255 Bytes
Start                   : 02:37:01 01-01-1970
End                     : N/A
Elapsed time            : 00:09:49
Tx packets              : 54

Capture rule #1
-----
Pass All

Capture rule #2
-----
Pass All
```

Packet Capture Status

Parameter	Value	Description
Capture State	in progress ready disabled	• In progress – Wireshark session is running • Ready – Wireshark session is completed • Disabled – capturing is administratively disabled in
Destination Address	<ip-address>	
Encapsulation protocol	RPCAP	
TCP port	<tcp-port-number>	
UDP port	<udp-port-number>	

Parameter	Value	Description
Capture port	<port name/number>	
Capture packet size	First X bytes Whole packet	Shows the number of bytes in case of of truncated packets or writes about whole packet
Start	hh:mm:ss dd-mm-yyyy	Start time of session
End	hh:mm:ss dd-mm-yyyy	End time of session
Elapsed time	hh:mm:ss	Time period from the session start
Tx packets	<packets number>	Number of packets processed during the session
Capture rule #1		See filters in the table above
Capture rule #2		See filters in the table above

11.7 Ping

MiNID can initiate a ping to test whether the host is reachable across the IP network.

Applicability and Scaling

This feature is applicable to all MiNID options.

Benefits

Can verify communication with the host and other devices.

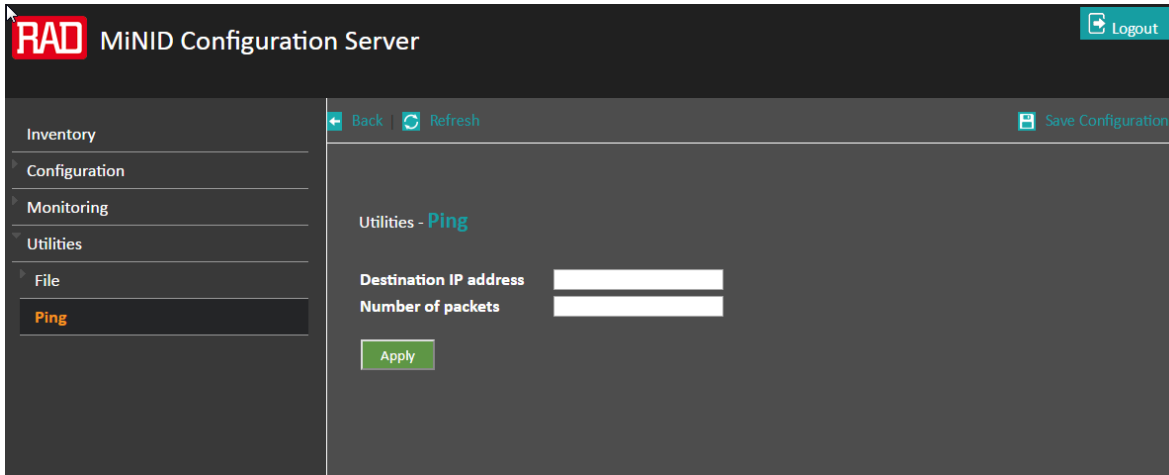
Factory Defaults

The default number of packets sent for the ping test is 5.

Configuring Ping Test

➤ To initiate a ping using the web interface:

1. Navigate to > **Utilities**> **Ping**.



The screenshot displays the MiNID Configuration Server web interface. The top header features the RAD logo and the text "MiNID Configuration Server" on the left, and a "Logout" button on the right. A left sidebar contains a menu with items: Inventory, Configuration, Monitoring, Utilities, File, and Ping (highlighted in orange). The main content area has a top bar with "Back" and "Refresh" buttons on the left, and a "Save Configuration" button on the right. Below this, the page is titled "Utilities - Ping". It contains two input fields: "Destination IP address" and "Number of packets". An "Apply" button is positioned below these fields.

2. In the **Destination IP address** field, type in the IP address.
3. In the **Number of packets** field, type in the number of packets you want the MiNID to send.
4. Click <**Apply**>.

The screenshot shows the MiNID Configuration Server web interface. On the left is a sidebar menu with options: Inventory, Configuration, Monitoring, Utilities, File, and Ping (highlighted in orange). The main content area is titled 'Utilities - Ping'. It features a green status message 'Info: The Ping was stopped' and a 'Stop ping' button. Below this is a text box displaying the results of a ping test to 192.168.205.1, showing 6 successful packets with 0% loss and a round-trip time of 0/0/0 ms. At the top right of the main area are 'Back', 'Refresh', and 'Save Configuration' buttons. The top header includes the 'RAD' logo and a 'Logout' button.

5. Click <Stop ping> to stop the ping test.

► To initiate a ping using the CLI:

- In the command line, type `ping <ip address> [number-of-packets <packets>]`.

Example

```
MiNID# ping 162.35.158.2 number-of-packets 6
Reply from 162.35.158.2: bytes=32, packet number=1, time<1ms
Reply from 162.35.158.2: bytes=32, packet number=2, time<1ms
Reply from 162.35.158.2: bytes=32, packet number=3, time<1ms
Reply from 162.35.158.2: bytes=32, packet number=4, time<1ms
Reply from 162.35.158.2: bytes=32, packet number=5, time<1ms
Reply from 162.35.158.2: bytes=32, packet number=6, time<1ms
6 packets transmitted. 6 packets received. 0% packet loss
Round Trip <ms> min/avg/max= 0/0/0
```

11.8 RFC-2544 Responder

MiNID provides an RFC-2544 responder that allows you to test the link at full capacity, and handles up to 1Gbps of traffic.

Applicability and Scaling

This feature is applicable to all MiNID options.

Standards Compliance

RFC-2544, Benchmarking Methodology for Carrier Ethernet Networks

Benefits

You can evaluate the performance of network devices to provide performance metrics of the Ethernet network and validate the SLA.

Functional Description

The RFC-2544 responder uses OAM CFM loopback messages (LBM) and loopback replies (LBR).

Configuring RFC-2544 Responder

➤ **To use the RFC-2544 responder:**

1. In MiNID, create a MEP with the appropriate MD level, MA ID, and VLAN (refer to **OAM (CFM)** for details on configuring MEPs).
2. To start the RFC-2544 test, send LBM frames towards the MEP in the desired rate, with the MD level, MA ID, and VLAN that correspond to the MEP.
3. To start RFC-2544 latency tests, send DMM frames towards the MEP, with the MD level, MA ID, and VLAN that correspond to the MEP.

11.9 Syslog

MiNID uses the Syslog protocol to generate and transport event notification messages over IP networks to Syslog servers.

Applicability and Scaling

This feature is applicable to all MiNID options.

Standards Compliance

RFC 3164, RFC 5674

Benefits

Syslog protocol collects heterogeneous data into a single data repository. It provides system administrators with a single point of management for collecting, distributing and processing audit data. Syslog standardizes log file formats, making it easier to examine log data with various standard tools. Data logging can be used for:

- Long-term auditing
- Intrusion detection
- Tracking user and administrator activity
- Product operation management

Functional Description

The Syslog protocol provides an instrument for generating and transporting event notification messages from MiNID to servers across IP networks.

Elements

Typical Syslog topology includes message senders (devices) and message receivers (servers). MiNID supports up to five Syslog servers. The receiver displays, stores, or forwards logged information. The standard designates two types of receivers:

<i>Relay</i>	Forwards messages
<i>Collector</i>	Displays and stores messages

Transport Protocol

Syslog uses User Datagram Protocol (UDP) for its transport. The UDP port assigned to Syslog is 514, but devices and servers can be defined to use any port for communication.

Message Format

The length of a Syslog message is 1024 bytes or less. It contains the following information:

- Facility and severity (see below)
- Host name or IP address of the device
- Timestamp
- Message content

A typical Syslog message looks like this:
<145>Jan 15 13:24:07 172.17.160.69 Eth 1: Loss of signal (LOS)

Facilities and Severities

Facility designates a device or application that sends a message. The standard includes some predefined facilities in the 0–15 range. For originator identification, MiNID uses facilities local1– local7.

Severity is assigned to a message to specify its importance. MiNID uses the following severity designations:

Syslog Severities

Code	Syslog Type	Description
0	Emergency	Emergency message
1	Alert	Critical alarm

Code	Syslog Type	Description
2	Critical	Major alarm
3	Error	Minor alarm
4	Warning	Event
5	Notice	Cleared alarm
6	Informational	Informational message, not in use
7	Debug	Debug-level messages, not in use

Note *The Syslog operation is not supported if uses a loaned IP address.*

Factory Defaults

By default, Syslog operation is disabled. When enabled, the default parameters are as follows:

Parameter	Default Value
facility	local1
port	514
severity-level	informational

Configuring Syslog Parameters

When configuring Syslog parameters, it is necessary to define Syslog device and servers.

- **To configure Syslog device in the CLI:**
- 1. Navigate to configure system syslog device.
The **config>system>syslog(device)#** prompt is displayed.
 - 2. Enter the necessary commands according to the tasks listed below.

Syslog Configuration Parameters

Task	Command	Comments
Defining a facility from which Syslog messages are sent	facility { local1 local2 local3 local4 local5 local6 local7 }	
Defining Syslog device UDP port for communication	port <udp-port-number>	Possible values: 1–65535 Port configuration is allowed only if a Syslog device is administratively disabled.
Defining severity level	severity-level { emergency alert critical error warning notice informational debug }	The log messages that contain severity level above or equal to the specified level are transmitted.
Administratively enabling Syslog device	no shutdown	shutdown administratively disables the Syslog device.
Displaying statistics	show statistics	
Clearing statistics	clear-statistics	

➤ **To configure a Syslog server in the CLI:**

1. Navigate to configure system).
The **config>system#** prompt is displayed.
2. At the **config>system#** prompt, enter **syslog server** <server-ID> to specify the server to receive Syslog messages, where <server-ID> is 1 to 5.
The **config>system>syslog(<server-ID>#** prompt is displayed.
3. Enter the necessary commands according to the tasks listed below.

Syslog Server Parameters

Task	Command	Comments
Enabling logging of command entries	accounting commands	To disable command logging, enter no accounting commands
Defining Syslog server IP address	address <ip-address>	Possible values: 0.0.0.0–255.255.255.255
Defining Syslog server UDP port for communication	port <udp-port-number>	Possible values: 1–65535

Task	Command	Comments
Administratively enabling Syslog server	no shutdown	shutdown administratively disables Syslog server.

► To configure Syslog device in the web interface:

1. Go to Configuration > System > Syslog.

Configuration – System - Syslog

Device Configuration:

Facility	Local1
Severity Level	informational
Port	514
Admin Status	Disable

Apply

Server Configuration

Server ID	Address	Accounting Commands	Port	Admin Status
1	FE80:B5BA:59E4:1A3D:F3CD	ENABLED	514	ENABLED
2	192.168.205.55	ENABLED	514	ENABLED
3	192.168.205.2	ENABLED	514	ENABLED
4	0.0.0.0	DISABLED	514	DISABLED
5	0.0.0.0	DISABLED	514	DISABLED

Statistics:

Total Tx Messages	356
Non-queued Dropped Messages	10

Clear Statistics

Syslog Configuration

2. Set the device configuration according to **Syslog Server Parameters** table.
3. Click <Apply>.

➤ To configure a Syslog server in the web interface:

- 1. Under **Server Configuration**, click the ID number of the server to receive Syslog messages.

Configuration – System - Syslog - Server Update

Server ID	2
Address	192.168.205.55
Accounting Commands	Enable
Port	514
Admin Status	Enable

Apply

- 1. Enter the necessary commands according to **Syslog Statistic Parameters** table.
- 2. Click <Apply>.

Viewing Syslog Statistics

➤ To display Syslog statistics in the CLI:

- 1. Navigate to configure system syslog device.
The **config>system>syslog(device)#** prompt is displayed.
- 2. At the config>system>syslog(device)#, enter show statistics.
Syslog statistics appear as shown below. The counters are described in the table below.

```
MinID>config>system>syslog(device)# show statistics
Total Tx Messages           : 356
Non-queued Dropped Messages : 265
```

Syslog Statistic Parameters

Parameter	Description
Total Tx Messages	The total number of Syslog messages transmitted

Parameter	Description
Non-queued Dropped Messages	The total number of Syslog messages that were dropped before being queued

► **To clear Syslog statistics in the CLI:**

1. Navigate to configure system syslog device.
The **config>system>syslog(device)#** prompt is displayed.
2. At the config>system>syslog(device)# prompt, enter clear-statistics.
The Syslog statistic counters are set to **0**.

► **To display Syslog statistics in the web interface:**

1. Go to Configuration > System > Syslog.
2. Syslog statistics are displayed at the bottom. The counters are described in the table above.

► **To clear Syslog statistics in the web interface:**

1. Go to Configuration > System > Syslog.
2. Click Clear Statistics.
The Syslog statistic counters are set to **0**.

Example

- Server IP address: 178.16.173.152
- UDP port: 155

```
exit all
configure system
  syslog device
    no shutdown
  exit
  syslog server 1
    address 178.16.173.152
    port 155
    no shutdown
  save
exit all
```

Configuration Errors

The table below lists messages generated by MiNID when a configuration error is detected.

<i>Configuration Error Messages</i>	
Message	Description
Syslog Port is out of range	Selected UDP port value is out of allowed range (1–65535).
Port is illegal or Device Port is already in use	Selected UDP port is already in use.
Parameter cannot be changed if Logging Status/Server Access is enabled	Device/server UDP port or server IP address cannot be changed while Syslog server is enabled.
Illegal Severity	Invalid severity value
Illegal Facility	Invalid facility value
Illegal Server IP Address	Invalid server IP address

11.10 Detecting Problems

Alarms and Traps

Alarms serve as notification of a fault in the device, and are indicated by an entry in the alarm and event history log, and/or an SNMP trap to a management station. See [Handling Alarms and Events](#) for further details on alarms, events, and traps.

Statistic Counters

Statistic counters provide information on possible abnormal behavior and failures. You can collect statistics on the following:

- Flows
- OAM CFM/EFM
- SFP/MSA (sleeve) ports

For further information, refer to the relevant sections in **Chapter 6 –10** and the relevant sections in the troubleshooting chart.

11.11 Handling Alarms and Events

This section describes how to display and clear the event log, and lists the events and traps.

Defining Alarm Types

The alarm type can be defined as the “legacy” alarms or the new SOAM alarms supported by MEF 36.

➤ **To define the alarm type using the web interface:**

1. Navigate to Configuration > OAM > CFM.

Configuration>OAM>CFM

Alarm Type

Legacy/SOAM

Add New MD

Apply

1. Select the relevant alarm type:
 - Legacy: Previous alarm set
 - SOAM: New MEF 36 alarm set. This mode supports alarm suppression.

➤ **To define the alarm type using the CLI:**

1. Navigate to MiNID>config>oam>cfm#.
2. Use the following command to select alarm type:

- **alarm-type** {legacy | soam}

Working with the Event Log

The event log contains the events that have occurred since the last startup of . The event log can hold up to 300 events. If it reaches that limit, further events are stored in a wraparound fashion, i.e. the later events overwrite the earlier events. The file entries are time-stamped with the time since the last startup.

➤ **To display the event log:**

Do one of the following:

- In the web interface, go to **Monitoring > System**, and click **Events Log**:

MiNID

Monitoring - System

Device Management MAC address	00-20-D2-51-7E-2D
Router Int2,4 MAC address	00-20-D2-51-7E-2F
Device Services MAC address	00-20-D2-51-7E-2E
System Up Time (hh:mm:ss)	05:35:48
DIP Switches Mode	Normal
Current date	10 May 2025 14:15:51+00:00
Summer time	not configured

NTP

Events Log

MiNID

Monitoring>System>Events Log

Log Page	-----			1
Time	Event	Severity	Description	
00:04:38	LOC	MAJOR	OFF, MEP [2.MA1.1.55]	
00:04:20	LOC	MAJOR	ON, MEP [2.MA1.1.55]	
00:04:12	UPLOAD FILE	MINOR	Upload log-file success via SFTP	
		MINOR	CLI Parsing startup-config Success	
00:01:01	TELNET MESSAGE			
00:00:53	LINK UP	MAJOR	SFP Port (sleeve)	
00:00:46	LINK DOWN	MAJOR	SFP Port (sleeve)	
00:00:31	LOGIN SUCCESS	MAJOR	User: su (telnet)	

Clear table

Displaying Event Log

- In the CLI, go to the **configure reporting** level, and enter **show log**.

```
>config>report# show log
```

Time	Event	Severity	Description

00:04:38	LOC	MAJOR	OFF, MEP [2.MA1.1.55]
00:04:20	LOC	MAJOR	ON, MEP [2.MA1.1.55]
00:04:12	UPLOAD FILE	MINOR	Upload log file success via SFTP
00:01:01	TELNET MESSAGE	MINOR	CLI Parsing startup-config Success
00:00:53	LINK UP	MAJOR	SFP Port (sleeve)
00:00:46	LINK DOWN	MAJOR	SFP Port(sleeve)
00:00:31	LOGIN SUCCESS	MAJOR	User: su (telnet)

► To clear the event log:

Do one of the following:

- In the web interface, go to **Monitoring > System > Log File** as described in the above procedure to display the event log, and click **Clear table**.
- In the CLI, go to the **configure reporting** level, and enter **clear-log**.

```
>config>report# clear-log
```

```
>config>report#
```

Alarm and Event Lists

You can view the full lists of alarms and events supported by , along with the traps corresponding to each alarm or event.

➤ To view the alarms table:

- Double-click the paperclip image on the following line.



➤ To view the events table:

- Double-click the paperclip image on the following line.



11.12 Performance Monitoring

maintains performance management (PM) statistics that are collected into a file periodically, for retrieval by RADview, for display in the RADview PM portal (refer to the RADview System User's Manual for further details on the PM portal). The PM file can also be retrieved by a third-party network management system.

Applicability and Scaling

This feature is applicable to all MiNID options.

Benefits

The PM data is useful for analyzing service quality. The flexible statistics collection allows only the necessary data to be collected.

Functional Description

PM statistics collection is configured globally for MiNID. When it is enabled, collects PM data. When it is disabled, deletes any PM data previously collected and stops collecting PM data.

MiNID stores PM data at user-configurable intervals (1, 5, 10, or 15 minutes). Additionally the interval for collecting the PM data from can be configured to 1, 5, 10, or 15 minutes.

- Note**
 - If the PM file reaches 90% capacity, sends the PM SPACE OVERFLOW event; if PM data cannot be saved due to insufficient space, stops collecting PM data and sends the PM PROCESS DISABLED event*
 - If performance monitoring is enabled, ensure that the data is collected by RADview or a third-party NMS, otherwise one or both of the alarms is triggered. If it is not possible to collect the data, then disable performance monitoring.*

Factory Defaults

Parameter	Default	Remarks
pm	pm	PM statistics collection in device is globally enabled by default
interval-duration	15	PM interval is 15 minutes by default
file-interval-duration	5	PM file is collected every 5 minutes by default

Configuring Performance Monitoring

➤ To configure PM collection:

Do one of the following:

- In the web interface:
 - Go to Configuration > System > Performance Monitoring.
 - Set PM Mode to Enable or Disable.

- c. Set PM Interval Duration to 1 Min, 5 Min, 10 Min, or 15 Min.
- d. Set PM File Interval Duration to 10 Sec, 1 Min, 5 Min, 10 Min, or 15 Min.

Note *PM File Interval Duration must be less than or equal to PM Interval Duration.*

- e. Click <Apply> to implement the changes, and then click <Save Configuration>.

MiNID

Configuration – System - Performance Monitoring

PM Mode	_____	Enable
PM Interval Duration	_____	15 Min
PM File Interval Duration	_____	5 Min

- In the CLI:
 - a. Go to the **configure reporting** context.
 - b. Enter **pm** to enable PM collection, or no pm to disable PM collection.
 - c. Enter the following to set the PM interval:
interval-duration {1 | 5 | 10 | 15}
 - d. Enter the following to set the interval for collecting the PM data:
file-interval-duration {10s | 1 | 5 | 10 | 15}

Note *The value set by file-interval-duration must be less than or equal to the value set by interval-duration.*

11.13 Troubleshooting

This section contains a general troubleshooting chart that lists possible failures and provides workarounds.

- Use this chart to identify the cause of a problem that may arise during operation. For a detailed description of the LED indicator functions, refer to **Chapter 3**.

To correct the reported problem, perform the suggested corrective actions. If a problem cannot be resolved by performing the suggested action, contact your RAD distributor.

Fault/Problem	Probable Cause	Corrective Action
The unit is unreachable	Incorrect management settings	Using a local serial connection, verify the management settings
	Management path disconnected	In case of remote management, analyze this issue using a local serial connection Check network connectivity issues and firewall settings
MinID does not answer pings	Ping packets are larger than 128 bytes	Send ping packets that are no larger than 128 bytes
Physical link fails to respond	Link may be administratively disabled.	Administratively enable the link

11.14 Frequently Asked Questions

- Q

What do I need to configure in order to automatically distribute a configuration file to via zero touch?
- A

You need to configure the management VLAN and set DHCP to enable. You also need to set up the DHCP and TFTP servers as specified in the explanation of zero touch (for further details refer to [Chapter 3](#)).

11.15 Technical Support

For technical support of registered products, contact your local authorized RAD partner or go to [RADcare Online](#) (if you have a valid [RADcare service package](#)).

RAD would like your help in improving its product documentation. Please [send us an e-mail](#) with your comments.

Thank you for your assistance!

A Connection Data

A.1 Ethernet Connector

The MSA and SFP Ethernet connectors are wired in accordance with the table below

Pin	Name	Function
1	VeeT	Transmitter Ground
2	Tx Fault	Transmitter Fault
3	Tx Disable	Transmitter Disable
4	MOD-DEF2	Module Definition 2
5	MOD-DEF1	Module Definition 1
6	MOD-DEF0	Module Definition 0
7	Rate Select	Not used
8	LOS	Loss of Signal
9	VeeR	Receiver Ground
10	VeeR	Receiver Ground
11	VeeR	Receiver Ground
12	RD-	Inv. Received Data Out
13	RD+	Received Data Out
14	VeeR	Receiver Ground
15	VccR	Receiver Power
16	VccT	Transmitter Power
17	VeeT	Transmitter Ground
18	TD+	Transmit Data In
19	TD-	Inv. Transmit Data In
20	VeeT	Transmitter Ground

B Test Plan

B.1 Introduction

This document describes basic tests to verify MiNID functionality for the following features:

- Basic functionality (connectivity, passing traffic without errors)
- Flows
- OAM (CFM)
- OAM (EFM)
- Service Loopbacks
- RFC-2544
- Sync-E
- Device reset

Note *All tests should pass if the following procedures are performed precisely.*

B.2 Required Equipment

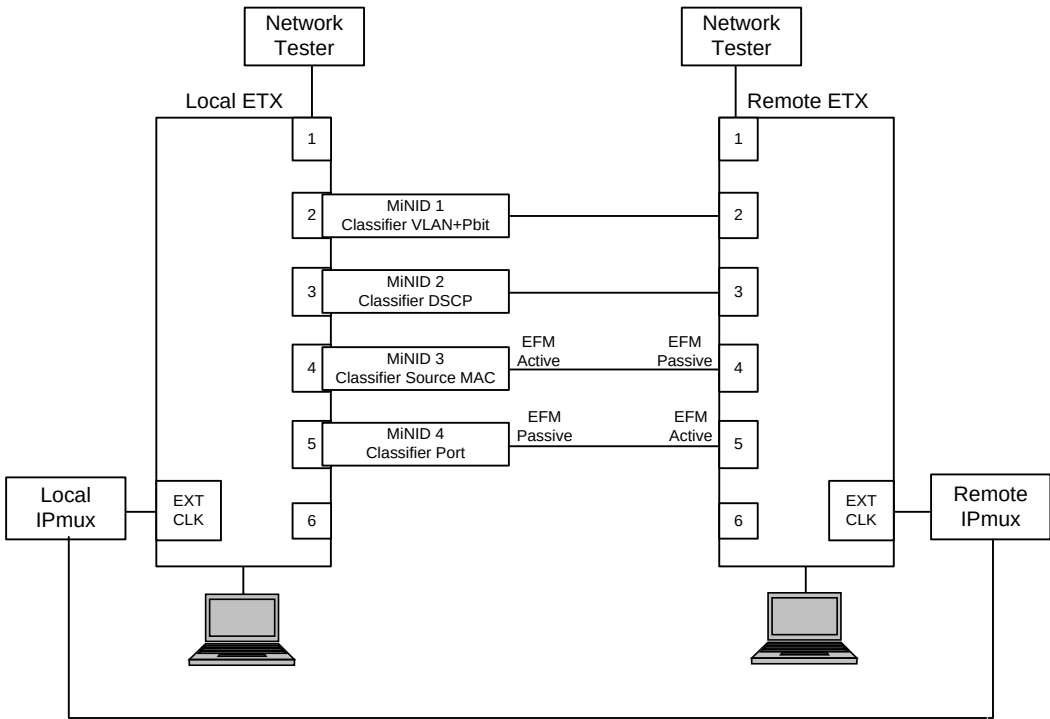
The table below lists devices required for conducting the tests.

Device	Description	Quantity
Unit under test (UUT)	MiNID	4
Local and remote ETX	Hosting device for MiNID: ETX-203AX or ETX-205A (must be ETX-205A with Sync-E option if you wish to perform Sync-E tests)	2

Device	Description	Quantity
Local and remote IPmux	Necessary for Sync-E tests	2
Network management station	PC or Unix workstation equipped with RADview	2
Network Tester	Network testing equipment with Ethernet interfaces such as IXIA 400T, Spirent Test Center, etc.	2

B.3 Preparing the Test Layout

The following diagram illustrates the test plan setup.



Connecting the Test Layout

The following tables list the connections between the devices, and the cable required for each connection.

Test Layout Connections

Device	Cable	Device
Local ETX port 1	Straight Ethernet cable	Local Network Tester
Local ETX port 2	Not applicable; MiNID is inserted	UUT-1
Local ETX port 3	Not applicable; MiNID is inserted	UUT-2
Local ETX port 4	Not applicable; MiNID is inserted	UUT-3
Local ETX port 5	Not applicable; MiNID is inserted	UUT-4
Local ETX EXT CLK port	Straight Ethernet cable	Local IPmux EXT CLK port
Local IPmux EXT CLK port	Straight Ethernet cable	Local ETX EXT CLK port
UUT-1	Straight Ethernet cable	Remote ETX port 2
UUT-2	Straight Ethernet cable	Remote ETX port 3
UUT-3	Straight Ethernet cable	Remote ETX port 4
UUT-4	Straight Ethernet cable	Remote ETX port 5
Remote ETX port 1	Straight Ethernet cable	Remote Network Tester
Remote ETX port 2	Straight Ethernet cable	UUT-1
Remote ETX port 3	Straight Ethernet cable	UUT-2
Remote ETX port 4	Straight Ethernet cable	UUT-3
Remote ETX port 5	Straight Ethernet cable	UUT-4
Remote ETX EXT CLK port	Straight Ethernet cable	Remote IPmux EXT CLK port
Remote IPmux EXT CLK port	Straight Ethernet cable	Remote ETX EXT CLK port
Local Network Tester	Straight Ethernet cable	Local ETX port 1
Remote Network Tester	Straight Ethernet cable	Remote ETX port 1

Configuring Devices

This section presents the device configurations. The data, such as VLAN IDs and IP addresses, were randomly selected for test purposes, and can be changed to suit your particular configuration.

The following table presents the configuration files for the MiNID and ETX devices. You can open or save the configuration files as follows:

- To open the file – Double-click the icon  in the table below.
- To save the file – Right-click the icon  in the table below and select Save Embedded File to Disk.

Device	File
UUT-1	
UUT-2	
UUT-3	
UUT-4	
Local ETX	
Remote ETX	

The following table presents the IPmux device configurations.

Parameter	Local	Remote
Host IP	192.168.205.32	192.168.205.33
E1 configuration	Unframed	Unframed
Peer IP	10.10.10.10	10.10.10.11
PW IP	10.10.10.11	10.10.10.10
Payload size	4	4
OAM	Disable	Disable

B.4 Testing Basic Functionality

This test verifies that the test setup functions properly.

Estimated Duration

The estimated duration of this test is one hour.

Setup and Configuration

See [Connecting the Test Layout](#) and [Configuring Devices](#).

Test Procedure

#	Action	Expected Result	Result
1	Ping the UUTs to verify network connectivity	The UUTs reply to ping	
2	Open Telnet sessions to the UUTs	Telnet and SSH sessions are established successfully	
3	Initiate SNMP management session from RADview NMS and verify management connectivity to all devices	Devices reply to ping and appear on RADview map	
4	Display UUT inventory	The inventory display is correct: Name: MinID-chassis Description: RAD.MinID.chassis HW Rev: xxx[FE/GE] FW Rev: yyy SW Rev: zzz SW Date\Time: date of SW	

#	Action	Expected Result	Result
5	Run Ethernet traffic for at least 15 minutes: From Network Tester port 1 and port 2, send packets with frame size 1518 bytes, VLAN 4040, P-bit 5, DSCP 16, at 98% line rate	The traffic passes without errors or packet loss	
6	Verify status of all MEPs	Verify that all MEPs are OK, e.g.: - Operational status is OK - No defects are displayed Remote MEP address matches the remote device MAC address.	
7	Clear alarm logs in all devices	Logs are empty with no active alarms	

B.5 Testing Flows

This test verifies flow functionality.

Estimated Duration

The estimated duration of this test is one hour.

Setup and Configuration

See [Connecting the Test Layout](#) and [Configuring Devices](#).

Test Procedure

#	Action	Expected Result	Result
1	In the remote ETX, administratively disable flows d6 and d61	Flows d6 and d1 are not active	

#	Action	Expected Result	Result
2	In UUT-1, verify classification Vlan range + Pbit range: Send traffic between Network Tester ports 1 and 2 with VLAN range 40–45 and p-bit 4, using a stream with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1 (stream #13 if using IXIA with the stream file from Configuring Devices)	Capture frames received in Network Tester port 2, and verify: • The traffic passes without errors or packet loss • In UUT-1, flow statistics display the same amount of frames sent by the Network Tester	
3	In UUT2-, verify classification DSCP: Send traffic between Network Tester ports 1 and 2 with VLAN range 100–4094 and DSCP 16, using a stream with line rate=100 pps ,length=100 bytes ,burst size=100 packets, number of bursts=1 (stream #5 if using IXIA with the stream file from Configuring Devices)	Capture frames received in Network Tester port 2, and verify: • The traffic passes without errors or packet loss • In UUT-2, flow statistics display the same amount of frames sent by the Network Tester	
4	In UUT-3, verify classification source MAC: Send traffic between Network Tester ports 1 and 2 with VLAN range 100–4094 and source MAC 00-00-00-00-00-BB, using a stream with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1 (stream #19 if using IXIA with the stream file from Configuring Devices)	Capture frames received in Network Tester port 2, and verify: • The traffic passes without errors or packet loss • In UUT-3, flow statistics display the same amount of frames sent by the Network Tester	
5	Verify pass-through action: Send traffic between Network Tester port 1 and Network Tester port 2 with VLAN range 101–4094 and DSCP 16, using a stream with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1 (stream #5 if using IXIA with the stream file from Configuring Devices)	Capture frames received in Network Tester port 2, and verify: • The traffic passes without errors or packet loss • Port statistics display the same amount of frames sent by the Network Tester	

#	Action	Expected Result	Result
6	Verify push VLAN action (fixed p-bit): Send traffic between Network Tester port 1 and Network Tester port 2 with VLAN range 101–4094 and DSCP 8, using a stream with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1, source MAC=00-00-00-00-00-02 (stream #2 if using IXIA with the stream file from Configuring Devices)	Capture frames received in Network Tester port 2, and verify: • The traffic passes without errors or packet loss • Flow statistics display the same amount of frames sent by the Network Tester • All frames with source MAC address 00-00-00-00-00-02 are received with VLAN 555 and p-bit 5, and Canonical Format Identifier (CFI) field changed to 1	
7	Verify push VLAN action (p-bit according to DSCP mapping): Send traffic from Network Tester port 1 to Network Tester port 2 with VLAN range 101–4094 and DSCP 32, using a stream with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1, source MAC=00-00-00-00-00-08 (stream #8 if using IXIA with the stream file from Configuring Devices)	Capture frames received in Network Tester port 2, and verify: • The traffic passes without errors or packet loss • Flow statistics display the same amount of frames sent by the Network Tester • All frames with source MAC address 00-00-00-00-00-08 are received with VLAN 888 and p-bit equal to DSCP mapping (5)	
8	Verify pop VLAN action: Send traffic from Network Tester port 1 to Network Tester port 2 with VLAN range 101–4094 and DSCP 12, using a stream with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1, source MAC=00-00-00-00-00-03 (stream #3 if using IXIA with the stream file from Configuring Devices)	Capture frames received in Network Tester port 2, and verify: • The traffic passes without errors or packet loss • Flow statistics display the same amount of frames sent by the Network Tester • All frames with source MAC address 00-00-00-00-00-03 are received without a VLAN	

#	Action	Expected Result	Result
9	Verify loop with MAC swap action: Send traffic between Network Tester port 1 and Network Tester port 2 with VLAN range 101–4094, DSCP 20, source MAC address 00-00-00-00-00-04, and destination MAC address 00-00-00-00-00-10; using a stream with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1 (stream #4 if using IXIA with the stream file from Configuring Devices)	Capture frames received in Network Tester port 2, and verify: • The traffic passes without errors or packet loss • Flow statistics display the same amount of frames sent by the Network Tester • All frames with source MAC address 00-00-00-00-00-10 and destination MAC address 00-00-00-00-00-04 are received by the port from which they were sent	
10	In the remote ETX, administratively enable flows d6 and d61	Flows d6 and d1 are active	
11	In UUT-4, verify classification Port Mode: Send traffic between Network Tester ports 1 and 2 with VLAN range 100–4094, using a stream with line rate=100 pps, length=100 bytes ,burst size=100 packets, number of bursts=1, source=aa-aa-aa-aa-aa-aa, des=bb-bb-bb-bb-bb-bb (stream #27 if using IXIA with the stream file from Configuring Devices)	Capture frames received in Network Tester port 2, and verify: • The traffic passes without errors or packet loss • In UUT-4, port statistics display the same amount of frames sent by the Network Tester	

B.6 Testing OAM (CFM)

This test verifies the functionality of the MEPs.

Estimated Duration

The estimated duration of this test is one hour.

Setup and Configuration

See [Connecting the Test Layout](#) and [Configuring Devices](#).

Test Procedure

#	Action	Expected Result	Result
1	In the remote ETX, administratively disable flows d6 and d61	Flows d6 and d1 are not active	
2	To verify MEP Status when remote MEP is down: In the remote ETX, set MEP 2 to administratively disabled (via command shutdown)	In UUT-1, display MEP 1 status and verify that operational status has changed to fail, and there is indication of loss of continuity	
3	To verify MEP Status when remote MEP returns to up: In the remote ETX, set MEP 2 to administratively enabled (via command no shutdown)	In UUT-1, display MEP 1 status and verify that operational status has returned to OK	
4	Run OAM link trace between the various pairs of MEPs and their corresponding remote MEPs	Link trace is successful	
5	Run OAM loopback between the various pairs of MEPs and their corresponding remote MEPs	Loopback is successful	
6	In the remote ETX, administratively enable flows d6 and d61	Flows d6 and d1 are active	

B.7 Testing OAM (EFM)

This test verifies the functionality of link OAM.

Estimated Duration

The estimated duration of this test is one hour.

Setup and Configuration

See [Connecting the Test Layout](#) and [Configuring Devices](#).

Test Procedure

#	Action	Expected Result	Result
1	In the remote ETX, verify that flows d6 and d61 are administratively enabled	Flows d6 and d1 are active	
2	Verify OAM (EFM) discovery between: • UUT-3 (active) to remote ETX port 4 (passive) • UUT-4 (passive) to remote ETX port 5 (active)	Verify OAM (EFM) status: In UUT-3: • Operational status is up • Local MAC address matches UUT3 MAC address • Remote MAC address matches remote ETX port 4 MAC address • UUT-3 Local is Active and ETX port 4 Remote is Passive In UUT-4: • Operational status is up • Local MAC address matches UUT-4 MAC address • Remote MAC address matches remote ETX port 5 MAC address • UUT-4 Local is Active and ETX port 5 Remote is Passive	

#	Action	Expected Result	Result
3	In remote ETX port 5, activate OAM EFM loopback: Run the loopback for 5 minutes Send traffic from Network Tester port 1 that matches the flow classification in UUT-3 and UUT-4, using one stream with MAC address 00-00-00-00-00-AA and one stream with MAC address AA-AA-AA-AA-AA-AA (stream #19 & 27 if using IXIA with the stream file from Configuring Devices)	Capture frames received in Network Tester port 1, and verify: - The traffic passes without errors or packet loss - In UUT-3 and UUT-4, flow statistics display the same amount of frames sent by the Network Tester - All frames with both source MAC addresses are received in Network Tester	
4	In remote ETX port 5, deactivate OAM EFM loopback: Run data for 5 minutes Send traffic from Network Tester port 1 that matches the flow classification in UUT-3 and UUT-4, using one stream with MAC address 00-00-00-00-00-AA and one stream with MAC address AA-AA-AA-AA-AA-AA (stream #19 & 27 if using IXIA with the stream file from Configuring Devices)	Capture frames received in Network Tester port 1, and verify: - The traffic passes without errors or packet loss - In UUT-3 and UUT-4, flow statistics display the same amount of frames sent by the Network Tester - All frames with both source MAC addresses are received in Network Tester	

B.8 Testing Service Loopbacks

This test verifies the functionality of service loopbacks.

Estimated Duration

The estimated duration of this test is one hour.

Setup and Configuration

See [Connecting the Test Layout](#) and [Configuring Devices](#).

Test Procedure

#	Action	Expected Result	Result
1	In the remote ETX, administratively disable flows d6 and d61	Flows d6 and d1 are not active	
2	Test Layer-2 loopback: Send traffic from Network Tester port 1 to Network Tester port 2 with VLAN range 101–4094 and DSCP 16, using streams with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1, source MAC=00-00-AA-AA-AA-AA to source MAC 00-00-DD-DD-DD-DD (stream #28–31 if using IXIA with the stream file from Configuring Devices)	Capture frames returned to Network Tester port 1, and verify: • The traffic passes without errors or packet loss • MAC address and IP address are swapped • Port statistics display the same amount of frames sent by the Network Tester	
3	Test Layer-2 loopback: Send traffic from Network Tester port 1 to Network Tester port 2 with VLAN range 101–4094 and DSCP 16, using streams with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1, source MAC=00-00-AA-AA-AA-AA to source MAC 00-00-DD-DD-DD-DD (stream #32–35 if using IXIA with the stream file from Configuring Devices)	Verify that frames are not returned to Network Tester port 1	
4	In the remote ETX, administratively enable flows d6 and d61	Flows d6 and d1 are active	

#	Action	Expected Result	Result
5	Test Layer-3/4 loopback by IP and UDP: Send traffic from Network Tester port 1 to Network Tester port 2 with VLAN range 101–4094 and DSCP 34, using streams with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1, source MAC=00-00-AA-AA-AA-AA to source MAC 00-00-DD-DD-DD-DD, dest IP 192.0.0.1, dest UDP 5000	Capture frames returned to Network Tester port 1, and verify: • The traffic passes without errors or packet loss • MAC address and IP address are swapped • Port statistics display the same amount of frames sent by the Network Tester	
6	Test Layer-3/4 loopback by IP and UDP: Send traffic from Network Tester port 1 to Network Tester port 2 with VLAN range 0–4094 and DSCP 34, using streams with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1, source MAC=00-00-00-00-00-0A to source MAC 00-00-00-00-00-0D, dest IP 192.0.0.1, dest UDP 1024	Verify that frames are not returned to Network Tester port 1	
7	Test Layer-3/4 loopback by IP: Send traffic from Network Tester port 1 to Network Tester port 2 with VLAN range 101–4094 and DSCP 34, using streams with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1, source MAC=00-00-AA-AA-AA-AA to source mac 00-00-DD-DD-DD-DD, dest IP 192.0.0.1, dest UDP 1024	Capture frames returned to Network Tester port 1, and verify: • The traffic passes without errors or packet loss • MAC address and IP address are swapped • Port statistics display the same amount of frames sent by the Network Tester	

#	Action	Expected Result	Result
8	Test Layer-3/4 loopback by IP: Send traffic from Network Tester port 1 to Network Tester port 2 with VLAN range 0–4094 and DSCP 34, using streams with line rate=100 pps, length=100 bytes, burst size=100 packets, number of bursts=1, source MAC=00-00-00-00-00-0A to source mac 00-00-00-00-00-0D, dest IP 192.0.0.2, dest UDP 1024	Verify that frames are not returned to Network Tester port 1	

B.9 Testing RFC-2544

This test verifies the functionality of the RFC-2544 benchmark testing.

Estimated Duration

The estimated duration of this test is one hour.

Setup and Configuration

See [Connecting the Test Layout](#) and [Configuring Devices](#).

Test Procedure

#	Action	Expected Result	Result
1	In the remote ETX, administratively disable flows d6 and d61	Flows d6 and d1 are not active	

#	Action	Expected Result	Result
2	Test throughput for 64-byte frame size: In the remote ETX, run the RFC-2544 throughput test that uses profile 1 (via command config test rfc2544 test 1 activate)	The test starts	
3	Check the test status (via command show status)	The command output shows that the test is in progress; after about four minutes, the command output shows that the test has completed	
4	Display the test results (via command show summary)	Verify that measured results match the expected results: Test result is successful Throughput (FPS) column displays a value such that the result of $[FPS \times \text{Frame Size} \times 8]$ is the following according to frame size: 64 – 76% 512 – 96% 1518 – 98%	

#	Action	Expected Result
1	In the remote ETX, administratively disable flows d6 and d61	Flows d6 and d1 are not active
2	Test throughput for 64-byte frame size: In the remote ETX, run the RFC-2544 throughput test that uses profile 1 (via command config test rfc2544 test 1 activate)	The test starts
3	Check the test status (via command show status)	The command output shows that the test is in progress; after about four minutes, the command output shows that the test has completed

#	Action	Expected Result
4	Display the test results (via command show summary)	Verify that measured results match the expected results: • Test result is successful • Throughput (FPS) column displays a value such that the result of $[FPS \times \text{Frame Size} \times 8]$ is the following according to frame size: ▪ 64 – 76% ▪ 512 – 96% ▪ 1518 – 98%
5	In the remote ETX, administratively enable flows d6 and d61	Flows d6 and d1 are active

B.10 Testing Sync-E

This test verifies the functionality of the Sync-E benchmark testing.

Estimated Duration

The estimated duration of this test is one hour.

Setup and Configuration

See [Connecting the Test Layout](#) and [Configuring Devices](#).

Test Procedure

#	Action	Expected Result	Result

#	Action	Expected Result	Result
1	Set local ETX as master clock		
2	Set remote ETX as slave clock with clock source Ethernet port 2	Verify the local and remote ETX are locked	
3	Set local IPmux to internal clock		
4	Set remote IPmux to loopback clock		
5	In UUT1, set the Sync-E clock source to the MSA port	After five minutes, verify that the connection statistics in the remote IPmux do not indicate errors	
6	Set local ETX as slave clock with clock source Ethernet port 2		
7	Set remote ETX as master clock	Verify the local and remote ETX are locked	
8	Set local IPmux to loopback clock		
9	Set remote IPmux to internal clock		
10	In UUT1, set the Sync-E clock source to the SFP port	After five minutes, verify that the connection statistics in the local IPmux do not indicate errors	

B.11 Testing Device Reset

This test verifies the correct execution of device reboot, and verifies that all configured entities operate properly after the reboot.

Estimated Duration

The estimated duration of this test is one hour.

Setup and Configuration

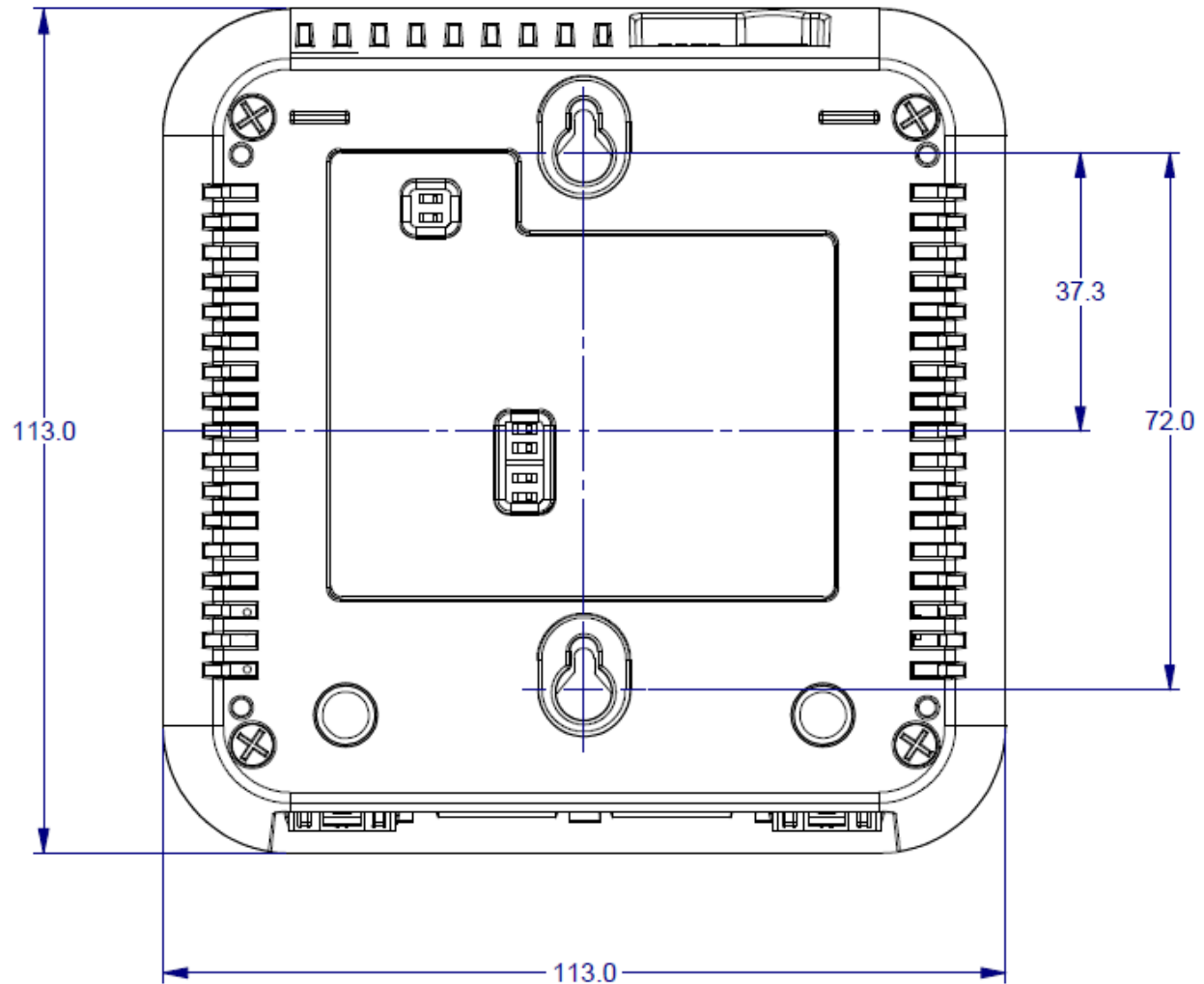
See [Connecting the Test Layout](#) and [Configuring Devices](#).

Test Procedure

#	Action	Expected Result	Result
1	Save the UUT configuration	File is saved successfully	
2	Upload configuration file	File is uploaded successfully	
3	Verify that traffic is running through all relevant ports	Traffic is running without errors or packet loss	
4	Reboot UUT	• UUT retains its configuration after the reboot • The device responds to pings • Management access (Telnet/SSH)has been restored Traffic is running error-free.	
5	Upload the device configuration file, and compare it to the configuration file uploaded before the reboot	The uploaded configuration files before and after the reboot are the same	

Supplement

Drilling Template for Wall Installation



SFP-CA-2

Configuration Tool for Smart SFPs

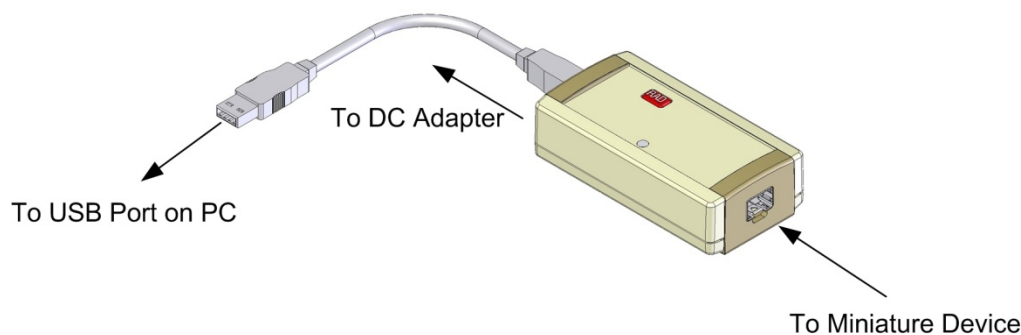
SFP-CA.2 is a desktop adapter that is used for configuration of miniature devices such as smart SFPs and MiNID. The adaptor operates under the following environmental conditions:

- Storage temperature: -40 to 85°C
- Operating temperature: 0 to 37°C
- Humidity: up to 90%

You can perform the following operations with the miniature device connected to your PC via the SFP-CA.2 unit:

- Assign a new IP address to the smart SFP/MiNID device
- Configure the smart SFP/MiNID device
- Download new software to the smart SFP/MiNID device

This figure illustrates the SFP-CA.2 unit and its connections.



Note All SFP-CA.2 modules have the same MAC address (00-00-E8-00-00-01).

For SFP-CA.2 to work, you must use one of the following supported operating systems:

- Windows XP
- Windows 7
- Windows 8
- Windows 8.1
- Windows 10
- Windows 11

Note Both 32 bit and 64 bit operating systems are supported.

You must perform the following before you can use SFP-CA.2:

1. Install SFP-CA.2 driver (see [Installing Driver for SFP-CA.2](#))
2. Configure PC network parameters for communication with SFP-CA.2 (see [Configuring PC Network Parameters for Communication with SFP-CA.2](#))
3. Configure HyperTerminal parameters (only required for downloading software to the miniature device) (see [Configuring HyperTerminal Parameters](#)).

INSTALLING DRIVER FOR SFP-CA.2

Before you can use SFP-CA.2, you must install its driver.

➤ **To install the SFP-CA.2 driver:**

1. Click the SFP-CA.2 driver links below:

- [USB UART drivers](#)
- [USB Ethernet drivers](#)

2. Log in to the site.

A dialog box is displayed requesting to save the file.

3. Click **Save**.

4. Open the .zip file to install the driver.

The SFP-CA.2 driver is installed in the background. No further action is required to install the driver.

CONFIGURING PC NETWORK PARAMETERS FOR COMMUNICATION WITH SFP-CA.2

You must configure the relevant network parameters of your PC to establish a communication link with SFP-CA.2 and the smart SFP/MiNID device.

Note

Before performing this procedure, ensure that you have installed the SFP-CA.2 driver, as described in [Installing Driver for SFP-CA.2](#).

➤ **To configure your PC for communication with SFP-CA.2 and smart SFP/MiNID device:**

1. Connect power to the SFP-CA.2 unit with smart SFP/MiNID plugged in.

The **RDY** LED on SFP-CA.2 lights up. The smart SFP DIP switch must be in DB INIT\CONFIG or NORMAL mode.

2. Plug the USB connector of SFP-CA.2 into a USB port on your PC.

New Hardware is Detected notice appears. SFP-CA.2 is identified as a new network.

3. Navigate to **My Network Places(WinXP)/Network and Sharing Center(Win7/Win8)**.

A new network connection appears in the list of network connections.

The dialog box closes and your settings are applied.

4. Type in the network properties.

5. In the **Internet Protocol (TCP/IP)** field, configure the **IP address**, subnet mask and default gateway:

- **IP Address:** 192.168.205.20
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.205.1

6. Click **OK**.

7. Close **My Network Places**.

The PC communication link with SFP-CA.2 and the miniature device is ready.

CONFIGURING HYPERTERMINAL PARAMETERS

You must establish a HyperTerminal connection, including configuring the serial (COM) port, in order to use the HyperTerminal application to download software to the smart SFP/MiNID device.

Note

- *Before performing this procedure, ensure that you have installed the SFP-CA.2 driver and configured the PC network parameters, as described in [Installing Driver for SFP-CA.2](#) and [Configuring PC Network Parameters for Communication with SFP-CA.2](#).*
- *Do not connect the smart SFP/MiNID device to SFP-CA.2 during this procedure.*

➤ **To configure the HyperTerminal parameters:**

1. If the SFP-CA.2 unit is not connected to your PC, connect power to SFP-CA.2 and plug the USB connector of SFP-CA.2 into a USB port on your PC.

2. Open the HyperTerminal application.

Configuration Tool for Smart SFPs

The **Connection Description** dialog box for a new connection appears.

3. Specify a name for the HyperTerminal connection, and click **OK**.

The **Connect To** dialog box appears.

4. Choose a virtual COM port, for example **COM8**, and click **OK**.

The COM Properties dialog box appears.

5. Specify the settings as listed below and then click **OK**.

- Bits Per Second: 115200
- Data Bits: 8
- Parity: None
- Stop Bits: 1
- Flow Control: None.

6. Click **Properties**.

The **Connection Properties** dialog box appears.

7. Select the Settings tab and under **Emulation**, choose **VT100**, and then click **OK**.

The COM port configuration is complete, and the HyperTerminal connection is ready for use.

8. Close the HyperTerminal application.

ASSIGNING A NEW IP ADDRESS TO SMART SFPs/MiNID

You can configure the smart SFP/MiNID via SFP-CA.2, including assigning it a new IP address.

Note

*Before performing this procedure, ensure that you have installed the SFP-CA.2 driver and configured the PC network parameters, as described in **Installing Driver for SFP-CA.2** and **Configuring PC Network Parameters for Communication with SFP-CA.2**.*

➤ To assign a new IP address to smart SFP/MiNID:

1. Select Configuration operation mode in the smart SFP/MiNID by setting its DIP switches as needed. Refer to the smart SFP/MiNID documentation for details on setting the operation mode.
2. If the SFP-CA.2 unit is not connected to your PC, connect power to SFP-CA.2 and plug the USB connector of SFP-CA.2 into a USB port on your PC.
3. Plug the smart SFP/MiNID into the SFP socket on the opposite side of the SFP-CA.2 unit.

The smart SFP/MiNID is ready for configuration.

4. Open the Web browser on your PC and type **http://192.168.205.1** into the Web browser's address field.

The Opening screen appears.

5. Click **Login**.

The Login screen appears.

6. Enter the default user name **su** and the default password **1234** for Superuser access, and then click **Submit**.

A menu appears to the left and you are able to configure smart SFP/MiNID.

7. Use the menus of the smart SFP/MiNID to assign a new IP address. You may continue specifying additional parameters or connect to the smart SFP/MiNID from any PC on your network at a later stage, using the newly assigned IP address.

Note

When Configuration mode is selected in smart SFP/MiNID via the DIP switches, the miniature device responds to only the default IP address 192.168.205.1, even if the device's IP address has been changed.

DOWNLOADING SOFTWARE TO SMART SFP/MiNID

You can upgrade the smart SFP/MiNID software by downloading software via SFP-CA.2. The HyperTerminal application is used for the software download.

Note

Before performing this procedure, ensure that you have installed the SFP-CA.2 driver, configured the PC network parameters, and configured the HyperTerminal parameters, as described in [Installing Driver for SFP-CA.2](#), [Configuring PC Network Parameters for Communication with SFP-CA.2](#), and [Configuring HyperTerminal Parameters](#).

► To download software to smart SFP/MiNID:

1. Verify that the upgrade image file is accessible from your PC.
2. Select SW Download (SW DNLD) mode in the smart SFP/MiNID by setting its DIP switches as needed. Refer to the smart SFP/MiNID documentation for details on setting the operation mode.
3. If the SFP-CA.2 unit is not connected to your PC, connect power to SFP-CA.2 and plug the USB connector of SFP-CA.2 into a USB port on your PC.
4. Plug the miniature device into the SFP socket on the SFP-CA.2 unit.
5. Open the HyperTerminal application and load the HyperTerminal connection that you established previously (refer to [Configuring HyperTerminal Parameters](#)).
6. Follow the smart SFP/MiNID device procedure for downloading software. Refer to the smart SFP/MiNID documentation for details.

REMOVING SMART SFP/MiNID AND SFP-CA.2 FROM PC**► To remove smart SFP/MiNID and SFP-CA.2 from the PC:**

1. Close all relevant management applications.
2. On your operation system, allocate the remove hardware icon.
3. Select the required USB port from the listed devices and click **Stop**.

Note

If you disconnect the smart SFP/MiNID and/or SFP-CA.2 before releasing it as described, your PC may stop responding.

4. Push the release button at the front of the smart SFP/MiNID device to disconnect it from SFP-CA.2.
5. Remove the smart SFP/MiNID from the SFP socket on SFP-CA.2.
6. Disconnect SFP-CA.2 from the PC and from the power.

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